Probability and Statistics

Marianopolis College 201-LCW

Professor C. Turner · Summer 2025

About these notes

These notes were compiled from handwritten class notes and annotated worksheets. Each section merges the key points of one worksheet with the lecture notes that go with it, and ends with worked examples taken from the worksheet. A star (⋆) after an example number marks an exercise that was left unworked in the original notes; its solution was written for this compilation. The formula sheet and the statistical tables are collected in the appendices.

Definitions · Results · Worked examples · Notes and cautions

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How to use these notes

The course is a first, calculus-based course in probability and statistics. It moves in one direction: describe data (Part I), learn to count (Part II), build probability on top of counting (Part III), package probability into random variables and their standard distributions (Parts IV and V), study how sample means behave (Part VI), and finally use all of it to draw conclusions about a population from a sample (Part VII).

How a section is organised

- **Theory first.** Definitions are in blue boxes, results (theorems, properties, formulas) in teal boxes, and remarks, warnings and study tips in purple boxes. Proofs are marked by a grey bar.
- **Worked examples last.** Every example states the question, then gives a complete solution below the dashed line. The tag in the top right corner says where the question comes from. Final answers are boxed.
- **Numbers.** Answers are exact or rounded to four decimal places, as the course requires. Normal, t and chi-squared probabilities are read from the tables in Appendix B, so z-values are normally rounded to two decimals before the look-up. Where a value falls between two table entries and the answer key interpolates, the example says so.

Part I

Descriptive Statistics

Statistics starts with data. This part shows how to say what kind of data you have, how to display it, and how to summarise it with a few numbers: where its centre is, how spread out it is, what shape it has, and where a single value sits relative to the rest.

1 Introduction to Statistics

Statistics is the science of collecting data, describing them and drawing conclusions from them. This first section sets up the vocabulary that the rest of the course relies on: what is being studied (population, sample, variable of interest), which numbers describe it (parameters, statistics), and the two kinds of statistical work (descriptive and inferential).

1.1 Populations and samples

Definition (population, sample, variable of interest)

- The *population* is the set of all observations relevant to the question asked. It does not have to be made up of people.
- A *sample* is a set of some, but not all, of these observations.
- The *sample size* is the number of observations in the sample. It is written n; the size of a population is written N.
- The *variable of interest* is the characteristic under study, that is, what is recorded for each member of the population or of the sample.

For example, to find out how long the batteries made in a factory last, a technician tests 40 of them. The population is the set of all batteries made in the factory, the sample is the 40 batteries that were tested ($n = 40$), and the variable of interest is the lifetime of a battery. Observing the whole population is usually impossible or too costly (here, testing a battery uses it up), which is why we work with samples.

1.2 Parameters and statistics

Definition (parameter, statistic)

- A *parameter* is a numerical measurement describing some characteristic of a *population*.
- A *statistic* is a numerical measurement describing some characteristic of a *sample*.

The same kind of measurement is written with a different symbol depending on whether it describes a population or a sample.

Measurement	Population parameter	Sample statistic
Mean (average)	μ	$\bar{x}$
Standard deviation	σ	s
Coefficient of correlation	ρ	r
Proportion	p	$\hat{p}$

The mean is studied in Section 4 and the standard deviation in Section 5; proportions come back in Section 29.

Note (keeping the two apart)

Parameter goes with **p**opulation and **s**tatistic goes with **s**ample. Parameters are mostly written with Greek letters (μ, σ, ρ) and statistics with Latin letters ($\bar{x}$, s, r). The proportion is the exception: p for the population and $\hat{p}$ ("p-hat") for the sample.

A parameter is one fixed number, but it is usually unknown, because we rarely get to see the whole population. A statistic can always be computed from the data in hand, but its value changes from one sample to the next (Section 27).

Caution (the word "statistic")

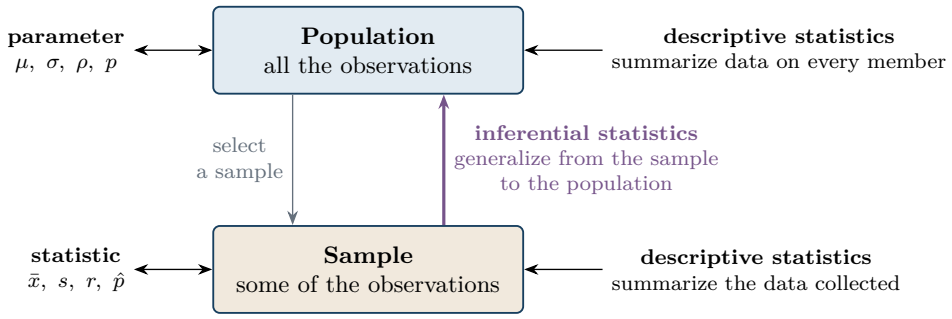
A *statistic*, in the singular, is one number computed from a sample, such as $\bar{x} = 55.35$. It is not the same thing as *statistics*, the subject, as in "descriptive statistics" or "inferential statistics".

1.3 Descriptive and inferential statistics

Definition (descriptive and inferential statistics)

- *Descriptive statistics* lists statistics based on the available data. The data in hand are organized and summarized, with numbers and with visual representations (tables and graphs), and nothing is claimed beyond those data.
- *Inferential statistics* uses a statistic to estimate the corresponding parameter, that is, it makes a generalization about a population based on a sample. We use the information from a sample to infer (predict) information about the general population.

Descriptive statistics can describe a sample, in which case the summary numbers are statistics, or a whole population, when data on every member are available, in which case the summary numbers are parameters. Inferential statistics is the step from one level to the other, from the sample up to the population.



Remark (probability and statistics ask opposite questions)

In *probability* we know the population and ask questions about a sample. A jar contains 60 red and 40 blue marbles: how likely is it that a handful of 10 contains exactly 7 red ones? In *inferential statistics* we know the sample and ask questions about the population. A handful of 10 marbles taken from a jar of unknown composition contains 7 red ones: what can be said about the proportion of red marbles in the jar?

Inference rests on probability, which is why probability (Sections 11–28) is developed before inference (Sections 29–32).

1.4 Worked examples

Each example below starts from a single sentence of the kind found in a news report. The task is always the same: decide whether the statement is descriptive or inferential and identify, as appropriate, the population, the sample,

the variable of interest, and the parameter or statistic.

How to classify a statement

1) *Who is the statement about?* Only the individuals who were actually observed (a sample), or everyone (a population)?
2) *Where can the number have come from?* If the statement is about a whole population but the number can only have been obtained from part of it, a generalization has been made: the statement is **inferential**. If the number simply summarizes the data that were collected (a statistic describing a sample, or a parameter computed from data on the entire population), the statement is **descriptive**.

Example 1.1 ▪ A claim about a whole city Worksheet 1, #1

Decide if the following statement is descriptive or inferential and describe (as appropriate) the underlying population, sample, variable of interest, parameter and/or statistic.

"Seventy-three percent of Montrealers take public transportation to get to work."

Solution. The statement is about *all* Montrealers, so it concerns a population, and a percentage of a group is a *proportion*.

- Population: the people who live in Montreal.
- Variable of interest: the mode of transport used to get to work.
- Parameter: the proportion p of all Montrealers who take public transportation to get to work. The statement gives it the value $p = 0.73$.
- Sample: no information is given.

Nobody can ask every Montrealer how they get to work. The figure of 73% must have been measured on a sample (it is really a sample proportion $\hat{p}$) and then extended to the whole city. A conclusion about a population drawn from a sample is a generalization, so the statement is inferential.

Example 1.2 ▪ Fifty shoppers Worksheet 1, #2

Descriptive or inferential? Describe the underlying population, sample, variable of interest, parameter and/or statistic.

"Fifty shoppers at Les Galeries d'Anjou were interviewed and it was found that they spent an average of \$55.35 that day."

Solution. The statement reports what was found for the fifty shoppers who were interviewed, and nothing more.

- Sample: the 50 shoppers who were interviewed, so $n = 50$.
- Underlying population: all the people who shop at Les Galeries d'Anjou.
- Variable of interest: the amount spent in one visit.
- Statistic: the sample mean $\bar{x} = \$55.35$. It is written $\bar{x}$, not μ, because it describes the sample and not the population.

Nothing is inferred about the population. The statement only summarizes the data that were collected, so it is descriptive.

It would become inferential if it went one step further and concluded, for instance, that shoppers at Les Galeries d'Anjou spend about \$55 per visit on average: that would be a claim about μ based on $\bar{x}$.

Example 1.3 ▪ A whole population on record Worksheet 1, #3

Descriptive or inferential? Describe the underlying population, sample, variable of interest, parameter and/or statistic.

"The average age of Marianopolis students is 18 years and 1 month."

Solution. The statement gives the average age of *all* the students of the college.

- Population: the students attending Marianopolis.
- Sample: there is none.
- Variable of interest: age.
- Parameter: the population mean $\mu = 18$ years and 1 month.

Like the statement in Example 1.1, this one is about a whole population, but here no sample is needed: the college has the date of birth of every one of its students on record, so the average can be computed from the complete data. A number computed from the entire population is a parameter, and stating it describes the population without generalizing from anything. The statement is descriptive.

2 Types of Data

The variable of interest of Section 1 can produce very different kinds of values: a colour, a count, a measurement. The type of data decides what can sensibly be done with them (eye colours cannot be averaged) and how they are displayed, so identifying it is the first step of any analysis.

2.1 Qualitative and quantitative data

Definition (qualitative and quantitative data)

- *Qualitative* (or *categorical*) data record a quality: each value is a category or a label, such as a hair colour.
- *Quantitative* data record a quantity: each value is a number that counts or measures something, so that arithmetic on the values is meaningful.

Note (the average test)

For qualitative data it does not make sense to talk about the average (μ or $\bar{x}$): there is no "average hair colour". This gives a quick test. Ask whether the average of the values would mean anything; if it would not, the data are qualitative.

Caution (numbers are not always quantitative)

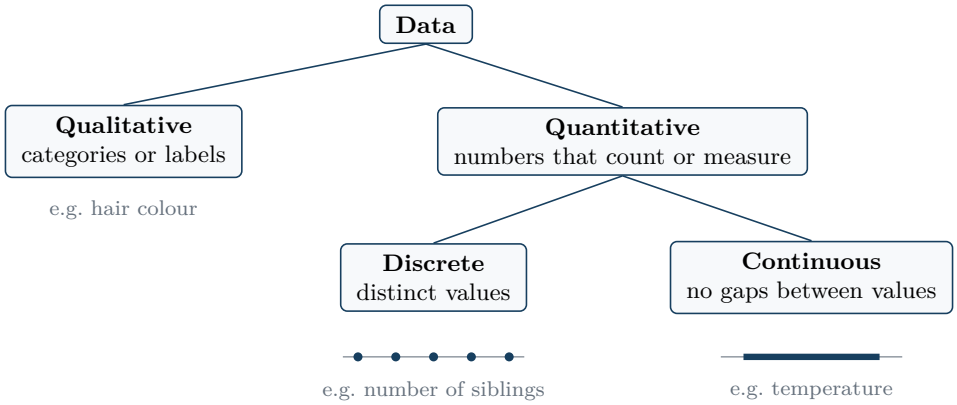
A value written with digits is not automatically quantitative. A phone number, a student ID number and the number on a player's jersey are labels: they identify, they do not count or measure, and averaging them is meaningless. They are qualitative.

2.2 Discrete and continuous data

Quantitative data are divided once more, according to which values are possible.

Definition (discrete and continuous data)

- Quantitative data are *discrete* if the possible values are distinct, separate values, with gaps between them. This is typically the result of *counting*: $0, 1, 2, 3, \ldots$
- Quantitative data are *continuous* if there are no gaps between the possible values: every value in some interval can occur. This is typically the result of *measuring* a length, a time, a temperature or a mass.



Remark (recorded values are always rounded)

A measurement is always written down to some precision (to the nearest kilometre, to a tenth of a degree), so a list of recorded measurements looks discrete. The classification refers to the variable itself, not to the way it was rounded: a distance could be 412 km, 412.3 km, 412.27 km, and so on, so distance is continuous.

Remark (two exceptions: money and population)

Money and population are exceptions to the simple "counted or measured" test. Strictly speaking both are discrete: an amount of money is a whole number of cents and a population is a whole number of people. But the steps are so small compared with the size of the values that both are routinely treated as continuous. So for these two variables each answer has a justification: discrete if we insist on the smallest unit, continuous if we look at how the values behave in practice.

Remark (why the type matters)

- *Summaries.* Qualitative data can only be counted category by category. Means and standard deviations (Sections 4 and 5) require quantitative data.
- *Displays.* Discrete data are drawn with one bar for each value; continuous data are first grouped into intervals (Section 3).
- *Probability.* Later in the course, discrete random variables are handled with sums (Section 15) and continuous ones with integrals (Section 21).

2.3 Worked examples

Example 2.1 ▪ Classifying fifteen variables Worksheet 2, #1–15

Decide whether the following variables are qualitative, quantitative discrete or quantitative continuous.

1. The number of heads obtained after flipping a coin five times
2. Nation of origin
3. The distance a Toyota Prius can go on a single battery charge
4. The gender of students at Marianopolis
5. The temperature at the top of Mount Royal
6. Phone number
7. Number of siblings
8. The number of cars that arrive at a Tim Horton's drive-through between 12pm and 1pm
9. Eye colour
10. Student ID number
11. Assessed value of a house
12. Number of moles of HCl in solution
13. Number on a football player's jersey
14. Population of a city
15. Number of unpopped kernels in a bag of microwave popcorn

Solution. Ask two questions, in this order.

1. *Would the average of the values mean anything?* If not, the variable is qualitative.
2. *If it is quantitative, is it counted or measured?* Counted (distinct values): discrete. Measured (no gaps between values): continuous.

	Variable	Type	Why
1	heads in five flips	quantitative discrete	a count: $0, 1, \ldots, 5$
2	nation of origin	qualitative	a category
3	distance on one charge	quantitative continuous	measured: any value in an interval
4	gender	qualitative	a category
5	temperature	quantitative continuous	measured
6	phone number	qualitative	a label made of digits
7	number of siblings	quantitative discrete	a count: $0, 1, 2, \ldots$
8	cars arriving in one hour	quantitative discrete	a count
9	eye colour	qualitative	a category
10	student ID number	qualitative	a label made of digits
11	assessed value of a house	quantitative continuous	money: treated as continuous
12	moles of HCl	quantitative continuous	an amount, measured not counted
13	jersey number	qualitative	a label made of digits
14	population of a city	quantitative discrete	a count of people
15	unpopped kernels	quantitative discrete	a count

Three kinds of item deserve a second look.

- *Labels made of digits (6, 10, 13).* The average of two phone numbers, two ID numbers or two jersey numbers means nothing, so these variables are qualitative even though their values are written with digits.
- *The word "number" (12).* "Number of moles" sounds like a count, but the amount of HCl in a solution is measured: it can be 0.1 mol, 0.125 mol or anything in between, so it is continuous. Compare with items 1, 7, 8 and 15, where individual objects really are counted.
- *Money and population (11, 14).* These are the two exceptions. The value of a house is classified as continuous, although strictly speaking it is a whole number of cents. The population of a city is classified as discrete because it is a count of people, although for a large city it is often treated as continuous.

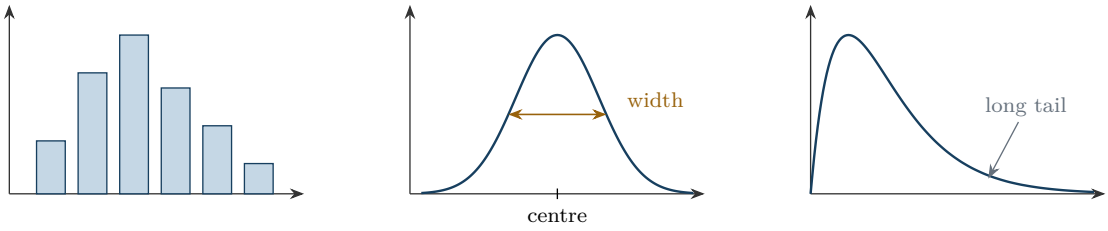
3 Displaying Data

A list of raw numbers says very little until it is organized. This section presents three standard ways of organizing a set of quantitative data so that its main features can be seen at a glance: the stem-and-leaf plot, the frequency table and the histogram. It also names the features to look for, which the next three sections turn into numbers.

3.1 Two ways to represent data

A data set can be represented in two ways: visually, with a picture, and numerically, with a few summary numbers.

Visually. What the picture looks like depends on the type of data (Section 2). Discrete data are drawn as a bar chart: one bar for each value, the height of the bar showing how often the value occurs, as in panel (a) below. Continuous data are first grouped into intervals and drawn as a histogram (Section 3.4). With a lot of data and narrow intervals the tops of the bars trace out a smooth curve, and this is how the distribution of a continuous variable is usually sketched, as in panel (b). Two features of such a picture stand out immediately: its *centre* and its *width*.



(a) discrete data (b) continuous data (c) skewed right

Numerical summaries. Three features of the picture are summarized by numbers or by a short description.

Feature	What it describes	Summarized by	Where
Centre	where the data are located	mode, median, mean	Section 4
Width (spread)	how spread out the data are	standard deviation (σ), range (max − min)	Section 5
Skewness	whether the data trail off farther on one side than on the other, as in panel (c)	a description such as "skewed right"	Section 6

3.2 Stem-and-leaf plots

Definition (stem-and-leaf plot)

In a *stem-and-leaf plot* every data value is split into a *stem* (its leading digit or digits) and a *leaf* (its last digit). The stems are listed once, in increasing order, to the left of a vertical line. Each data value then contributes its leaf to the row of its stem, to the right of the line.

To build one:

1. Sort the data in increasing order.
2. Write down every stem from the smallest to the largest, including any stem that has no leaves.
3. Write each leaf in the row of its stem, in increasing order. A value that occurs several times gets one leaf for each occurrence, and all leaves take up the same width.
4. Add a key that says how to put a stem and a leaf back together, for instance "2 | 4 means 24".

Note (like a horizontal histogram)

Because every leaf takes up the same width, the length of a row shows how many data values share that stem. A stem-and-leaf plot is therefore like a histogram (Section 3.4) lying on its side: turn the page a quarter turn and the rows become the bars. Unlike a histogram, it still shows every individual value. Example 3.1 shows a complete plot.

3.3 Frequency tables and class intervals

Definition (frequency and relative frequency)

The *frequency* f of a value is the number of times it occurs in the data. Its *relative frequency* is the fraction of the data that it represents,

$$\text{relative frequency} = \frac{f}{n},$$

where n is the total number of observations. A *frequency table* lists the values together with their frequencies, and often their relative frequencies as well. The frequencies add up to n and the relative frequencies add up to 1.

Definition (class intervals)

When the data take many different values, the values are first grouped into *class intervals* (or *classes*) such as

$$[0, 5), \quad [5, 10), \quad [10, 15), \ \ldots$$

and the table gives the frequency of each class, that is, the number of observations that fall in it. The class $[a, b)$ contains the values x with $a \le x < b$: the left endpoint is included and the right endpoint is not.

Class intervals must not overlap and must cover all the data, so that every observation is counted in exactly one class. The half-open notation takes care of the boundaries: with the classes above, an observation equal to 5 is counted in $[5, 10)$ and not in $[0, 5)$. Classes are normally given the same width.

Remark (three ways to read a relative frequency)

A relative frequency is a fraction, a percentage (a relative frequency of 0.20 means 20% of the data) and also a probability: if one observation is chosen at random from the data set, the probability that it falls in a given class is the relative frequency of that class.

3.4 Histograms

Definition (histogram)

A *histogram* is the picture of a frequency table with class intervals. The horizontal axis carries the variable, with the class boundaries marked on it. Above each class stands a bar whose height is the frequency (or the relative frequency) of the class. Neighbouring bars touch, because neighbouring classes share a boundary.

A complete histogram has a title and a label on each axis. Using relative frequencies instead of frequencies divides every height by n: the scale on the vertical axis changes, the shape does not.

Describing the shape of a distribution

When asked to describe the shape of a distribution, comment on two things.

1) **Symmetry.** Is the left half roughly the mirror image of the right half? If not, on which side is the longer tail?

2) **Overall shape.** For example *bell-shaped*: a single peak in the middle, with the frequencies falling away on both sides.

The vocabulary for shapes is developed in Section 6.

3.5 Worked examples

The three examples of this section use the same data set and build on one another.

Example 3.1 ▪ A stem-and-leaf plot Worksheet 3, #1

The following data set represents the number of grams of sugar in one serving (330 ml) of the leading soft drinks.

0, 8, 9, 12, 12, 14, 15, 16, 16, 19, 21, 22, 24, 24, 27, 28, 29, 31, 36, 39

Create a stem-and-leaf plot for this data with stems 0, 1, 2 and 3.

Solution. The $n = 20$ values are already sorted. With stems 0, 1, 2, 3, the stem of a value is its tens digit and the leaf is its units digit. The one-digit values 0, 8 and 9 are read as 00, 08 and 09, so they go in the row of stem 0.

stem	leaf
0	0 8 9
1	2 2 4 5 6 6 9
2	1 2 4 4 7 8 9
3	1 6 9

Key: 2 | 4 means 24 g

The row 2 | 1 2 4 4 7 8 9, for example, stands for the values $21, 22, 24, 24, 27, 28, 29$.

The shading shows why the plot is like a horizontal histogram: the rows have lengths 3, 7, 7 and 3, which are the numbers of drinks with 0 to 9, 10 to 19, 20 to 29 and 30 to 39 grams of sugar.

Example 3.2 ▪ A frequency table with class intervals Worksheet 3, #2

For the sugar data of Example 3.1, create a frequency table using the class intervals $[0, 5)$, $[5, 10)$, $[10, 15)$, etc. Include both the frequency and the relative frequency.

Solution. The largest value is 39, so the last class is $[35, 40)$ and there are eight classes. Go through the sorted data and count how many values fall in each class. A value on a boundary belongs to the class that starts there: 15 is counted in $[15, 20)$, not in $[10, 15)$. Each relative frequency is the frequency divided by $n = 20$.

Class	Values in the class	Frequency	Relative frequency
$[0, 5)$	0	1	$1/20 = 0.05$
$[5, 10)$	8, 9	2	$2/20 = 0.10$
$[10, 15)$	12, 12, 14	3	0.15
$[15, 20)$	15, 16, 16, 19	4	0.20
$[20, 25)$	21, 22, 24, 24	4	0.20
$[25, 30)$	27, 28, 29	3	0.15
$[30, 35)$	31	1	0.05
$[35, 40)$	36, 39	2	0.10
Total		20	1.00

As a check, the frequencies add up to $n = 20$ and the relative frequencies add up to 1.

Example 3.3 ▪ A histogram and its shape Worksheet 3, #3–4

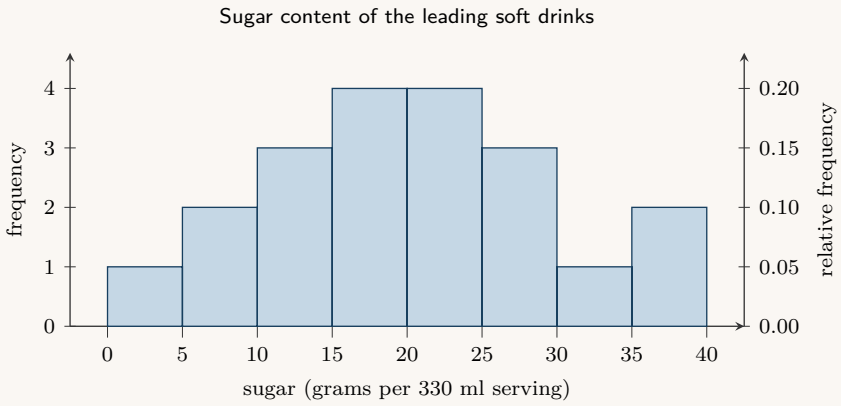
For the sugar data of Example 3.1:

a) Create a histogram. Be sure to label both axes and to include an appropriate title.

b) Describe the shape of this distribution.

Solution.

a) Use the frequency table of Example 3.2. The class boundaries $0, 5, 10, \ldots, 40$ go on the horizontal axis and each class gets a bar whose height is its frequency.



The vertical axis may show the frequency or the relative frequency. The right-hand scale shows the relative frequencies: the bars are the same, only the numbers on the axis change.

b) *Symmetry.* The histogram is fairly symmetric: the frequencies rise $1, 2, 3, 4$ on the left and fall $4, 3, 1, 2$ on the right, so the two halves are almost mirror images. Only the last two bars spoil the pattern.

Overall shape. There is a single peak in the middle, at 15 to 25 grams, and the frequencies fall away on both sides.

The distribution is fairly symmetric and approximately bell-shaped.

4 Measures of Centre

A measure of centre is a single number that says where a data set is located: a typical value around which the observations lie. This section defines the three measures of centre announced in Section 3, the mean, the median and the mode, and shows how to find them from a list of data, from a frequency table and, approximately, from a table of class intervals. It ends with the weighted average, of which the mean is a special case.

4.1 Mean, median and mode

Definition (mean)

The *mean* (or average) of a data set is the sum of the observations divided by the number of observations.

$$\text{Sample of size } n: \quad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \text{Population of size } N: \quad \mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

The computation is the same in both cases; only the symbols change. Write $\bar{x}$ and n when the data are a sample (the mean is then a statistic) and μ and N when the data are a whole population (the mean is then a parameter), as in Section 1.

Definition (median)

The *median* m is the middle value of the data once they have been put in increasing order, $x_1 \le x_2 \le \cdots \le x_n$:

$$m = \begin{cases} x_{\frac{n+1}{2}} & \text{if } n \text{ is odd (the one middle value),} \\ \dfrac{x_{\frac{n}{2}} + x_{\left(\frac{n}{2}+1\right)}}{2} & \text{if } n \text{ is even (the average of the two middle values).} \end{cases}$$

For a population the formulas are the same, with N in place of n.

Caution (sort first)

In the formulas for the median, x_i stands for the i-th *smallest* observation, so the data must be sorted before anything else. The subscript $\frac{n+1}{2}$ is a *position* in the sorted list, not the value of the median.

Definition (mode)

The *mode* is the value that occurs most often. There may be more than one mode, when several values are tied for the highest frequency. There may also be no mode at all, when no value occurs more often than the others.

Remark (which measure for which data)

- The mean needs quantitative data, and the median needs data that can at least be put in order. For qualitative data such as eye colour (Section 2) the mode is the only measure of centre available: one can speak of the most common eye colour, but not of the average eye colour.
- The mean uses every observation, so a single extreme value can move it a long way, whereas the median depends only on the middle of the sorted list. The data $1, 3, 5, 9$ have mean 4.5 and median 4. Replace 9 by 90: the median is still 4, but the mean jumps to 24.75. Section 6 comes back to the relative positions of the mean and the median.

4.2 Data given in a frequency table

When values are repeated many times it is more economical to list each distinct value once, together with its frequency (Section 3). Suppose that the distinct values, or *types*, are $x_1, x_2, \ldots$ and that the value x_j occurs f_j times. The subscript j now numbers the distinct values, not the individual observations.

Formula (mean from a frequency table)

$$\bar{x} = \frac{1}{n} \sum_{j=1}^{n_p} f_j \, x_j \qquad \text{or} \qquad \mu = \frac{1}{N} \sum_{j=1}^{N_P} f_j \, x_j,$$

where n_p (for a sample) or N_P (for a population) is the number of possible types and f_j is the frequency of type j. The number of observations is the sum of the frequencies: $n = \sum_j f_j$.

This is the basic formula for the mean, written more compactly: in the sum of all the observations the value x_j appears f_j times, so it contributes $f_j \, x_j$.

Definition (cumulative frequency and cumulative relative frequency)

List the rows of a frequency table (values or classes) in increasing order.

- The *cumulative frequency* of a row is the sum of its frequency and the frequencies of all the rows before it.
- The *cumulative relative frequency* of a row is obtained in the same way from the relative frequencies: add up the relative frequency of the row and those of all the preceding rows.

"Cumulative" means that the frequency keeps accumulating as we move down the table. It stops at the total: the last cumulative frequency is n and the last cumulative relative frequency is 1.

For instance, if the first four relative frequencies of a table are 0.2, 0.1, 0.2 and 0.01, then the first four cumulative relative frequencies are

$$0.2, \qquad 0.2+0.1=0.3, \qquad 0.3+0.2=0.5, \qquad 0.5+0.01=0.51.$$

The cumulative frequency of a row answers the question "how many observations lie in this row or in an earlier one?". That is exactly what is needed to find a *median* in a frequency table. Work out the position of the middle observation, $\frac{n+1}{2}$ (or the two positions $\frac{n}{2}$ and $\frac{n}{2}+1$ when n is even), then go down the cumulative frequencies until that position is reached. The *mode* is read directly from the table: it is the value with the highest frequency.

4.3 Data grouped into class intervals

A frequency table with class intervals no longer shows the individual observations, only how many of them fall in each class. The measures of centre can then only be *estimated*. The idea is to act as if every observation in a class were equal to the *midpoint* of that class,

$$m_j = \frac{\text{lower boundary} + \text{upper boundary}}{2},$$

for example $m_j = \frac{70+80}{2} = 75$ for the class $[70, 80)$.

Formula (estimated mean from class intervals)

$$\bar{x} \approx \frac{1}{n}\sum_{j=1}^{n_c} f_j\, m_j \qquad \text{or} \qquad \mu \approx \frac{1}{N}\sum_{j=1}^{N_C} f_j\, m_j,$$

where n_c (for a sample) or N_C (for a population) is the number of classes, f_j is the frequency of class j and m_j is the midpoint of class j.

Note (an estimation)

The result of this formula is only an estimate of the mean, not its exact value, which is why it is written with $\approx$. The exact mean can only be computed from the original data.

The *median* is estimated in the same spirit. Use the cumulative frequencies to find the class that contains the middle observation, and take the midpoint of that class as the estimate. When n is even and the two middle observations fall in two different classes, average the two midpoints.

Definition (modal class)

For data grouped into class intervals, the *modal class* is the class interval with the highest frequency. It takes over the role of the mode: when the data are known only through the table, it is the class interval taken to contain the mode. As with the mode, there may be more than one modal class.

Caution (two different m's)

The letter m without a subscript is the median; m_j with a subscript is the midpoint of class j.

4.4 The weighted average

Definition (weighted average)

If the values $x_1, x_2, \ldots$ are given the weights $w_1, w_2, \ldots$, their *weighted average* is

$$\frac{\sum w_i\, x_i}{\sum w_i},$$

where w_i is the weight of element x_i.

A weight says how much an element counts. The division is by the sum of the weights, not by the number of elements.

Remark (the mean is a weighted average)

- If all the weights are equal, the weighted average is the ordinary mean.
- The mean computed from a frequency table is a weighted average in which the weights are the frequencies:

$$\bar{x} = \frac{\sum f_j\, x_j}{\sum f_j}.$$

4.5 Worked examples

The data sets S_1, S_2 and S_3 introduced in Examples 4.1–4.3 come back in Section 5, where their spread is measured.

Example 4.1 ▪ A small data set Worksheet 4, #1

Consider the small set of data $S_1 = \{1, 3, 5, 9\}$. Calculate the mean, median and mode for S_1.

Solution. There are four observations and they are already in increasing order.

Mean. Add the observations and divide by their number:

$$\frac{1+3+5+9}{4} = \frac{18}{4} = \boxed{4.5}.$$

The mean would be written $\bar{x}$ if S_1 is a sample and μ if it is a population; the calculation is the same.

Median. The number of observations is even, so the median is the average of the two middle values, the ones in positions $\frac{4}{2} = 2$ and $\frac{4}{2} + 1 = 3$:

$$m = \frac{x_2 + x_3}{2} = \frac{3+5}{2} = \boxed{4}.$$

Mode. Every value occurs exactly once, so no value occurs more often than the others: there is $\boxed{\text{no mode}}$.

Example 4.2 ▪ Mean, median and mode from a frequency table Worksheet 4, #2

A small sample of students was surveyed to find out how many textbooks they carry around with them in their book-bag. The results (S_2) are given by the table below.

No. books	0	1	2	3	4	5
No. people	2	6	7	5	3	2

Calculate the mean, median and mode for S_2.

Solution. This is a frequency table: the values x_j are the numbers of books and the frequencies f_j are the numbers of people. The students are a *sample*, so the mean is written $\bar{x}$ (not μ) and the sample size is $n = 2+6+7+5+3+2 = 25$.

Number of books x_j	0	1	2	3	4	5	**Total**
Frequency f_j	2	6	7	5	3	2	25
$f_j\,x_j$	0	6	14	15	12	10	57
Cumulative frequency	2	8	15	20	23	25	

Mean. Each value is multiplied by its frequency:

$$\bar{x} = \frac{2\cdot 0 + 6\cdot 1 + 7\cdot 2 + 5\cdot 3 + 3\cdot 4 + 2\cdot 5}{2+6+7+5+3+2} = \frac{57}{25} = \boxed{2.28 \text{ books}}.$$

Median. Since $n = 25$ is odd, the median is the observation in position $\frac{25+1}{2} = 13$ of the sorted list. The cumulative frequencies show that positions 1 to 2 hold the value 0, positions 3 to 8 the value 1 and positions 9 to 15 the value 2. Position 13 is among the 2's, so

$$m = x_{13} = \boxed{2 \text{ books}}.$$

Mode. The highest frequency is 7, for the value 2: the mode is $\boxed{2 \text{ books}}$.

Example 4.3 ▪ Estimating from class intervals

Worksheet 4, #3

Marks for a certain Calculus class (S_3) are given below.

38 38 41 43 50 50 50 50 50 51 51 52 54 55 56 57 60 60 60 60 60 61 61 61 62 62 62
63 63 63 63 63 64 64 65 66 66 66 66 66 66 66 66 66 67 67 67 67 67 67 67 67 68 68
68 68 68 69 69 69 69 70 70 70 71 71 71 71 72 72 72 72 72 72 72 72 72 72 73 73
73 73 73 73 74 74 74 74 75 75 75 75 75 75 75 76 76 77 77 77 77 77 77 77 77 78 78
78 78 78 78 78 78 79 79 79 79 79 79 79 79 80 80 80 80 80 81 81 81 81 82 82 82
82 82 82 83 83 83 83 83 83 83 83 84 84 84 84 84 84 84 84 85 85 85 85 85 85 86
86 86 86 86 86 86 86 87 87 87 87 87 87 87 88 88 88 88 88 89 89 89 89 90 90 90 90
91 91 91 91 91 91 92 92 92 93 94 94 94 95 95 95 95 96 96 97 97 97 97 97 97 97 98
98 99 99 99

Complete a frequency table with the class intervals $[30, 40)$, $[40, 50)$, …, $[90, 100)$, giving for each class the number of students, the relative frequency and the cumulative relative frequency.

a) The actual average grade is 76.79. Use the midpoints of the class intervals to estimate the average.

b) The actual median grade is 78. Use the midpoints of the class intervals to estimate the median.

c) Is there a modal class?

d) Draw the histogram of this distribution using the class intervals on the horizontal axis and the relative frequency on the vertical axis.

Solution. *The table.* Counting the marks class by class gives the frequencies f_j below; they add up to $N = 220$. The marks of the whole class are given, so the data are treated as a population: we write N and μ. Each relative frequency is $f_j/220$, and the cumulative columns are running totals.

Mark	Midpoint m_j	No. students f_j	Cumulative frequency	Relative frequency	Cumul. rel. freq.
$[30, 40)$	35	2	2	0.0091	0.0091
$[40, 50)$	45	2	4	0.0091	0.0182
$[50, 60)$	55	12	16	0.0545	0.0727
$[60, 70)$	65	45	61	0.2045	0.2773
$[70, 80)$	75	62	123	0.2818	0.5591
$[80, 90)$	85	62	185	0.2818	0.8409
$[90, 100)$	95	35	220	0.1591	1.0000
Total		220		1	

The relative frequencies are rounded to four decimals, so the rounded values add up to 0.9999 instead of exactly 1. To keep such rounding errors from piling up, each cumulative relative frequency was computed as cumulative frequency divided by 220; for example $61/220 \approx 0.2773$.

a) Replace every mark by the midpoint of its class:

$$\mu \approx \frac{1}{220}\big(2 \cdot 35 + 2 \cdot 45 + 12 \cdot 55 + 45 \cdot 65 + 62 \cdot 75 + 62 \cdot 85 + 35 \cdot 95\big)$$
$$= \frac{70 + 90 + 660 + 2925 + 4650 + 5270 + 3325}{220} = \frac{16\,990}{220} \approx \boxed{77.227}.$$

This is close to the actual average, 76.79, but not equal to it: it is only an estimate.

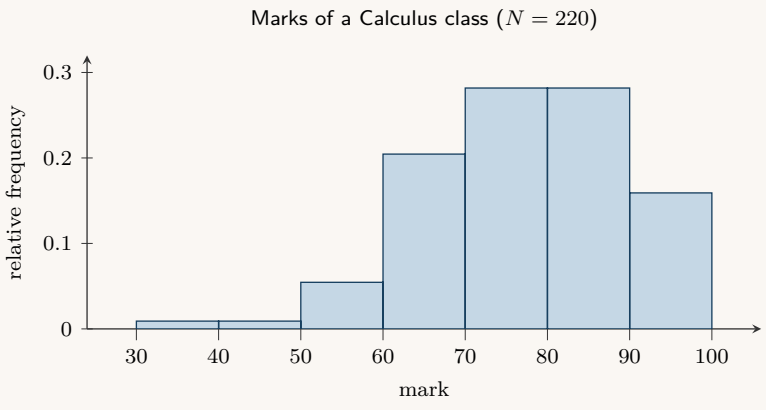
b) $N = 220$ is even, so the median is the average of the marks in positions 110 and 111 of the sorted list. The cumulative frequencies show that 61 marks are below 70 and 123 marks are below 80, so positions 62 to 123 are all in the class $[70, 80)$. Both middle marks are in that class and each is replaced by its midpoint, 75:

$$m \approx \frac{x_{110} + x_{111}}{2} = \frac{75 + 75}{2} = \boxed{75}.$$

The cumulative relative frequencies tell the same story: they pass 0.5 in the class $[70, 80)$, going from 0.2773 to 0.5591. The actual median is 78, which does lie in this class.

c) The modal class is the class with the highest frequency. Here the highest frequency, 62, is reached twice, so there are two modal classes: $\boxed{[70, 80) \text{ and } [80, 90)}$.

d) Each class gets a bar whose height is its relative frequency.



Example 4.4 ▪ An R-score for the term: a weighted average Worksheet 4, #4

Calculate the R-score for the term for the student whose marks are given below.

Subject	No. credits	R-score
English	2.33	30
French	2	28
Phys. Ed.	1	27
History	2	29
Math	2.66	31
Chemistry	2.66	33
Biology	2.66	34

Solution. The courses do not all count equally: each one counts in proportion to its number of credits. The R-score for the term is therefore the weighted average of the course R-scores x_i, with the numbers of credits as the weights w_i:

$$\frac{\sum w_i\, x_i}{\sum w_i} = \frac{2.33(30) + 2(28) + 1(27) + 2(29) + 2.66(31) + 2.66(33) + 2.66(34)}{2.33 + 2 + 1 + 2 + 2.66 + 2.66 + 2.66}$$

$$= \frac{471.58}{15.31} \approx \boxed{30.8}.$$

The plain average of the seven R-scores is $\frac{212}{7} \approx 30.3$. The weighted average is higher because the three best R-scores (31, 33 and 34) belong to the courses with the most credits.

5 Measures of Spread

A measure of centre (Section 4) says where a dataset sits, but not how tightly the data are packed around that place: two classes can have the same average mark while one has every student close to the average and the other has marks all over the scale. A *measure of spread*, also called *width* or *variability*, puts a number on this. The course uses two kinds: measures based on the *position* of the ordered data (range, quartiles, interquartile range) and measures based on the *distances of the data from the mean* (variance and standard deviation).

5.1 Range, quartiles and the interquartile range

In this subsection the data are always put in increasing order first: $x_1 \le x_2 \le \cdots \le x_n$.

Definition (range)

The *range* is the distance between the largest and the smallest data point:

$$R = \max - \min\,.$$

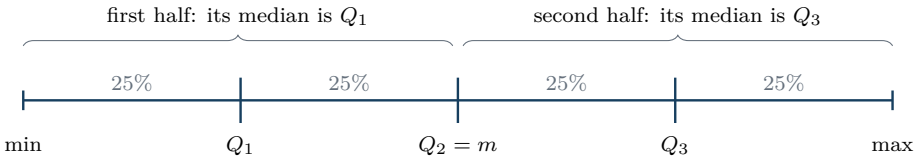
Recall (median)

The median m is the middle value of the ordered data: $m = x_{\frac{n+1}{2}}$ if n is odd, and $m = \frac{1}{2}\left(x_{\frac{n}{2}} + x_{\frac{n}{2}+1}\right)$ if n is even (Section 4). Half of the data lie on each side of the median.

Definition (quartiles)

The *quartiles* Q_1, Q_2, Q_3 are three numbers that cut the ordered data into four parts with the same number of data points in each part.

- $Q_2 = m$ is the median of the whole dataset.
- Q_1 is the median of the *first half* of the data, the part from the minimum up to Q_2.
- Q_3 is the median of the *second half* of the data, the part from Q_2 up to the maximum.



Note (which data points make up each half)

- If n is *even*, the median falls between two data points. The first half is the lowest $\frac{n}{2}$ data points and the second half is the highest $\frac{n}{2}$. For $\{1, 3, 5, 9\}$: $Q_2 = \frac{3+5}{2} = 4$, the halves are $\{1, 3\}$ and $\{5, 9\}$, so $Q_1 = 2$ and $Q_3 = 7$.
- If n is *odd*, the median is itself a data point. Leave it out: the halves are the data points below it and the data points above it. For $\{2, 4, 5, 7, 8, 10, 13\}$: $Q_2 = 7$, the halves are $\{2, 4, 5\}$ and $\{8, 10, 13\}$, so $Q_1 = 4$ and $Q_3 = 10$.

Definition (five-number summary)

The *five-number summary* of a dataset is the list

$$\{\min,\ Q_1,\ Q_2,\ Q_3,\ \max\}.$$

Definition (interquartile range)

The *interquartile range* is the distance between the first and the third quartile:

$$\mathrm{IQR} := Q_3 - Q_1.$$

A quarter of the data lie below Q_1 and a quarter lie above Q_3. So *the middle 50% of the data lie between* Q_1 *and* Q_3, and the IQR is the width of the interval that holds this middle half. This is true for every dataset, whatever its shape.

Remark (range or IQR?)

The range uses only the two most extreme data points, so a single unusual value can change it completely. The IQR ignores the lowest quarter and the highest quarter of the data, so a few extreme values do not affect it. This makes the IQR the safer description of how spread out the bulk of the data is.

Definition (outlier)

An *outlier* is any data point outside the interval

$$\big[\,Q_1 - 1.5\,\mathrm{IQR},\quad Q_3 + 1.5\,\mathrm{IQR}\,\big],$$

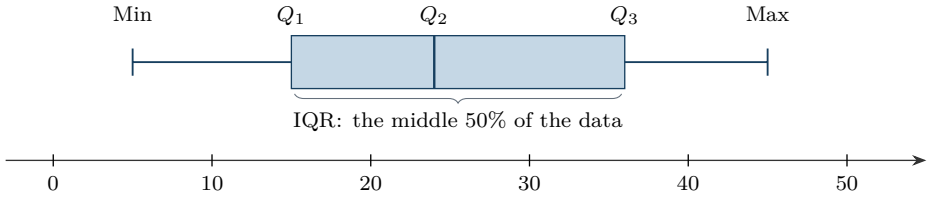
that is, a point lying more than 1.5 times the interquartile range below Q_1 or above Q_3.

5.2 Box-and-whisker plots

A *box-and-whisker plot* (or box plot) is a picture of the five-number summary drawn above a number line.

How to draw a box-and-whisker plot

1. Give the plot a title, and draw an axis with a scale that covers all the data.
2. Find the five-number summary $\{\min, Q_1, Q_2, Q_3, \max\}$.
3. Compute IQR $= Q_3 - Q_1$; this is the length of the box.
4. Draw a box from Q_1 to Q_3 and a line across it at Q_2. Then draw the two "whiskers", from the box out to the minimum and out to the maximum.



Each of the four pieces of the plot (left whisker, left part of the box, right part of the box, right whisker) contains a quarter of the data. A long piece does not mean more data: it means that this quarter of the data is more spread out.

5.3 Variance and standard deviation

The range and the IQR are built from a few of the ordered values. The variance and the standard deviation use every data point: they measure how far the data are from the mean.

The *deviation* of a data point from the mean is $x_i - \mu$ (or $x_i - \bar{x}$ in a sample). Averaging the deviations themselves is useless, because the positive and the negative ones always cancel: $\sum(x_i - \mu) = \sum x_i - N\mu = 0$. Squaring each deviation removes the signs, and this is what the definition does.

Definition (variance and standard deviation)

- For a *population* $x_1, \ldots, x_N$ with mean μ, the *variance* σ^2 and the *standard deviation* σ are

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2, \qquad \sigma = \sqrt{\sigma^2}.$$

- For a *sample* $x_1, \ldots, x_n$ with mean $\bar{x}$, the variance s^2 and the standard deviation s are

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

The variance is, in essence, the average squared distance of the data from the mean. Its units are the square of the units of the data (marks^2, dollars^2). Taking the square root brings back the original units, which is why the standard deviation is the quantity that gets quoted and interpreted: roughly, it is the typical distance between a data point and the mean.

Caution (sample or population?)

The two definitions do not divide by the same number: the population variance divides by N, the sample variance divides by $n - 1$. Decide which kind of dataset you have before you compute anything.

The reason for $n - 1$ is that a sample is used to estimate the spread of the population it was drawn from. The

deviations are measured from $\bar{x}$, the centre of the sample itself, instead of from the unknown population mean μ, and this makes them slightly too small on average. Dividing by $n-1$ rather than n compensates for that.

The definition is not always the quickest way to get the number. Expanding the square leads to a second formula that only needs the sum of the data and the sum of their squares.

Theorem (computational formula for the variance of a population)

For a population $x_1, \dots, x_N$ with mean μ,

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 - \mu^2 = \frac{1}{N}\left(\sum_{i=1}^{N} x_i^2 - \frac{1}{N}\Big(\sum_{i=1}^{N} x_i\Big)^2\right). \qquad (*)$$

In words: the variance is the mean of the squares minus the square of the mean.

Proof (deriving the computational formula from the definition)

Start from the definition and expand the square:

$$\begin{aligned}\sigma^2 &= \frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i^2 - 2\mu x_i + \mu^2)\\ &= \frac{1}{N}\sum_{i=1}^{N} x_i^2 \;-\; 2\mu\cdot\frac{1}{N}\sum_{i=1}^{N} x_i \;+\; \frac{1}{N}\sum_{i=1}^{N}\mu^2.\end{aligned}$$

In the middle term, $\frac{1}{N}\sum x_i = \mu$ by the definition of the mean. In the last term, μ^2 is a constant that is added N times, so $\sum_{i=1}^{N}\mu^2 = N\mu^2$. Therefore

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 - 2\mu\cdot\mu + \frac{1}{N}\cdot N\mu^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 - 2\mu^2 + \mu^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 - \mu^2.$$

Finally, replacing μ by $\frac{1}{N}\sum x_i$ gives $\mu^2 = \frac{1}{N^2}\big(\sum x_i\big)^2$, and so

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 - \frac{1}{N^2}\Big(\sum_{i=1}^{N} x_i\Big)^2 = \frac{1}{N}\left(\sum_{i=1}^{N} x_i^2 - \frac{1}{N}\Big(\sum_{i=1}^{N} x_i\Big)^2\right).$$

□

The same expansion works for a sample: $\sum(x_i-\bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = \sum x_i^2 - \frac{1}{n}\big(\sum x_i\big)^2$, and this sum is then divided by $n-1$. All four formulas side by side:

Formula (variance: definition and computational formulas)

	Definition	Computational
sample	$s^2 = \dfrac{1}{n-1}\sum(x_i-\bar{x})^2$	$s^2 = \dfrac{1}{n-1}\left(\sum x_i^2 - \dfrac{1}{n}\Big(\sum x_i\Big)^2\right)$
population	$\sigma^2 = \dfrac{1}{N}\sum(x_i-\mu)^2$	$\sigma^2 = \dfrac{1}{N}\left(\sum x_i^2 - \dfrac{1}{N}\Big(\sum x_i\Big)^2\right) = \dfrac{1}{N}\sum x_i^2 - \mu^2$

In every case the standard deviation is the square root of the variance: $s = \sqrt{s^2}$, $\sigma = \sqrt{\sigma^2}$.

Note (data given in a frequency table)

If a table lists the different values x_i together with their frequencies f_i, then each row stands for f_i equal data points. The number of data points is $\sum f_i$, and in the formulas above

- the sum of the data becomes $\sum f_i\, x_i$,
- the sum of the squares becomes $\sum f_i\, x_i^2$,
- the sum of the squared deviations becomes $\sum f_i\,(x_i - \bar{x})^2$.

If the table has class intervals, use the midpoint of each class as x_i. The answer is then only approximate, because every data point has been replaced by the midpoint of its class.

Remark (which formula to use)

Both formulas give exactly the same number. The definition shows what the variance means and is pleasant when the mean is a round number. The computational formula is quicker when the mean is an awkward decimal, because it needs only the two totals $\sum x_i$ and $\sum x_i^2$. Keep full calculator precision until the end: the computational formula subtracts two large numbers that are close to each other, so a mean that has been rounded too early can spoil the answer.

5.4 Worked examples

Example 5.1 ▪ Five-number summary, IQR and box plot Worksheet 5, #1

The marks of the 220 students of a Calculus class (dataset S_3) are listed in increasing order, 16 per row; the first column gives the positions of the marks in that row.

1–16	38	38	41	43	50	50	50	50	50	51	51	52	54	55	56	57
17–32	60	60	60	60	60	61	61	61	62	62	62	63	63	63	63	63
33–48	64	64	65	66	66	66	66	66	66	66	66	67	67	67	67	67
49–64	67	67	67	67	68	68	68	68	68	69	69	69	69	70	70	70
65–80	71	71	71	71	72	72	72	72	72	72	72	72	72	72	72	73
81–96	73	73	73	73	73	74	74	74	74	75	75	75	75	75	75	75
97–112	76	76	77	77	77	77	77	77	77	77	78	78	78	78	78	78
113–128	78	78	78	79	79	79	79	79	79	79	79	80	80	80	80	80
129–144	81	81	81	81	82	82	82	82	82	82	83	83	83	83	83	83
145–160	83	83	83	84	84	84	84	84	84	84	84	85	85	85	85	85
161–176	85	86	86	86	86	86	86	86	86	86	87	87	87	87	87	87
177–192	87	88	88	88	88	89	89	89	89	90	90	90	90	91	91	91
193–208	91	91	91	92	92	92	93	94	94	94	95	95	95	95	96	96
209–220	97	97	97	97	97	97	97	98	98	99	99	99				

Give the five-number summary, give the interquartile range and draw a box and whisker plot for S_3. What is the interpretation of the interquartile range?

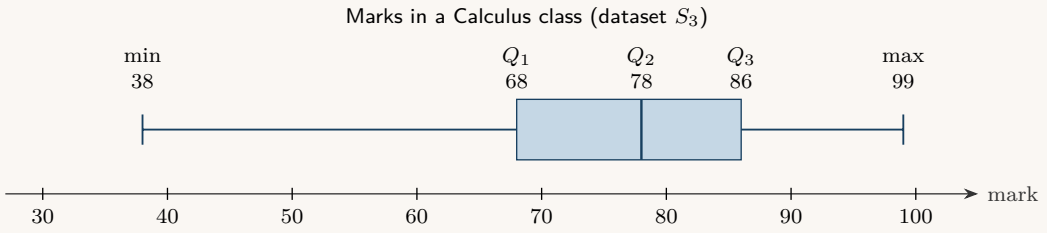
Solution. There are $n = 220$ marks, an even number, and they are already in order.

- *Minimum and maximum.* $\min = x_1 = 38$ and $\max = x_{220} = 99$.
- *Median.* With n even, Q_2 is the average of the two middle marks, x_{110} and x_{111}. Both are 78 (row 97–112), so $Q_2 = 78$.
- *First quartile.* The first half is $x_1, \ldots, x_{110}$. It has 110 marks, so its median is the average of its 55th and 56th marks: $Q_1 = \frac{1}{2}(x_{55} + x_{56}) = \frac{1}{2}(68 + 68) = 68$.
- *Third quartile.* The second half is $x_{111}, \ldots, x_{220}$. Its 55th and 56th marks are x_{165} and x_{166}, so $Q_3 = \frac{1}{2}(x_{165} + x_{166}) = \frac{1}{2}(86 + 86) = 86$.

The five-number summary and the interquartile range are

$$\{\min, Q_1, Q_2, Q_3, \max\} = \boxed{\{38,\ 68,\ 78,\ 86,\ 99\}}, \qquad \mathrm{IQR} = Q_3 - Q_1 = 86 - 68 = \boxed{18}.$$

(For comparison, the range is $R = 99 - 38 = 61$.)



Interpretation of the IQR. The middle 50% of the class, that is 110 of the 220 students (the 56th to the 165th marks), have marks inside the 18-point interval from 68 to 86.

Two more things the summary shows. The interval used to detect outliers is

$$\big[\,Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}\,\big] = [\,68 - 27,\ 86 + 27\,] = [\,41,\ 113\,],$$

so the two marks of 38 are outliers and no other mark is. Also, the left whisker (38 to 68) is more than twice as long as the right one (86 to 99): the lowest quarter of the marks is far more spread out than the highest quarter. The distribution has a long tail on the left; it is *skewed left* (Section 6).

Example 5.2 ▪ Sample variance from a frequency table

Worksheet 5, #2

A small sample of students was surveyed to find out how many textbooks they carry around with them in their book-bag. The results (dataset S_2) are given in the table.

No. books	0	1	2	3	4	5
No. people	2	6	7	5	3	2

Use the definition to calculate the variance and the standard deviation of S_2. Remember that S_2 is a sample.

Solution. The sample size is $n = \sum f_i = 2 + 6 + 7 + 5 + 3 + 2 = 25$, and the mean is

$$\bar{x} = \frac{1}{n}\sum f_i\, x_i = \frac{2(0) + 6(1) + 7(2) + 5(3) + 3(4) + 2(5)}{25} = \frac{57}{25} = 2.28.$$

Now build the table one column at a time: the deviation of each value from the mean, its square, and the square multiplied by the frequency.

x_i	f_i	$\bar{x}$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i \cdot (x_i - \bar{x})^2$
0	2	2.28	−2.28	5.1984	10.3968
1	6	2.28	−1.28	1.6384	9.8304
2	7	2.28	−0.28	0.0784	0.5488
3	5	2.28	0.72	0.5184	2.5920
4	3	2.28	1.72	2.9584	8.8752
5	2	2.28	2.72	7.3984	14.7968
	25				47.0400

In the first row, for example, two people carry 0 books, and each of them contributes $(-2.28)^2$, which gives $2(-2.28)^2 = 10.3968$. Adding the last column therefore adds the squared deviation of every one of the 25 people: its total is exactly the sum $\sum(x_i - \bar{x})^2$ that the definition asks for.

Because S_2 is a sample, this sum is divided by $n-1=24$:

$$s^2=\frac{1}{n-1}\sum f_i\,(x_i-\bar{x})^2=\frac{47.04}{24}=\boxed{1.96},\qquad s=\sqrt{1.96}=\boxed{1.4\text{ books}}.$$

Check with the computational formula. Here $\sum f_i\,x_i=57$ and $\sum f_i\,x_i^2=2(0)+6(1)+7(4)+5(9)+3(16)+2(25)=177$, so

$$s^2=\frac{1}{n-1}\left(\sum f_i\,x_i^2-\frac{1}{n}\left(\sum f_i\,x_i\right)^2\right)=\frac{1}{24}\left(177-\frac{57^2}{25}\right)=\frac{177-129.96}{24}=\frac{47.04}{24}=1.96.$$

Example 5.3 ▪ Approximate variance from class intervals Worksheet 5, #3

The 220 Calculus marks of Example 5.1 (dataset S_3) are grouped into class intervals as follows.

Mark	$[30,40)$	$[40,50)$	$[50,60)$	$[60,70)$	$[70,80)$	$[80,90)$	$[90,100)$
No. students	2	2	12	45	62	62	35

Use the computational formula to calculate the (approximate) variance and the standard deviation of S_3 from this table.

Solution. The individual marks are not used. Every mark is replaced by the midpoint x_i of its class, so the result is an approximation. The computational formula $(*)$ needs the number of data points, the sum of the data and the sum of their squares; the columns f_i, $f_i\,x_i$ and $f_i\,x_i^2$ of the following table supply exactly these three totals.

Interval	midpoint x_i	f_i	$f_i\,x_i$	$f_i\,x_i^2$
$[30,40)$	35	2	70	2 450
$[40,50)$	45	2	90	4 050
$[50,60)$	55	12	660	36 300
$[60,70)$	65	45	2 925	190 125
$[70,80)$	75	62	4 650	348 750
$[80,90)$	85	62	5 270	447 950
$[90,100)$	95	35	3 325	315 875
total		220	16 990	1 345 500

The 220 marks are the marks of the whole class, so S_3 is treated as a population with $N=220$:

$$\sigma^2\approx\frac{1}{N}\left(\sum f_i\,x_i^2-\frac{1}{N}\left(\sum f_i\,x_i\right)^2\right)=\frac{1}{220}\left(1\,345\,500-\frac{16\,990^2}{220}\right)$$
$$=\frac{1\,345\,500-1\,312\,091.36}{220}=\frac{33\,408.64}{220}\approx\boxed{151.86},$$

and $\sigma\approx\sqrt{151.86}\approx\boxed{12.32\text{ marks}}$.

The second form of $(*)$ gives the same value: the mean is $\mu\approx 16\,990/220\approx 77.227$, and $\sigma^2\approx 1\,345\,500/220-\mu^2\approx 6115.91-5964.05=151.86$. Use the unrounded mean here: with μ rounded to 77.2 the variance would come out as 156.07.

For comparison, the 220 individual marks give $\mu\approx 76.79$ and $\sigma\approx 12.66$, so the grouped table produces a good estimate with much less work.

6 Shapes of Distributions

The measures of centre and spread of Sections 4 and 5 compress a dataset into a few numbers. A picture of the data shows something those numbers do not: the *shape* of the distribution. For a large dataset the tops of the bars of a histogram (Section 3) are smoothed into a curve, called a *distribution curve*. The curve is high above the values that occur often and low above the values that are rare, and the area under the curve between two values is the proportion of the data lying between them. This section gives the vocabulary for describing such curves and shows how the mean, the median, the mode and the standard deviation can be read off a curve without any calculation.

6.1 Describing a shape

To describe a distribution in words, answer two questions.

1) **Shape or form.** How many peaks does the curve have?
2) **Symmetry.** Is the curve symmetric, or is one of its tails longer than the other?

Definition (unimodal, bimodal, bell-shaped)

A peak of a distribution curve marks a value around which the data pile up; it is called a *mode* of the distribution.

- A distribution with one peak is *unimodal*. Its curve rises to the peak and falls away on both sides, like the outline of a bell, so it is also called *bell-shaped*.
- A distribution with two peaks is *bimodal*.

In this course "bell-shaped" and "unimodal" mean the same thing: one hump. If the hump is lopsided, the curve is called *approximately* bell-shaped.

Definition (symmetric and skewed distributions)

- A distribution is *symmetric* if the left half of its curve is the mirror image of the right half.
- A distribution is *skewed right* if it has a fat right tail: the curve falls away slowly on the right and quickly on the left.
- A distribution is *skewed left* if it has a fat left tail: the curve falls away slowly on the left and quickly on the right.

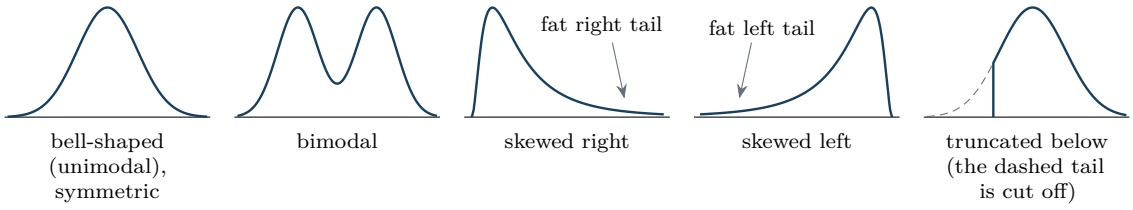
Caution

The skew is named after the side of the long tail, not after the side of the peak. A curve whose peak is on the left is skewed *right*.

Definition (truncated distributions)

Ordinarily a distribution curve tails off gradually and meets the horizontal axis smoothly. A distribution is *truncated* if instead its curve is cut off sharply at one end. It is *truncated below* if the cut is at the lower end: there are no data below a certain value, and the curve starts abruptly there. If the cut is at the upper end, the distribution is *truncated above*.

Truncation below is typical of a quantity that cannot be smaller than some bound (for instance a quantity that cannot be negative) when a lot of the data sit close to that bound.



6.2 Mean, median and mode on a distribution curve

The three measures of centre can be located on a distribution curve.

- The **mode** is the value under the highest point of the curve.
- The **median** m is the value that splits the area under the curve into two equal parts: 50% of the data lie to its left and 50% to its right.
- The **mean** μ is the balance point: if the region under the curve were cut out of cardboard, it would balance on a pivot placed at μ.

Property (symmetric distributions)

In a symmetric distribution the mean and the median are both at the centre of symmetry, so $\mu = m$. If the distribution is also bell-shaped, its peak is at the centre as well, and

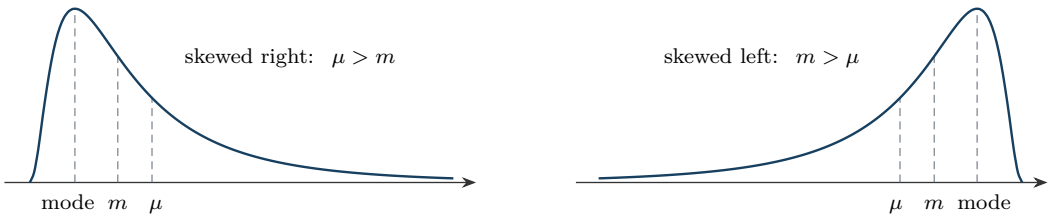
$$\mu = m = \text{mode}.$$

A symmetric bimodal distribution still has $\mu = m$ at its centre, halfway between the two peaks, but its modes are at the peaks and not at the centre.

Property (skewed distributions)

In a skewed bell-shaped distribution the three measures of centre separate, and they do so in a typical order.

- Skewed right: $\text{mode} < m < \mu$, in particular $\mu > m$.
- Skewed left: $\mu < m < \text{mode}$, in particular $m > \mu$.



The reason is the way the two numbers are computed. The median only counts how many data points lie on each side of it, so it does not matter how far away the points in the long tail are. The mean uses the size of every data point, so the few extreme values in the tail drag it toward the tail. The mode stays under the peak. Hence, for a skewed bell-shaped distribution, the mean is normally the measure that lies farthest out in the direction of the long tail. (This is a rule of thumb for bell-shaped data, not a theorem: unusual shapes can break it.)

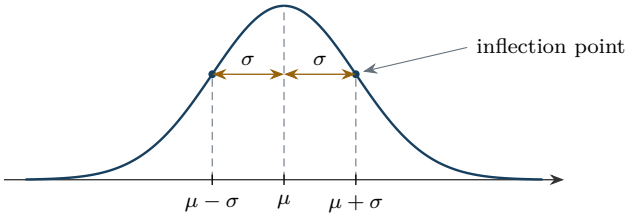
Caution (the median always splits the data 50–50)

Skewness changes where the mean falls relative to the median. It does not change what the median is: in every distribution, symmetric or skewed, half of the data lie below the median and half lie above it.

6.3 The standard deviation on a bell-shaped curve

Property (reading σ off the graph)

On a perfectly bell-shaped, symmetric curve, the standard deviation σ is the horizontal distance from the middle of the curve to its *inflection point*, the point on the side of the bell where the curve stops bending downward and starts bending upward.



So the two parameters of a bell-shaped curve can be read off separately. The position of the middle gives μ, and the width gives σ: a wide, flat bell has a large standard deviation, and a narrow, pointed bell has a small one. Sliding a curve along the axis without changing its shape changes μ but not σ.

Why?

The perfect bell shape is the normal curve of Section 24, whose height at x is proportional to $y = e^{-(x-\mu)^2/(2\sigma^2)}$. Differentiating twice,

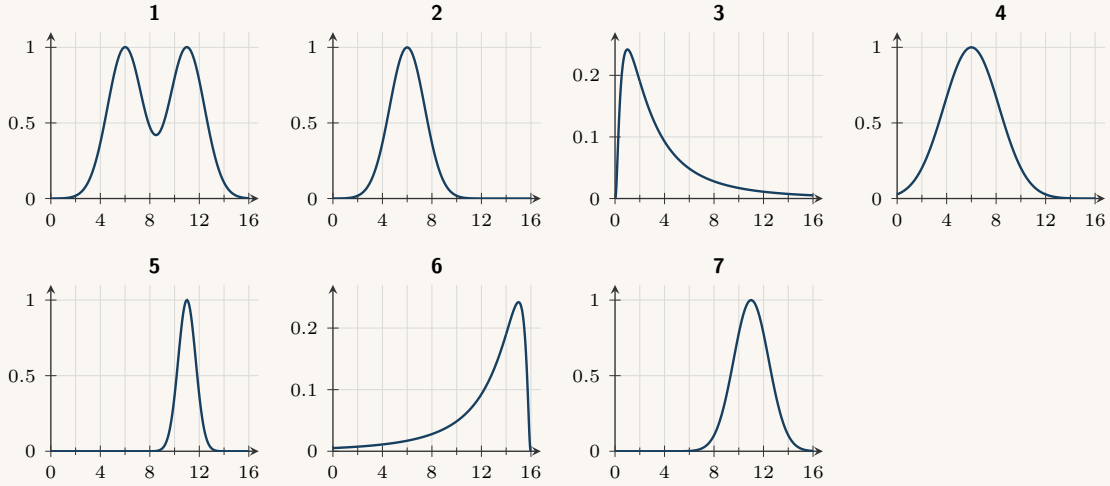
$$y' = -\frac{x-\mu}{\sigma^2}\,y, \qquad y'' = \Big(\frac{(x-\mu)^2}{\sigma^4} - \frac{1}{\sigma^2}\Big)y,$$

and $y'' = 0$ exactly when $(x-\mu)^2 = \sigma^2$, that is at $x = \mu \pm \sigma$. The curve changes the direction of its bending one standard deviation on either side of the mean.

6.4 Worked example

Example 6.1 ⋆ ▪ **Describing seven distributions** Worksheet 6, part I

Consider the following seven distributions. Describe the shape of each of them in words.



Solution. For each curve, count the peaks, check the symmetry, and look at how the curve ends.

	Description	What shows it
1	bimodal, symmetric	two peaks of the same height, at $x = 6$ and $x = 11$; the left half mirrors the right half about $x = 8.5$
2	bell-shaped, symmetric	one peak, at $x = 6$; the curve falls away in the same way on both sides
3	approximately bell-shaped, truncated below, skewed right	one peak, near $x = 1$; the curve starts abruptly at $x = 0$ and has a long, fat tail on the right
4	bell-shaped, symmetric	one peak, at $x = 6$; wider than curve 2
5	bell-shaped, symmetric	one narrow peak, at $x = 11$
6	approximately bell-shaped, truncated above, skewed left	the mirror image of curve 3: a long, fat tail on the left and a sharp cut-off at the upper end, $x = 16$
7	bell-shaped, symmetric	one peak, at $x = 11$; the same shape as curve 2, moved 5 units to the right

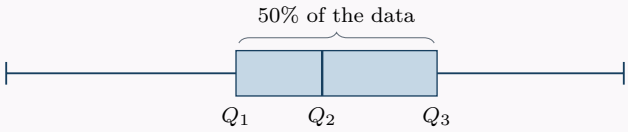
7 The Empirical Rule, Chebyshev's Theorem, z-Scores and Percentiles

The standard deviation of Section 5 becomes much more useful once we know how much of a dataset lies within one, two or three standard deviations of its mean. Two results answer this question: the empirical rule, for bell-shaped distributions, and Chebyshev's theorem, for all distributions. Measuring distances from the mean in units of the standard deviation also leads to the z-score, which makes values from different datasets comparable. The section ends with percentiles, which describe the position of a data point by its rank.

7.1 The empirical rule

Recall (the IQR)

The quartiles already give one statement of the form "this interval contains this percentage of the data": the box of a box-and-whisker plot, from Q_1 to Q_3, holds the middle 50% of the data, and its length is the IQR (Section 5).



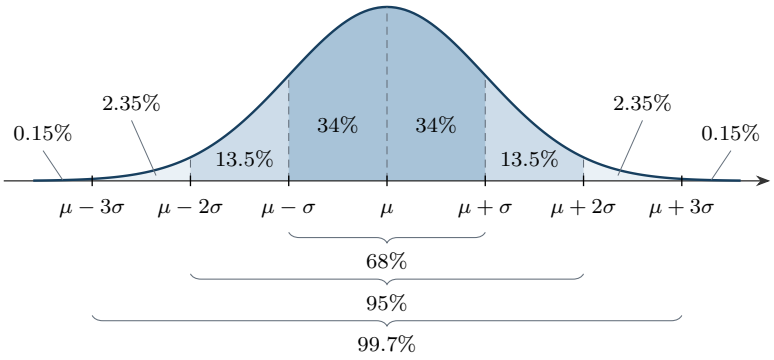
The empirical rule makes the same kind of statement, with the mean and the standard deviation in place of the quartiles.

Empirical rule

If a distribution is unimodal (bell-shaped) and reasonably symmetrical, then roughly 68% of the distribution lies within one standard deviation of the mean, 95% within two standard deviations of the mean, and 99.7% within three standard deviations of the mean.

Distance from the mean	Interval	Share of the distribution
within 1 standard deviation	$[\mu - \sigma,\ \mu + \sigma]$	about 68%
within 2 standard deviations	$[\mu - 2\sigma,\ \mu + 2\sigma]$	about 95%
within 3 standard deviations	$[\mu - 3\sigma,\ \mu + 3\sigma]$	about 99.7%

For a sample, use $\bar{x}$ and s in place of μ and σ.



Note (splitting the percentages)

A bell-shaped curve is symmetric about μ, so each percentage of the rule is shared equally between the two sides of the mean. Subtracting and halving gives the percentage in each band of the picture:

- between μ and $\mu+\sigma$: $\frac{68\%}{2} = 34\%$;
- between $\mu+\sigma$ and $\mu+2\sigma$: $\frac{95\%-68\%}{2} = 13.5\%$;
- between $\mu+2\sigma$ and $\mu+3\sigma$: $\frac{99.7\%-95\%}{2} = 2.35\%$;
- beyond $\mu+3\sigma$: $\frac{100\%-99.7\%}{2} = 0.15\%$,

and the same on the left of the mean.

7.2 Chebyshev's theorem

The empirical rule needs a bell shape. For a skewed, bimodal or unknown shape there is a weaker statement that is always true.

Theorem (Chebyshev's theorem)

For *any* distribution, at least

$$1-\frac{1}{k^2}$$

of the distribution lies within k standard deviations of the mean, that is on the interval $[\,\mu-k\sigma,\ \mu+k\sigma\,]$ (or $[\,\bar{x}-ks,\ \bar{x}+ks\,]$ for a sample). Here k can be any number greater than 1.

Share of the data within	**Empirical rule** (bell-shaped only)	**Chebyshev's theorem** (any distribution)
1 standard deviation of the mean	about 68%	no information ($1-\frac{1}{1^2}=0$)
2 standard deviations	about 95%	at least $1-\frac{1}{2^2}=75\%$
3 standard deviations	about 99.7%	at least $1-\frac{1}{3^2}\approx 88.9\%$

Caution ("at least")

Chebyshev's theorem gives a guaranteed minimum, not an estimate and not a maximum. "At least 75% of the data lie within two standard deviations of the mean" allows 80%, 95% or even 100%; the only thing it rules out is less than 75%. The price for a statement that holds for every shape is that it is usually far below the true percentage.

Why?

Take a population $x_1, \dots, x_N$, and suppose M of its data points lie at a distance of $k\sigma$ or more from the mean. Each of these points contributes at least $(k\sigma)^2$ to the sum of squared deviations, so

$$N\sigma^2 = \sum_{i=1}^{N}(x_i - \mu)^2 \;\geq\; M\,k^2\sigma^2, \qquad \text{which gives} \qquad \frac{M}{N} \leq \frac{1}{k^2}.$$

At most $\frac{1}{k^2}$ of the data are k or more standard deviations away from the mean. The rest, at least $1 - \frac{1}{k^2}$ of the data, are closer than that.

7.3 z-scores and standardized datasets

Definition (z-score)

The *z-score* of a value is

$$z = \frac{\text{value} - \text{mean}}{\text{st. dev.}}.$$

For the data point x_i of a dataset with mean μ and standard deviation σ,

$$z_i = \frac{x_i - \mu}{\sigma}$$

(for a sample, $z_i = \dfrac{x_i - \bar{x}}{s}$).

The z-score shows how far a data point is from the mean *in terms of standard deviations*. A z-score of 1.5 means "one and a half standard deviations above the mean"; a negative z-score belongs to a data point below the mean; and the mean itself has $z = 0$. The units cancel in the division, so a z-score is a pure number.

Definition (standardized dataset)

Replacing every data point of a dataset $X = \{x_1, \dots, x_n\}$ by its z-score gives the collection

$$Z = \{z_1, \dots, z_n\},$$

called the *standardized dataset* of X. Computing it is called *standardizing* X.

Proposition (mean and standard deviation of a standardized dataset)

Let $x_1, \dots, x_N$ be a dataset with mean μ and standard deviation $\sigma > 0$, and let $z_i = \dfrac{x_i - \mu}{\sigma}$. Then the standardized dataset $z_1, \dots, z_N$ has mean 0 and standard deviation 1:

$$\mu_z = \frac{1}{N}\sum_{i=1}^{N} z_i = 0 \qquad \text{and} \qquad \sigma_z = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(z_i - \mu_z)^2} = 1.$$

Proof that $\mu_z = 0$

By the definition of the mean, applied to the dataset $z_1, \dots, z_N$,

$$\mu_z = \frac{1}{N}\sum_{i=1}^{N} z_i = \frac{1}{N}\sum_{i=1}^{N}\frac{x_i - \mu}{\sigma} = \frac{1}{N\sigma}\sum_{i=1}^{N}(x_i - \mu) = \frac{1}{N\sigma}\Big(\sum_{i=1}^{N} x_i - \sum_{i=1}^{N}\mu\Big).$$

The first sum is $\sum x_i = N\mu$, because $\mu = \frac{1}{N}\sum x_i$. The second sum adds the constant μ exactly N times, so it

is also $N\mu$. Therefore

$$\mu_z = \frac{1}{N\sigma}\left(N\mu - N\mu\right) = 0.$$

□

Proof that $\sigma_z = 1$

Start with the variance of the z-scores. Since $\mu_z = 0$ by the first part,

$$\sigma_z^2 = \frac{1}{N}\sum_{i=1}^{N}(z_i - \mu_z)^2 = \frac{1}{N}\sum_{i=1}^{N} z_i^2 = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \mu}{\sigma}\right)^2 = \frac{1}{\sigma^2}\cdot \underbrace{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}_{=\,\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1.$$

The constant $\frac{1}{\sigma^2}$ was taken out of the sum, and what remains is exactly the definition of the variance σ^2 of the original data. Hence $\sigma_z = \sqrt{\sigma_z^2} = \sqrt{1} = 1$. □

Remark

The same is true for a sample that is standardized with $\bar{x}$ and s: the z-scores have mean 0, and their standard deviation, computed with the divisor $n - 1$, is 1.

Note (why standardize? comparing across datasets)

Whatever the original data were (marks out of 100, incomes in dollars, weights in grams), the standardized dataset always has mean 0 and standard deviation 1. So z-scores from different datasets are on a common scale and can be compared with each other.

When a larger value is the better one, a higher z-score indicates a better *relative* performance, that is, a better position within one's own dataset. "Higher" includes negative values: -2 is higher than -3.

Both results of the previous subsections can be restated with z-scores. In a bell-shaped distribution, about 68% of the z-scores are between -1 and 1, about 95% are between -2 and 2, and about 99.7% are between -3 and 3. In any distribution, at least $1 - \frac{1}{k^2}$ of the z-scores are between $-k$ and k.

7.4 Percentiles

Definition (percentile rank)

A data point has *percentile rank* n if n% of the observations are less than or equal to it. The data point is then said to be at the n-th *percentile*.

Finding the percentile rank of a data point

1. Count the observations that are less than *or equal to* the data point.
2. Divide by the total number of observations and write the result as a percentage.
3. Round *down* to a whole number.

Percentiles round down: every percentage from n% up to, but not including, $(n+1)$% gives the percentile rank n. For example, if 82.67% of the observations are less than or equal to a data point, the data point is at the 82nd percentile, not the 83rd.

The question can also be asked the other way round: which observation has percentile rank n? Compute n% of the number of observations. That many observations have to be less than or equal to the answer, so if the result is not a whole number, go up to the next whole position and read off the observation at that position in the ordered data (Example 7.3).

Remark (percentiles, quartiles and the empirical rule)

The quartiles are percentiles: Q_1, Q_2 and Q_3 sit at the 25th, 50th and 75th percentiles. In a bell-shaped distribution, the percentages of the empirical rule give the percentile ranks of the points μ, $\mu \pm \sigma$, $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$: add up the bands to the left of the point.

data point	$\mu - 3\sigma$	$\mu - 2\sigma$	$\mu - \sigma$	μ	$\mu + \sigma$	$\mu + 2\sigma$	$\mu + 3\sigma$
z-score	-3	-2	-1	0	1	2	3
% of the data at or below it	0.15	2.5	16	50	84	97.5	99.85

For example, a data point one standard deviation above the mean has $z = 1$, and $50\% + 34\% = 84\%$ of the data lie at or below it: its percentile rank is 84.

7.5 Worked examples

Example 7.1 * ▪ **Empirical rule or Chebyshev?** Worksheet 7, #1

If the mean weight of 1000 boxes of chopped broccoli is 450 grams with a standard deviation of 50 grams, at least how many boxes will weigh between 350 and 550 grams?

a) Assume a bell-shaped distribution of weights.

b) Answer for any distribution.

Solution. First measure the interval in standard deviations. With $\mu = 450$ and $\sigma = 50$,

$$350 = 450 - 2(50) = \mu - 2\sigma, \qquad 550 = 450 + 2(50) = \mu + 2\sigma,$$

so the question is about the boxes within $k = 2$ standard deviations of the mean.

a) For a bell-shaped distribution the empirical rule applies: about 95% of the boxes are within two standard deviations of the mean. That is about $0.95 \times 1000 =$ 950 boxes.

b) For an arbitrary distribution only Chebyshev's theorem is available. With $k = 2$, at least

$$1 - \frac{1}{2^2} = \frac{3}{4} = 75\%$$

of the boxes are within two standard deviations of the mean, so at least 750 boxes weigh between 350 and 550 grams.

The guarantee in (b) is weaker because it has to hold for every possible shape of distribution.

Example 7.2 ▪ **Light bulbs and the empirical rule** Worksheet 7, #2

LongLive light bulbs are known to have a mean life of 800 hours, with a standard deviation of 100 hours. Suppose that we have reasons to believe that the distribution of the lifetimes of these bulbs is unimodal and roughly symmetrical.

a) What percentage of the light bulbs will have a life of 700 to 900 hours?

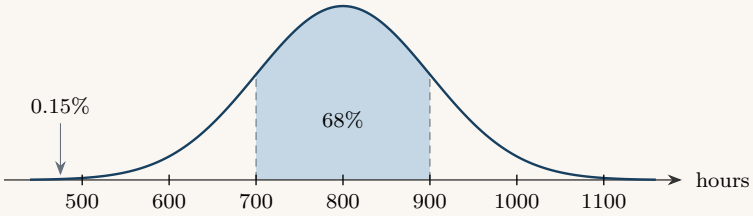
b) What percentage of the light bulbs will last less than 500 hours?

Solution. The distribution is unimodal and roughly symmetrical, so the empirical rule applies, with $\mu = 800$ and $\sigma = 100$.

a) $700 = \mu - \sigma$ and $900 = \mu + \sigma$. The interval $[700, 900]$ is "within one standard deviation of the mean", so it contains about 68% of the bulbs.

b) $500 = 800 - 3(100) = \mu - 3\sigma$. The bulbs that last less than 500 hours are those in the interval $(0, 500) = (0,\ \mu - 3\sigma)$, more than three standard deviations below the mean. About 99.7% of the bulbs are inside $[\mu - 3\sigma,\ \mu + 3\sigma]$, which leaves $100\% - 99.7\% = 0.3\%$ outside. By symmetry, half of these are below $\mu - 3\sigma$ and half are above $\mu + 3\sigma$:

$$\frac{100\% - 99.7\%}{2} = \boxed{0.15\%}.$$



Example 7.3 ▪ Deer count: standard deviation and percentiles Worksheet 7, #3

The following are the numbers of deer observed in a sample of 75 sections of land in a wildlife count. The data have been ordered, 25 per row; the first column gives the positions of the observations in that row.

1–25	1	3	5	6	6	7	7	8	8	9	9	9	10	10	10	10	11	11	11	11	11	12	12	12	12
26–50	12	12	13	13	13	13	13	13	14	14	14	14	14	14	14	15	15	15	15	15	15	15	16	16	16
51–75	16	16	16	17	17	17	17	17	18	18	18	18	19	19	19	19	20	20	20	21	21	22	23	25	27

Note that $\sum x_i = 1054$ and $\sum x_i^2 = 16\,576$.

a) What percentage of the observations lie within one standard deviation of the mean?

b) What is the percentile rank of a section of land with 18 deer?

c) Which observation has a percentile rank of 67?

Solution. The observations are counts, already in increasing order, so positions in the list can be read off directly.

a) The data are a sample of $n = 75$ sections, so we need $\bar{x}$ and s. The two given sums are exactly what the computational formula of Section 5 uses:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{1054}{75} \approx 14.0533,$$

$$s^2 = \frac{1}{n-1}\left(\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2\right) = \frac{1}{74}\left(16\,576 - \frac{1054^2}{75}\right)$$

$$\approx \frac{16\,576 - 14\,812.2133}{74} = \frac{1763.7867}{74} \approx 23.8350,$$

so $s \approx \sqrt{23.8350} \approx 4.8821$. "Within one standard deviation of the mean" is the interval

$$[\,\bar{x} - s,\ \bar{x} + s\,] \approx [\,14.0533 - 4.8821,\ 14.0533 + 4.8821\,] = [\,9.1712,\ 18.9354\,].$$

The observations are whole numbers (the data are discrete), so the ones inside this interval are those from 10 to 18 inclusive. In the ordered list the first 10 is at position 13 and the last 18 is at position 62, so there are $62 - 12 = 50$ of them:

$$\frac{50}{75} \approx \boxed{66.67\%}.$$

This is close to the 68% of the empirical rule, as it should be for data whose distribution is roughly bell-shaped.

b) Count the observations that are less than or equal to 18. The last 18 is at position 62, so 62 of the 75 observations are at most 18. (Equivalently, 13 observations are greater than 18, and $75 - 13 = 62$.) Then

$$\frac{62}{75} \approx 0.8267 = 82.67\%.$$

Percentiles round down, so a section with 18 deer has percentile rank $\boxed{82}$: it is at the 82nd percentile.

c) We need the observation such that 67% of the 75 observations are less than or equal to it:

$$0.67 \times 75 = 50.25 \text{ observations.}$$

Fifty observations are not enough to cover 50.25, so we go up to the 51st observation. It is the first entry of the third row: $\boxed{16 \text{ deer}}$.

Check. 53 of the 75 observations are at most 16, which is 70.67% and covers the required 67%; only 47 observations, 62.67%, are at most 15, which is not enough. Because 16 occurs six times, every percentile rank from 63 to 70 leads to this same observation. In a small dataset with repeated values a percentile rank cannot single out a data point more finely than that.

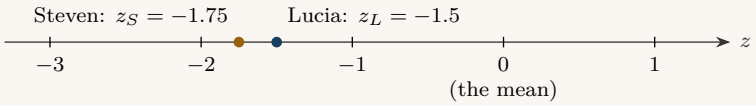
Example 7.4 ⋆ ▪ Who is relatively better off? Worksheet 7, #4

Steven lives in a neighbourhood where the average yearly gross income is \$120 000 with a standard deviation of \$20 000. Steven makes \$85 000. Lucia lives in a neighbourhood where the average yearly income is \$45 000 with a standard deviation of \$10 000. Lucia makes \$30 000 a year. Which of the two is relatively better off?

Solution. "Relatively" means compared with the people around them. So each income is turned into a z-score inside its own neighbourhood, and the two z-scores are compared:

$$z_S = \frac{85\,000 - 120\,000}{20\,000} = \frac{-35\,000}{20\,000} = -1.75, \qquad z_L = \frac{30\,000 - 45\,000}{10\,000} = \frac{-15\,000}{10\,000} = -1.5.$$

Both z-scores are negative: both of them earn less than the average of their neighbourhood. Steven is 1.75 standard deviations below his neighbourhood's average, Lucia only 1.5 standard deviations below hers.



Since $-1.5 > -1.75$, Lucia has the higher z-score, so $\boxed{\text{Lucia is relatively better off}}$, even though Steven's income is almost three times hers in dollars.

Example 7.5 ⋆ ▪ Standardizing a dataset Worksheet 7, #5

Consider the dataset $S_1 = \{1, 3, 5, 9\}$. Its mean is $\mu = 4.5$ and its standard deviation is $\sigma = \sqrt{8.75} \approx 2.958$ (Section 5). Standardize this dataset, then verify by direct calculation that $\mu_z = 0$ and $\sigma_z = 1$.

Solution. Each data point is replaced by $z_i = \dfrac{x_i - \mu}{\sigma} = \dfrac{x_i - 4.5}{\sqrt{8.75}}$.

x_i	1	3	5	9	total
$x_i - \mu$	−3.5	−1.5	0.5	4.5	0
$z_i = \dfrac{x_i - \mu}{\sigma}$	−1.1832	−0.5071	0.1690	1.5213	0.0000
z_i^2	1.4000	0.2571	0.0286	2.3143	4.0000

The standardized dataset is $Z \approx \boxed{\{-1.1832,\ -0.5071,\ 0.1690,\ 1.5213\}}$.

Mean. The z-scores add up to 0, so

$$\mu_z = \frac{1}{4}\sum z_i = \frac{-1.1832 - 0.5071 + 0.1690 + 1.5213}{4} = \frac{0}{4} = \boxed{0}.$$

Standard deviation. Since $\mu_z = 0$, the deviations $z_i - \mu_z$ are the z-scores themselves, and their squares add up to 4:

$$\sigma_z^2 = \frac{1}{4}\sum (z_i - \mu_z)^2 = \frac{1}{4}\sum z_i^2 = \frac{1.4000 + 0.2571 + 0.0286 + 2.3143}{4} = \frac{4}{4} = 1,$$

and therefore $\sigma_z = \sqrt{1} = \boxed{1}$.

The totals are exact, not just correct to four decimals: $\sum z_i = \frac{1}{\sigma}\sum (x_i - \mu) = 0$ and $\sum z_i^2 = \frac{1}{\sigma^2}\sum (x_i - \mu)^2 = \frac{35}{8.75} = 4$. These are the two steps of the general proofs, carried out with numbers.

Part II

Counting

When all outcomes are equally likely, a probability is a ratio of two counts. This part builds the counting toolkit: the multiplication and addition rules, permutations, combinations and arrangements with repeated objects.

8 Counting Basics: the Multiplication and Addition Rules

Probability, which starts in Section 11, very often comes down to a count: when all the outcomes of an experiment are equally likely, the probability of an event is the number of outcomes in the event divided by the total number of outcomes. Listing the outcomes one by one stops being practical almost immediately, so this section and the next two develop ways of counting without listing. Everything rests on two rules: *multiply* the numbers of choices for steps that are carried out one after another, and *add* the numbers of choices for alternatives that exclude each other.

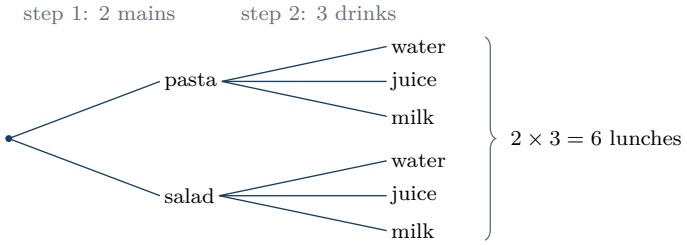
8.1 The multiplication rule

Multiplication rule

Suppose a task is carried out in k steps, one after another. If step 1 can be done in n_1 ways, step 2 in n_2 ways whatever was done in step 1, and so on up to step k, which can be done in n_k ways whatever was done before, then the whole task can be done in

$$n_1 \times n_2 \times \cdots \times n_k \quad \text{ways.}$$

For example, a lunch special consists of one of 2 mains (pasta or salad) and one of 3 drinks (water, juice or milk). Each main can be combined with each drink, so there are $2 \times 3 = 6$ different lunches. A *tree diagram* shows why the numbers multiply: every branch of the first step splits into the same number of branches at the second step, and each path from left to right is one outcome.



The rule needs the *number* of choices at each step to be the same whatever happened earlier; the choices themselves may change. If two different digits are picked one after the other, which digits are still available for the second pick depends on the first pick, but there are always 9 of them.

The slot method

Draw one blank (a "slot") for each step, write the number of choices in each slot, and multiply. Two patterns come up constantly when k positions are filled from n available symbols or objects.

- **Repetition allowed.** Every slot has n choices, so there are $n \times n \times \cdots \times n = n^k$ possibilities. For a 4-digit PIN:

$$\underset{\text{1st}}{10} \cdot \underset{\text{2nd}}{10} \cdot \underset{\text{3rd}}{10} \cdot \underset{\text{4th}}{10} = 10^4 = 10\,000.$$

- **Repetition not allowed.** Each choice uses up one object, so the number of choices drops by one from slot to slot. For a 4-digit PIN made of four different digits:

$$\underset{\text{1st}}{10} \cdot \underset{\text{2nd}}{9} \cdot \underset{\text{3rd}}{8} \cdot \underset{\text{4th}}{7} = 5040.$$

8.2 The addition rule

Addition rule

Suppose the possibilities to be counted fall into k cases, and no possibility belongs to two cases at once (the cases are *mutually exclusive*). If the cases contain $n_1, n_2, \dots, n_k$ possibilities, then altogether there are

$$n_1 + n_2 + \cdots + n_k \quad \text{possibilities.}$$

Note ("and" multiplies, "or" adds)

- Choices that are *all* made, one after another ("choose this **and then** that"): multiply.
- Alternatives of which *exactly one* happens ("either this **or** that"): add.

Most problems use both rules: split the possibilities into cases, count each case with the multiplication rule, then add the cases.

Caution

The addition rule requires cases that do not overlap. If two cases share some possibilities, the sum counts the shared ones twice, so they must be subtracted once. For example, among the integers from 1 to 20 there are 10 multiples of 2 and 4 multiples of 5, but only $10 + 4 - 2 = 12$ integers that are multiples of 2 or of 5, because 10 and 20 belong to both groups. Section 11 states this correction in general, and Section 12 gives the corresponding rule for probabilities.

Note (counting the opposite)

Every possibility either has a given property or does not have it, so by the addition rule

$$\text{(number with the property)} = \text{(total number)} - \text{(number without the property)}.$$

Use this whenever the opposite is easier to count, typically for "at least one". For example, the number of 4-digit PINs in which at least one digit is repeated is $10^4 - 10 \cdot 9 \cdot 8 \cdot 7 = 10\,000 - 5040 = 4960$.

8.3 Factorials and permutations

Products of consecutive whole numbers counting down, such as $10 \cdot 9 \cdot 8 \cdot 7$, appear every time objects are chosen one after another without repetition. Factorials are the notation for them.

Definition (factorial)

For a positive integer n, n *factorial* is the product of all the integers from n down to 1:

$$n! = n\,(n-1)\,(n-2)\cdots 3 \cdot 2 \cdot 1, \qquad \text{and by convention} \quad 0! = 1.$$

So $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, $5! = 120$, $6! = 720$, $7! = 5040$ and already $10! = 3\,628\,800$. By the slot method, $n!$ is the number of ways of arranging n different objects in a row: there are n choices for the first place, $n-1$ for the second, and so on down to a single choice for the last place.

Since $n! = n \cdot (n-1)!$, a quotient of factorials simplifies by cancellation:

$$\frac{8!}{5!} = \frac{8 \cdot 7 \cdot 6 \cdot 5!}{5!} = 8 \cdot 7 \cdot 6 = 336.$$

Always cancel before computing. Factorials grow so fast that $70! > 10^{100}$, which is more than most calculators can display.

Definition (permutation)

A *permutation* of k objects chosen from n different objects is an *ordered* selection: k of the n objects are chosen, no object more than once, and the order in which they are chosen (the position that each one receives) matters. The number of such permutations is written ${}_nP_k$.

Formula (number of permutations)

For integers $0 \le k \le n$,

$${}_nP_k = \underbrace{n\,(n-1)\,(n-2)\cdots(n-k+1)}_{k \text{ factors}} = \frac{n!}{(n-k)!}.$$

In particular ${}_nP_n = n!$: all n objects can be arranged in a row in $n!$ ways.

The product comes from the slot method: there are n choices for the first position, $n-1$ for the second, and so on, and the k-th position is filled from the $n-k+1$ objects that are left. It equals $n!/(n-k)!$ because the factors $(n-k)(n-k-1)\cdots 1$ of $n!$ cancel against $(n-k)!$. For example,

$${}_7P_3 = 7 \cdot 6 \cdot 5 = 210 = \frac{7!}{4!} \qquad \text{and} \qquad {}_nP_2 = n\,(n-1) = \frac{n!}{(n-2)!}.$$

The convention $0! = 1$ is what makes the factorial form work when $k = n$: ${}_nP_n = n!/0! = n!$.

8.4 Counting and probability

Probability by counting

Several of the problems below end with a probability. Whenever an outcome is chosen *at random*, meaning that all the possible outcomes are equally likely, the probability of an event A is

$$P(A) = \frac{N(A)}{N(S)} = \frac{\text{number of outcomes in which } A \text{ happens}}{\text{total number of possible outcomes}},$$

where S is the set of all possible outcomes and $N(\,\cdot\,)$ means "the number of outcomes in". Such a probability is two counting problems followed by a division. Section 11 develops this definition; until then it is used only in this form.

8.5 Worked examples

Example 8.1 ⋆ ▪ **Licence plates** Worksheet 8, #1

Licence plate numbers in Quebec are typically of the form LLL ### or ### LLL, where L represents a letter from A to Z and # represents a digit from 0 to 9.

a) How many licence plate numbers in total of these forms are possible?

b) How many licence plate numbers contain the word "CAT"?

c) If a licence plate number is selected at random, what is the probability that it contains the word "CAT"? (Assume no combinations are banned.)

Solution.

a) Count one form at a time. A plate of the form LLL### is built by filling six slots, and letters and digits may repeat:

$$\underset{\text{L}}{26}\cdot\underset{\text{L}}{26}\cdot\underset{\text{L}}{26}\cdot\underset{\#}{10}\cdot\underset{\#}{10}\cdot\underset{\#}{10}=26^3\cdot10^3=17\,576\,000.$$

The form ###LLL has the same six slots in another order, so it also gives 17 576 000 plates. A plate has one form *or* the other and never both (it starts either with a letter or with a digit), so the addition rule applies:

$$26^3\cdot10^3+10^3\cdot26^3=2\cdot26^3\cdot10^3=\boxed{35\,152\,000\text{ plates}}.$$

b) The three letters must be C, A, T in that order, so each letter slot has only one choice. The digits are free:

$$\underset{\text{C}}{1}\cdot\underset{\text{A}}{1}\cdot\underset{\text{T}}{1}\cdot\underset{\#}{10}\cdot\underset{\#}{10}\cdot\underset{\#}{10}=1000$$

plates of the form CAT###, and in the same way 1000 plates of the form ###CAT. In total $1000+1000=\boxed{2000\text{ plates}}$.

c) "Selected at random" means that the 35 152 000 plates are equally likely, so

$$P(\text{contains CAT})=\frac{2000}{35\,152\,000}=\boxed{\frac{1}{17\,576}}\approx0.000057.$$

The answer is $1/26^3$, as it should be: the digits and the form play no role, and each of the three letters has one chance in 26 of being the right one.

Example 8.2 ▪ An equation with permutations Worksheet 8, #6

Solve for n: $\quad 3\left({}_nP_5\right)=8\left({}_{n-1}P_5\right)$.

Solution. Write each side as a product of five factors counting down:

$${}_nP_5=\frac{n!}{(n-5)!}=n\,(n-1)(n-2)(n-3)(n-4),$$
$${}_{n-1}P_5=\frac{(n-1)!}{(n-6)!}=(n-1)(n-2)(n-3)(n-4)(n-5).$$

The equation becomes

$$3\,n\,(n-1)(n-2)(n-3)(n-4)=8\,(n-1)(n-2)(n-3)(n-4)(n-5).$$

The symbol ${}_{n-1}P_5$ only makes sense when $n-1\geq5$, that is $n\geq6$. For such n the common factors $(n-1)(n-2)(n-3)(n-4)$ are not zero and can be cancelled from both sides:

$$3n=8\,(n-5)\quad\Longrightarrow\quad3n=8n-40\quad\Longrightarrow\quad5n=40\quad\Longrightarrow\quad\boxed{n=8}.$$

Check: $3\left({}_8P_5\right)=3\cdot6720=20\,160$ and $8\left({}_7P_5\right)=8\cdot2520=20\,160$.

9 Permutations and Combinations

Section 8 counted *ordered* selections. In many problems the order is irrelevant: a committee, a hand of cards or a sample is the same whatever the order in which its members were picked. What must be counted then is *unordered* selections, called combinations. This section sets permutations and combinations side by side, extends the count of arrangements to objects that are not all different, and ends with a guide for choosing the right formula.

9.1 Permutations: order matters

Recall (permutations)

A *permutation* is a choice of k objects from n different objects *with ordering*: the same k objects taken in a different order form a different permutation. The number of permutations is

$$ {}_nP_k = \frac{n!}{(n-k)!} = n\,(n-1)\cdots(n-k+1), $$

and all n objects can be arranged in a row in ${}_nP_n = n!$ ways.

The order matters whenever the chosen objects receive different positions or roles: places in a row or in a ranking, seats, time slots, the digits of a code, titles such as president and secretary.

9.2 Combinations: order does not matter

Definition (combination)

A *combination* is a choice of k objects from n different objects *without ordering*: only which objects are chosen matters, not the order in which they are chosen. The number of combinations is written

$$ {}_nC_k \qquad \text{or} \qquad \binom{n}{k} \quad \text{(read “}n\text{ choose }k\text{”).} $$

Theorem (number of combinations)

For integers $0 \le k \le n$,

$$ {}_nC_k = \binom{n}{k} = \frac{n!}{(n-k)!\,k!}. $$

Note (how permutations and combinations are related)

For all integers $0 \le k \le n$,

$$ {}_nP_k = k!\;{}_nC_k. $$

An ordered selection can be made in two steps: first choose *which* k objects are taken (${}_nC_k$ ways), then put these k objects in order ($k!$ ways). By the multiplication rule ${}_nP_k = {}_nC_k \cdot k!$. Dividing by $k!$ gives the formula of the theorem:

$$ {}_nC_k = \frac{{}_nP_k}{k!} = \frac{n!}{(n-k)!\,k!}. $$

In words: every combination can be ordered in $k!$ ways, so there are $k!$ times fewer combinations than permutations.

For example, take $k = 3$ letters from the $n = 4$ letters A, B, C, D. There are ${}_4P_3 = 4 \cdot 3 \cdot 2 = 24$ permutations, but they come in groups of $3! = 6$ that use the same three letters, so there are only ${}_4C_3 = 24/6 = 4$ combinations:

Combination	**Its $3! = 6$ permutations**					
$\{A, B, C\}$	ABC	ACB	BAC	BCA	CAB	CBA
$\{A, B, D\}$	ABD	ADB	BAD	BDA	DAB	DBA
$\{A, C, D\}$	ACD	ADC	CAD	CDA	DAC	DCA
$\{B, C, D\}$	BCD	BDC	CBD	CDB	DBC	DCB

Property (useful facts about ${}_nC_k$)

- ${}_nC_0 = {}_nC_n = 1$ and ${}_nC_1 = n$.
- **Symmetry:** ${}_nC_k = {}_nC_{n-k}$, because choosing the k objects to take is the same as choosing the $n-k$ objects to leave behind. For example ${}_{10}C_6 = {}_{10}C_4$.
- Calculators have an nCr key. To evaluate by hand, cancel the larger factorial of the denominator first:

$$\binom{8}{3} = \frac{8!}{5!\,3!} = \frac{8\cdot 7\cdot 6}{3\cdot 2\cdot 1} = 56.$$

9.3 Arrangements with identical objects: the multinomial coefficient

The formula $n!$ counts the arrangements of n *different* objects. When some of the objects cannot be told apart, fewer arrangements can be distinguished.

Theorem (arrangements with identical objects)

Suppose that n objects consist of n_1 identical objects of a first kind, n_2 identical objects of a second kind, …, and n_k identical objects of a k-th kind, where $n_1 + n_2 + \cdots + n_k = n$. The number of distinguishable arrangements of these n objects in a row is

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}.$$

This number is called a *multinomial coefficient*; it is also written $\binom{n}{n_1,\, n_2,\, \ldots,\, n_k}$.

Why divide?

Take the three letters A, L, L. Make the two L's different for a moment by labelling them L_1 and L_2. Three different letters have $3! = 6$ arrangements, and removing the labels makes them coincide in pairs:

$$\begin{array}{ccc} AL_1L_2,\ AL_2L_1 & L_1AL_2,\ L_2AL_1 & L_1L_2A,\ L_2L_1A \\ \downarrow & \downarrow & \downarrow \\ ALL & LAL & LLA \end{array}$$

Each distinguishable arrangement was counted $2!$ times, once for every way of permuting the identical letters among themselves, so there are $3!/2! = 3$ of them.

In general, labelling all the objects gives $n!$ arrangements, and each distinguishable arrangement is obtained $n_1!\,n_2!\cdots n_k!$ times: permute the objects of the first kind among themselves, those of the second kind among themselves, and so on. Dividing $n!$ by this number gives the formula.

For example, the word STATISTICS has $n = 10$ letters: three S, three T, two I, one A and one C. Its letters can be put in

$$\frac{10!}{3!\,3!\,2!\,1!\,1!} = \frac{3\,628\,800}{72} = 50\,400$$

distinguishable orders. Kinds that occur only once contribute $1! = 1$ and may be left out of the denominator.

Remark (link with combinations)

With only two kinds of objects ($k = 2$, so $n_2 = n - n_1$) the multinomial coefficient is a combination:

$$\frac{n!}{n_1!\,(n-n_1)!} = \binom{n}{n_1}.$$

This makes sense: an arrangement of n_1 objects of one kind and $n - n_1$ of another is known as soon as we choose which n_1 of the n positions the first kind occupies. It is the reason why $\binom{n}{x}$ appears in the binomial

distribution (Section 17): it counts the sequences of n trials that contain exactly x successes and $n-x$ failures.

With k kinds the positions can be chosen one kind after the other, which gives the multinomial coefficient a second time:

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n_k}{n_k} = \frac{n!}{n_1!\,n_2!\cdots n_k!}.$$

9.4 Choosing the right tool

Which tool when

Two questions decide which formula applies when k objects are taken from n different objects.

1) **Does the order matter?** Would the same objects in a different order (in different positions, or with different roles) count as a different outcome?
2) **Is repetition allowed?** Can the same object be used more than once?

	Repetition allowed	**No repetition**
Order matters	n^k (multiplication rule)	${}_nP_k = \dfrac{n!}{(n-k)!}$ (permutations)
Order does not matter	not covered in this course	${}_nC_k = \dfrac{n!}{(n-k)!\,k!}$ (combinations)

Arranging *all* n objects in a row can be done in $n!$ ways if the objects are all different, and in $n!/(n_1!\,n_2!\cdots n_k!)$ distinguishable ways if they come in k groups of identical objects.

Typical words when the order matters: arrange, line up, seat, schedule, rank, assign to different positions or titles. Typical words when it does not: choose, select, a group, a committee, a sample, a hand of cards.

Arrangements with restrictions: four standard moves

- **A restricted position** (a fixed first letter, a reserved seat or section): fill the restricted positions first, then arrange what is left in the free positions (Example 9.1).
- **Objects that must stay together:** glue them into a single block. Arrange the blocks, then arrange the objects inside each block, and multiply (Example 9.2).
- **Two types that must alternate:** choose which type comes first, then arrange each type in its own places (Example 9.2).
- **Objects that must *not* be next to each other:** count all the arrangements and subtract those in which the objects are together (an example is worked in Section 10).

9.5 Worked examples

Example 9.1 * ▪ **Arranging the letters of a word** Worksheet 9, #1

a) How many distinct permutations can be made from the letters of the word *columns*?

b) If one of these permutations is chosen at random, what is the probability that it will begin with the letter *m*?

Solution.

a) The word has 7 letters, c, o, l, u, m, n, s, and they are all different. Arranging all of them in a row can be done in ${}_7P_7 = 7! = \boxed{5040 \text{ ways}}$.

b) The first position is restricted, so fill it first: it must receive the m (1 choice). The other 6 letters are then arranged freely in the remaining 6 positions:

$$\underset{\text{m}}{\underline{1}} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 6! = 720.$$

All 7! permutations are equally likely, so

$$P(\text{begins with m}) = \frac{6!}{7!} = \frac{720}{5040} = \boxed{\frac{1}{7}} \approx 0.1429.$$

This agrees with common sense: each of the 7 letters is equally likely to come first.

Example 9.2* ▪ Seating couples in a row

Worksheet 9, #4

Four married opposite-sex couples have bought all of the 8 seats in a same row for a concert. In how many different ways can they be seated

a) with no restrictions?
b) if each couple is to sit together?
c) if all the men sit to the right of all the women?
d) if they alternate genders?

Solution.

a) Eight different people are arranged in eight seats: $8! = \boxed{40\,320 \text{ ways}}$.

b) Glue each couple into a block. The 4 blocks can be arranged in 4! ways. Inside each block the two partners can sit in 2! orders, and this choice is made for each of the 4 blocks:

$$4! \cdot (2!)^4 = 24 \cdot 16 = \boxed{384 \text{ ways}}.$$

c) The women must take the four seats on the left and the men the four seats on the right. Arrange the women in their seats (4! ways), then the men in theirs (4! ways):

$$4! \cdot 4! = \boxed{576 \text{ ways}}.$$

d) The row must read $MWMWMWMW$ or $WMWMWMWM$: 2 patterns. Once the pattern is chosen, the men are arranged in the men's seats (4! ways) and the women in the women's seats (4! ways):

$$2 \cdot 4! \cdot 4! = \boxed{1152 \text{ ways}}.$$

b) M_1 W_1 | M_2 W_2 | M_3 W_3 | M_4 W_4 — 4! orders of the blocks, 2! orders inside each block

c) W W W W M M M M — 4! for the women, 4! for the men

d) M W M W M W M W
W M W M W M W M — 2 patterns, then 4! · 4! for each pattern

Example 9.3 ▪ Trees of the same kind

Worksheet 9, #5

In how many ways can 3 oaks, 4 pines and 2 maples be arranged along a property line if one does not distinguish between trees of the same kind?

Solution. There are $3+4+2=9$ trees, and an arrangement looks like $O\,P\,P\,M\,O\,P\,O\,P\,M$. If the nine trees were all different, they could be arranged in ${}_9P_9 = 9!$ ways. But trees of the same kind are identical: permuting the 3 oaks among themselves (3! ways), the 4 pines among themselves (4! ways) or the 2 maples among themselves (2! ways) leaves the row looking exactly the same. Each distinguishable arrangement is therefore counted $3!\,4!\,2!$ times in 9!, and the number of arrangements is the multinomial coefficient with $n=9$, $n_1=3$, $n_2=4$, $n_3=2$:

$$\frac{{}_9P_9}{3!\,4!\,2!} = \frac{9!}{3!\,4!\,2!} = \frac{362\,880}{6\cdot 24\cdot 2} = \boxed{1260 \text{ ways}}.$$

Check by choosing positions: pick 3 of the 9 positions for the oaks, then 4 of the remaining 6 for the pines; the maples take the last 2:

$$\binom{9}{3}\binom{6}{4}\binom{2}{2} = 84\cdot 15\cdot 1 = 1260.$$

Example 9.4 ⋆ ▪ Selections from two groups

Worksheet 9, #9

A car-rental agency has 10 foreign cars and 15 domestic cars waiting to be serviced on a given morning, but the service department only has time to service 6 cars. Cars are chosen at random to be serviced.

a) In how many ways can they choose the cars to be serviced?

b) What is the probability that they service the same number of foreign cars as domestic cars?

c) In how many orders can they end up servicing the same number of foreign cars as domestic cars?

d) What is the probability that they service at least one foreign car?

e) What is the probability that they service at least four domestic cars?

Solution.

a) Only *which* 6 of the $10+15=25$ cars are chosen matters, so this is a combination:

$$\binom{25}{6} = \frac{25!}{19!\,6!} = \boxed{177\,100 \text{ ways}}.$$

These 177 100 selections are equally likely; they are the denominator of every probability below.

b) The same number of each means 3 foreign *and* 3 domestic cars. Choose the 3 foreign cars among the 10, then the 3 domestic cars among the 15, and multiply:

$$\binom{10}{3}\binom{15}{3} = 120\cdot 455 = 54\,600 \text{ selections},$$

$$P(\text{3 foreign and 3 domestic}) = \frac{54\,600}{177\,100} = \boxed{\frac{78}{253}} \approx 0.3083.$$

c) Now the order of servicing matters as well. Each of the 54 600 selections of part b) can be serviced in $6! = 720$ different orders, so there are

$$\binom{10}{3}\binom{15}{3}\cdot 6! = 54\,600\cdot 720 = \boxed{39\,312\,000 \text{ orders}}.$$

d) "At least one foreign car" covers 1, 2, …, 6 foreign cars: six cases. Its opposite is a single case, "no foreign car", in which all 6 cars are domestic: $\binom{15}{6} = 5005$ selections. Counting the opposite,

$$P(\text{at least one foreign}) = 1 - \frac{\binom{15}{6}}{\binom{25}{6}} = \frac{177\,100 - 5005}{177\,100} = \frac{172\,095}{177\,100} = \boxed{\frac{447}{460}} \approx 0.9717.$$

e) "At least four domestic" means exactly 4, 5 or 6 domestic cars, the others being foreign. The three cases do not overlap, so their counts are added:

$$\begin{aligned}\binom{15}{4}\binom{10}{2} + \binom{15}{5}\binom{10}{1} + \binom{15}{6}\binom{10}{0} &= 1365\cdot 45 + 3003\cdot 10 + 5005\cdot 1 \\ &= 61\,425 + 30\,030 + 5005 = 96\,460,\end{aligned}$$

$$P(\text{at least four domestic}) = \frac{96\,460}{177\,100} = \boxed{\frac{689}{1265}} \approx 0.5447.$$

Probabilities of this type, for a sample drawn without replacement from a population made of two groups, return in Section 19 as the hypergeometric distribution.

Caution (a tempting wrong count for "at least one")

In Example 9.4 d) it is tempting to argue: "choose one foreign car to make sure there is one (10 ways), then choose any 5 of the other 24 cars ($\binom{24}{5} = 42\,504$ ways)", for a total of $10 \cdot 42\,504 = 425\,040$. This is more than the 177 100 selections that exist, so it cannot be right. The construction produces the same selection several times: a selection with three foreign cars is obtained three times, once for each of its foreign cars playing the role of the car chosen first. For "at least one", count the opposite ("none") and subtract, or add the cases "exactly 1", "exactly 2", and so on.

10 Counting: Mixed Problems

The problems of this section need nothing beyond Sections 8 and 9. What makes them harder is that a problem no longer announces which tool it calls for, and that one count often combines several tools. The plan below turns the rules of the last two sections into a routine; the worked examples then apply it.

10.1 A plan for counting problems

A plan for any counting problem

1. **Describe one outcome.** Write down a typical example of what is being counted (one plate, one seating, one committee, one hand of cards) and think of how it could be built step by step.
2. **Does the order matter?** If rearranging the same objects gives a different outcome, count arrangements (slots, $n!$, ${}_nP_k$). If not, count selections (${}_nC_k$).
3. **Is repetition allowed?** With repetition the number of choices is the same in every slot; without repetition it drops by one each time.
4. **Restrictions first.** Deal with the most restricted positions or objects before the free ones, and glue together objects that must stay together.
5. **"And" multiplies, "or" adds.** Multiply the numbers of choices of successive steps; add the counts of cases that do not overlap.
6. **"Not", "at least one": count the opposite.** Subtract the outcomes that are not wanted from the total whenever they are easier to count.
7. **For a probability,** count the favourable outcomes and all the possible outcomes in the same way (both with order, or both without) and divide.

Caution (count every outcome exactly once)

Once a count is built, ask whether the same outcome could have been produced in two different ways. If it could, it has been counted more than once. The usual causes are cases that overlap, and objects that play the *same* role being chosen one after the other instead of all at once with a combination (the members of a committee, or the three unpaired cards in Example 10.3). A reliable safeguard is to count in two different ways, for instance directly and through the opposite, and compare.

10.2 Worked examples

Example 10.1 ⋆ ▪ **A group that must avoid one pair** Worksheet 10, #1

a) In how many ways can a hostess select six guests for lunch from 10 women?

b) Two of the ten women refuse to see each other. If the hostess selects her six guests at random, what is the probability that she selects a "safe" group, i.e., a group that does not contain both women together?

Solution.

a) A group of guests has no order, so the hostess has $\binom{10}{6} = \boxed{210 \text{ ways}}$.

b) The 210 groups are equally likely. It is easiest to count the opposite: an *unsafe* group contains both of the two women. Both of them are in the group (1 way), and the other 4 guests are chosen among the remaining 8 women:

$$\binom{2}{2}\binom{8}{4} = 1 \cdot 70 = 70 \text{ unsafe groups.}$$

So there are $210 - 70 = 140$ safe groups, and

$$P(\text{safe group}) = \frac{\binom{10}{6} - \binom{2}{2}\binom{8}{4}}{\binom{10}{6}} = \frac{140}{210} = \boxed{\frac{2}{3}} \approx 0.6667.$$

Check by cases: a safe group contains neither of the two women, or exactly one of them:

$$\binom{2}{0}\binom{8}{6} + \binom{2}{1}\binom{8}{5} = 28 + 2 \cdot 56 = 140.$$

Example 10.2 ⋆ ▪ **Two objects that must not be neighbours** Worksheet 10, #7

In a set of 10 different exams, 2 are Mathematics (NYA & NYB). In how many different orders can the exams be given if the two mathematics exams are not consecutive?

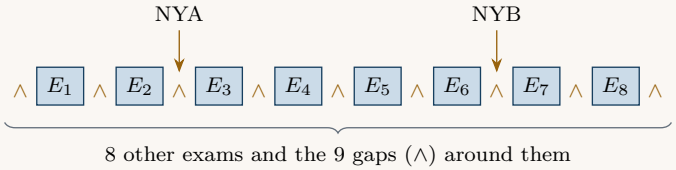
Solution. "Not consecutive" is awkward to count directly, so count the opposite.

- *All orders:* 10 different exams can be given in $10! = 3\,628\,800$ orders.
- *Orders in which the two mathematics exams are consecutive:* glue NYA and NYB into one block. There are then 9 objects to order (9! ways), and inside the block the two exams can come in 2! orders: $2! \cdot 9! = 725\,760$ orders.

Subtracting,

$$10! - 2! \cdot 9! = 3\,628\,800 - 725\,760 = \boxed{2\,903\,040 \text{ orders}}.$$

Check by a direct count: first order the 8 other exams ($8! = 40\,320$ ways). They leave 9 gaps: one before the first exam, 7 between two exams and one after the last. The mathematics exams are not consecutive exactly when they go into two *different* gaps: 9 choices for NYA, then 8 for NYB. This gives $8! \cdot 9 \cdot 8 = 40\,320 \cdot 72 = 2\,903\,040$ orders again.



Example 10.3 ⋆ ▪ **Poker hands** Worksheet 10, #8

A standard deck of cards has 52 cards divided into 4 suits (spades, hearts, diamonds & clubs). Each suit has 13 different cards from 2–10 with Jack, Queen, King and Ace. A player in the game of poker is dealt a hand of five cards randomly.

a) How many different poker hands are there?

b) What's the probability of getting a full house (a pair + three of a kind)?

c) What's the probability of getting exactly one pair?

Solution. The deck has 13 *ranks* $(2, 3, \ldots, 10$, Jack, Queen, King, Ace) and 4 cards of each rank, one per suit. A hand is a set of 5 cards: the order in which they are dealt does not matter.

a) There are $\binom{52}{5} = \boxed{2\,598\,960 \text{ hands}}$, all equally likely.

b) A full house consists of two cards of one rank and three cards of another rank, for example two kings and three fives. Build it step by step: choose the rank of the pair, then 2 of the 4 cards of that rank; choose the rank of the triple among the 12 other ranks, then 3 of the 4 cards of that rank.

$$\underbrace{\binom{13}{1}}_{\text{rank of the pair}} \underbrace{\binom{4}{2}}_{\text{its 2 cards}} \underbrace{\binom{12}{1}}_{\text{rank of the triple}} \underbrace{\binom{4}{3}}_{\text{its 3 cards}} = 13 \cdot 6 \cdot 12 \cdot 4 = 3744 \text{ hands},$$

$$P(\text{full house}) = \frac{3744}{2\,598\,960} = \boxed{\frac{6}{4165}} \approx 0.0014.$$

The two ranks are chosen one after the other ($13 \cdot 12$ ways) because they play different roles: two kings with three fives is not the same hand as two fives with three kings.

c) "Exactly one pair" means two cards of one rank, and three more cards that neither match the pair nor each other (otherwise the hand would hold three of a kind, two pairs or a full house). So the three other cards have 3 *different* ranks, taken among the 12 ranks not used by the pair. These three ranks play the same role, so they are chosen all at once, with a combination; each of them then contributes one of its 4 cards.

$$\underbrace{\binom{13}{1}}_{\text{rank of the pair}} \underbrace{\binom{4}{2}}_{\text{its 2 cards}} \underbrace{\binom{12}{3}}_{\text{3 other ranks}} \underbrace{\binom{4}{1}\binom{4}{1}\binom{4}{1}}_{\text{one card of each}} = 13 \cdot 6 \cdot 220 \cdot 64 = 1\,098\,240 \text{ hands},$$

$$P(\text{exactly one pair}) = \frac{1\,098\,240}{2\,598\,960} = \boxed{\frac{352}{833}} \approx 0.4226.$$

Part III

Probability

Events are sets, and probability is a way of measuring them. This part sets up the language of sample spaces and events, then the rules for combining probabilities: the complement and addition rules, conditional probability, the multiplication rule, Bayes' theorem and independence.

11 Sets and Probability

Probability is written in the language of sets: the possible results of an experiment form a set, the things whose probability we want are subsets of it, and the words "not", "or" and "and" become operations on those subsets. This section sets up that language, pictures it with Venn diagrams, and uses the diagrams to count. When a set is too large to list, its size comes from the counting techniques of Sections 8–10.

11.1 Experiments, sample spaces and events

Definition (experiment, sample space)

An *experiment* is any process which generates data, for example: toss a coin three times and record the results (heads or tails). An experiment of this kind is often called a *probability experiment*.

The *sample space* of an experiment is the collection of all its possible outcomes. It is denoted S, Ω or U (for *universal set*).

Definition (event)

An *event* is any subset of the sample space, that is, any collection of possible outcomes. An event which contains only a single outcome is called a *simple event*, whereas an event which contains several outcomes is called a *compound event*.

Take the experiment of tossing a coin three times, and write H for heads and T for tails. An outcome is a string such as HTH, and the sample space is

$$S = \{\text{HHH},\ \text{HHT},\ \text{HTH},\ \text{THH},\ \text{HTT},\ \text{THT},\ \text{TTH},\ \text{TTT}\}.$$

"All three tosses are heads" is the simple event {HHH}, and "exactly two heads" is the compound event {HHT, HTH, THH}. An event *occurs* when the outcome of the experiment is one of its elements.

Definition (cardinality)

The *cardinality* or *size* of a set is the number of elements contained in the set. The size of a set A is written $N(A)$; for example, if the set S contains 8 elements, we write $N(S) = 8$. When S is the sample space, we often write N for $N(S)$.

Definition (probability of an event)

The *probability* of an event A is the ratio of the size of A to the size of the sample space:

$$P(A) = \frac{N(A)}{N(S)} = \frac{N(A)}{N}.$$

For three tosses of a fair coin, $N = 8$ and three of the outcomes have exactly two heads, so the probability of getting exactly two heads is $\frac{3}{8}$.

Property (range of a probability)

Any probability p satisfies $0 \le p \le 1$. Moreover $P(S) = 1$.

The reason is that an event A is a part of S, so $0 \le N(A) \le N$. An event of probability 0 never occurs, and an event of probability 1, such as S itself, is certain to occur.

Caution (equally likely outcomes)

The formula $P(A) = N(A)/N$ treats all the outcomes in S alike, so it is valid only when the outcomes are *equally likely*: a fair coin, a balanced die, a person or an object "selected at random". In the three-toss experiment the eight strings are equally likely. The four possible *numbers of heads* $0, 1, 2, 3$ are not, so counting them would give the wrong value $\frac{1}{4}$ for the probability of two heads.

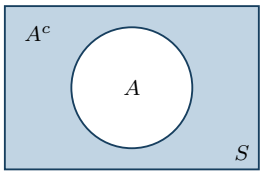
11.2 Combining events: complement, union and intersection

Definition (complement, union, intersection)

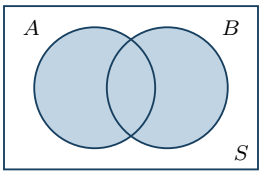
Let A and B be events in a sample space S.

- The *complement* of A is written A^c (also A' or $\overline{A}$) and consists of all the elements of S which are *not* in A.
- The *union* of A and B is written $A \cup B$, is read "A union B" or "A *or* B", and is the collection of all the elements which are in A, in B or *in both*.
- The *intersection* of A and B is written $A \cap B$, is read "A intersect B" or "A *and* B", and is the collection of all the elements which are in both A and B *at the same time*.

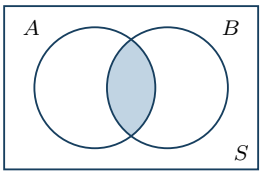
A *Venn diagram* shows these sets as regions: the sample space is a rectangle and each event is a circle inside it.



A^c: "not A"



$A \cup B$: "A or B"



$A \cap B$: "A and B"

Note ("or" includes "both")

In everyday speech "or" often means one or the other but not both. In probability, "A or B" always includes the case where both happen: $A \cup B$ is the event that *at least one* of A and B occurs.

Definition (empty set)

The *empty set* (or *empty event*) is written $\emptyset$ and is the set which contains no elements. It has

$$N(\emptyset) = 0, \qquad P(\emptyset) = 0.$$

Definition (mutually exclusive events)

Two events A and B are called *mutually exclusive* or *disjoint* if their intersection is empty:

$$A \cap B = \emptyset.$$

They have no outcome in common, so they cannot occur at the same time. More than two events are mutually

exclusive if every pair of them is disjoint, so that at most one of the events can occur.

An event and its complement are always mutually exclusive: $A \cap A^c = \emptyset$ for any event A.

Definition (partition)

Events $A_1, \ldots, A_k$ are called a *partition* of S if all the A_i are non-empty and disjoint, and if

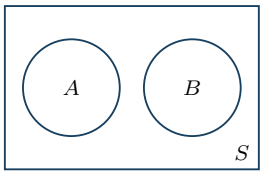
$$A_1 \cup A_2 \cup \cdots \cup A_k = S.$$

Events whose union is the whole sample space are called *exhaustive*. A partition is therefore a collection of non-empty events that is both mutually exclusive and exhaustive: it cuts S into pieces so that every outcome lies in exactly one piece. For example, A and A^c form a partition of S whenever both are non-empty.

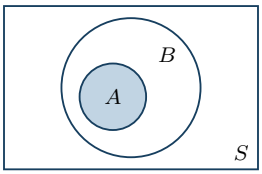
Definition (containment)

Event A is said to be *contained in* event B if *all* the elements of A are also elements of B. We write $A \subseteq B$. In particular, $A \subseteq S$ for any event A.

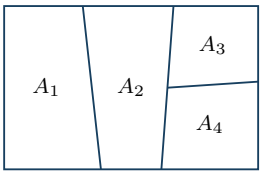
If $A \subseteq B$, then B occurs whenever A does, and $N(A) \leq N(B)$, so $P(A) \leq P(B)$.



mutually exclusive: $A \cap B = \emptyset$



$A \subseteq B$



a partition $A_1, \ldots, A_4$ of S

11.3 Properties of set operations

Property (basic identities)

For any events A, B and C in a sample space S:

$$\begin{array}{lll} A \cap \emptyset = \emptyset & A \cap A^c = \emptyset & S^c = \emptyset \\ A \cup \emptyset = A & A \cup A^c = S & \emptyset^c = S \end{array}$$

$$(A^c)^c = A, \qquad C \cap (A \cup B) = (C \cap A) \cup (C \cap B).$$

Each identity can be checked on a Venn diagram or straight from the definitions. For instance, $A \cup A^c = S$ says that every outcome is either in A or not in A. The last identity is the *distributive law*: the part of C that lies in $A \cup B$ is made of the part of C in A together with the part of C in B.

Theorem (De Morgan's laws)

For any events A and B,

$$(A \cap B)^c = A^c \cup B^c, \qquad (A \cup B)^c = A^c \cap B^c.$$

The laws extend to any number of events $A_1, A_2, \ldots, A_n$:

$$\Big(\bigcup_i A_i\Big)^c = \bigcap_i A_i^c, \qquad \Big(\bigcap_i A_i\Big)^c = \bigcup_i A_i^c.$$

In words: taking a complement turns a union into an intersection and an intersection into a union.

- $(A \cup B)^c = A^c \cap B^c$: "not ($A$ or B)" is the same as "neither A nor B".
- $(A \cap B)^c = A^c \cup B^c$: "not both A and B" is the same as "at least one of the two does not occur".

Proof

An outcome is in $(A \cup B)^c$ exactly when it is not in $A \cup B$, that is, when it is in neither A nor B. This says that it is in A^c and in B^c, that is, in $A^c \cap B^c$. So the two sets have the same elements. In the same way, an outcome is in $(A \cap B)^c$ exactly when it is missing from at least one of A and B, that is, when it is in A^c or in B^c. The argument for n events is identical. □

Example 11.2 checks both laws with Venn diagrams.

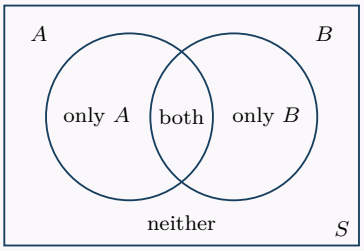
Definition (symmetric difference)

The *symmetric difference* of A and B, written $A \triangle B$, is the set of elements which belong to *exactly one* of A and B:

$$A \triangle B = (A \cap B^c) \cup (A^c \cap B).$$

Translating words into set notation

In words	In symbols
A does not occur	A^c
A or B; at least one of A, B	$A \cup B$
A and B; both	$A \cap B$
A but not B; only A	$A \cap B^c$
neither A nor B	$A^c \cap B^c = (A \cup B)^c$
not both	$A^c \cup B^c = (A \cap B)^c$
exactly one of A, B	$(A \cap B^c) \cup (A^c \cap B)$



11.4 Counting with Venn diagrams

Two events A and B cut the sample space into four regions that do not overlap: only A, both, only B, and neither (see the diagram in the box above). Three events cut it into eight regions. Because the regions do not overlap, the size of any event is the sum of the sizes of the regions that make it up. Two consequences are used all the time.

Theorem (counting rules)

For any events A and B in a sample space S with N elements,

$$N(A \cup B) = N(A) + N(B) - N(A \cap B),$$
$$N(A) + N(A^c) = N, \qquad \text{or} \qquad N(A) = N - N(A^c).$$

Why?

In the sum $N(A) + N(B)$ every element of the overlap $A \cap B$ is counted twice, once as an element of A and once as an element of B. Subtracting $N(A \cap B)$ removes the extra count. For the second rule, every element of S is in exactly one of A and A^c.

If A and B are mutually exclusive, then $N(A \cap B) = 0$ and the first rule reduces to $N(A \cup B) = N(A) + N(B)$, the addition rule for counting of Section 8.

Dividing both rules by N turns them into rules for probabilities, because $P(\,\cdot\,) = N(\,\cdot\,)/N$:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \qquad P(A) = 1 - P(A^c).$$

These are the addition rule and the complement rule, the subject of Section 12.

Solving a counting problem with a Venn diagram

1. **Name** the sample space and the events with letters.
2. **Translate** each piece of given information into set notation, for example "27 have both" becomes $N(A \cap B) = 27$ and "34 have neither" becomes $N\big((A \cup B)^c\big) = 34$.
3. **Fill in** the regions of the diagram from the inside out: start with the innermost region (the intersection) and subtract to get the regions around it. A number given for a whole circle has to be shared among all the regions of that circle. If a region cannot be found directly, call it x and write an equation from one of the totals.
4. **Read off** the answer by adding the regions that make up the event. For a probability, divide by N.

11.5 Worked examples

Example 11.1 ▪ Tossing a coin four times Lecture notes

A fair coin is tossed 4 times and the result of each toss is recorded (H: heads, T: tails).

a) Describe the sample space S and find its cardinality.

b) Let A be the event "get 4 heads". Is A a simple or a compound event? Find $P(A)$.

c) Let B be the event "get exactly 3 heads". Is B a simple or a compound event? Find $P(B)$.

d) Are A and B mutually exclusive?

e) Find the probability of getting at least 3 heads, and the probability of getting at least one tail.

Solution.

a) An outcome is a string of four letters, one for each toss. Each toss has 2 possible results, so by the multiplication rule (Section 8) there are $2 \cdot 2 \cdot 2 \cdot 2 = 16$ outcomes: $N(S) = 16$. Sorted by the number of heads, they are:

Heads	Outcomes	How many
4	HHHH	1
3	THHH, HTHH, HHTH, HHHT	4
2	HHTT, HTHT, HTTH, THHT, THTH, TTHH	6
1	HTTT, THTT, TTHT, TTTH	4
0	TTTT	1

The coin is fair, so the 16 outcomes are equally likely. (The last column lists the combinations ${}_4C_k$ of Section 9: choose which k of the 4 tosses are heads.)

b) $A = \{\text{HHHH}\}$ contains a single outcome, so it is a simple event. Its cardinality is $N(A) = 1$, and

$$P(A) = \frac{N(A)}{N(S)} = \boxed{\frac{1}{16}}.$$

c) $B = \{\text{THHH, HTHH, HHTH, HHHT}\}$ contains several outcomes, so it is a compound event. $N(B) = 4$ (there are 4 places for the single tail), and

$$P(B) = \frac{N(B)}{N(S)} = \frac{4}{16} = \boxed{\frac{1}{4}}.$$

d) $\boxed{\text{Yes}}$. No outcome has 4 heads and exactly 3 heads at the same time, so $A \cap B = \emptyset$: the events are disjoint, that is, mutually exclusive.

e) "At least 3 heads" means 3 heads or 4 heads, which is the event $A \cup B$. By the counting rule,

$$N(A \cup B) = N(A) + N(B) - N(A \cap B) = 1 + 4 - 0 = 5, \qquad P(A \cup B) = \boxed{\frac{5}{16}}.$$

"At least one tail" is the complement of "4 heads", so it is A^c:

$$N(A^c) = N(S) - N(A) = 16 - 1 = 15, \qquad P(A^c) = \boxed{\frac{15}{16}}.$$

Example 11.2 ▪ De Morgan's laws in pictures Worksheet 11, #1

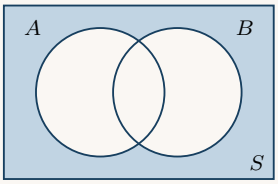
a) In a Venn diagram of two events A and B, shade the region that corresponds to each of the expressions $A^c \cap B^c$, $(A \cup B)^c$, $A^c \cup B^c$, $(A \cap B)^c$.

b) What do you notice about the regions in (a)?

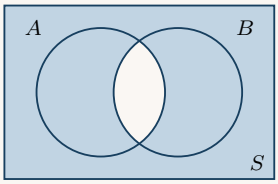
Solution.

a) Read each expression as a sentence about where a point lies.

- $A^c \cap B^c$: outside A *and* outside B. This is the part of S outside both circles.
- $(A \cup B)^c$: $A \cup B$ is everything inside at least one circle, so its complement is again everything outside both circles.
- $A^c \cup B^c$: outside A *or* outside B. The only points left out are those inside both circles, so this is everything except the overlap.
- $(A \cap B)^c$: $A \cap B$ is the overlap, so its complement is again everything except the overlap.



$A^c \cap B^c = (A \cup B)^c$



$A^c \cup B^c = (A \cap B)^c$

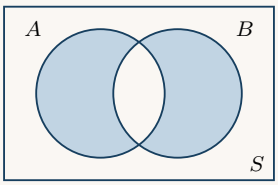
b) The four regions agree in pairs:

$$A^c \cap B^c = (A \cup B)^c \qquad \text{and} \qquad A^c \cup B^c = (A \cap B)^c.$$

These are De Morgan's laws.

Example 11.3 ▪ Naming a shaded region Worksheet 11, #2

Express the shaded region in terms of unions, intersections and complements of A and B.



Solution. The shaded region has two separate pieces, so describe each piece and take the union.

- The left piece is inside A but outside B: it is $A \cap B^c$.
- The right piece is inside B but outside A: it is $A^c \cap B$.

The region is therefore

$$\boxed{(A^c \cap B) \cup (A \cap B^c)}.$$

It contains the outcomes that are in exactly one of A and B, so it is the symmetric difference $A \triangle B$. Another correct description is $(A \cup B) \cap (A \cap B)^c$: in at least one of the two events, but not in both.

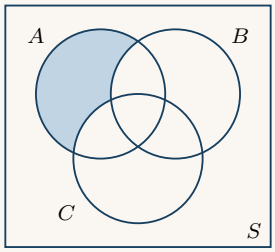
Example 11.4 ▪ Shading a region for three events Worksheet 11, #3

In a Venn diagram of three events A, B and C, shade in the region that corresponds to $A \cap (B \cup C)^c$.

Solution. Work from the inside of the expression outwards.

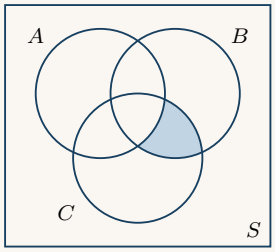
- $B \cup C$ is everything inside circle B or circle C.
- $(B \cup C)^c$ is everything outside both of these circles. (By De Morgan's law it equals $B^c \cap C^c$.)
- Intersecting with A keeps only the points of A that are outside B and outside C.

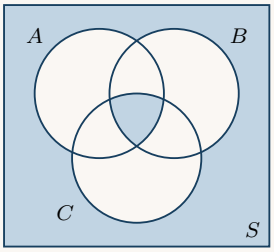
So $A \cap (B \cup C)^c$ is the part of A that belongs to neither B nor C: the outcomes that are in A *only*.



Example 11.5 ▪ Describing regions for three events Worksheet 11, #4

For each of the Venn diagrams below, describe the shaded region in English and in terms of intersections and unions of the events A, B and C.

a)


b)


Solution.

a) The shaded region is inside B and inside C, but outside A. In English: the elements that are in both B and C but not in A. In symbols:

$$\boxed{(B \cap C) \cap A^c}.$$

b) There are two separate pieces. The small piece in the centre is where all three circles overlap, $A \cap B \cap C$. The large piece is everything outside all three circles, $A^c \cap B^c \cap C^c$, which by De Morgan's law is also $(A \cup B \cup C)^c$. The region is the union of the two pieces:

$$\boxed{(A \cap B \cap C) \cup (A^c \cap B^c \cap C^c)} = (A \cap B \cap C) \cup (A \cup B \cup C)^c.$$

In English: the elements that are in all three of A, B and C, or in none of them.

Example 11.6 ▪ Insurance policies: filling in a Venn diagram Worksheet 11, #5

Out of 115 clients of an insurance company who have a basic home insurance policy, 58 also have an extended fire protection policy, 27 have both extended fire protection and extended flood protection, and 34 have neither extension.

a) How many clients have only extended flood protection (but not extended fire protection)?

b) How many have exactly one of the extended policies?

Draw a Venn diagram.

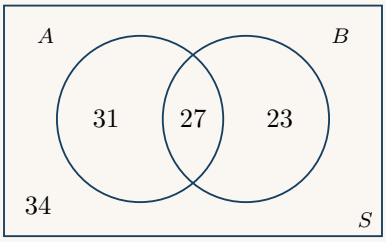
Solution. *Name the sets.* The sample space S is the set of the 115 clients who have the basic policy. Let A be the set of clients with the fire extension and B the set of clients with the flood extension.

Translate the given information.

$$N(S) = 115, \qquad N(A) = 58, \qquad N(A \cap B) = 27, \qquad N\big((A \cup B)^c\big) = 34.$$

Solve for the Venn diagram, from the inside out.

- Both extensions: $N(A \cap B) = 27$ goes in the overlap.
- Fire only: the 58 clients of A include the 27 in the overlap, so $58 - 27 = 31$ have fire only.
- Neither: 34 goes outside both circles.
- Flood only: the four regions add up to 115, so the last region is $115 - 31 - 27 - 34 = 23$.



A: fire extension, B: flood extension

a) "Only flood protection" is the region $A^c \cap B$: $\boxed{23 \text{ clients}}$.

b) "Exactly one of the extended policies" is fire only or flood only: $31 + 23 = \boxed{54 \text{ clients}}$.

Example 11.7 ▪ Cats and dogs: unknown regions

Worksheet 11, #7

In a group of 35 people, 14 do not have a pet, 12 have at least one cat, and 14 have at least one dog. A person from this group is selected at random. What is the probability that

a) they own both at least one cat and at least one dog?

b) they do not own any cats?

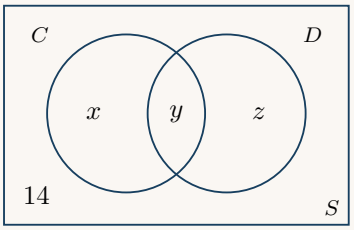
Solution. Let S be the group of 35 people, and let

$$C = \{\text{people with at least one cat}\}, \qquad D = \{\text{people with at least one dog}\}.$$

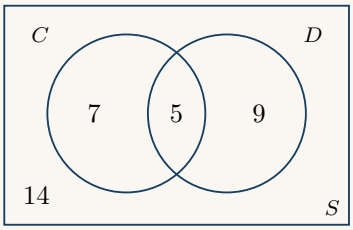
The only pets in this problem are cats and dogs, so the 14 people without a pet are the ones outside both circles. None of the three regions inside the circles is given directly, so call them x (cat only), y (cat and dog) and z (dog only). The data give three equations:

$$\begin{cases} x + y = 12 & \text{(cat owners)} \\ y + z = 14 & \text{(dog owners)} \\ x + y + z = 35 - 14 = 21 & \text{(people with a pet)} \end{cases}$$

Subtracting the first equation from the third gives $z = 21 - 12 = 9$, subtracting the second from the third gives $x = 21 - 14 = 7$, and then $y = 12 - 7 = 5$.



set-up



solved: $7+5+9+14=35$

a) "Both" is the event $C \cap D$, which has $y = 5$ people:

$$P(C \cap D) = \frac{N(C \cap D)}{N} = \frac{5}{35} = \boxed{\frac{1}{7}}.$$

The counting rule gives the same number without any unknowns: 21 people have a pet, so $N(C \cap D) = N(C) + N(D) - N(C \cup D) = 12 + 14 - 21 = 5$.

b) "No cats" is the complement of C, and $N(C^c) = N - N(C) = 35 - 12 = 23$ (in the diagram, $9 + 14$):

$$P(C^c) = \boxed{\frac{23}{35}}.$$

Example 11.8 ⋆ ▪ **Three languages: a three-circle diagram** Worksheet 11, #8

The following information is known about the languages spoken by the delegates at a diplomatic convention.

- There are 24 delegates.
- All speak either German, Italian or Cantonese.
- 6 speak German.
- 12 speak Italian.
- 15 speak Cantonese.
- 1 speaks all three languages.
- 2 speak German and Italian but not Cantonese.
- 2 speak German and Cantonese, but not Italian.

What is the probability that a randomly selected delegate speaks only Cantonese? (Draw a Venn diagram.)

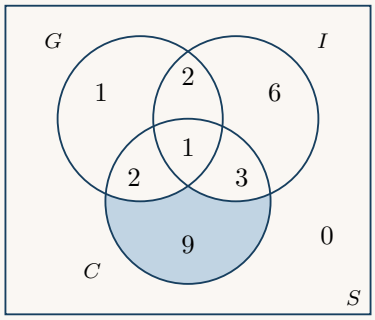
Solution. Let G, I and C be the sets of delegates who speak German, Italian and Cantonese. Three circles make eight regions. Fill them in from the centre outwards.

- All three languages: 1. German and Italian only: 2. German and Cantonese only: 2.
- German only: the 6 German speakers include these three regions, so $6 - 1 - 2 - 2 = 1$.
- Italian and Cantonese but not German: not given, so call it x.
- Italian only: $12 - 1 - 2 - x = 9 - x$. Cantonese only: $15 - 1 - 2 - x = 12 - x$.
- Outside all three circles: 0, because every delegate speaks at least one of the languages.

The eight regions must add up to the 24 delegates:

$$1 + 2 + 2 + 1 + x + (9 - x) + (12 - x) + 0 = 24 \quad \Longrightarrow \quad 27 - x = 24 \quad \Longrightarrow \quad x = 3.$$

So 6 delegates speak only Italian and $12 - 3 = 9$ speak only Cantonese.



A delegate is selected at random, so all 24 are equally likely and

$$P(\text{only Cantonese}) = P(G^c \cap I^c \cap C) = \frac{9}{24} = \boxed{\frac{3}{8}}.$$

Check with the circle totals: Italian $6+2+1+3=12$ and Cantonese $9+2+1+3=15$, as given.

12 The Addition Rules

Section 11 ended with two counting rules, one for the size of a union and one for the size of a complement. Divided by the size of the sample space, they become the two most used rules of probability: the addition rule for $P(A \cup B)$ and the complement rule for $P(A^c)$. This section states both rules, shows how they combine with De Morgan's laws, and extends the addition rule to three events.

12.1 The addition rule for two events

Theorem (general addition rule)

For any two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

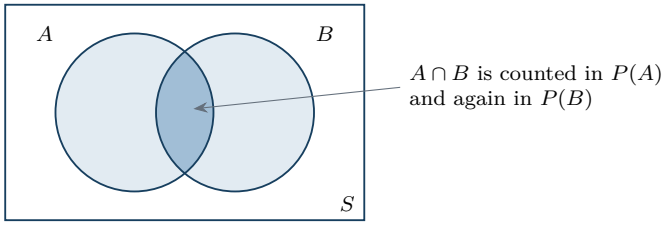
In words: the probability that A or B occurs is the probability of A, plus the probability of B, minus the probability that both occur.

Proof

By the counting rule of Section 11, $N(A \cup B) = N(A) + N(B) - N(A \cap B)$. Dividing every term by $N = N(S)$ gives

$$P(A \cup B) = \frac{N(A \cup B)}{N} = \frac{N(A)}{N} + \frac{N(B)}{N} - \frac{N(A \cap B)}{N} = P(A) + P(B) - P(A \cap B).$$

The subtraction is there for the same reason as in the counting rule: the sum $P(A) + P(B)$ includes the overlap $A \cap B$ twice, so it has to be taken away once. □



Remark (beyond equally likely outcomes)

The proof above counts equally likely outcomes, but the rule itself holds for every way of assigning probabilities. Think of P as an area in a Venn diagram whose rectangle has total area 1: the argument about the overlap is unchanged. All the rules of this section may therefore be used with probabilities that are given directly as decimals or percentages.

Recall (mutually exclusive events)

Two events A and B are *mutually exclusive* (or *disjoint*) if $A \cap B = \emptyset$: they have no outcome in common, so they cannot occur at the same time. In that case $P(A \cap B) = P(\emptyset) = 0$.

Corollary (addition rule for mutually exclusive events)

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

More generally, if the events $A_1, A_2, \ldots, A_k$ are mutually exclusive (no two of them can occur together), then

$$P(A_1 \cup A_2 \cup \cdots \cup A_k) = P(A_1) + P(A_2) + \cdots + P(A_k).$$

Probabilities may be added without any correction only when the events are mutually exclusive. If the events can occur together, the plain sum counts the overlap twice; it can even come out larger than 1. The same was true of the addition rule for counting in Section 8, which required cases that do not overlap; the general addition rule is that rule with the overlap corrected.

Caution (finding $P(A \cap B)$)

The product formula

$$P(A \cap B) = P(A) \cdot P(B)$$

is true only if A and B are *independent* of each other (Section 14). It is not a general rule. When nothing is said about independence, find $P(A \cap B)$ another way. The addition rule links the four numbers $P(A)$, $P(B)$, $P(A \cup B)$ and $P(A \cap B)$, so any three of them give the fourth; in particular

$$P(A \cap B) = P(A) + P(B) - P(A \cup B).$$

12.2 The complement rule

Theorem (complement rule)

For any event A,

$$P(A^c) = 1 - P(A), \qquad \text{or equivalently} \qquad P(A) = 1 - P(A^c).$$

Proof

Every outcome is in exactly one of A and A^c, so $N(A) = N - N(A^c)$. Dividing by N gives

$$P(A) = \frac{N(A)}{N} = \frac{N}{N} - \frac{N(A^c)}{N} = 1 - P(A^c).$$

Moving $P(A^c)$ to the other side gives $P(A^c) = 1 - P(A)$. □

The complement rule is also a special case of the addition rule: A and A^c are mutually exclusive and $A \cup A^c = S$, so $P(A) + P(A^c) = P(S) = 1$.

Combined with De Morgan's laws, the complement rule gives the probabilities of "neither" and "not both".

Corollary (neither, not both)

For any events A and B,

$$P(A^c \cap B^c) = P\big((A \cup B)^c\big) = 1 - P(A \cup B) \qquad \text{(neither } A \text{ nor } B\text{)},$$
$$P(A^c \cup B^c) = P\big((A \cap B)^c\big) = 1 - P(A \cap B) \qquad \text{(not both } A \text{ and } B\text{)}.$$

Note (when to use the complement)

Use the complement rule whenever the opposite event is easier to handle than the event itself. The standard case is "at least one": its complement is "none", so $P(\text{at least one}) = 1 - P(\text{none})$.

12.3 Splitting an event: "only" and "exactly one"

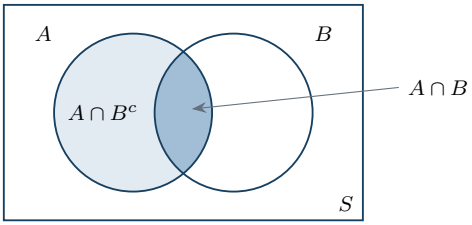
Property (splitting an event)

For any events A and B,

$$P(A \cap B^c) = P(A) - P(A \cap B), \qquad P(A^c \cap B) = P(B) - P(A \cap B).$$

The probability that *exactly one* of A and B occurs is

$$P\big[(A \cap B^c) \cup (A^c \cap B)\big] = P(A \cap B^c) + P(A^c \cap B) = P(A \cup B) - P(A \cap B).$$



The event A splits into the part outside B and the part inside B: $P(A) = P(A \cap B^c) + P(A \cap B)$.

Proof

The event A is made of two pieces that cannot occur together: the part of A outside B and the part of A inside B. That is, $A = (A \cap B^c) \cup (A \cap B)$ with the two pieces mutually exclusive, so $P(A) = P(A \cap B^c) + P(A \cap B)$. This is the first formula, and the second is the same with the roles of A and B exchanged. Finally, "only A" and "only B" are mutually exclusive, so their probabilities add, and since $A \cup B$ consists of the three mutually exclusive pieces "only A", "both" and "only B", the sum equals $P(A \cup B) - P(A \cap B)$. □

Note (a Venn diagram of probabilities)

The four regions of a two-event Venn diagram (only A, both, only B, neither) are mutually exclusive and together make up S, so their probabilities add up to 1. Writing the probability of each region into the diagram, exactly as counts were written into it in Section 11, reduces every question about A and B to adding up regions (Examples 12.1 and 12.2).

12.4 The addition rule for three events

Theorem (addition rule for three events)

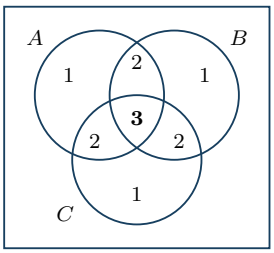
For any three events A, B and C,

$$\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) \\ & - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ & + P(A \cap B \cap C). \end{aligned}$$

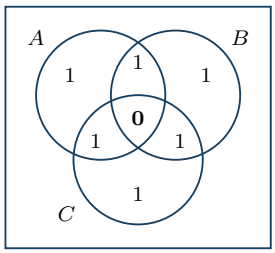
In words: add the three single events, subtract the three pairs, add back the triple.

Derivation from the Venn diagram

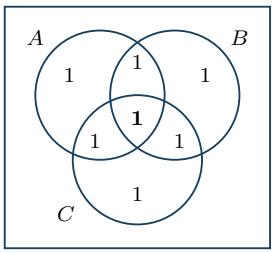
Three circles cut $A \cup B \cup C$ into seven regions. The goal is to count the probability of each region exactly once. The diagrams show how many times each region has been counted after each step.



after adding $P(A)$, $P(B)$, $P(C)$



after subtracting the three pairs



after adding $P(A \cap B \cap C)$

1. The sum $P(A) + P(B) + P(C)$ counts each region once for every circle that contains it: regions in one circle only are counted once, regions in exactly two circles twice, and the centre $A \cap B \cap C$ three times.
2. Subtracting $P(A \cap B)$, $P(A \cap C)$ and $P(B \cap C)$ removes one count from each region that lies in exactly two circles, which is what is needed. But the centre lies in all three of these intersections, so it loses three counts and is now not counted at all.
3. Adding $P(A \cap B \cap C)$ puts the centre back once.

Every region of $A \cup B \cup C$ is now counted exactly once, which proves the formula. □

Remark

The same formula comes out of algebra alone: apply the two-event rule to the union of $A \cup B$ and C, and expand $P\big((A \cup B) \cap C\big)$ with the distributive law $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ and the two-event rule again. If A, B and C are mutually exclusive, every intersection is empty and the formula reduces to $P(A \cup B \cup C) = P(A) + P(B) + P(C)$.

12.5 Worked examples

Example 12.1 ▪ Two traffic lights Worksheet 12, #2

The route used by a certain motorist in commuting to work contains two intersections with traffic lights. The probability that she must stop at the first light (A) is 0.4 and 0.5 at the second (B). The probability that she must stop at at least one of the two lights ($A \cup B$) is 0.6. What is the probability that she must stop:

a) at both lights?

b) at the first light but not at the second?

c) at exactly one light?

Solution. The given information is $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.6$.

a) "Both lights" is the event $A \cap B$. Solve the addition rule for the intersection:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.4 + 0.5 - 0.6 = \boxed{0.3}.$$

Multiplying $P(A) \cdot P(B) = 0.4 \times 0.5 = 0.2$ would be wrong here: nothing says that the two lights are independent of each other, and the answer $0.3 \neq 0.2$ shows that they are not.

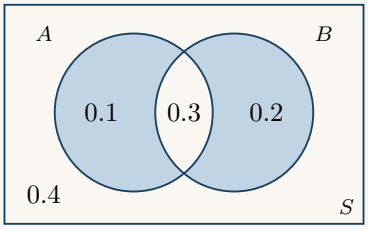
b) "First but not second" is the event $A \cap B^c$, the part of A outside B:

$$P(A \cap B^c) = P(A) - P(A \cap B) = 0.4 - 0.3 = \boxed{0.1}.$$

c) "Exactly one light" means the first but not the second, or the second but not the first: $(A \cap B^c) \cup (A^c \cap B)$. The second piece has probability $P(A^c \cap B) = P(B) - P(A \cap B) = 0.5 - 0.3 = 0.2$. Now apply the addition rule to the two pieces:

$$\begin{aligned} P\big[(A \cap B^c) \cup (A^c \cap B)\big] &= P(A \cap B^c) + P(A^c \cap B) - P\big[(A \cap B^c) \cap (A^c \cap B)\big] \\ &= 0.1 + 0.2 - 0 = \boxed{0.3}. \end{aligned}$$

The last term is 0 because the two pieces are mutually exclusive: no outcome is in A and in A^c at the same time.



The four regions add up to 1. Shaded: exactly one light, $0.1 + 0.2 = 0.3$, which is also $P(A \cup B) - P(A \cap B) = 0.6 - 0.3$.

Example 12.2* ▪ Three telephone options

Worksheet 12, #4

The three most popular options on a domestic land line offered by Belle Telephone are A: voice mail, B: call-waiting, and C: caller-id. 70% of all subscribers request A, 80% request B, 75% request C, 85% request A or B (remember "or" also implies "or both"), 90% request A or C, 95% request B or C and 98% request A or B or C. Calculate the probability that the next subscriber will request:

a) at least one of the three options.

b) none of the three options.

c) only caller-id.

d) exactly one of the three options.

Solution. In set notation, the given information is

$$P(A) = 0.70, \quad P(B) = 0.80, \quad P(C) = 0.75,$$
$$P(A \cup B) = 0.85, \quad P(A \cup C) = 0.90, \quad P(B \cup C) = 0.95, \quad P(A \cup B \cup C) = 0.98.$$

a) "At least one of the three" is the union $A \cup B \cup C$, which is given: $P(A \cup B \cup C) = \boxed{0.98}$.

b) "None of the three" is $A^c \cap B^c \cap C^c = (A \cup B \cup C)^c$ by De Morgan's law, so by the complement rule its probability is $1 - 0.98 = \boxed{0.02}$.

For (c) and (d) we need the individual regions of the Venn diagram, and for that we need the intersections.

Pairs. Solve the two-event addition rule for each intersection:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.70 + 0.80 - 0.85 = 0.65,$$
$$P(A \cap C) = P(A) + P(C) - P(A \cup C) = 0.70 + 0.75 - 0.90 = 0.55,$$
$$P(B \cap C) = P(B) + P(C) - P(B \cup C) = 0.80 + 0.75 - 0.95 = 0.60.$$

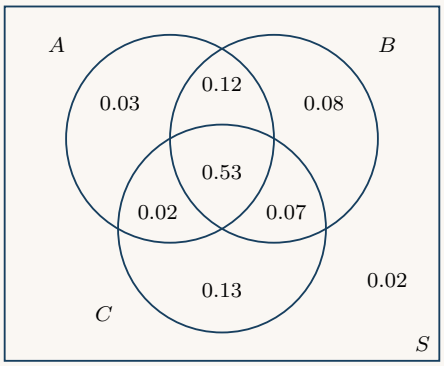
Triple. Put everything into the three-event addition rule and solve for $P(A \cap B \cap C)$:

$$\begin{aligned} 0.98 &= (0.70 + 0.80 + 0.75) - (0.65 + 0.55 + 0.60) + P(A \cap B \cap C) \\ &= 0.45 + P(A \cap B \cap C), \end{aligned}$$

so $P(A \cap B \cap C) = 0.53$.

Regions. Fill in the diagram from the centre outwards. Each pair loses the centre: $0.65 - 0.53 = 0.12$ (A and B only), $0.55 - 0.53 = 0.02$ (A and C only), $0.60 - 0.53 = 0.07$ (B and C only). Each circle then loses the three regions already filled in:

$$\begin{aligned} \text{only } A &: \ 0.70 - 0.12 - 0.02 - 0.53 = 0.03, \\ \text{only } B &: \ 0.80 - 0.12 - 0.07 - 0.53 = 0.08, \\ \text{only } C &: \ 0.75 - 0.02 - 0.07 - 0.53 = 0.13. \end{aligned}$$



c) "Only caller-id" is the region of C outside A and B: $\boxed{0.13}$.

d) "Exactly one of the three options" is only A, or only B, or only C. These regions are mutually exclusive, so their probabilities add: $0.03 + 0.08 + 0.13 = \boxed{0.24}$.

Check. The eight regions add up to $0.03 + 0.08 + 0.13 + 0.12 + 0.02 + 0.07 + 0.53 + 0.02 = 1$. There is also a shortcut for (c): a subscriber who requests only caller-id is in $A \cup B \cup C$ but not in $A \cup B$, so the probability is $P(A \cup B \cup C) - P(A \cup B) = 0.98 - 0.85 = 0.13$.

13 Conditional Probability and Bayes' Theorem

In Sections 11 and 12 every probability was measured against the whole sample space. Often, though, we learn something about the outcome before we are asked for a probability, and that information changes the answer. This section defines conditional probability, turns the definition into the multiplication rule, organizes experiments that happen in stages with tree diagrams, and ends with the law of total probability and Bayes' theorem, which reverses the direction of a conditional probability.

13.1 Conditional probability

The probability of an event changes with more contextual information. Roll a fair die once: the probability of a six is $\frac{1}{6}$. Now suppose you are told that the roll was an even number. Only the outcomes 2, 4 and 6 are still possible,

they are still equally likely, and one of them is a six, so the probability of a six has become $\frac{1}{3}$.

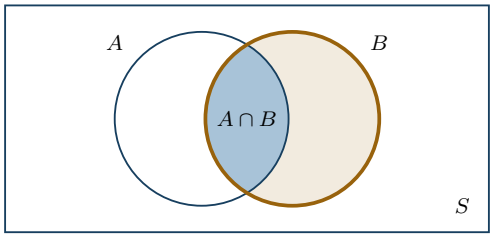
Definition (conditional probability)

Let A and B be two events of a sample space S (also written Ω), with $P(B) > 0$. The *conditional probability of A given B*, written $P(A \mid B)$ and read "A given B", is the probability of A after knowing that B has occurred:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

When the outcomes of the sample space are equally likely, this is a ratio of counts:

$$P(A \mid B) = \frac{N(A \cap B)}{N(B)}.$$



Once B is known to have occurred, B (outlined) is the new sample space, and A can only occur through $A \cap B$ (shaded).

In the die example $A = \{6\}$ and $B = \{2, 4, 6\}$, so $A \cap B = \{6\}$ and $P(A \mid B) = \dfrac{N(A \cap B)}{N(B)} = \dfrac{1}{3}$. Here the condition raised the probability of A; in other problems it lowers it, and sometimes it leaves it unchanged (that special case is the subject of Section 14).

Remark (the sample space is reduced to B)

Knowing that B has occurred rules out every outcome outside B. The sample space is *no longer* S: it has been reduced to B. Inside this smaller sample space, A can only happen through the outcomes that also lie in B, that is, through $A \cap B$. So $P(A \mid B)$ compares $A \cap B$ with B, not with S.

For equally likely outcomes the two formulas of the definition agree, because the size $N = N(S)$ of the original sample space cancels:

$$\frac{P(A \cap B)}{P(B)} = \frac{N(A \cap B)/N}{N(B)/N} = \frac{N(A \cap B)}{N(B)}.$$

Caution

- Divide by $P(B)$, not by the probability of the whole sample space. Since $P(S) = 1$, dividing by $P(S)$ would simply give back $P(A \cap B)$, and in general

  $$P(A \mid B) \neq \frac{P(A \cap B)}{P(S)} = P(A \cap B).$$

  $P(A \cap B)$ is the probability that both events occur, measured against all of S; $P(A \mid B)$ is measured against B only.
- The order matters. $P(A \mid B)$ and $P(B \mid A)$ have the same numerator $P(A \cap B)$ but different denominators, so in general $P(A \mid B) \neq P(B \mid A)$ (see Example 13.1, parts (c) and (d)).

Note (recognizing a condition)

In a word problem the condition is announced by phrases such as "given that", "knowing that", "if", "of those who…" or "among the…". Whatever follows the phrase is the event B, the reduced sample space; the event whose probability is asked for is A.

Property (a conditional probability is a probability)

For a fixed condition B with $P(B) > 0$, the numbers $P(A \mid B)$ obey all the usual rules of probability, with B playing the role of the sample space. In particular

$$0 \le P(A \mid B) \le 1, \qquad P(B \mid B) = 1, \qquad P(A^c \mid B) = 1 - P(A \mid B).$$

The complement rule applies to the event *in front of* the bar. Changing the condition changes the sample space, so $P(A \mid B^c)$ is a different quantity altogether; it is *not* $1 - P(A \mid B)$.

13.2 The multiplication rule

Multiplying both sides of the definition $P(A \mid B) = P(A \cap B)/P(B)$ by $P(B)$ gives a formula for the probability that two events *both* occur. Exchanging the roles of A and B (remember that $A \cap B = B \cap A$) gives a second one.

Theorem (multiplication rule of probability)

For any two events A and B with positive probabilities,

$$P(A \cap B) = P(A \mid B) \cdot P(B) = P(B \mid A) \cdot P(A).$$

In words: for both events to happen, one of them must happen, and then the other must happen *given that the first one did*. Both forms are always true; use the one whose conditional probability you know. When the events occur one after the other, that is usually the form that conditions on the earlier event.

Remark

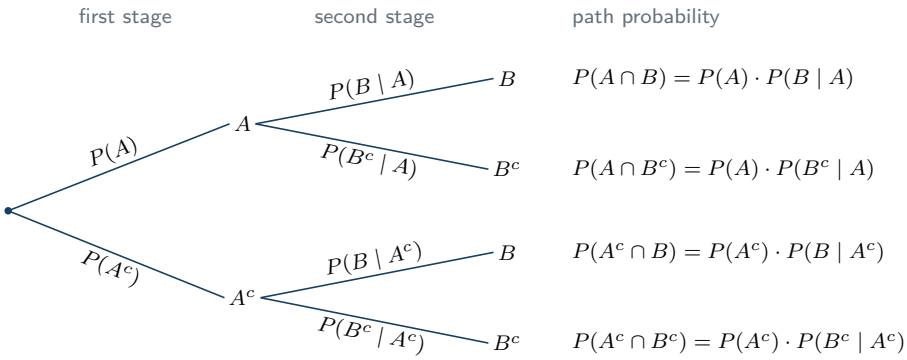
The definition of $P(A \mid B)$ and the multiplication rule are the *general* rules for $P(A \mid B)$ and $P(A \cap B)$: they hold for every pair of events, with no extra assumption. The shortcut $P(A \cap B) = P(A) \cdot P(B)$ is a special case, valid only for independent events (Section 14).

The rule extends to more events by conditioning each one on everything that came before it. For three events,

$$P(A \cap B \cap C) = P(A) \cdot P(B \mid A) \cdot P(C \mid A \cap B).$$

13.3 Tree diagrams

When an experiment happens in stages (draw a ball, then draw another; have a disease or not, then test positive or negative), a *tree diagram* keeps track of the multiplication rule. Each stage adds a level of branches, and each branch is labelled with the probability of taking it.



How to read a probability tree

- The branches of the first stage carry ordinary probabilities, such as $P(A)$ and $P(A^c)$.
- The branches of a later stage carry *conditional* probabilities, given the branch they grow out of, such as $P(B \mid A)$.
- The probabilities on the branches leaving any one point add up to 1.
- **Multiply along a path.** The probability of a complete path is the product of the probabilities on its branches. This is the multiplication rule: $P(A \cap B) = P(A) \cdot P(B \mid A)$.
- **Add across paths.** Different paths are mutually exclusive, so the probability of an event is the sum of the probabilities of all the paths that lead to it. The probabilities of all the paths of the tree add up to 1.

13.4 The law of total probability and Bayes' theorem

"Add across paths" is a theorem in its own right. Recall from Section 11 that events $A_1, \ldots, A_k$ form a *partition* of the sample space S if they are non-empty, mutually exclusive, and their union is S; in other words, exactly one of them occurs. The simplest partition is an event together with its complement, A and A^c.

Theorem (law of total probability)

If $A_1, A_2, \ldots, A_k$ form a partition of the sample space and $P(A_i) > 0$ for every i, then for any event B

$$P(B) = P(B \mid A_1)\,P(A_1) + P(B \mid A_2)\,P(A_2) + \cdots + P(B \mid A_k)\,P(A_k).$$

In particular, for an event A and its complement,

$$P(B) = P(B \mid A)\,P(A) + P(B \mid A^c)\,P(A^c).$$

Proof

The events $B \cap A_1, \ldots, B \cap A_k$ are mutually exclusive pieces of B, and together they make up all of B. So, by the addition rule for mutually exclusive events,

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \cdots + P(B \cap A_k),$$

and by the multiplication rule each term is $P(B \cap A_i) = P(B \mid A_i)\,P(A_i)$. □

Sometimes we know $P(B \mid A)$ but need the conditional probability the other way round, $P(A \mid B)$. Bayes' theorem makes the switch.

Theorem (Bayes' theorem)

For two events A and B with positive probabilities,

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}.$$

Proof

Start from the definition of $P(A \mid B)$ and rewrite its numerator:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)\,P(A)}{P(B)},$$

because $P(A \cap B) = P(B \mid A)\,P(A)$ by the multiplication rule. □

In most problems $P(B)$ is not given and has to be found first with the law of total probability. Putting it into the denominator gives the form of Bayes' theorem that is used in practice.

Formula (Bayes' theorem with the denominator written out)

For an event A and its complement,

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B \mid A)\,P(A) + P(B \mid A^c)\,P(A^c)}.$$

More generally, if $A_1, \dots, A_k$ form a partition of the sample space with $P(A_i) > 0$ for every i, then for each j

$$P(A_j \mid B) = \frac{P(B \mid A_j)\,P(A_j)}{P(B \mid A_1)\,P(A_1) + \cdots + P(B \mid A_k)\,P(A_k)}.$$

Note (Bayes' theorem on a tree)

A tree is drawn in the order in which the experiment happens: first $A_1, \dots, A_k$, then B or B^c. Bayes' theorem answers a question about the first stage once the outcome of the second stage is known:

$$P(A_j \mid B) = \frac{\text{probability of the path through } A_j \text{ that ends in } B}{\text{sum of the probabilities of all the paths that end in } B}.$$

13.5 Worked examples

Example 13.1 ▪ Eye and hair colour: a two-way table Worksheet 13, #1

The following table shows the eye and hair colour of a sample of 150 Marianopolis students.

	Blond hair	Brown hair	Black hair
Blue eyes	20	8	7
Hazel eyes	10	20	15
Brown eyes	5	40	25

Of this sample, one student is selected randomly.

a) What is the probability that the student has brown hair?
b) What is the probability that the student has blue eyes?
c) What is the probability that the student has blue eyes knowing that the student has brown hair?
d) What is the probability that the student has brown hair knowing that the student has blue eyes?
e) Are the events having blue eyes and having brown hair mutually exclusive?

Solution. Every student is equally likely to be selected, so each probability is a ratio of counts. Start by adding the row and column totals.

	Blond hair	Brown hair	Black hair	*Total*
Blue eyes	20	**8**	7	*35*
Hazel eyes	10	20	15	*45*
Brown eyes	5	40	25	*70*
Total	*35*	*68*	*47*	*150*

a) 68 of the 150 students have brown hair: $P(\text{brown hair}) = \dfrac{68}{150} = \boxed{\dfrac{34}{75}} \approx 0.4533$.

b) 35 of the 150 students have blue eyes: $P(\text{blue eyes}) = \dfrac{35}{150} = \boxed{\dfrac{7}{30}} \approx 0.2333$.

c) "Knowing that the student has brown hair" is a condition. The sample space is reduced from all 150 students to the 68 students of the brown-hair column, and 8 of them have blue eyes:

$$P(\text{blue eyes} \mid \text{brown hair}) = \frac{N(\text{blue eyes} \cap \text{brown hair})}{N(\text{brown hair})} = \frac{8}{68} = \boxed{\frac{2}{17}} \approx 0.1176.$$

d) Now the condition is "blue eyes". The sample space is the blue-eyes row, 35 students, and 8 of them have brown hair:

$$P(\text{brown hair} \mid \text{blue eyes}) = \frac{N(\text{brown hair} \cap \text{blue eyes})}{N(\text{blue eyes})} = \boxed{\frac{8}{35}} \approx 0.2286.$$

e) $\boxed{\text{No}}$. Mutually exclusive events cannot occur together: their intersection is empty. Here 8 students have both blue eyes and brown hair, so the intersection is not empty.

Parts (c) and (d) use the same 8 students in the numerator but different denominators, because the condition decides which column or row becomes the new sample space. The formula with probabilities gives the same values; for instance, in (c),

$$\frac{P(\text{blue eyes} \cap \text{brown hair})}{P(\text{brown hair})} = \frac{8/150}{68/150} = \frac{8}{68}.$$

Example 13.2 ▪ Two boxes of balls: a tree diagram Worksheet 13, #4

One box contains six red balls and four green balls, and a second box contains seven red balls and three green balls. A ball is randomly chosen from the first box and placed in the second box. Then a ball is randomly selected from the second box and placed in the first box.

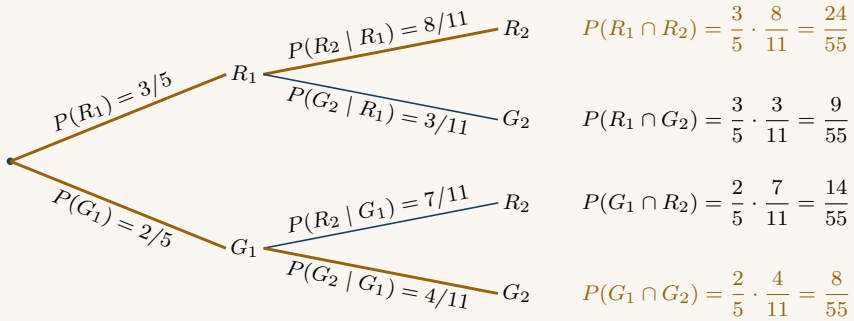
a) What is the probability that a red ball is selected from the first box and then a red ball is selected from the second box?

b) At the conclusion of the process, what is the probability that the number of each coloured ball in each box has remained the same?

Solution. Let R_1 and G_1 be the events that the ball taken from the first box is red, respectively green, and let R_2 and G_2 be the same events for the ball then taken from the second box. The first box holds 10 balls, so

$$P(R_1) = \frac{6}{10} = \frac{3}{5}, \qquad P(G_1) = \frac{4}{10} = \frac{2}{5}.$$

What the second box contains depends on the first draw, so the probabilities of the second stage are conditional. After the transfer the second box holds 11 balls: 8 red and 3 green if a red ball was added, 7 red and 4 green if a green ball was added.



The four path probabilities add up to 1. On the two highlighted paths the ball that comes back has the colour of the ball that left.

a) "Red, then red" is the top path. Multiplying along it (the multiplication rule),

$$P(R_1 \cap R_2) = P(R_2 \mid R_1) \cdot P(R_1) = \frac{8}{11} \cdot \frac{3}{5} = \boxed{\frac{24}{55}} \approx 0.4364.$$

b) The contents of both boxes are unchanged exactly when the ball that comes back has the same colour as the ball that left: red then red, or green then green. These two paths are mutually exclusive, so their probabilities add:

$$P(R_1 \cap R_2) + P(G_1 \cap G_2) = \frac{24}{55} + \frac{2}{5} \cdot \frac{4}{11} = \frac{24}{55} + \frac{8}{55} = \boxed{\frac{32}{55}} \approx 0.5818.$$

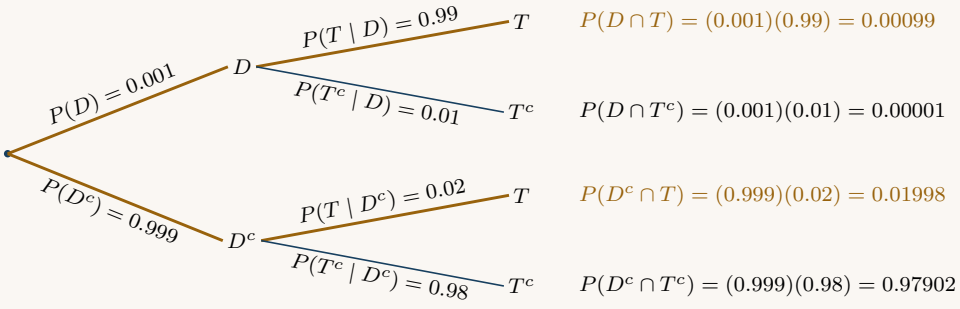
Example 13.3 ⋆ ▪ **A rare disease and a positive test** Worksheet 13, #6

Only 1 in 1000 adults is afflicted with a rare disease for which a diagnostic test has been developed. The test is such that, when an individual actually has the disease, a positive result will occur 99% of the time, while an individual without the disease will show a false positive result only 2% of the time. If a randomly selected individual is tested and the result is positive, what is the probability that the individual actually has the disease?

Solution. Let D be the event that the individual has the disease and T the event that the test is positive. The statement gives

$$P(D) = 0.001, \qquad P(D^c) = 0.999, \qquad P(T \mid D) = 0.99, \qquad P(T \mid D^c) = 0.02.$$

The test's reliability is described *given the person's true state*, but the question asks for the true state *given the test result*, $P(D \mid T)$. This is the situation Bayes' theorem is made for.



The two highlighted paths end in a positive test: a true positive (top) and a false positive.

First find the probability of a positive test with the law of total probability (add the two paths that end in T):

$$\begin{aligned} P(T) &= P(T \mid D)\, P(D) + P(T \mid D^c)\, P(D^c) \\ &= (0.99)(0.001) + (0.02)(0.999) = 0.00099 + 0.01998 = 0.02097. \end{aligned}$$

Then, by Bayes' theorem,

$$P(D \mid T) = \frac{P(T \mid D)\,P(D)}{P(T)} = \frac{0.00099}{0.02097} \approx \boxed{0.0472}.$$

Even after a positive result, the probability that the individual has the disease is under 5%.

Why so small? Picture 100 000 adults. About 100 of them have the disease, and 99 of those test positive. The other 99 900 do not have the disease, yet 2% of them, 1998 people, test positive anyway. Of the $99 + 1998 = 2097$ positive results, only 99 are genuine: $\frac{99}{2097} \approx 0.0472$. The disease is so rare that the false positives from the very large healthy group swamp the true positives. Notice also how different the two conditional probabilities are: $P(T \mid D) = 0.99$, but $P(D \mid T) \approx 0.047$.

14 Independence

Conditional probability (Section 13) measures how the knowledge that B has occurred changes the probability of A. This section is about the special case in which it does not change it at all. Such events are called independent, and for them the multiplication rule becomes a plain product. Independence is the assumption behind the binomial distribution (Section 17) and most of the models that come after it.

14.1 Independent events

Definition (independent events)

Two events A and B with positive probabilities are *independent* if

$$P(A \mid B) = P(A),$$

that is, if knowing that B has occurred does not change the probability of A. Two events that are not independent are called *dependent*.

The definition looks lopsided, because it conditions on B and not on A, but the two events can trade places:

Proposition (independence is symmetric)

If $P(A \mid B) = P(A)$, then we also have $P(B \mid A) = P(B)$.

Proof

By Bayes' theorem,

$$P(B \mid A) = \frac{P(A \mid B)\,P(B)}{P(A)} = \frac{P(A)\,P(B)}{P(A)} = P(B),$$

where the second step uses the assumption $P(A \mid B) = P(A)$. □

So "A is independent of B" and "B is independent of A" say the same thing, and we simply say that A and B are independent.

Theorem (multiplication rule for independent events)

If A and B are independent, then

$$P(A \cap B) = P(A) \cdot P(B).$$

This is the general multiplication rule of Section 13 with $P(A \mid B)$ replaced by $P(A)$:

$$P(A \cap B) = P(A \mid B) \cdot P(B) = P(A) \cdot P(B).$$

The argument also runs backwards. If $P(A \cap B) = P(A) \cdot P(B)$, then

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A),$$

so the product rule is not just a consequence of independence: it characterizes it.

Three equivalent ways to say that A and B are independent

$$P(A \mid B) = P(A) \quad \Longleftrightarrow \quad P(B \mid A) = P(B) \quad \Longleftrightarrow \quad P(A \cap B) = P(A) \cdot P(B)$$

If one of the three holds, all three hold; if one fails, all three fail. The product form has the advantage of making sense even when an event has probability 0, and many books take it as the definition.

Note (two ways independence is used)

- **Testing for independence.** When a problem asks whether A and B are independent, compute both sides of one of the three conditions from the data (for example, check if $P(A \mid B) = P(A)$) and compare. Equal means independent; not equal means dependent. The story told in the problem does not settle the matter: independence has to be verified with the numbers.
- **Using independence.** When a problem tells you that the events are independent ("operate independently", "independent trials"), you may multiply: $P(A \cap B) = P(A) \cdot P(B)$.

Property (complements of independent events)

If A and B are independent, then so are A and B^c, A^c and B, and A^c and B^c.

Proof

The event A splits into the part inside B and the part outside B, so

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A) \cdot P(B) = P(A)\big(1 - P(B)\big) = P(A) \cdot P(B^c).$$

This shows that A and B^c are independent. The other two cases follow by applying the same result again with the roles of the events exchanged. □

14.2 Independent is not the same as mutually exclusive

These two notions are easy to confuse, because both describe how two events are related. They say very different things.

	Mutually exclusive	**Independent**
Meaning	A and B cannot occur together	the occurrence of one does not change the probability of the other
Defining condition	$A \cap B = \emptyset$	$P(A \mid B) = P(A)$
$P(A \cap B)$	0	$P(A) \cdot P(B)$
$P(A \cup B)$	$P(A) + P(B)$	$P(A) + P(B) - P(A) \cdot P(B)$
$P(A \mid B)$	0	$P(A)$

Caution

Mutually exclusive events with positive probabilities are *never* independent. If A and B are mutually exclusive, then knowing that B has occurred tells you that A has *not* occurred:

$$P(A \mid B) = 0 \neq P(A).$$

This is an extreme form of dependence: B rules A out completely. Conversely, independent events with positive probabilities always overlap, since $P(A \cap B) = P(A) \cdot P(B) > 0$.

Being mutually exclusive is a property of the *sets*, and it shows on a Venn diagram: the two regions do not overlap. Being independent is a property of the *probabilities*. It cannot be seen on a Venn diagram and has to be checked by calculation.

14.3 More than two events: mutual independence

Definition (mutual independence)

Events $A_1, A_2, \ldots, A_n$ are *mutually independent* if the multiplication rule for independent events holds for every selection of two or more of them: the probability that all the selected events occur is the product of their probabilities.

Property (product rule for mutually independent events)

If $A_1, \ldots, A_n$ are mutually independent, then

$$P\Big(\bigcap_{i=1}^{n} A_i\Big) = \prod_{i=1}^{n} P(A_i), \qquad \text{that is,} \qquad P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2) \cdots P(A_n).$$

The same is true when some of the events are replaced by their complements.

Remark

When a problem says that several components "operate independently", or that the trials of an experiment are independent, mutual independence is what is meant.

Checking the events two at a time is not enough. Toss a fair coin twice and let A be "the first toss is heads", B be "the second toss is heads" and C be "the two tosses show the same face". Each event has probability $\frac{1}{2}$, and any two of them occur together only when both tosses are heads, with probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$. So the events are independent two at a time. But $A \cap B \cap C$ is again "both tosses are heads", with probability $\frac{1}{4}$ and not $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$: the three events are not mutually independent.

14.4 Independence and sampling

Selecting several individuals from a group is normally done *without replacement*: the same person cannot be picked twice. Each selection then changes the make-up of the group that is left, so successive selections are *dependent*, and the general multiplication rule with conditional probabilities is needed.

The large population independence assumption

When the population is very large compared with the number of individuals selected, removing a few individuals changes the proportions in the population by a negligible amount. The selections are then treated as *independent*, each with the same probability, and their probabilities are simply multiplied.

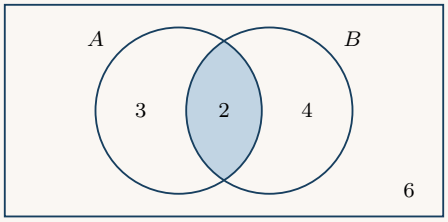
14.5 Worked examples

Example 14.1 ▪ Choir and theatre Worksheet 14, #1

Out of 15 students, 5 are in the choir, 6 do theatre, and 6 do neither. Let A be the event that a randomly chosen student is in the choir and B be the event that a randomly chosen student does theatre. Are A and B independent?

Solution. We check whether $P(A \mid B) = P(A)$. For $P(A \mid B)$ we need $N(A \cap B)$, the number of students who do both activities, and a Venn diagram finds it. Since 6 of the 15 students do neither, $N(A \cup B) = 15 - 6 = 9$. The sum $N(A) + N(B) = 5 + 6 = 11$ counts the students who do both twice, so

$$N(A \cap B) = N(A) + N(B) - N(A \cup B) = 5 + 6 - 9 = 2.$$



Number of students in each region (A: choir, B: theatre).

Then

$$P(A \mid B) = \frac{N(A \cap B)}{N(B)} = \frac{2}{6} = \frac{1}{3} \qquad \text{and} \qquad P(A) = \frac{N(A)}{N} = \frac{5}{15} = \frac{1}{3}.$$

Since $P(A \mid B) = P(A)$, yes, A and B are independent. One third of all the students are in the choir, and one third of the theatre students are in the choir: learning that a student does theatre does not change the probability that the student is in the choir.

The product rule gives the same verdict: $P(A \cap B) = \frac{2}{15}$ and $P(A) \cdot P(B) = \frac{5}{15} \cdot \frac{6}{15} = \frac{2}{15}$.

Example 14.2 ▪ Brown hair and brown eyes Worksheet 14, #2

In a group of 19 people, 7 have brown hair, 8 have brown eyes, and 3 have both brown hair and brown eyes. A person is randomly selected from the group. Let A be the event that the person has brown hair and B be the event that that person has brown eyes. Are A and B independent?

Solution. The question to settle is again: is $P(A \mid B) = P(A)$? This time $N(A \cap B) = 3$ is given directly.

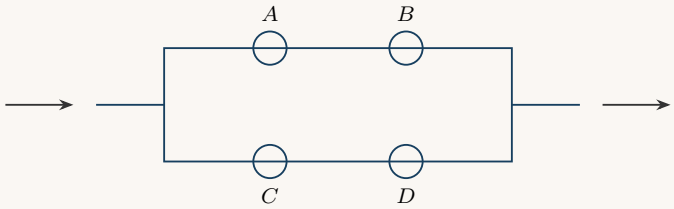
$$P(A \mid B) = \frac{N(A \cap B)}{N(B)} = \frac{3}{8} = 0.375 \qquad \text{but} \qquad P(A) = \frac{N(A)}{N} = \frac{7}{19} \approx 0.3684.$$

Since $P(A \mid B) \neq P(A)$, no, A and B are not independent. Knowing that the person has brown eyes makes brown hair slightly more likely. (Equivalently, $P(A \cap B) = \frac{3}{19} \approx 0.1579$, whereas $P(A) \cdot P(B) = \frac{7}{19} \cdot \frac{8}{19} \approx 0.1551$.)

The two values are close, but independence is an exact condition: here the whole group is known, so any difference at all means that the events are dependent. Deciding from a *sample* whether two characteristics are independent in the population it came from is a different problem, taken up in Section 32.

Example 14.3 ▪ A network of pipes Worksheet 14, #4

Water is meant to flow through the following network of pipes, where the circles represent taps that can be closed.



If the taps operate independently and each has a 90% chance of being open at any given time, what is the probability that there is water flowing out of the system?

Solution. Call the taps A, B (upper pipe) and C, D (lower pipe), as marked in the figure, and use the same letters for the events "tap A is open", and so on. Each event has probability 0.9, and since the taps operate independently the four events are mutually independent.

Water gets through the upper pipe only if A *and* B are both open, and through the lower pipe only if C *and* D are both open. Water flows out of the system if it gets through the upper pipe *or* the lower pipe:

$$\{\text{water flows}\} = (A \cap B) \cup (C \cap D).$$

Both pipes can be open at the same time, so $A \cap B$ and $C \cap D$ are not mutually exclusive and the general addition rule (Section 12) is needed:

$$P\big[(A \cap B) \cup (C \cap D)\big] = P(A \cap B) + P(C \cap D) - P(A \cap B \cap C \cap D).$$

By mutual independence each of these intersections is a product:

$$P(A \cap B) = P(C \cap D) = (0.9)^2 = 0.81, \qquad P(A \cap B \cap C \cap D) = (0.9)^4 = 0.6561.$$

Therefore

$$P(\text{water flows}) = 0.9^2 + 0.9^2 - 0.9^4 = 0.81 + 0.81 - 0.6561 = \boxed{0.9639}.$$

Check with the complement. No water flows only if the upper pipe *and* the lower pipe are both blocked. Each pipe is blocked with probability $1 - 0.81 = 0.19$, and the two pipes depend on different taps, so they are independent of each other: $P(\text{no water}) = (0.19)^2 = 0.0361$ and $P(\text{water flows}) = 1 - 0.0361 = 0.9639$.

Note ("and" in a row, "or" side by side)

Components placed one after the other must *all* work: this is an intersection, so for independent components the probabilities are multiplied. Alternative routes placed side by side need only *one* to work: this is a union, so use the addition rule or, often faster, $1 - P(\text{every route fails})$. Simply adding $0.81 + 0.81 = 1.62$ in Example 14.3 would give a "probability" greater than 1, because the two routes are not mutually exclusive.

Part IV

Discrete Random Variables

A random variable attaches a number to each outcome of an experiment. This part introduces probability mass functions, expected value and variance, and then the standard discrete distributions: binomial, negative binomial, geometric, hypergeometric and Poisson.

15 Random Variables and Probability Mass Functions

So far, probabilities have been attached to events described in words: "at least one head", "both students have blue eyes". In most experiments, however, what we finally record is a *number*: how many heads, how many defective parts, how many draws. A random variable is that number. Once the outcomes of an experiment are numbers, the tools of descriptive statistics become available again: tables, histograms, shapes and, in Section 16, means and standard deviations.

15.1 Random variables and their support

Definition (random variable)

A *random variable* is a variable whose value depends on chance: it attaches a number to each outcome of a probability experiment. Random variables are written with capital letters (X, Y, Z, ...), and the particular values they can take with the matching lower-case letters (x, y, z, ...).

For example, roll two dice and let M be the larger of the two numbers showing. Each outcome of the experiment fixes the value of M: the outcome $(1, 5)$ gives $M = 5$ and the outcome $(3, 3)$ gives $M = 3$. In other words, a random variable is a function defined on the sample space S. Before the experiment is run its value is unknown; afterwards it has exactly one value.

Because the value of X is decided by the outcome, a statement such as "$X = 2$" or "$X \leq 2$" is an *event*: it stands for the set of all outcomes for which the statement is true. So it has a probability, written $P(X = 2)$ or $P(X \leq 2)$.

Definition (support)

The *support* of a random variable X, written $\text{Supp}(X)$, is the set of all possible values of X.

Some books call this set the *range* of X; these notes always say *support*.

Definition (discrete and continuous random variables)

- X is *discrete* if its support is a discrete set, that is, at most countable: its values can be written out as a list $x_1, x_2, x_3, \ldots$, which may stop or go on forever.
- X is *continuous* if its support is not countable; typically the support is an interval of real numbers.

Random variable	Support	Type
X = number of heads in 3 tosses of a coin	$\{0, 1, 2, 3\}$	discrete (finite list)
M = larger number when two dice are rolled	$\{1, 2, 3, 4, 5, 6\}$	discrete (finite list)
N = number of tosses until the first head	$\{1, 2, 3, \ldots\}$	discrete (infinite list)
T = waiting time (minutes) for the next bus	an interval, e.g. $[0, 15]$	continuous

As a rule of thumb, discrete random variables come from *counting* and continuous random variables from *measuring*. Sections 15 to 20 deal with discrete random variables; continuous ones are taken up in Section 21.

15.2 The probability mass function

Definition (probability mass function)

The *probability mass function* (pmf) of a discrete random variable X is the function p that gives the probability of each value in $\text{Supp}(X)$:

$$p(x) := P(X = x), \qquad x \in \text{Supp}(X).$$

Keep the two letters apart: lower-case p is the name of the function (the pmf), while capital P means "the probability of the event". So $p(2)$ is shorthand for $P(X = 2)$. For a number x outside the support, $P(X = x) = 0$. When several random variables are in play, a subscript says whose pmf is meant: $p_X(x)$, $p_Z(z)$. The pmf shows how the total probability 1 is shared out among the possible values of X; it is also called the *probability distribution* of X.

The pmf of a discrete random variable is usually given as a *table* that lists each value x with its probability $p(x)$. It can also be given by a *formula*, or drawn as a *graph*:

- a *point-mass graph*: above each value x, a vertical line of height $p(x)$ topped by a dot;
- a *probability histogram*: above each value x, a bar of height $p(x)$.

Example 15.1 ▪ Three tosses of a coin Lecture notes

A fair coin is tossed 3 times and X is the number of heads. Find the support and the pmf of X, and graph the pmf.

Solution. The sample space has $2^3 = 8$ equally likely outcomes:

$$S = \{HHH,\ HHT,\ HTH,\ THH,\ HTT,\ THT,\ TTH,\ TTT\}.$$

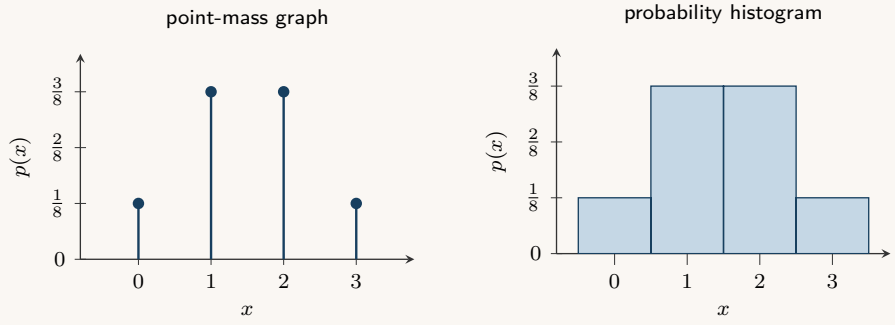
Counting the heads in each outcome, X can be 0, 1, 2 or 3, so $\text{Supp}(X) = \{0, 1, 2, 3\}$. Each outcome has probability $\frac{1}{8}$, so $p(x) = P(X = x)$ is found by counting the outcomes that give the value x. For instance,

$$p(0) = P(X = 0) = P(\{TTT\}) = \tfrac{1}{8}, \qquad p(2) = P(X = 2) = P(\{HHT, HTH, THH\}) = \tfrac{3}{8}.$$

The complete pmf, as a table:

x	0	1	2	3
outcomes with $X = x$	TTT	$HTT,\ THT,\ TTH$	$HHT,\ HTH,\ THH$	HHH
$p(x) = P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

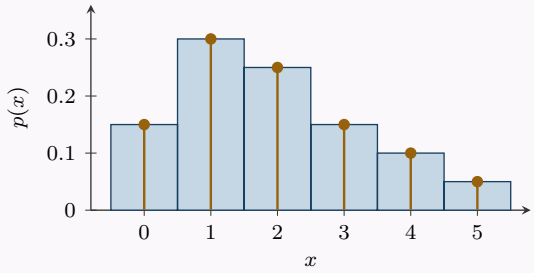
The same pmf as a graph, in both styles:



Remark (the language of descriptive statistics carries over)

Everything that was said about data sets in descriptive statistics also applies to probability mass functions. A pmf plays the role of a relative-frequency table: if the experiment is repeated a very large number of times, the relative frequency of the value x settles down to $p(x)$. So a pmf has a *shape* (symmetric, bell-shaped, skewed; see Section 6), a *centre* and a *spread* (Section 16).

For instance, the pmf drawn below is roughly bell-shaped (it has a single peak) but it is not symmetric: its tail is longer on the right, so it is right-skewed. The pmf of Example 15.1, by contrast, is symmetric.



One pmf drawn both ways: point-mass graph (lines and dots) and probability histogram (bars).

You can always redraw a point-mass graph as a histogram. When the values of X are consecutive whole numbers, each bar is given width 1 and is centred on its value. The area of a bar then equals its height $p(x)$, the total area of the histogram is 1, and the probability of an event is the total area of the bars that make it up.

15.3 Valid pmfs and probabilities of events

Not every table of numbers can be a pmf.

Property (the two conditions for a valid pmf)

For p to be a valid pmf for a discrete random variable X, it must satisfy both of the following conditions.

1) Every value is a probability: $p(x) \in [0,1]$, that is, $0 \le p(x) \le 1$, for all $x \in \text{Supp}(X)$.
2) The values add up to one: $\displaystyle\sum_{x \in \text{Supp}(X)} p(x) = 1$.

Why both conditions are necessary

(1) By definition, $p(x) = P(X = x)$ is the probability of an event, and every probability lies between 0 and 1.

(2) Every time the experiment is run, X takes exactly one of the values in its support. Hence the events "$X = x$", one for each $x \in \text{Supp}(X)$, are mutually exclusive and together they make up the whole sample space S. Their probabilities therefore add up to $P(S)$:

$$\sum_{x \in \text{Supp}(X)} p(x) = \sum_{x \in \text{Supp}(X)} P(X = x) = P(S) = 1,$$

by the addition rule for mutually exclusive events (Section 12). □

Note (using the two conditions)

The two conditions are also sufficient: any function on a list of values that satisfies both of them is the pmf of some random variable. So checking them is all there is to deciding whether a proposed pmf is legitimate. They are used in two ways.

- *To test a candidate pmf.* A single negative value, or a total different from 1, is enough to reject it.

- *To find an unknown constant* in a formula for $p(x)$: set the sum of all the values equal to 1 and solve for the constant, then make sure that no value is negative.

Once the values add up to 1, it is enough to check that none of them is negative: non-negative numbers with total 1 cannot exceed 1.

Property (probabilities of events from the pmf)

To find the probability that X falls in a set A of values, add the pmf over the values of the support that lie in A:

$$P(X \in A) = \sum_{x \in A} p(x).$$

For example, if the possible values of X are whole numbers, then

$$P(2 \le X \le 4) = P(X = 2) + P(X = 3) + P(X = 4) = p(2) + p(3) + p(4).$$

This works because the events $X = 2$, $X = 3$ and $X = 4$ are disjoint (X cannot take two values at once), so the addition rule for mutually exclusive events applies. The complement rule is often quicker: for instance, $P(X \neq 0) = 1 - P(X = 0) = 1 - p(0)$.

Caution ($<$ is not $\le$)

For a discrete random variable the endpoints matter: $P(X \le 2)$ and $P(X < 2)$ differ by $p(2)$. Before adding anything, write out exactly which values of the support satisfy the inequality.

15.4 The cumulative distribution function

Definition (cumulative distribution function)

The *cumulative distribution function* (cdf) of a random variable X is the function

$$F(x) := P(X \le x), \qquad \text{defined for every real number } x.$$

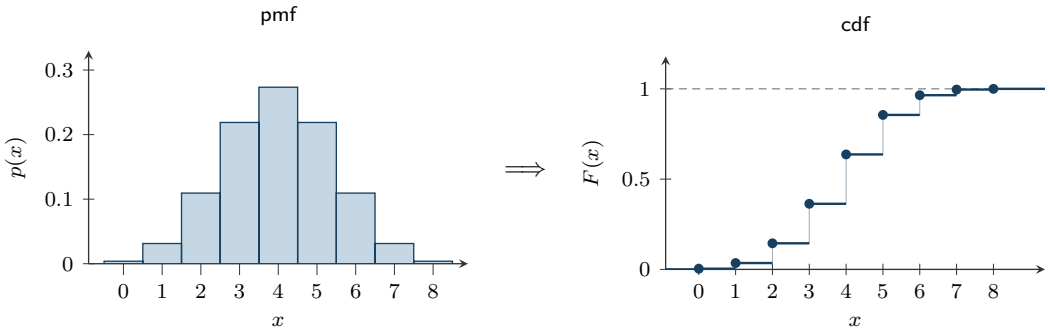
For a discrete random variable it is a running total of the pmf:

$$F(x) = \sum_{\substack{t \in \mathrm{Supp}(X) \\ t \le x}} p(t).$$

$F(x)$ collects all the probability sitting at x or to its left, in the same way as a cumulative relative frequency does in a frequency table. The number x does not have to be a possible value of X: for the three coin tosses of Example 15.1, $F(1.5) = P(X \le 1.5) = p(0) + p(1) = \frac{1}{2}$.

The cdf of a discrete random variable is a step function

- $F(x) = 0$ for every x to the left of the smallest value of the support, and $F(x) = 1$ from the largest value of the support onwards. (If the values go on forever to the right, as in $1, 2, 3, \ldots$, then $F(x)$ gets closer and closer to 1 without reaching it.)
- F never decreases as x increases.
- F stays constant between two consecutive values of the support and jumps up at each value x of the support. The height of the jump at x is exactly $p(x)$.
- At a jump, F takes the upper value, because $F(x) = P(X \le x)$ includes the probability at x itself.



A bell-shaped pmf and its cdf. The cdf climbs from 0 to 1 in an S-shape: slowly where the pmf is small, quickly where the pmf is large. Each step of the cdf is as high as the bar of the pmf above the same value.

The pmf and the cdf carry the same information, and each can be recovered from the other.

Formula (from the cdf back to the pmf; probabilities of intervals)

Let X be a discrete random variable whose possible values are whole numbers. Then

$$p(x) = F(x) - F(x-1) \qquad \text{(the height of the jump of } F \text{ at } x\text{)},$$

and, for whole numbers $a \le b$,

$$P(a < X \le b) = F(b) - F(a), \qquad P(a \le X \le b) = F(b) - F(a-1), \qquad P(X > a) = 1 - F(a).$$

The first formula holds because the event $X \le x$ splits into the two disjoint events $X \le x-1$ and $X = x$, so $F(x) = F(x-1) + p(x)$. The others follow in the same way; for example, $X \le b$ splits into $X \le a$ and $a < X \le b$. For a support that does not consist of whole numbers the idea is the same: the support is the set of points where F jumps, and $p(x)$ is the value of F at x minus its value just before x.

15.5 Worked examples

Example 15.2 ▪ Underinflated tires: checking and using a pmf Worksheet 15, #2

Let X = the number of tires on a randomly selected car that are underinflated.

a) Which of the following three functions is a valid pmf for X, and why are the other two not legitimate?

x	0	1	2	3	4
$p_1(x)$	0.3	0.2	0.1	0.05	0.05
$p_2(x)$	0.4	0.1	0.1	0.1	0.3
$p_3(x)$	0.4	0.2	0.2	−0.1	0.3

b) For the legitimate pmf, compute $P(2 \le X \le 4)$, $P(X \le 2)$ and $P(X \ne 0)$.

c) If $p_4(x) = c \cdot (5 - x)$ is a valid pmf, find c.

Solution.

a) Test each candidate against the two conditions for a valid pmf.

- p_1: every value lies in $[0,1]$, but $\sum p_1(x) = 0.3 + 0.2 + 0.1 + 0.05 + 0.05 = 0.7 \ne 1$. Not valid (condition 2 fails).
- p_3: $p_3(3) = -0.1 < 0$, and a probability cannot be negative. Not valid (condition 1 fails), even though the values of p_3 add up to 1.
- p_2: every value lies in $[0,1]$ and $\sum p_2(x) = 0.4 + 0.1 + 0.1 + 0.1 + 0.3 = 1$. So p_2 is the valid pmf.

b) Work with $p = p_2$. The events $X = 2$, $X = 3$ and $X = 4$ are disjoint, so their probabilities add:

$$P(2 \le X \le 4) = P(X = 2) + P(X = 3) + P(X = 4) = 0.1 + 0.1 + 0.3 = \boxed{0.5},$$
$$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2) = 0.4 + 0.1 + 0.1 = \boxed{0.6},$$
$$P(X \neq 0) = 1 - P(X = 0) = 1 - 0.4 = \boxed{0.6}.$$

The last one uses the complement rule; adding $p(1) + p(2) + p(3) + p(4)$ gives the same result.

c) A car has four tires, so the support is still $\{0, 1, 2, 3, 4\}$. The values of a valid pmf add up to 1:

$$\sum_{x=0}^{4} c\,(5 - x) = c\,(5 + 4 + 3 + 2 + 1) = 15c = 1 \quad \Longrightarrow \quad c = \boxed{\tfrac{1}{15}}.$$

With this c the values are $\frac{5}{15}, \frac{4}{15}, \frac{3}{15}, \frac{2}{15}, \frac{1}{15}$, all between 0 and 1, so p_4 is indeed a valid pmf.

Example 15.3 * ▪ **From the cdf back to the pmf** Worksheet 15, #3

The following function is a cdf for a discrete random variable:

x	$x < 0$	$0 \le x < 1$	$1 \le x < 2$	$2 \le x < 3$	$3 \le x < 4$	$4 \le x < 5$	$5 \le x < 6$	$6 \le x$
$F(x)$	0	0.06	0.19	0.39	0.67	0.92	0.97	1

Find the pmf $p(x)$.

Solution. The cdf of a discrete random variable jumps exactly at the values of the support, and the height of each jump is the probability of that value. Here F changes value at $x = 0, 1, 2, 3, 4, 5, 6$, so $\text{Supp}(X) = \{0, 1, 2, 3, 4, 5, 6\}$ and $p(x) = F(x) - F(x - 1)$:

$$p(0) = 0.06 - 0 = 0.06, \quad p(1) = 0.19 - 0.06 = 0.13, \quad p(2) = 0.39 - 0.19 = 0.20,$$
$$p(3) = 0.67 - 0.39 = 0.28, \quad p(4) = 0.92 - 0.67 = 0.25, \quad p(5) = 0.97 - 0.92 = 0.05,$$
$$p(6) = 1 - 0.97 = 0.03.$$

x	0	1	2	3	4	5	6
$p(x)$	0.06	0.13	0.20	0.28	0.25	0.05	0.03

Check: the seven values add up to 1, as they must, since the jumps take F all the way from 0 to 1.

The cdf also gives probabilities of intervals directly. For example,

$$P(2 \le X \le 4) = F(4) - F(1) = 0.92 - 0.19 = 0.73,$$

which agrees with $p(2) + p(3) + p(4) = 0.20 + 0.28 + 0.25$.

16 Expected Value and Variance of a Discrete Random Variable

The pmf describes a discrete random variable completely (Section 15). Often, though, two numbers are enough, just as for a data set: one that says where the distribution is centred and one that says how spread out it is. The centre is measured by the expected value, the spread by the variance and the standard deviation. Throughout this section, X is a discrete random variable with pmf $p(x) = P(X = x)$, and every sum $\sum_x$ runs over all the values x in $\text{Supp}(X)$.

16.1 Expected value

Definition (expected value)

The *expected value* (or *mean*) of a discrete random variable X is

$$\mu_X = E(X) := \sum_{x \in \mathrm{Supp}(X)} x \cdot p(x),$$

where $p(x) = P(X = x)$ is the probability of the value x.

In words: multiply each possible value by its probability and add up the results. For example, if X is the number of heads in three tosses of a fair coin, then $p(0) = \frac{1}{8}$, $p(1) = \frac{3}{8}$, $p(2) = \frac{3}{8}$, $p(3) = \frac{1}{8}$ and

$$E(X) = 0 \cdot \tfrac{1}{8} + 1 \cdot \tfrac{3}{8} + 2 \cdot \tfrac{3}{8} + 3 \cdot \tfrac{1}{8} = \tfrac{12}{8} = 1.5.$$

Note (a weighted average)

The expected value of X is a weighted average of the values in the support of X, with the probabilities $p(x)$ as weights. Recall that the weighted average of values x_i with weights w_i is $\sum w_i x_i / \sum w_i$ (Section 4). Taking the probabilities as weights,

$$\mu_X = \frac{\sum w_i x_i}{\sum w_i} = \frac{\sum_x x\, p(x)}{\sum_x p(x)} = \sum_{x \in \mathrm{Supp}(X)} x\, p(x),$$

because $\sum_x p(x) = 1$ for a valid pmf. A likely value therefore pulls the mean towards itself more strongly than an unlikely one.

Remark (what $E(X)$ means)

$E(X)$ is the long-term average of X: if the experiment were repeated infinitely many times and the value of X recorded each time, the average of the recorded values would be $E(X)$. The reason is that in a large number N of repetitions the value x comes up about $N\,p(x)$ times, so the average of the N results is about $\frac{1}{N}\sum_x x \cdot N\,p(x) = \sum_x x\,p(x)$. "Mean", "expected value" and "long-term average" are three names for the same number.

- "Expected" does not mean "likely". $E(X)$ need not even be a possible value of X: nobody gets 1.5 heads in three tosses.
- On a graph, μ_X is the balance point of the probability histogram. In particular, if the pmf is symmetric, the mean is at the centre of symmetry.

16.2 Functions of a random variable and LOTUS

Definition (function of a random variable)

A *function of a random variable* X is another random variable H whose value depends on X; we write $H = H(X)$. Whenever X takes the value x, H takes the value $H(x)$.

For instance X^2, $2X + 1$ and $|X - 3|$ are functions of X. Because H is a random variable in its own right, it has its own support, its own pmf p_H and its own expected value. There are two ways to calculate $E(H)$.

1) **By the definition.** First find the support and the pmf of H, then

$$E(H) = \sum_{h \in \mathrm{Supp}(H)} h \cdot p_H(h).$$

Several values of X may be sent to the same value of H. To get $p_H(h)$, add the probabilities of all the values x for which $H(x) = h$ (these are disjoint events):

$$p_H(h) = P(H = h) = \sum_{x\,:\,H(x)=h} p_X(x).$$

2) **By the Law of the Unconscious Statistician (LOTUS).** Stay with the pmf of X and average the values $H(x)$ directly, as in the theorem below.

Theorem (Law of the Unconscious Statistician, LOTUS)

If $H = H(X)$ is a function of a discrete random variable X with pmf $p(x)$, then

$$E(H) = \sum_{x \in \text{Supp}(X)} H(x) \cdot p(x).$$

LOTUS is the useful one in practice, because there is no need to work out the support and the pmf of H. The formula looks so natural that it tends to be used as if it were the definition of $E(H)$, without noticing that it is a theorem; hence the name.

Why LOTUS agrees with the definition

Sort the terms of the sum $\sum_x H(x)\,p(x)$ according to the value $h = H(x)$ that they produce. All the values x in the same group share the factor h, so

$$\sum_{x \in \text{Supp}(X)} H(x)\,p(x) = \sum_{h \in \text{Supp}(H)} \sum_{x\,:\,H(x)=h} h\,p(x) = \sum_{h \in \text{Supp}(H)} h\Big(\sum_{x\,:\,H(x)=h} p(x)\Big)$$
$$= \sum_{h \in \text{Supp}(H)} h\,p_H(h),$$

and the last sum is $E(H)$ by the definition of expected value. □

Example 16.1 ▪ Heads against tails: $E(H)$ two ways Lecture notes

Flip a fair coin 3 times and let X be the number of heads, so that $\text{Supp}(X) = \{0, 1, 2, 3\}$. Now define H as the magnitude of the difference between the number of heads and the number of tails. Find $E(H)$.

Solution. With X heads there are $3 - X$ tails, so H is a function of X:

$$H(X) = \big|X - (3 - X)\big| = |2X - 3|.$$

Evaluate H at each value of X and keep the pmf of X (Section 15) underneath:

x	0	1	2	3
$H(x) = \lvert 2x - 3\rvert$	3	1	1	3
$p_X(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Way 1: by the definition. We need the pmf of H. From the table, $\text{Supp}(H) = \{1, 3\}$. The value $H = 1$ comes from $x = 1$ and from $x = 2$, and the value $H = 3$ comes from $x = 0$ and from $x = 3$, so

$$p_H(1) = p_X(1) + p_X(2) = \tfrac{3}{8} + \tfrac{3}{8} = \tfrac{6}{8} = \tfrac{3}{4}, \qquad p_H(3) = p_X(0) + p_X(3) = \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.$$

Therefore

$$E(H) = \sum_h h\,p_H(h) = 1 \cdot \tfrac{3}{4} + 3 \cdot \tfrac{1}{4} = \boxed{1.5}.$$

Way 2: by LOTUS. Multiply each value $H(x)$ by the probability $p_X(x)$ below it and add:

$$E(H) = \sum_x H(x)\,p_X(x) = 3\cdot\tfrac{1}{8} + 1\cdot\tfrac{3}{8} + 1\cdot\tfrac{3}{8} + 3\cdot\tfrac{1}{8} = \tfrac{12}{8} = \boxed{1.5}.$$

Both ways give the same answer, as they must; LOTUS skips the step of finding p_H.

Caution (the mean of a function is not the function of the mean)

In general $E\big(H(X)\big) \neq H\big(E(X)\big)$. In Example 16.1, $E(X) = 1.5$ and $H(1.5) = |2(1.5) - 3| = 0$, whereas $E(H) = 1.5$. In the same way, $E(X^2)$ is not $[E(X)]^2$. Linear functions are the exception, as the next subsection shows.

16.3 Linearity of the expected value

Property (linearity of the expected value)

For any discrete random variable X and any constants $a, b \in \mathbb{R}$,

$$E(aX + b) = a\,E(X) + b.$$

Proof

The random variable $aX + b$ is a function of X: it is $H(X)$ with $H(x) = ax + b$. By LOTUS,

$$\begin{aligned} E(aX+b) &= \sum_x (ax+b)\,p(x) = \sum_x a\,x\,p(x) + \sum_x b\,p(x) \\ &= a\sum_x x\,p(x) + b\sum_x p(x) = a\,E(X) + b\cdot 1 = a\,E(X) + b. \end{aligned}$$

Here $\sum_x x\,p(x) = E(X)$ by the definition of the expected value, and $\sum_x p(x) = 1$ because p is a valid pmf. □

Remark

- Special cases: $E(b) = b$ (the mean of a constant is that constant), $E(aX) = a\,E(X)$ and $E(X+b) = E(X)+b$.
- The same computation works term by term for a sum of several functions of X; for example, $E(X^2-6X+9) = E(X^2) - 6\,E(X) + 9$.
- Taking $a = 1$ and $b = -\mu_X$ gives $E(X - \mu_X) = 0$: on average, the deviations from the mean cancel out. This is why spread has to be measured with *squared* deviations.

16.4 Variance and standard deviation

Definition (variance and standard deviation)

Let X be a discrete random variable with mean $\mu_X = E(X)$. The *variance* of X is the expected value of the squared deviation of X from its mean:

$$\sigma_X^2 = V(X) := E\big[(X - \mu_X)^2\big] = \sum_{x \in \mathrm{Supp}(X)} (x - \mu_X)^2\,p(x).$$

The *standard deviation* of X is $\sigma_X = \sqrt{\sigma_X^2} = \sqrt{V(X)}$.

The sum on the right is LOTUS applied to the function $(X - \mu_X)^2$. Notation and meaning are the same as for a population of data: we write σ_X^2 for the variance and still call σ_X the standard deviation, and σ_X is still a measure

of width. The larger σ_X is, the more the probability is spread out around μ_X. A variance can never be negative, since it is a sum of squares multiplied by probabilities. The standard deviation is in the same units as X, whereas the variance is in squared units.

Theorem (computational formula for the variance)

For any discrete random variable X,

$$V(X) = E(X^2) - \big[E(X)\big]^2, \qquad \text{where} \quad E(X^2) = \sum_{x \in \mathrm{Supp}(X)} x^2\, p(x)$$

by LOTUS (applied to $Y = X^2$).

Proof

Write $\mu = \mu_X = E(X)$. Start from the definition of the variance and expand the square:

$$\begin{aligned} V(X) &= \sum_x (x-\mu)^2\, p(x) = \sum_x \big(x^2 - 2\mu x + \mu^2\big)\, p(x) \\ &= \sum_x x^2\, p(x) \;-\; 2\mu \sum_x x\, p(x) \;+\; \mu^2 \sum_x p(x) \\ &= E(X^2) - 2\mu \cdot \mu + \mu^2 \cdot 1 \\ &= E(X^2) - \mu^2 = E(X^2) - \big[E(X)\big]^2. \end{aligned}$$

The third line uses $\sum_x x^2 p(x) = E(X^2)$ (LOTUS), $\sum_x x\, p(x) = \mu$ (definition of the mean) and $\sum_x p(x) = 1$ (valid pmf). □

This is the same calculation as the one that gives the computational formula for the variance of a data set (Section 5); the probabilities $p(x)$ simply take over the role of the relative frequencies.

Note (using the computational formula)

The definition needs the deviations $x - \mu_X$, which are awkward when μ_X is a fraction. With the computational formula you only need two sums taken from the same table: $E(X) = \sum_x x\, p(x)$ and $E(X^2) = \sum_x x^2\, p(x)$. Square the values, not the probabilities, and remember to subtract $[E(X)]^2$ at the end. Where a formula shows $E(X)^2$, it means $[E(X)]^2$.

Property (variance of a linear function)

For any discrete random variable X and any constants $a, b \in \mathbb{R}$,

$$V(aX + b) = a^2\, V(X), \qquad \text{and therefore} \qquad \sigma_{aX+b} = |a|\, \sigma_X.$$

Proof

Write $\mu = E(X)$. By linearity, the mean of $aX + b$ is $E(aX + b) = a\mu + b$. Apply the definition of the variance to the random variable $aX + b$ and evaluate the expected value with LOTUS:

$$\begin{aligned} V(aX+b) &= E\Big[\big((aX+b) - (a\mu+b)\big)^2\Big] = \sum_x \big(ax + b - a\mu - b\big)^2\, p(x) \\ &= \sum_x a^2 (x-\mu)^2\, p(x) = a^2 \sum_x (x-\mu)^2\, p(x) = a^2\, V(X). \end{aligned}$$

Taking square roots gives $\sigma_{aX+b} = \sqrt{a^2}\, \sigma_X = |a|\, \sigma_X$. □

Alternative proof: expand the brackets in the computational formula

Again write $\mu = E(X)$. By LOTUS and by linearity,

$$E\big[(aX+b)^2\big] = \sum_x \big(a^2x^2 + 2abx + b^2\big)\,p(x) = a^2E(X^2) + 2ab\mu + b^2,$$

$$\big[E(aX+b)\big]^2 = (a\mu + b)^2 = a^2\mu^2 + 2ab\mu + b^2.$$

Subtracting the second line from the first, the computational formula gives

$$V(aX+b) = a^2E(X^2) - a^2\mu^2 = a^2\big(E(X^2) - \mu^2\big) = a^2\,V(X),$$

where the last step is the computational formula for X itself. □

Remark (what the property says)

Adding a constant b slides the whole distribution along the axis without changing its width, so b disappears from the variance. Multiplying by a stretches every distance from the mean by the factor $|a|$, so the standard deviation is multiplied by $|a|$ and the variance by a^2. In particular, $V(X+b) = V(X)$ and $V(-X) = V(X)$.

Caution (the variance is not linear)

Compare the two properties: in $E(aX+b) = a\,E(X)+b$ both constants survive unchanged, but in $V(aX+b) = a^2\,V(X)$ the constant b drops out, the factor a comes out *squared*, and a minus sign disappears. For example, $V(2-3X) = 9\,V(X)$, not $2-3\,V(X)$. A negative variance is always a sign of a mistake.

Expected value and variance at a glance

Mean	$\mu_X = E(X) = \sum_x x\,p(x)$ (expected value = long-term average)
Function $Y = Y(X)$	$\mu_Y = E(Y) = \sum_y y\,p_Y(y) = \sum_x y(x)\,p(x)$ (the last sum is LOTUS)
Linearity	$E(aX+b) = a\,E(X) + b$
Variance	$\sigma_X^2 = V(X) = E\big[(X-\mu_X)^2\big] = E(X^2) - [E(X)]^2$
Standard deviation	$\sigma_X = \sqrt{V(X)}$
Linear function	$V(aX+b) = a^2\,V(X)$, $\sigma_{aX+b} = \lvert a\rvert\,\sigma_X$

16.5 Worked examples

Example 16.2 ▪ A pmf with an unknown constant Worksheet 16, #1

Let X be a discrete random variable with pmf $p(x) = \dfrac{k}{x}$ and support $\{2,3,4\}$.

a) Find k.

b) Calculate $\mu_X = E(X)$, the expected value for X.

c) What is $P(X = \mu_X)$?

d) Let $Z = |X-3|$.

   i) What is the support of Z?

   ii) Find the pmf for Z.

   iii) Calculate $E(Z)$ two ways: (A) directly, using the definition of expected value and the pmf of Z; (B) using the pmf for X and the Law of the Unconscious Statistician (LOTUS).

Solution.

a) The values of a valid pmf add up to 1 (Section 15):

$$\sum_x p(x) = 1 \implies \frac{k}{2} + \frac{k}{3} + \frac{k}{4} = 1 \implies \frac{13k}{12} = 1 \implies k = \boxed{\frac{12}{13}}.$$

Hence $p(2) = \frac{6}{13}$, $p(3) = \frac{4}{13}$ and $p(4) = \frac{3}{13}$, which all lie between 0 and 1.

b) Multiply each value by its probability and add:

$$\mu_X = E(X) = \sum_x x \cdot p(x) = 2 \cdot \tfrac{6}{13} + 3 \cdot \tfrac{4}{13} + 4 \cdot \tfrac{3}{13} = \boxed{\tfrac{36}{13}} \approx 2.77.$$

c) The only possible values of X are 2, 3 and 4. Since $\frac{36}{13} \notin \text{Supp}(X)$,

$$P(X = \mu_X) = \boxed{0}.$$

The mean is the long-term average of X; it does not have to be a value that X can actually take.

d) Evaluate $Z = |X - 3|$ at each value of X:

x	2	3	4
$z = \lvert x - 3 \rvert$	1	0	1
$p_X(x)$	$\frac{6}{13}$	$\frac{4}{13}$	$\frac{3}{13}$

i) The only values that occur are 0 and 1, so $\text{Supp}(Z) = \boxed{\{0, 1\}}$.

ii) $Z = 0$ happens only when $X = 3$, and $Z = 1$ happens when $X = 2$ or $X = 4$. These two events are disjoint, so their probabilities add:

$$P(Z = 0) = P(X = 3) = \tfrac{4}{13},$$
$$P(Z = 1) = P\big((X = 2) \cup (X = 4)\big) = P(X = 2) + P(X = 4) = \tfrac{6}{13} + \tfrac{3}{13} = \tfrac{9}{13}.$$

z	0	1
$p_Z(z)$	$\frac{4}{13}$	$\frac{9}{13}$

iii) (A) By the definition, with the pmf of Z:

$$E(Z) = \sum_z z \cdot p_Z(z) = 0 \cdot \tfrac{4}{13} + 1 \cdot \tfrac{9}{13} = \boxed{\tfrac{9}{13}}.$$

(B) By LOTUS, with the pmf of X:

$$E(Z) = \sum_x |x - 3| \cdot p_X(x) = |2 - 3| \cdot \tfrac{6}{13} + |3 - 3| \cdot \tfrac{4}{13} + |4 - 3| \cdot \tfrac{3}{13} = \tfrac{6}{13} + 0 + \tfrac{3}{13} = \boxed{\tfrac{9}{13}}.$$

Example 16.3 ▪ Drawing marbles until the red one appears Worksheet 16, #2

We put three green marbles and one red marble in a bag. Then we draw out marbles one by one until we draw the red one. Let X represent the number of marbles we draw out of the bag.

a) What is the support of X?

b) Give the pmf for X.

c) Find $E(X)$.

d) Explain in words how to interpret $E(X)$ in the context of this probability experiment.

e) Once we have drawn the red marble, let Z be the magnitude of the difference between the number of marbles outside of the bag and the number of marbles remaining inside the bag.

i) Express Z as a function of X.

ii) Find $E(Z)$ using LOTUS.

Solution.

a) The red marble can come out first, second, third or last, so $\text{Supp}(X) = \boxed{\{1, 2, 3, 4\}}$.

b) The marbles are not put back, so the probabilities change from draw to draw. $X = x$ means that the first $x - 1$ marbles are green and the next one is red; by the multiplication rule (Section 13),

$$p(1) = \tfrac{1}{4}, \qquad p(2) = \tfrac{3}{4} \cdot \tfrac{1}{3} = \tfrac{1}{4}, \qquad p(3) = \tfrac{3}{4} \cdot \tfrac{2}{3} \cdot \tfrac{1}{2} = \tfrac{1}{4}, \qquad p(4) = \tfrac{3}{4} \cdot \tfrac{2}{3} \cdot \tfrac{1}{2} \cdot 1 = \tfrac{1}{4}.$$

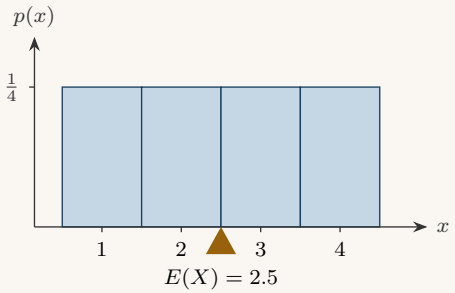
x	1	2	3	4
$p(x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

This makes sense: the red marble is equally likely to be in any of the four positions of the drawing order. The four values add up to 1, as they should. (A pmf like this one, in which all the values are equally likely, is called a discrete uniform distribution; see Section 17.)

c) By the definition of expected value,

$$E(X) = \sum_x x \cdot p(x) = 1 \cdot \tfrac{1}{4} + 2 \cdot \tfrac{1}{4} + 3 \cdot \tfrac{1}{4} + 4 \cdot \tfrac{1}{4} = \tfrac{10}{4} = \boxed{2.5}.$$

The pmf is symmetric, and 2.5 is its centre: the point where the histogram balances.



d) $E(X)$ is the long-term average from repeating the process. If we ran this experiment a very large number of times (refill the bag, draw until the red marble appears, record how many marbles were drawn) and averaged the recorded numbers, the average would tend to 2.5 marbles as the number of repetitions tends to infinity. No single run can give 2.5 draws.

e) i) When the red marble appears, X marbles are outside the bag and $4 - X$ are still inside. So

$$Z = \big|X - (4 - X)\big| = \boxed{|2X - 4|}.$$

ii) By LOTUS there is no need for the pmf of Z:

$$E(Z) = \sum_x Z(x)\, p(x) = |2(1) - 4| \cdot \tfrac{1}{4} + |2(2) - 4| \cdot \tfrac{1}{4} + |2(3) - 4| \cdot \tfrac{1}{4} + |2(4) - 4| \cdot \tfrac{1}{4}$$

$$= \tfrac{2+0+2+4}{4} = \boxed{2}.$$

Example 16.4 ▪ Variance and standard deviation from a pmf Worksheet 16, #4

Calculate the variance and the standard deviation for X in Examples 16.2 and 16.3.

Solution. In both cases we use the computational formula $V(X) = E(X^2) - [E(X)]^2$. The mean is known from the earlier example, and $E(X^2) = \sum_x x^2\, p(x)$ by LOTUS (with $Y = X^2$).

1) X of Example 16.2: $p(2) = \frac{6}{13}$, $p(3) = \frac{4}{13}$, $p(4) = \frac{3}{13}$ and $E(X) = \frac{36}{13}$.

$$E(X^2) = 2^2 \cdot \tfrac{6}{13} + 3^2 \cdot \tfrac{4}{13} + 4^2 \cdot \tfrac{3}{13} = \tfrac{24+36+48}{13} = \tfrac{108}{13},$$
$$V(X) = \tfrac{108}{13} - \left(\tfrac{36}{13}\right)^2 = \tfrac{1404}{169} - \tfrac{1296}{169} = \boxed{\tfrac{108}{169}} \approx 0.639,$$
$$\sigma_X = \sqrt{V(X)} = \sqrt{\tfrac{108}{169}} = \tfrac{\sqrt{108}}{13} = \boxed{\tfrac{6\sqrt{3}}{13}} \approx 0.799.$$

2) X of Example 16.3: $p(x) = \frac{1}{4}$ for $x = 1, 2, 3, 4$ and $E(X) = \frac{5}{2}$.

$$E(X^2) = \left(1^2 + 2^2 + 3^2 + 4^2\right) \cdot \tfrac{1}{4} = \tfrac{30}{4} = \tfrac{15}{2},$$
$$V(X) = \tfrac{15}{2} - \left(\tfrac{5}{2}\right)^2 = \tfrac{30}{4} - \tfrac{25}{4} = \boxed{\tfrac{5}{4}},$$
$$\sigma_X = \sqrt{\tfrac{5}{4}} = \boxed{\tfrac{\sqrt{5}}{2}} \approx 1.118.$$

As a check, the definition gives the same variance here: $\sum_x (x - 2.5)^2 \cdot \frac{1}{4} = \frac{2.25+0.25+0.25+2.25}{4} = \frac{5}{4}$.

Example 16.5 ▪ The variance of a linear function Worksheet 16, #5

For a discrete random variable X with variance 5, find $V(2 - 3X)$.

Solution. $2 - 3X$ has the form $aX + b$ with $a = -3$ and $b = 2$. By the property for the variance of a linear function, $V(aX + b) = a^2\, V(X)$:

$$V(2 - 3X) = (-3)^2\, V(X) = 9\, V(X) = 9(5) = \boxed{45}.$$

The constant 2 has no effect on the variance, and the factor -3 is squared. Compare with the mean, for which the same function gives $E(2 - 3X) = 2 - 3\, E(X)$.

Example 16.6 ▪ A standard deviation without the pmf Worksheet 16, #6

If $E(X) = 4$ and $E(Z^2) = 89$, where $Z = 2X + 1$, find σ_Z.

Solution. We want $\sigma_Z = \sqrt{V(Z)}$, and the computational formula $V(Z) = E(Z^2) - [E(Z)]^2$ needs two things. $E(Z^2) = 89$ is given, and $E(Z)$ follows from $E(X)$ by linearity:

$$E(Z) = E(2X + 1) = 2\, E(X) + 1 = 2(4) + 1 = 9.$$

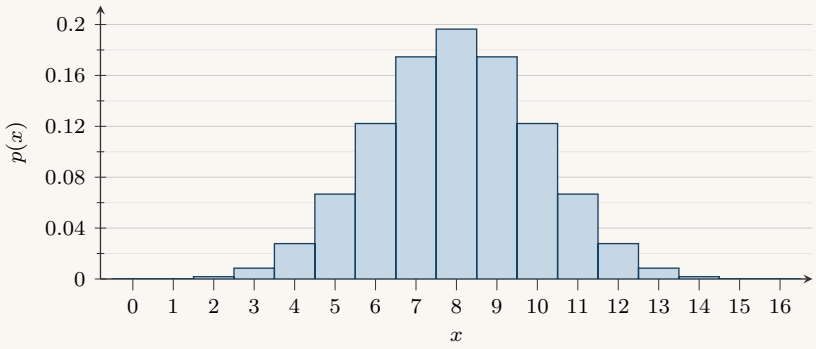
Therefore

$$V(Z) = E(Z^2) - \left[E(Z)\right]^2 = 89 - 9^2 = 8, \qquad \sigma_Z = \sqrt{8} = \boxed{2\sqrt{2}} \approx 2.83.$$

The same information also gives the spread of X: since $V(Z) = V(2X + 1) = 4\, V(X)$, we get $V(X) = 2$ and $\sigma_X = \sqrt{2}$.

Example 16.7 ▪ Within one standard deviation of the mean Worksheet 16, #7

Below is the pmf histogram for a variable X. You are told that $E(X) = 8$ and $V(X) = 4$.



a) Use the histogram to estimate $P(\mu - \sigma < X \leq \mu + \sigma)$.

b) Shade the area of the histogram that corresponds to the probability in (a) and label μ and σ on the histogram.

Solution.

a) We are given the variance, so first take its square root: $\sigma = \sqrt{V(X)} = \sqrt{4} = 2$. With $\mu = 8$, the interval runs from $\mu - \sigma = 6$ to $\mu + \sigma = 10$. X takes whole-number values, 6 is excluded ($<$) and 10 is included ($\leq$), so

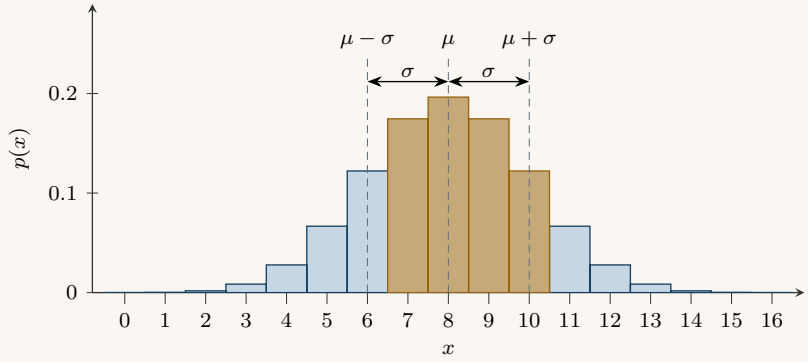
$$\begin{aligned} P(\mu - \sigma < X \leq \mu + \sigma) &= P(6 < X \leq 10) \\ &= P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10). \end{aligned}$$

Reading the heights of these four bars from the histogram,

$$p(7) + p(8) + p(9) + p(10) \approx 0.175 + 0.196 + 0.175 + 0.122 \approx \boxed{0.67}.$$

The heights are read from a graph, so this is an estimate.

b) The probability in (a) is the total area of the four bars above 7, 8, 9 and 10 (each bar has width 1, so its area equals its height).



The bar above 6 is left unshaded because $X = 6$ is not part of the event. About two thirds of the probability lies within one standard deviation of the mean, which is what the empirical rule (Section 7) predicts for a bell-shaped distribution: μ and σ are read for a random variable exactly as they are for a data set.

17 The Binomial Distribution

Sections 15 and 16 dealt with discrete random variables in general: you were handed a pmf, or you built one from scratch. In practice the same few situations come up again and again, and each of them has a ready-made pmf, mean and variance. This section lists those *standard discrete distributions* and then studies the most important one,

the binomial distribution, which counts successes in a fixed number of independent trials.

17.1 The standard discrete distributions: a roadmap

The course covers six standard discrete distributions. The main skill in using them is *recognition*: read the story, decide which distribution fits, then use its formulas.

Distribution	The situation it describes	Where
Discrete uniform	one trial, n equally likely outcomes	this subsection
Binomial	two outcomes, *fixed number of trials*, count the successes	this section
Negative binomial	two outcomes, *fixed number of successes*, count the trials	Section 18
Geometric	two outcomes, count the trials until the *first* success	Section 18
Hypergeometric	two outcomes, fixed number of draws *without replacement*	Section 19
Poisson	number of events in a fixed interval of time or space	Section 20

Definition (discrete uniform distribution)

A random variable X has a *discrete uniform distribution* if every value in its support is equally likely. For example, if $\text{Supp}(X) = \{1, 2, \ldots, n\}$, then

$$p(x) = \frac{1}{n} \qquad \text{for all } x \in \text{Supp}(X).$$

For this support, $E(X) = \dfrac{n+1}{2}$ and $V(X) = \dfrac{n^2-1}{12}$.

Remark (telling the distributions apart)

- **Uniform:** only *one* trial, but possibly more than two outcomes (rolling a die once).
- **Binomial / negative binomial:** each trial has only *two* outcomes, and the trials are *independent*. In the binomial the number of trials n is fixed; in the negative binomial the number of trials is unknown and is the thing we count.
- **Hypergeometric:** a binomial-type count, but sampling is done *without replacement*, so the trials are *dependent*.

17.2 Binomial experiments

Definition (binomial experiment)

A *binomial experiment* is an experiment that satisfies all four of the following conditions.

1) It consists of n repeated trials, where n is fixed in advance.
2) Each trial has only two possible outcomes: *success* (S) or *failure* (F).
3) The probabilities $P(S)$ and $P(F)$ are the same for every trial. We write
$$p = P(S), \qquad q = P(F) = 1 - p.$$
4) The trials are independent.

Note

Check all four criteria every time before you use a binomial formula. "Success" is only a label for the outcome being counted; it does not have to be a good thing (a phone that has to be replaced can be the "success").

Definition (binomial random variable)

X is a *binomial random variable* if

$$X = \text{the number of successes in the } n \text{ trials of a binomial experiment.}$$

In this case we write $X \sim \text{Bin}(n, p)$, where n is the number of trials and $p = P(S)$ is the probability of success on a single trial. Its support is $\text{Supp}(X) = \{0, 1, 2, \ldots, n\}$.

17.3 The pmf, mean and variance

Theorem (binomial pmf)

If $X \sim \text{Bin}(n, p)$ and $q = 1 - p$, then

$$p(x) = P(X = x) = \binom{n}{x} p^x q^{n-x}, \qquad x = 0, 1, \ldots, n,$$

where $\binom{n}{x} = {}_nC_x = \dfrac{n!}{x!\,(n-x)!}$.

Why?

Think of one particular run of the experiment in which the first x trials are successes and the remaining $n - x$ are failures:

$$\underbrace{S\,S\,S\cdots S}_{x \text{ successes}}\ \underbrace{F\,F\cdots F}_{n-x \text{ failures}}$$

Because the trials are independent, this exact sequence has probability $p^x q^{n-x}$. Every other arrangement of x successes among the n trials has the same probability, and there are $\binom{n}{x}$ such arrangements (choose which x of the n trials are the successes). Adding them up gives $\binom{n}{x} p^x q^{n-x}$.

Remark (where the name comes from)

$p(x)$ is exactly the term containing p^x in the binomial expansion of $(p + q)^n$:

$$(p + q)^n = \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x}.$$

Since $p + q = 1$, this also shows that $\sum_x p(x) = 1^n = 1$, so the binomial pmf is a valid pmf.

Notation

Some books and worksheets write $b(x; n, p)$ for the binomial pmf. It means the same thing:

$$b(x; n, p) = P(X = x) \quad \text{where } X \sim \text{Bin}(n, p).$$

Theorem (mean and variance of a binomial random variable)

If $X \sim \text{Bin}(n, p)$, then

$$\mu_X = E(X) = np, \qquad \sigma_X^2 = V(X) = npq, \qquad \sigma_X = \sqrt{npq}.$$

Proof that $E(X) = np$

By the definition of expected value (the $x = 0$ term is 0, so the sum may start at $x = 1$),

$$\begin{aligned} E(X) = \sum_x x\,p(x) &= \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x} = \sum_{x=1}^{n} x\,\frac{n!}{x!\,(n-x)!}\,p^x q^{n-x} \\ &= \sum_{x=1}^{n} x\,\frac{n\,(n-1)!}{x\,(x-1)!\,\big((n-1)-(x-1)\big)!}\,p\cdot p^{x-1} q^{(n-1)-(x-1)} \\ &= np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!\,\big((n-1)-(x-1)\big)!}\,p^{x-1} q^{(n-1)-(x-1)}. \end{aligned}$$

Now substitute $y = x - 1$ and $m = n - 1$:

$$E(X) = np \underbrace{\sum_{y=0}^{m} \binom{m}{y} p^y q^{m-y}}_{\text{all of the pmf of } \text{Bin}(m,p)} = np \cdot 1 = np.$$

The sum equals 1 because it adds up the whole pmf of a $\text{Bin}(m, p)$ random variable. □

Remark

The formula $V(X) = npq$ can be proved with the same trick. First compute $E\big(X(X-1)\big) = n(n-1)p^2$ by cancelling $x(x-1)$ against $x!$. Then

$$\begin{aligned} V(X) = E(X^2) - E(X)^2 &= E\big(X(X-1)\big) + E(X) - E(X)^2 \\ &= n(n-1)p^2 + np - n^2p^2 = np(1-p) = npq. \end{aligned}$$

A shorter proof appears in Section 26, where X is written as a sum of n independent trials.

Binomial distribution at a glance

Conditions	(1) n repeated trials, n fixed; (2) two outcomes per trial; (3) $P(S)$ and $P(F)$ fixed; (4) independent trials
Variable	X = number of successes in the n trials; $X \sim \text{Bin}(n, p)$
pmf	$p(x) = {}_nC_x\,p^x q^{n-x}$ for $x = 0, 1, \ldots, n$
Mean, variance	$\mu_X = np$, $\sigma_X^2 = npq$, $\sigma_X = \sqrt{npq}$

17.4 Worked examples

Example 17.1 ▪ Reading the notation Worksheet 17, #1

Evaluate a) $b(7; 11, 0.35)$ b) $P(2 < X \le 5)$ if $X \sim \text{Bin}(8, 0.3)$.

Solution.

a) $b(7; 11, 0.35)$ is $P(X = 7)$ where $X \sim \text{Bin}(11, 0.35)$ counts the number of successes. So

$$P(X = 7) = \binom{11}{7}(0.35)^7(0.65)^4 \approx \boxed{0.0379}.$$

b) X only takes integer values, so $2 < X \le 5$ means $X \in \{3, 4, 5\}$. With $n = 8$, $p = 0.3$, $q = 0.7$:

$$\begin{aligned} P(2 < X \le 5) &= p(3) + p(4) + p(5) \\ &= \binom{8}{3}(0.3)^3(0.7)^5 + \binom{8}{4}(0.3)^4(0.7)^4 + \binom{8}{5}(0.3)^5(0.7)^3 \\ &\approx 0.2541 + 0.1361 + 0.0467 \approx \boxed{0.4369}. \end{aligned}$$

Watch the inequality signs: 2 is excluded ($<$) and 5 is included ($\le$).

Example 17.2 ▪ Passenger cars on the bridge

Worksheet 17, #2

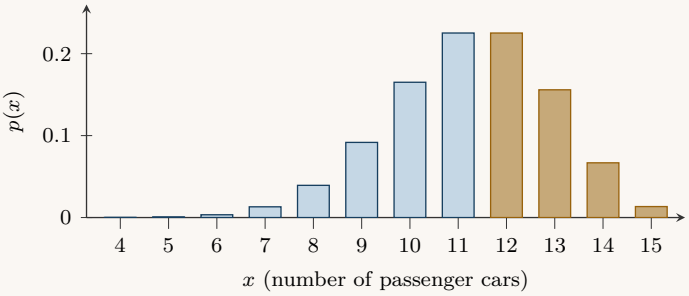
Passenger cars make up 3 times the amount of other traffic on the Jake Carter bridge. If a police officer stops 15 vehicles at random for an inspection, what is the probability that at least 12 of the 15 are passenger cars?

Solution. *Find p.* Passenger cars and other vehicles are in the ratio $3 : 1$, so a randomly stopped vehicle is a passenger car with probability $p = \frac{3}{4}$, and $q = \frac{1}{4}$.

Identify the distribution. There are $n = 15$ trials (vehicles), each one is a passenger car or not, p is the same each time, and the vehicles are stopped independently. So if X is the number of passenger cars among the 15, then $X \sim \text{Bin}\left(15, \frac{3}{4}\right)$.

Compute. "At least 12" means $X \ge 12$:

$$\begin{aligned} P(X \ge 12) &= P(X = 12) + P(X = 13) + P(X = 14) + P(X = 15) \\ &= \binom{15}{12}\left(\frac{3}{4}\right)^{12}\left(\frac{1}{4}\right)^3 + \binom{15}{13}\left(\frac{3}{4}\right)^{13}\left(\frac{1}{4}\right)^2 + \binom{15}{14}\left(\frac{3}{4}\right)^{14}\left(\frac{1}{4}\right)^1 + \binom{15}{15}\left(\frac{3}{4}\right)^{15} \\ &\approx 0.2252 + 0.1559 + 0.0668 + 0.0134 \approx \boxed{0.4613}. \end{aligned}$$



The pmf of $\text{Bin}(15, \frac{3}{4})$. The dark bars are the event $X \ge 12$; their heights add up to 0.4613.

Example 17.3 ▪ Paying by plastic: mean, spread and an expected cost

Worksheet 17, #4

At a given gas station, 62% of customers pay using plastic (debit or credit) and the remaining pay cash. The station typically gets 150 customers per day. Customers pay plastic or cash independently of each other.

a) What is the expected number of customers who will pay using plastic on a typical day?

b) What range of number of customers is within one standard deviation of the number in (a)?

c) If the station pays bank fees of \$0.25 for the first 90 plastic transactions per day and \$0.10 for every

transaction thereafter, what is the expected daily cost for the station to offer payment by plastic? Give your answer in summation notation. [Hint: use LOTUS.]

Solution. Let X be the number of customers, out of $n = 150$, who pay with plastic. Each customer is a trial with $p = 0.62$ (plastic) and $q = 0.38$ (cash), so $X \sim \text{Bin}(150, 0.62)$.

a) "Expected number" means the long-run average, μ_X:

$$\mu_X = E(X) = np = 150(0.62) = \boxed{93 \text{ customers}}.$$

b) $\sigma_X = \sqrt{V(X)} = \sqrt{npq} = \sqrt{150(0.62)(0.38)} \approx 5.94$, so

$$[\mu_X - \sigma_X,\ \mu_X + \sigma_X] \approx [87.06,\ 98.94].$$

A number of customers is a whole number, so the range is $\boxed{88 \text{ to } 98 \text{ customers}}$.

c) The cost is itself a random variable that depends on X. If x customers pay with plastic, the bank fees are

$$C(x) = \begin{cases} 0.25\,x, & x \le 90, \\ 0.25(90) + 0.10\,(x - 90), & x > 90. \end{cases}$$

We do not need the pmf of C. By LOTUS (Section 16),

$$E(C) = \sum_x C(x)\,p(x) \qquad \text{with} \qquad p(x) = \binom{150}{x}(0.62)^x(0.38)^{150-x},$$

and the sum splits where the formula for C changes:

$$E(C) = \sum_{x=0}^{90} 0.25\,x \binom{150}{x}(0.62)^x(0.38)^{150-x} + \sum_{x=91}^{150} \left[0.25(90) + 0.10\,(x - 90)\right]\binom{150}{x}(0.62)^x(0.38)^{150-x}.$$

Evaluating the sum with software gives $E(C) \approx \boxed{\$22.62}$ per day.

Example 17.4 ▪ A binomial inside a binomial

Worksheet 17, #5

The Bogus Reed polling company has employed 10 telephonists to take a phone survey about the announced tuition fee increases in Quebec. At the beginning of their shift, each telephonist is given a list of 25 numbers to call. Historically, 60% of calls result in a completed survey. What is the probability that 2 or more of the telephonists complete between 14 and 16 surveys, inclusive?

Solution. There are two binomial experiments here, one nested inside the other, so we need two random variables.

Second (inner) binomial: one telephonist. Let X be the number of surveys one telephonist completes out of 25 calls. Then $X \sim \text{Bin}(25, 0.6)$, and the chance that a given telephonist completes between 14 and 16 surveys is

$$p^* = P(14 \le X \le 16) = \binom{25}{14}(0.6)^{14}(0.4)^{11} + \binom{25}{15}(0.6)^{15}(0.4)^{10} + \binom{25}{16}(0.6)^{16}(0.4)^{9}$$
$$\approx 0.1465 + 0.1612 + 0.1511 \approx 0.4588.$$

First (outer) binomial: the ten telephonists. Now each telephonist is a trial, and a "success" is a telephonist who completes between 14 and 16 surveys. Let Y be the number of such telephonists. Then $Y \sim \text{Bin}(10, p^*)$ with $q^* = 1 - p^* \approx 0.5412$.

"2 or more" is easiest through the complement, since only two values are left out:

$$P(Y \geq 2) = 1 - P(Y = 0) - P(Y = 1) = 1 - \binom{10}{0}(p^*)^0(q^*)^{10} - \binom{10}{1}(p^*)^1(q^*)^9$$
$$\approx 1 - 0.0022 - 0.0183 \approx \boxed{0.9796}.$$

Note (complement trick)

For "at least" questions, compare how many terms each route needs. $P(Y \geq 2)$ directly would take nine terms, but $1 - P(Y \leq 1)$ takes two. In Example 17.2 the direct sum $P(X \geq 12)$ had only four terms, so there the direct route was the shorter one.

18 The Negative Binomial and Geometric Distributions

A binomial random variable (Section 17) fixes the number of trials and counts the successes. This section turns the question around: fix the number of successes we want, and count how many trials it takes to get them. The result is the negative binomial distribution. Its simplest case, waiting for the *first* success, is the geometric distribution.

18.1 Negative binomial experiments

Definition (negative binomial experiment)

A *negative binomial experiment* is an experiment that satisfies all four of the following conditions.

1) Each trial has only two possible outcomes: success (S) or failure (F).
2) The probabilities $p = P(S)$ and $q = P(F) = 1 - p$ are constants: they are the same for every trial.
3) The trials are independent, and they are repeated one after another.
4) The number of successes we want, k, is fixed in advance. The number of trials is *not* fixed: the trials go on until the k-th success occurs, and then they stop.

Definition (negative binomial random variable)

X is a *negative binomial random variable* if

$$X = \text{the number of trials needed to have } k \text{ successes.}$$

In this case we write $X \sim \mathrm{NB}(k, p)$, where k is the number of successes wanted and $p = P(S)$ is the probability of success on a single trial. Its support is

$$\mathrm{Supp}(X) = \{k,\ k+1,\ k+2,\ \dots\}.$$

The smallest possible value is k (every trial is a success). There is no largest value, because the run of failures before the k-th success can be as long as you like; the support is infinite.

Conditions 1 to 3 are the same as for a binomial experiment. Only the fourth one changes, and it swaps the roles of "fixed" and "random":

	Binomial, $X \sim \mathrm{Bin}(n, p)$	**Negative binomial,** $X \sim \mathrm{NB}(k, p)$
Fixed in advance	the number of trials, n	the number of successes, k
Random (what X counts)	the number of successes	the number of trials
Support of X	$\{0, 1, \dots, n\}$	$\{k,\ k+1,\ k+2, \dots\}$
A typical event	"3 malfunctions in 5 days"	"the 3rd malfunction happens on day 5"

18.2 The pmf, mean and variance

Theorem (negative binomial pmf)

If $X \sim \mathrm{NB}(k, p)$ and $q = 1 - p$, then

$$p(x) = P(X = x) = \binom{x-1}{k-1} p^k q^{x-k}, \qquad x = k,\ k+1,\ k+2,\ \dots$$

Why?

The event $X = x$ says that the k-th success arrives exactly on trial x. The trials stop as soon as the k-th success arrives, so the last trial, trial x, must be a success. The question therefore becomes: in how many ways can the other $k-1$ successes be arranged among the first $x-1$ trials?

$$\underbrace{F\ S\ F\ F\ S\ \cdots\ S\ F}_{\substack{\text{first } x-1 \text{ trials:}\\ \text{exactly } k-1 \text{ successes, in any order}}} \qquad \underbrace{S}_{\substack{\text{trial } x:\\ \text{the } k\text{-th success}}}$$

The first $x-1$ trials form a binomial experiment, so by the binomial pmf (Section 17) the probability that they contain exactly $k-1$ successes is $\binom{x-1}{k-1} p^{k-1} q^{(x-1)-(k-1)}$. Trial x is a success with probability p, independently of the earlier trials. Multiplying the two,

$$p(x) = \underbrace{\binom{x-1}{k-1} p^{k-1} q^{(x-1)-(k-1)}}_{k-1 \text{ successes in the first } x-1 \text{ trials}} \cdot \underbrace{p}_{\text{trial } x} = \binom{x-1}{k-1} p^k q^{x-k}.$$

Caution

The coefficient is $\binom{x-1}{k-1}$, not $\binom{x}{k}$. The last trial is forced to be a success, so only the first $x-1$ trials can be arranged freely. The exponents are easy to check: k successes and $x-k$ failures make x trials in total.

Theorem (mean and variance of a negative binomial random variable)

If $X \sim \mathrm{NB}(k, p)$, then

$$\mu_X = E(X) = \frac{k}{p}, \qquad \sigma_X^2 = V(X) = \frac{kq}{p^2}, \qquad \sigma_X = \frac{\sqrt{kq}}{p},$$

where k is the number of successes wanted.

The mean is what common sense suggests. If $p = 0.2$, one trial in five succeeds in the long run, so one success costs $\frac{1}{p} = 5$ trials on average and k successes cost $\frac{k}{p}$ trials.

Remark (the pmf adds up to 1 over the whole support)

Like every pmf, the negative binomial pmf adds up to 1, provided that the sum runs over the *whole* support, starting at $x = k$:

$$\sum_{x=k}^{\infty} \binom{x-1}{k-1} p^k q^{x-k} = 1.$$

For example,

$$\sum_{x=3}^{\infty} \binom{x-1}{2} (0.25)^3 (0.75)^{x-3} = 1,$$

because the terms are the pmf of $\mathrm{NB}(3,\ 0.25)$ and the sum covers all of its support $\{3, 4, 5, \dots\}$. The same sum started at $x = 4$ leaves out $p(3) = (0.25)^3$, so it equals $1 - 0.015625 \approx 0.9844$, not 1.

The algebra behind this is the binomial series with a *negative* exponent, valid for $|q| < 1$:

$$(1-q)^{-k} = \sum_{x=k}^{\infty} \binom{x-1}{k-1} q^{x-k} \qquad \Longrightarrow \qquad \sum_{x=k}^{\infty} p(x) = p^k (1-q)^{-k} = p^k \, p^{-k} = 1.$$

This series is where the name "negative binomial" comes from.

18.3 The geometric distribution

The most common special case is $k = 1$: we wait for a single success.

Definition (geometric random variable)

The *geometric distribution* is the negative binomial distribution with $k = 1$. It describes an experiment in which

1) we count the trials until 1 success occurs (the first success ends the experiment);
2) $p = P(S)$ and $q = P(F) = 1 - p$ are constants;
3) the trials are independent.

The random variable is

$$X = \text{the number of trials needed for 1 success}$$

(the trial on which the success occurs is counted). We write $X \sim \text{Geom}(p)$. Its support is $\text{Supp}(X) = \{1, 2, 3, \dots\}$.

Theorem (pmf, mean and variance of a geometric random variable)

If $X \sim \text{Geom}(p)$ and $q = 1 - p$, then

$$p(x) = P(X = x) = p \, q^{x-1}, \qquad x = 1, 2, 3, \dots$$

and

$$\mu_X = E(X) = \frac{1}{p}, \qquad \sigma_X^2 = V(X) = \frac{1-p}{p^2} = \frac{q}{p^2}.$$

The pmf can be read off directly: $X = x$ means that the first $x - 1$ trials are failures and trial x is a success,

$$\underbrace{F \, F \, F \cdots F}_{x-1 \text{ failures}} \, S,$$

and by independence this sequence has probability $q^{x-1}p$. There is only one such sequence, so no binomial coefficient is needed.

Remark (a special case of the negative binomial)

The geometric distribution can be viewed as a special case of the negative binomial distribution:

$$\text{Geom}(p) = \text{NB}(1, p).$$

Putting $k = 1$ in the negative binomial formulas gives back everything in the theorem above:

$$\binom{x-1}{0} p^1 q^{x-1} = p \, q^{x-1}, \qquad \frac{k}{p} = \frac{1}{p}, \qquad \frac{kq}{p^2} = \frac{q}{p^2}.$$

The name comes from the probabilities $p,\ pq,\ pq^2,\ pq^3, \dots$, which form a geometric sequence with common ratio q.

The two proofs below rest on one fact from calculus.

Recall (geometric series)

If $|r| < 1$, then

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \cdots = \frac{1}{1-r}.$$

Proof that $\sum p(x) = 1$ for a geometric random variable

Let $X \sim \text{Geom}(p)$ with $0 < p \leq 1$. Factor p out of the sum and recognise a geometric series with ratio $r = q$:

$$\sum_{x=1}^{\infty} p(x) = \sum_{x=1}^{\infty} p\, q^{\,x-1} = p\left(1 + q + q^2 + q^3 + \cdots\right) = p \cdot \frac{1}{1-q} = p \cdot \frac{1}{p} = 1.$$

The series converges because $0 \leq q < 1$. □

Proof that $E(X) = \frac{1}{p}$ for a geometric random variable

Start from the definition of expected value and replace p by $1 - q$:

$$E(X) = \sum_{x} x\, p(x) = \sum_{x=1}^{\infty} x\, q^{\,x-1} p = \sum_{x=1}^{\infty} x\, q^{\,x-1}(1-q) = \sum_{x=1}^{\infty} \left(x\, q^{\,x-1} - x\, q^{\,x}\right)$$
$$= (1-q) + (2q - 2q^2) + (3q^2 - 3q^3) + (4q^3 - 4q^4) + \cdots.$$

Now combine the second term of each bracket with the first term of the next bracket: $-q + 2q = q$, then $-2q^2 + 3q^2 = q^2$, then $-3q^3 + 4q^3 = q^3$, and so on. This leaves a geometric series:

$$E(X) = 1 + q + q^2 + q^3 + \cdots = \sum_{x=1}^{\infty} q^{\,x-1} = \frac{1}{1-q} = \frac{1}{1-(1-p)} = \frac{1}{p}.$$

(The regrouping is legitimate: the first n brackets add up to $1 + q + \cdots + q^{\,n-1} - n\, q^{\,n}$, and $n\, q^{\,n} \to 0$ as $n \to \infty$ because $0 \leq q < 1$.) □

Property (tail of a geometric distribution)

If $X \sim \text{Geom}(p)$, then for $n = 0, 1, 2, \ldots$

$$P(X > n) = q^{\,n}, \qquad \text{equivalently} \qquad P(X \geq n) = q^{\,n-1} \quad (n \geq 1).$$

The reason is that $X > n$ says exactly that the first n trials are all failures. This shortcut replaces an infinite sum (or a long complement) by a single power.

Remark (a negative binomial variable is a sum of geometric waiting times)

Split the wait for k successes into k stages: the trials up to the first success, then the further trials up to the second success, and so on. Each stage is a fresh geometric wait, so

$$X = X_1 + X_2 + \cdots + X_k, \qquad X_i \sim \text{Geom}(p) \text{ independent.}$$

By the rules for sums of independent random variables (Section 26), the means add and the variances add, which explains the factor k in the negative binomial formulas:

$$E(X) = k \cdot \frac{1}{p}, \qquad V(X) = k \cdot \frac{q}{p^2}.$$

Negative binomial and geometric distributions at a glance

	Negative binomial	Geometric
Conditions	two outcomes per trial; p, q constant; independent trials; stop at the k-th success	the same, with $k = 1$
Variable	X = number of trials needed for k successes; $X \sim \text{NB}(k, p)$	X = number of trials needed for the first success; $X \sim \text{Geom}(p)$
Support	$\{k, k+1, k+2, \dots\}$	$\{1, 2, 3, \dots\}$
pmf	$p(x) = \binom{x-1}{k-1} p^k q^{x-k}$	$p(x) = p\, q^{x-1}$
Mean	$\mu_X = \dfrac{k}{p}$	$\mu_X = \dfrac{1}{p}$
Variance	$\sigma_X^2 = \dfrac{kq}{p^2}$	$\sigma_X^2 = \dfrac{q}{p^2}$

18.4 Worked examples

Example 18.1 ⋆ ▪ **The telemarketer's quota** Worksheet 18, #1

A telemarketer must interview 5 people for a long survey. Historically, the probability is $p = 0.2$ that a randomly chosen person agrees to answer such a time-consuming survey.

a) What is the probability that the telemarketer has to call 15 people in order to satisfy her quota?

b) How many people should she expect to have to call?

c) What is the standard deviation of the number in (b)?

Solution. *Identify the distribution.* Each call is a trial with two outcomes: the person agrees (success, $p = 0.2$) or refuses (failure, $q = 0.8$). The calls are independent and p is the same each time. The number of successes wanted is fixed, $k = 5$, while the number of calls is not. So if

$$X = \text{the number of people she has to call to find 5 who take the survey,}$$

then $X \sim \text{NB}(5, 0.2)$.

a) "Has to call 15 people" means that the fifth success comes exactly on the 15th call: among the first 14 calls exactly 4 people agree, and the 15th person agrees.

$$P(X = 15) = \binom{15-1}{5-1}(0.2)^5(0.8)^{15-5} = \binom{14}{4}(0.2)^5(0.8)^{10}$$
$$= 1001\,(0.2)^5(0.8)^{10} \approx \boxed{0.0344}.$$

b) $\mu_X = E(X) = \dfrac{k}{p} = \dfrac{5}{0.2} = \boxed{25 \text{ people}}$.

c) $\sigma_X^2 = V(X) = \dfrac{kq}{p^2} = \dfrac{5(0.8)}{(0.2)^2} = 100$, so $\sigma_X = \sqrt{100} = \boxed{10 \text{ people}}$.

Example 18.2 ▪ **Third malfunction on the fifth day** Worksheet 18, #2

On any given day, a certain machine has either no malfunctions or exactly one malfunction. The probability of malfunction on any given day is 0.40. Machine malfunctions on different days are mutually independent. Calculate the probability that the machine has its third malfunction on the fifth day, given that the machine has not had three malfunctions in the first three days.

Solution. *Identify the distribution.* Each day is a trial. Call a malfunction a "success" (it is only a label): $p = 0.4$, $q = 0.6$, and the days are independent. Let

$$X = \text{the number of days needed to get 3 malfunctions.}$$

Then $X \sim \text{NB}(3,\, 0.4)$ with $\text{Supp}(X) = \{3, 4, 5, \dots\}$.

Translate the two events. "The third malfunction is on the fifth day" is the event $X = 5$. Three malfunctions in the first three days would mean $X = 3$, so "has not had three malfunctions in the first three days" is the event $X \neq 3$.

Use the definition of conditional probability (Section 13):

$$P(X = 5 \mid X \neq 3) = \frac{P(X = 5 \ \cap \ X \neq 3)}{P(X \neq 3)} = \frac{P(X = 5)}{1 - P(X = 3)}.$$

The intersection simplifies because $X = 5$ already guarantees $X \neq 3$; the denominator uses the complement rule. With the negative binomial pmf,

$$P(X = 5) = \binom{4}{2}(0.4)^3(0.6)^2 = 6\,(0.064)(0.36) = 0.13824,$$
$$P(X = 3) = \binom{2}{2}(0.4)^3(0.6)^0 = 0.064,$$

so

$$P(X = 5 \mid X \neq 3) = \frac{0.13824}{1 - 0.064} = \frac{0.13824}{0.936} = \frac{48}{325} \approx \boxed{0.1477}.$$

Example 18.3 ▪ Accidents in the home: total probability

Worksheet 18, #4

A researcher wants to study accidents in the home and has assembled a group of 28 accident-prone people. 17 of them have fallen down the stairs at least once in the past month, and 19 of them have cut themselves while cooking at least once in the past month. Nobody in the group has been accident-free in the last month, and these are the only two kinds of accidents the subjects have experienced. Accidents are known to occur in a mutually independent fashion.

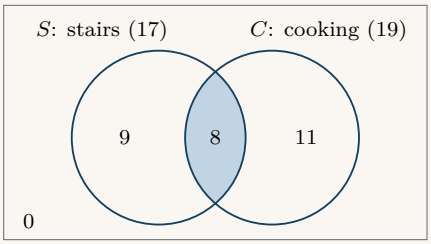
It turns out that if an individual in the group has not cut themselves cooking, they have had exactly one accident (falling down the stairs) in the last month. If an individual has both fallen down the stairs and cut themselves while cooking, they have had either 2 or 3 accidents total in the last month, with equal probability. And if the individual has not fallen down the stairs, then the number of times they have cut themselves follows a geometric distribution with mean 2.5.

What is the probability that a randomly selected subject has had a total of 3 accidents in the last month?

Solution. *Step 1: sort the 28 people into three groups.* Let S be the set of people who fell down the stairs and C the set of people who cut themselves cooking. Nobody was accident-free, so everyone is in $S \cup C$ and $N(S \cup C) = 28$. By the addition rule (Section 12),

$$N(S \cap C) = N(S) + N(C) - N(S \cup C) = 17 + 19 - 28 = 8.$$

So 8 people had both kinds of accident, $17 - 8 = 9$ only fell down the stairs, and $19 - 8 = 11$ only cut themselves; these 11 are the individuals who did not fall.



28 people in all;
nobody outside $S \cup C$

Step 2: the chance of exactly 3 accidents inside each group.

- **Stairs only** (has not cut themselves): exactly one accident, so the chance of three is 0.
- **Both:** 2 or 3 accidents with equal probability, so the chance of three is 0.5.
- **Cooking only** (has not fallen): all of this person's accidents are cuts, and the number of cuts X is geometric with mean 2.5. From $\mu_X = 1/p = 2.5$ we get $p = 0.4$ and $q = 0.6$, so

$$P(X = 3) = p\,q^{3-1} = (0.4)(0.6)^2 = 0.144.$$

Each of these three numbers is a *conditional* probability: it holds only given that the selected person belongs to that group. In particular 0.144 is not yet the answer; it still has to be weighted by the chance of picking someone from the cooking-only group.

Step 3: combine with the law of total probability (Section 13). Let A be the event that the selected subject had a total of 3 accidents. A randomly selected subject falls in the three groups with probabilities $\frac{9}{28}$, $\frac{8}{28}$ and $\frac{11}{28}$.

Group	**People**	$P(\text{group})$	$P(A \mid \text{group})$
stairs only	9	9/28	0
both	8	8/28	0.5
cooking only	11	11/28	$(0.6)^2(0.4) = 0.144$

$$P(A) = \frac{9}{28}\cdot 0 + \frac{8}{28}\cdot 0.5 + \frac{11}{28}\cdot (0.6)^2(0.4) = 0 + \frac{1}{7} + \frac{99}{1750} = \frac{349}{1750} \approx \boxed{0.1994}.$$

19 The Hypergeometric Distribution

The binomial distribution needs independent trials with the same p every time, which is what happens when we sample *with* replacement. When a sample is drawn *without* replacement from a small population, every draw changes what is left, the trials are no longer independent, and the binomial formulas do not apply. The number of successes in such a sample follows the hypergeometric distribution.

19.1 Hypergeometric experiments

Definition (hypergeometric experiment)

A *hypergeometric experiment* has the following features.

1) The population is finite and is made up of two groups of items: a *group of interest* (the "successes") and a second group (the "failures").
2) We are concerned with the group of interest: the question is how many of its items get picked.
3) A fixed number of items is picked at random **without replacement**.
4) As a consequence, the trials (the individual picks) are **not** independent.

Three numbers describe the experiment:

$$N = \text{population size}, \qquad M = \text{number of successes in the population}, \qquad n = \text{sample size},$$

so that the second group contains $N - M$ items.

Definition (hypergeometric random variable)

X is a *hypergeometric random variable* if

$$X = \text{the number of items in the sample that come from the group of interest}$$

(the number of successes among the n items picked). We write $X \sim \mathrm{H}(N, M, n)$.

Caution (the trials are not independent)

Take a bag with 5 green and 7 red marbles and draw two marbles without replacement. The first marble is green with probability $\frac{5}{12}$. For the second marble the probability depends on what happened first:

$$P(\text{2nd green} \mid \text{1st green}) = \frac{4}{11}, \qquad P(\text{2nd green} \mid \text{1st red}) = \frac{5}{11}.$$

The chance of success on a pick depends on what was picked before, so the picks are not independent: condition 4 of a binomial experiment (Section 17) fails.

Binomial or hypergeometric?

Both count the successes in a fixed number n of trials, each with two possible outcomes.

- Sampling *with* replacement, or trials that do not affect one another: independent trials, constant p. Use the **binomial** distribution.
- Sampling *without* replacement from a small population split into two groups: dependent trials. Use the **hypergeometric** distribution.

The wording that gives it away is "without replacement", or any description of taking several items out of one finite collection: "select 15 of these 20 instruments", "grade the first 15 reports of the pile".

19.2 Building the pmf from a counting argument

The pmf of a hypergeometric random variable is not new: it is a counting problem of the kind solved in Section 9. The next example builds it step by step.

Example 19.1 ⋆ ▪ **Green marbles: from one probability to the pmf** Worksheet 19, #1

A bag contains 5 green marbles and 7 red marbles. We draw 4 marbles without replacement.

a) What is the probability that we draw exactly one green marble?

b) Let X be the number of green marbles we draw. What is $P(X = x)$? (Said another way, find an expression for the probability that we draw x green marbles.)

c) What is the support of X?

d) What is the probability that we draw at least one green marble?

e) If we label N as the population size (the total number of marbles), M the number of successes (here, green marbles), and n our sample size (the number of marbles we draw), re-express the pmf for X from part (b) in terms of N, M, n and x.

f) If you are given that $E(X) = n \cdot \dfrac{M}{N}$ and $V(X) = \dfrac{N-n}{N-1} \cdot n \cdot \dfrac{M}{N} \cdot \left(1 - \dfrac{M}{N}\right)$, find the mean and the standard

deviation of the number of green marbles we draw.

Solution. Drawing 4 marbles without replacement amounts to choosing a group of 4 of the 12 marbles, and every group of 4 is equally likely. There are $\binom{12}{4} = 495$ such groups, so each probability below is

$$\frac{\text{number of favourable groups of 4}}{495}.$$

a) "Exactly one green" means 1 green *and* 3 red. The green marble can be chosen in $\binom{5}{1}$ ways and the three red ones in $\binom{7}{3}$ ways, so by the multiplication rule

$$P(\text{exactly one green}) = \frac{\binom{5}{1}\binom{7}{3}}{\binom{12}{4}} = \frac{5 \cdot 35}{495} = \frac{35}{99} \approx \boxed{0.3535}.$$

b) Repeat the count with x in place of 1: choose x of the 5 green marbles, and the other $4 - x$ marbles from the 7 red ones.

$$P(X = x) = p(x) = \boxed{\frac{\binom{5}{x}\binom{7}{4-x}}{\binom{12}{4}}}$$

c) We draw 4 marbles, and the bag holds at least 4 marbles of each colour, so every number of green marbles from 0 to 4 can occur: $\mathrm{Supp}(X) = \boxed{\{0, 1, 2, 3, 4\}}$. The whole distribution is

x	0	1	2	3	4
favourable groups $\binom{5}{x}\binom{7}{4-x}$	35	175	210	70	5
$p(x)$	7/99	35/99	42/99	14/99	1/99

and the counts add up to 495, so the probabilities add up to 1.

d) "At least one" is the complement of "none":

$$P(X \geq 1) = 1 - P(X < 1) = 1 - p(0) = 1 - \frac{\binom{5}{0}\binom{7}{4}}{\binom{12}{4}} = 1 - \frac{35}{495} = \frac{92}{99} \approx \boxed{0.9293}.$$

e) Replace the numbers by what they stand for: $12 = N$, $5 = M$, $7 = N - M$ and $4 = n$.

$$P(X = x) = p(x) = \boxed{\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}}$$

f) With $N = 12$, $M = 5$ and $n = 4$,

$$\mu_X = E(X) = n \cdot \frac{M}{N} = 4 \cdot \frac{5}{12} = \frac{5}{3} \approx \boxed{1.6667 \text{ green marbles}},$$
$$V(X) = \frac{N-n}{N-1} \cdot n \cdot \frac{M}{N} \cdot \left(1 - \frac{M}{N}\right) = \frac{8}{11} \cdot 4 \cdot \frac{5}{12} \cdot \frac{7}{12} = \frac{70}{99},$$
$$\sigma_X = \sqrt{V(X)} = \sqrt{\frac{70}{99}} \approx \boxed{0.8409 \text{ green marbles}}.$$

As a check, the table in part (c) gives the same mean directly:

$$E(X) = \sum_x x\,p(x) = \frac{0(35) + 1(175) + 2(210) + 3(70) + 4(5)}{495} = \frac{825}{495} = \frac{5}{3}.$$

19.3 The pmf, mean and variance

Part (e) of Example 19.1 is the general formula.

Theorem (hypergeometric pmf)

If $X \sim \mathrm{H}(N, M, n)$, then

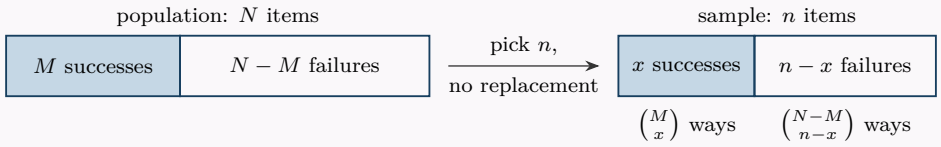
$$p(x) = P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

for every whole number x with $0 \le x \le M$ and $0 \le n - x \le N - M$. These are the values in the support of X; written as one chain of inequalities,

$$\max\bigl(0,\ n-(N-M)\bigr) \ \le\ x\ \le\ \min(n,\ M).$$

Why?

All $\binom{N}{n}$ possible samples of n items are equally likely. A sample with exactly x successes is built in two steps: choose which x of the M successes are in it, and choose which $n - x$ of the $N - M$ failures fill the remaining places. By the multiplication rule there are $\binom{M}{x}\binom{N-M}{n-x}$ such samples.



Caution (check the support)

The sample cannot contain more successes than exist ($x \le M$), and it cannot contain more failures than exist ($n - x \le N - M$). The support is $\{0, 1, \ldots, n\}$ only when both groups are at least as large as the sample, as in Example 19.1. If one group is smaller than the sample, some values are impossible: in Example 19.2 the support starts at 5, not at 0.

Theorem (mean and variance of a hypergeometric random variable)

If $X \sim \mathrm{H}(N, M, n)$, then

$$\mu_X = E(X) = n \cdot \frac{M}{N}, \qquad \sigma_X^2 = V(X) = \frac{N-n}{N-1} \cdot n \cdot \frac{M}{N} \cdot \Bigl(1 - \frac{M}{N}\Bigr), \qquad \sigma_X = \sqrt{V(X)}.$$

Note (use the version on the formula sheet)

Work with the hypergeometric formulas in the form printed on the formula sheet (Appendix A). They are the formulas above, except that the sheet writes k where this section writes M for the number of successes in the population:

$$p(x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}, \qquad \mu = n \cdot \frac{k}{N}, \qquad \sigma^2 = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} \cdot \Bigl(1 - \frac{k}{N}\Bigr).$$

This k is a number of items in the population. It has nothing to do with the k of the negative binomial distribution (Section 18), which is a number of successes wanted.

Remark (comparison with the binomial distribution)

Write $p = M/N$ for the proportion of successes in the population and $q = 1 - p$. Then

$$E(X) = np, \qquad V(X) = npq \cdot \frac{N-n}{N-1}.$$

The mean is the same as for $\mathrm{Bin}(n, p)$. The variance is the binomial variance multiplied by the factor $(N-n)/(N-1)$, which is at most 1: sampling without replacement is less variable, because every item picked tells us something about what is left. For $n = 1$ the factor is 1 (a single pick is the same with or without replacement); for $n = N$ it is 0 (the whole population is picked, so $X = M$ for certain). When N is much larger than n the factor is close to 1, and the hypergeometric distribution is practically the same as $\mathrm{Bin}(n, p)$.

Remark (the same formulas in terms of the two group sizes)

Some books describe the population by the sizes of its two groups instead of by N and M. With

r = the size of the group of interest, b = the size of the second group,
n = the number of items picked, X = the number of items from the first group,

they write $X \sim \mathrm{H}(r, b, n)$ and

$$\mu_X = E(X) = \frac{nr}{r+b}, \qquad \sigma_X = \sqrt{\frac{r\,b\,n\,(r+b-n)}{(r+b)^2(r+b-1)}}, \qquad V(X) = \sigma_X^2.$$

This is the same distribution: substitute $M = r$ and $N = r + b$ (so $N - M = b$) in the theorems above and the two versions agree.

Hypergeometric distribution at a glance

Conditions	(1) finite population made of two groups; (2) we count the items picked from the group of interest; (3) picks made without replacement; (4) so the trials are not independent
Variable	X = number of successes among the n items picked; $X \sim \mathrm{H}(N, M, n)$ with N = population size, M = successes in the population, n = sample size
pmf	$p(x) = \dfrac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$ for $\max\big(0,\, n-(N-M)\big) \le x \le \min(n, M)$
Mean, variance	$\mu_X = n \cdot \dfrac{M}{N}$, $\sigma_X^2 = \dfrac{N-n}{N-1} \cdot n \cdot \dfrac{M}{N}\Big(1 - \dfrac{M}{N}\Big)$

19.4 Worked examples

Example 19.2* ▪ Stocks and options: watch the support Worksheet 19, #2

A portfolio manager has chosen 10 stocks and 10 options for a particular fund. The portfolio manager instructs a co-op student to randomly select 15 of these final instruments for in-depth analysis. Let X represent the number of options selected for analysis.

a) What is the support of X?
b) Give the pmf for X.
c) What is the probability that all 10 options are selected for analysis?
d) What is the probability that all instruments of one of the two types are selected for analysis?
e) Find $\mu_X = E(X)$.

f) Give $P(X = \mu_X)$.

g) What is the probability that the number of options selected for analysis is within one standard deviation of its mean value?

Solution. *Identify the distribution.* The 15 instruments are selected from a population of $N = 20$ without replacement (the same instrument cannot be selected twice). The group of interest is the options, $M = 10$; the second group is the $N - M = 10$ stocks; the sample size is $n = 15$. So $X \sim \mathrm{H}(20, 10, 15)$.

a) If x options are selected, the other $15 - x$ instruments are stocks. There are only 10 options, so $x \le 10$; there are only 10 stocks, so $15 - x \le 10$, that is $x \ge 5$. Hence $\mathrm{Supp}(X) = \boxed{\{5, 6, 7, 8, 9, 10\}}$.

b) Choose x of the 10 options and $15 - x$ of the 10 stocks, out of $\binom{20}{15} = 15\,504$ equally likely selections:

$$p(x) = \boxed{\frac{\binom{10}{x}\binom{10}{15-x}}{\binom{20}{15}}} \qquad x = 5, 6, \ldots, 10.$$

x	5	6	7	8	9	10
$\binom{10}{x}\binom{10}{15-x}$	252	2100	5400	5400	2100	252
$p(x)$	0.0163	0.1354	0.3483	0.3483	0.1354	0.0163

c) $p(10) = \dfrac{\binom{10}{10}\binom{10}{5}}{\binom{20}{15}} = \dfrac{252}{15\,504} = \dfrac{21}{1292} \approx \boxed{0.0163}$.

d) Either all 10 options are selected ($X = 10$), or all 10 stocks are selected, which leaves room for exactly 5 options ($X = 5$). The two events cannot happen together, so their probabilities add:

$$p(10) + p(5) = \frac{\binom{10}{10}\binom{10}{5}}{\binom{20}{15}} + \frac{\binom{10}{5}\binom{10}{10}}{\binom{20}{15}} = \frac{504}{15\,504} = \frac{21}{646} \approx \boxed{0.0325}.$$

e) $\mu_X = E(X) = n \cdot \dfrac{M}{N} = 15 \cdot \dfrac{10}{20} = \boxed{7.5 \text{ options}}$.

f) $P(X = 7.5) = \boxed{0}$, because 7.5 is not in the support: X counts whole instruments. The mean of a random variable does not have to be one of its possible values.

g) First the standard deviation:

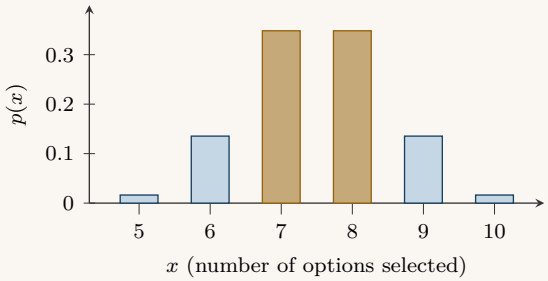
$$\sigma_X = \sqrt{V(X)} = \sqrt{\frac{20-15}{20-1} \cdot 15 \cdot \frac{10}{20} \cdot \frac{10}{20}} = \sqrt{\frac{5}{19} \cdot 15 \cdot \frac{1}{2} \cdot \frac{1}{2}} = \sqrt{\frac{75}{76}} \approx 0.993.$$

"Within one standard deviation of the mean" is the interval

$$[\mu_X - \sigma_X,\ \mu_X + \sigma_X] \approx [7.5 - 0.993,\ 7.5 + 0.993] \approx [6.51,\ 8.49],$$

and the only values of the support inside it are 7 and 8. Therefore

$$P(6.51 \le X \le 8.49) = p(7) + p(8) = \frac{5400 + 5400}{15\,504} = \frac{225}{323} \approx \boxed{0.6966}.$$



The pmf of $\mathrm{H}(20, 10, 15)$ is symmetric about $\mu_X = 7.5$, a value that X never takes. The dark bars are the values within one standard deviation of the mean.

20 The Poisson Distribution

The binomial, negative binomial and hypergeometric distributions all come from an experiment made of trials. The Poisson distribution is different: no experiment of that kind lies behind it. It is a model for the number of times an event happens during a fixed interval of time, or inside a fixed region of space, when the events occur at a constant average rate.

20.1 Poisson random variables

Definition (Poisson random variable)

Suppose that events of some kind occur at a constant average rate, and let

$$X = \text{the number of events in a given interval (of time or of space)},$$
$$\lambda = \text{the constant average rate of the events in that interval}$$

(that is, the average number of events in an interval of that size, with $\lambda > 0$). Then X is a *Poisson random variable* with parameter λ, written $X \sim \text{Pois}(\lambda)$. Its support is

$$\text{Supp}(X) = \{0, 1, 2, \dots\}.$$

Remark (a model, not an experiment)

- There is no natural probability experiment that corresponds to the Poisson distribution: there are no trials, no n and no p. The rate λ is the only parameter.
- The distribution is useful for modelling the probability of events occurring through time: aircraft arriving at an airport in an hour, customers reaching a drive-through window in half an hour, calls coming into an emergency centre in a minute.

The model is reasonable when the events happen independently of one another and the average rate stays the same throughout the interval.

Note (objects per area also fit a Poisson model)

The "interval" does not have to be an interval of time. Objects scattered over a region are counted in exactly the same way: elm trees distributed in a forest at 80 trees per acre, grasshoppers distributed in a field at 2 per square metre. The rate is then a density (objects per unit of area) and the interval is a region of land.

20.2 The pmf, mean and variance

Theorem (Poisson pmf)

If $X \sim \text{Pois}(\lambda)$, then

$$p(x) = P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \dots$$

Theorem (mean and variance of a Poisson random variable)

If $X \sim \text{Pois}(\lambda)$, then

$$\mu_X = E(X) = \lambda, \qquad \sigma_X^2 = V(X) = \lambda, \qquad \sigma_X = \sqrt{\lambda}.$$

The parameter is the mean. This is why a statement such as "aircraft arrive at an average rate of 8 per hour" hands you the parameter directly: for a one-hour interval, $\lambda = 8$.

Caution (the support never ends)

$\text{Supp}(X) = \{0, 1, 2, 3, \dots\}$ has no largest value. A probability such as $P(X \geq 5)$ is therefore an infinite sum and cannot be added up term by term. Use the complement:

$$P(X \geq a) = 1 - P(X \leq a-1) = 1 - \sum_{x=0}^{a-1} \frac{e^{-\lambda}\lambda^x}{x!}.$$

A probability such as $P(X \leq 10)$ is a finite sum and is computed directly.

The proofs of the next two facts use the series for the exponential function.

Recall (exponential series)

For every real number λ,

$$e^{\lambda} = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \cdots.$$

Proof that $\sum p(x) = 1$ for a Poisson random variable

The factor $e^{-\lambda}$ does not depend on x, so it comes out of the sum, and what is left is the series for e^{λ}:

$$\sum_{x=0}^{\infty} p(x) = \sum_{x=0}^{\infty} \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} \cdot e^{\lambda} = 1.$$

So the Poisson pmf is a valid pmf. □

Proof that $E(X) = \lambda$ for a Poisson random variable

By the definition of expected value (the $x = 0$ term is 0, so the sum may start at $x = 1$),

$$E(X) = \sum_{x=0}^{\infty} x\, p(x) = \sum_{x=0}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x!} = \sum_{x=1}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x\,(x-1)!}$$
$$= \sum_{x=1}^{\infty} \frac{e^{-\lambda}\lambda^{x-1}}{(x-1)!} \cdot \lambda = \lambda \sum_{x=1}^{\infty} \frac{e^{-\lambda}\lambda^{x-1}}{(x-1)!}.$$

Now substitute $y = x - 1$; as x runs through $1, 2, 3, \dots$, y runs through $0, 1, 2, \dots$:

$$E(X) = \lambda \underbrace{\sum_{y=0}^{\infty} \frac{e^{-\lambda}\lambda^y}{y!}}_{\text{all of the pmf of Pois}(\lambda)} = \lambda \cdot 1 = \lambda.$$

The sum equals 1 because it adds up the whole Poisson pmf. □

Remark

The formula $V(X) = \lambda$ is proved with the same trick. Cancelling $x(x-1)$ against $x!$ and shifting the index by 2 gives $E\big(X(X-1)\big) = \lambda^2$. Then

$$V(X) = E(X^2) - E(X)^2 = E\big(X(X-1)\big) + E(X) - E(X)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

20.3 Changing the size of the interval

A rate is always a rate *per* something: per hour, per acre, per square metre. Questions often ask about an interval that is longer or shorter than that unit.

Property (rescaling the rate)

Suppose that λ is a *unit rate*, that is, the rate per unit interval, and that $X \sim \text{Pois}(\lambda)$ counts the number of events in one unit. Let

$$Y = \text{the number of events in an interval of size } t \text{ units.}$$

Then

$$Y \sim \text{Pois}(\lambda_t), \qquad \text{where} \quad \lambda_t = \lambda \cdot t.$$

The average number of events is proportional to the size of the interval: twice the time, or twice the area, means twice as many events on average. The count is still a Poisson random variable; only its parameter changes.

Setting out a Poisson solution

Begin every solution by naming the random variable and stating its distribution:

$$\text{“Let } X = \dots \quad \text{Then } X \sim \text{Pois}(\dots)\text{.”}$$

When the question moves to an interval of a different size, do not reuse X. Introduce a new variable with its own parameter:

$$\text{“Let } Y = \dots \quad \text{Then } Y \sim \text{Pois}(\dots)\text{.”}$$

Each description must say what is counted and in which interval.

Caution (units)

Measure t in the unit of the rate. With a rate per hour, a 90-minute period is $t = 1.5$, not $t = 90$. The factor t can be smaller than 1 (a quarter-acre plot is $t = \frac{1}{4}$ acre), and it does not have to be a time (a disc of radius r metres has area $t = \pi r^2$ square metres).

Poisson distribution at a glance

Use it for	counting events that occur at a constant average rate through time, or objects scattered at a constant average density over an area; there are no trials
Variable	X = number of events in a given interval; $X \sim \text{Pois}(\lambda)$, where λ = average number of events in that interval
Support	$\{0, 1, 2, \dots\}$
pmf	$p(x) = \dfrac{e^{-\lambda}\lambda^x}{x!}$
Mean, variance	$\mu_X = \lambda, \quad \sigma_X^2 = \lambda, \quad \sigma_X = \sqrt{\lambda}$
Rescaling	an interval of t units has parameter $\lambda_t = \lambda \cdot t$

20.4 Worked examples

Example 20.1 ▪ Aircraft arrivals Worksheet 20, #1

Suppose that aircraft arrive at an airport at an average rate of 8 per hour.

a) What is the probability that exactly 5 aircraft arrive during a given 1-hour period?

b) What is the probability that at least 5 arrive? At most 10?

c) What is the probability that 14 aircraft arrive in a 90-minute period?

d) In a given 2.5-hour period, 30 aircraft arrive. Is this unusual? Justify your answer.

Solution. Let X be the number of aircraft that arrive in a 1-hour period. Arrivals are events that occur through time at a constant average rate of 8 per hour, so $X \sim \text{Pois}(8)$ and

$$p(x) = \frac{e^{-8}\,8^x}{x!}, \qquad x = 0, 1, 2, \dots$$

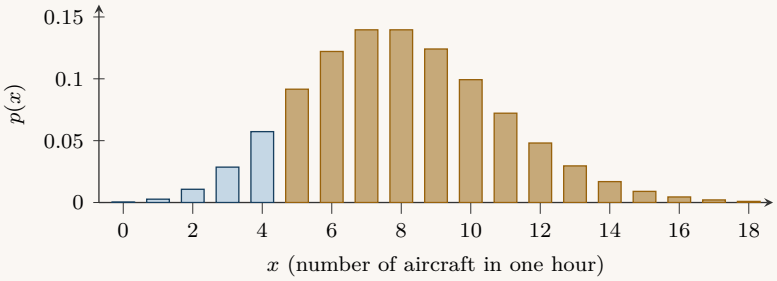
a) $P(X = 5) = \dfrac{e^{-8}\,8^5}{5!} \approx \boxed{0.0916}$.

b) $\text{Supp}(X) = \{0, 1, 2, 3, \dots\}$ has no largest value, so "at least 5" would be an infinite sum. Use the complement, which has only five terms:

$$\begin{aligned} P(X \geq 5) &= 1 - P(X \leq 4) = 1 - \sum_{x=0}^{4} \frac{e^{-8}\,8^x}{x!} \\ &= 1 - e^{-8}\left(1 + 8 + \frac{8^2}{2!} + \frac{8^3}{3!} + \frac{8^4}{4!}\right) \approx 1 - 0.0996 = \boxed{0.9004}. \end{aligned}$$

"At most 10" is a finite sum of eleven terms, so it is computed directly:

$$P(X \leq 10) = \sum_{x=0}^{10} \frac{e^{-8}\,8^x}{x!} \approx \boxed{0.8159}.$$



The pmf of Pois(8). The dark bars, which continue forever to the right, are the event $X \geq 5$. It is easier to subtract the five light bars from 1.

c) The interval has changed, so a new variable is needed. Let Y be the number of aircraft that arrive in a 90-minute period. Ninety minutes is $t = 1.5$ hours, so $\lambda_t = 8(1.5) = 12$ and $Y \sim \text{Pois}(12)$:

$$P(Y = 14) = \frac{e^{-12}\,12^{14}}{14!} \approx \boxed{0.0905}.$$

d) Let W be the number of aircraft that arrive in a 2.5-hour period. Then $\lambda_t = 8(2.5) = 20$ and $W \sim \text{Pois}(20)$, so

$$\mu_W = 20, \qquad \sigma_W = \sqrt{20} \approx 4.47.$$

The z-score of the observed value 30 (Section 7) is

$$z = \frac{30 - \mu_W}{\sigma_W} = \frac{30 - 20}{\sqrt{20}} \approx 2.24,$$

so 30 lies more than 2 standard deviations above the mean. A value more than 2 standard deviations away from the mean is considered unusual; in other words, 30 is outside the interval $[\mu_W - 2\sigma_W,\ \mu_W + 2\sigma_W] \approx [11.06,\ 28.94]$. So $\boxed{\text{yes, 30 arrivals is unusual}}$.

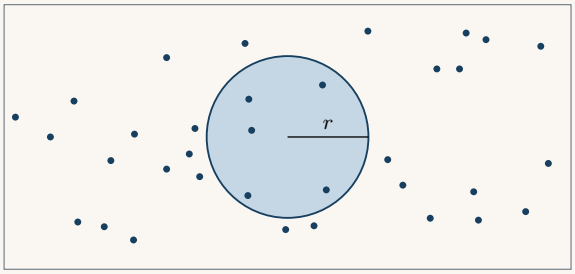
Example 20.2 ▪ Grasshoppers: solving for the size of the region Worksheet 20, #3

Grasshoppers are distributed in a large field with an average density of 2 per square metre. How large must the radius of a circular sampling region be so that the probability of finding at least one grasshopper in the region is 0.99?

Solution. Let X be the number of grasshoppers in one square metre. The density is $\lambda = 2$ grasshoppers per m^2, so $X \sim \text{Pois}(2)$.

The sampling region is not one square metre, so the rate has to be rescaled. Let Y be the number of grasshoppers in a circular region of radius r metres. The region has area $t = \pi r^2$ square metres, so

$$\lambda_t = \lambda \cdot t = 2\pi r^2 \qquad \text{and} \qquad Y \sim \text{Pois}(2\pi r^2).$$



field: on average 2 grasshoppers per m^2

sampling region: area πr^2 m^2

We want $P(Y \geq 1) = 0.99$. By the complement rule this is the same as

$$p(0) = P(Y = 0) = 0.01.$$

The pmf of Y gives $p(0)$ in terms of r:

$$p(0) = \frac{e^{-2\pi r^2}(2\pi r^2)^0}{0!} = e^{-2\pi r^2} = 0.01.$$

Now solve for r. Taking reciprocals, $e^{2\pi r^2} = 100$; taking square roots, $e^{\pi r^2} = 10$. Hence

$$\pi r^2 = \ln 10 \quad \Longrightarrow \quad r^2 = \frac{\ln 10}{\pi} \quad \Longrightarrow \quad r = \sqrt{\frac{\ln 10}{\pi}} \approx \boxed{0.8561 \text{ m}}.$$

As a check, a disc of this radius has area $\ln 10 \approx 2.303$ m^2 and contains $2(2.303) \approx 4.605$ grasshoppers on average, and $e^{-4.605} \approx 0.01$.

Part V

Continuous Random Variables

When a random variable can take every value in an interval, sums become integrals and the probability mass function becomes a density. This part covers densities and cumulative distribution functions, then the uniform, exponential and normal distributions and the normal approximation to the binomial.

21 Continuous Random Variables

The random variables of Sections 15–20 were discrete: their possible values could be listed, and a pmf gave the probability of each one. A waiting time, a volume or a height is different: it can take *any* value in an interval. For such a variable the pmf is replaced by a *density*, and every sum becomes an integral. The ideas themselves (cdf, expected value, variance, LOTUS) stay the same.

21.1 Probability density functions

Definition (continuous random variable)

A random variable X is *continuous* if its support is infinite and *uncountable*: its possible values cannot be written out as a list, not even an endless one. In practice the support of a continuous random variable is an interval of real numbers (or a union of intervals), for example $\text{Supp}(X) = [1, 2]$ or $\text{Supp}(X) = (0, \infty)$.

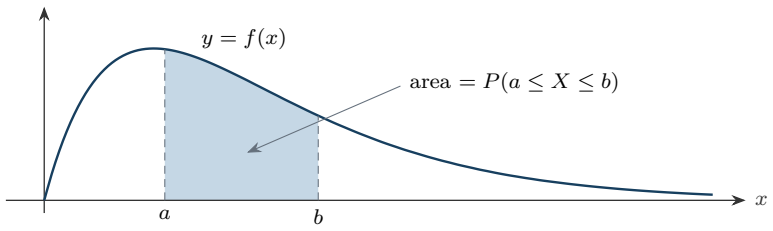
As a rule of thumb, quantities that are *counted* are discrete, and quantities that are *measured* (time, length, volume, mass) are continuous.

Definition (probability density function)

For a continuous random variable X, probability is determined by a *probability density function* (pdf): a function $f(x)$ such that, for all numbers $a \le b$,

$$P(a \le X \le b) = \int_a^b f(x)\,\mathrm{d}x.$$

In words: the probability that X lands in an interval is the *area under the graph of* f above that interval. The pdf is defined for every real number x; outside the support it is simply 0.



Property (the two criteria for a valid pdf)

A function f is a valid pdf if and only if

1) $f(x) \ge 0$ for every real number x, and

2) $\displaystyle\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1.$

Criterion 1 says that no interval can have a negative probability. Criterion 2 says that X certainly takes *some* value: $P(-\infty < X < \infty) = 1$. They are the continuous versions of the two conditions for a valid pmf, $0 \le p(x) \le 1$ and $\sum_x p(x) = 1$ (Section 15), with one difference: nothing forces $f(x) \le 1$. When a pdf contains an unknown constant, criterion 2 is the equation that determines it (Example 21.1).

Caution (a density is not a probability)

The number $f(x)$ is *not* $P(X = x)$. It is a density, that is, probability *per unit of* x, and only *areas* under f are probabilities. In particular, we can have $f(x) > 1$. For example,

$$f(x) = \begin{cases} 2, & 0 \le x \le \frac{1}{2}, \\ 0, & \text{otherwise} \end{cases}$$

is a valid pdf: it is never negative, and the area under it is $2 \cdot \frac{1}{2} = 1$.

Property (a single point has probability zero)

If X is a continuous random variable, then for every number a

$$P(X = a) = \int_a^a f(x)\,\mathrm{d}x = 0.$$

Consequently it makes no difference whether the endpoints of an interval are included:

$$P(a \le X \le b) = P(a < X \le b) = P(a \le X < b) = P(a < X < b).$$

$P(X = a) = 0$ does not mean that the value a is impossible. It means that a single point is too thin to carry any area: for a continuous random variable, probability belongs to intervals, not to points.

Caution (not equal if X is discrete)

Replacing $<$ by $\le$ (or $>$ by $\ge$) is allowed *only* for continuous random variables. If X is discrete, single values do carry probability, and the two signs give different answers. For the roll of a fair die, for example, $P(X \le 3) = \frac{3}{6}$ but $P(X < 3) = \frac{2}{6}$.

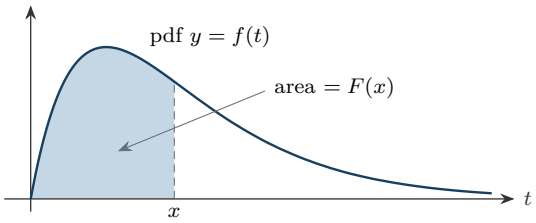
21.2 The cumulative distribution function

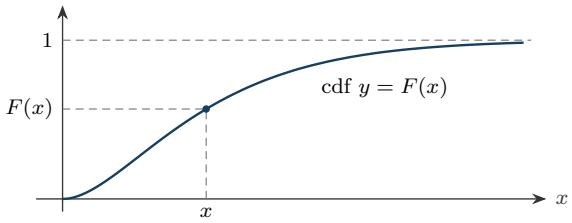
Definition (cumulative distribution function)

The *cumulative distribution function* (cdf) of a continuous random variable X with pdf f is

$$F(x) := P(X \le x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t.$$

The cdf tells us what portion of the total probability lies *before* x, that is, at or to the left of x. The letter t is only the variable of integration; x is already in use as the upper limit.





Property (the cdf is an antiderivative of the pdf)

By the Fundamental Theorem of Calculus,

$$F'(x) = f(x) \qquad \text{at every } x \text{ where } f \text{ is continuous.}$$

So integrating the pdf gives the cdf, and differentiating the cdf gives back the pdf. Moreover, F is continuous and never decreases, with

$$F(-\infty) = \lim_{x \to -\infty} F(x) = 0 \qquad \text{and} \qquad F(+\infty) = \lim_{x \to +\infty} F(x) = 1.$$

Formula (probabilities from the cdf)

Let X be a continuous random variable with pdf f and cdf F, and let $a \le b$. Then

$$\begin{aligned} P(a \le X \le b) &= F(b) - F(a) = \int_a^b f(t)\,\mathrm{d}t, \\ P(X \le b) &= F(b) = P(X < b), \\ P(X > a) &= 1 - P(X \le a) = 1 - F(a) = P(X \ge a). \end{aligned}$$

The last equality in each of the second and third lines holds because a single point has probability zero. These two equalities are *not* valid if X is discrete.

Note (two ways to compute a probability)

A probability such as $P(c \le X \le d)$ can therefore be evaluated in two ways:

1) by integrating the pdf: $\displaystyle\int_c^d f(x)\,\mathrm{d}x$;
2) by using the cdf as the antiderivative: $F(d) - F(c)$.

Once F is known, the second way needs no further integration.

Remark (using $F(+\infty) = 1$)

The computation of $F(x)$ can sometimes be simplified with the fact that $F(+\infty) = 1$. If $f(x)$ vanishes on $[a, +\infty)$, then $F(a) = 1$ automatically, and $F(x) = 1$ for every $x \ge a$: all of the area has already been collected. In the same way, $F(x) = 0$ everywhere to the left of the support. So when a pdf is zero outside an interval $[c, d]$, only one integral has to be computed:

$$F(x) = \begin{cases} 0, & x < c, \\ \displaystyle\int_c^x f(t)\,\mathrm{d}t, & c \le x \le d, \\ 1, & x > d. \end{cases}$$

The middle formula must give 0 at $x = c$ and 1 at $x = d$, which is a quick check on the integration.

21.3 Expected value, variance and LOTUS

The definitions are those of Section 16, with the pmf replaced by the pdf and the sum replaced by an integral.

Definition (expected value, variance and standard deviation)

Let X be a continuous random variable with pdf f.

- The *expected value* (or *mean*) of X is
$$\mu_X = E(X) = \int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x.$$
- The *variance* of X is
$$\sigma_X^2 = V(X) = E\big[(X-\mu_X)^2\big] = \int_{-\infty}^{\infty} (x-\mu_X)^2 f(x)\, \mathrm{d}x.$$
- The *standard deviation* of X is $\sigma_X = \sqrt{V(X)}$.

As before, $E(X)$ is the long-run average of X (the balance point of the region under the pdf), and σ_X measures the width of the distribution in the units of X. Because f is zero outside the support, each of these integrals only has to be taken over the support.

Theorem (LOTUS for a continuous random variable)

Let X be a continuous random variable with pdf f, and let $Y = Y(X)$ be a function of X, where $y(x)$ denotes the value of Y when $X = x$. Then
$$\mu_Y = E(Y) = \int_{-\infty}^{\infty} y(x)\, f(x)\, \mathrm{d}x.$$

LOTUS (the law of the unconscious statistician) says that the average of a function of X is found by weighting the values of that function with the density of X itself; the pdf of Y is never needed. Two uses appear all the time. The integral in the definition of the variance is LOTUS applied to $y(x) = (x-\mu_X)^2$, and
$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\, \mathrm{d}x$$
is LOTUS applied to $y(x) = x^2$.

Formula (computational formula for the variance)
$$V(X) = E(X^2) - \big[E(X)\big]^2.$$

Proof

Write $\mu = \mu_X$ and expand the square in the definition of $V(X)$:
$$\begin{aligned} V(X) &= \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, \mathrm{d}x \\ &= \int_{-\infty}^{\infty} x^2 f(x)\, \mathrm{d}x - 2\mu \int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x + \mu^2 \int_{-\infty}^{\infty} f(x)\, \mathrm{d}x \\ &= E(X^2) - 2\mu \cdot \mu + \mu^2 \cdot 1 = E(X^2) - \mu^2. \end{aligned}$$
The last line uses $\int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \mu$ (the definition of the mean) and $\int_{-\infty}^{\infty} f(x)\, \mathrm{d}x = 1$ (the pdf is valid). □

Note

Use the computational formula in calculations: it needs the two plain integrals $E(X)$ and $E(X^2)$ instead of the integral of a squared difference. It also shows that $E(X^2)$ and $[E(X)]^2$ are different numbers (they differ by $V(X)$). More generally, $E\big(Y(X)\big)$ is *not* obtained by plugging $E(X)$ into the function: the function has to be averaged with LOTUS.

The rules $E(aX+b) = aE(X)+b$ and $V(aX+b) = a^2V(X)$ of Section 16 hold for continuous random variables as well; the proofs are the same, with integrals in place of sums.

21.4 Median and percentiles

Definition (median and percentiles of a continuous distribution)

Let X be a continuous random variable with cdf F, and let $0 < p < 1$.

- The $(100p)$*th percentile* of the distribution is the value x for which
$$F(x) = P(X \le x) = p.$$
- The *median* m of the distribution is its 50th percentile:
$$F(m) = P(X \le m) = 0.5.$$

Percentile questions are answered with the cdf, because the cdf is exactly the function that says what portion of the probability lies before x: set $F(x)$ equal to the required portion and solve for x. For the 85th percentile, for instance, solve $F(x) = 0.85$. If the equation has more than one solution, keep the one that lies in the support.

Continuous random variables at a glance (next to the discrete case)

	Discrete X	**Continuous X**
Described by	pmf $p(x) = P(X = x)$	pdf $f(x)$ (not a probability)
Valid if	$0 \le p(x) \le 1$, $\sum_x p(x) = 1$	$f(x) \ge 0$, $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$
$P(a \le X \le b)$	$\sum_{a \le x \le b} p(x)$	$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$
$P(X = a)$	$p(a)$	0
$P(X \le b)$	$F(b) = P(X < b) + p(b)$	$F(b) = P(X < b)$
$P(X > a)$	$1 - F(a) = P(X \ge a) - p(a)$	$1 - F(a) = P(X \ge a)$
cdf $F(x)$	$P(X \le x) = \sum_{t \le x} p(t)$	$P(X \le x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$, and $F' = f$
$\mu_X = E(X)$	$\sum_x x\,p(x)$	$\int_{-\infty}^{\infty} x\,f(x)\,\mathrm{d}x$
$E(Y)$, $Y = Y(X)$	$\sum_x y(x)\,p(x)$	$\int_{-\infty}^{\infty} y(x)\,f(x)\,\mathrm{d}x$
$\sigma_X^2 = V(X)$	$E\big[(X-\mu_X)^2\big] = E(X^2) - [E(X)]^2$	in both cases

21.5 Worked examples

The four examples below follow one random variable through every technique of this section.

Example 21.1 ⋆ ▪ **A valid pdf; probabilities as areas** Worksheet 21, #1–3

The weekly demand for propane gas (in thousands of litres) in a certain region is a random variable X with pdf

$$f(x) = \begin{cases} k\left(1 - \dfrac{1}{x^2}\right), & 1 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

a) Find k.
b) Find $P(X = 1.5)$.
c) Find $P(X \ge 1.5)$.

Solution.

a) A pdf must have total area 1. Since f is zero outside $[1, 2]$, the integral over the whole line reduces to an integral from 1 to 2:

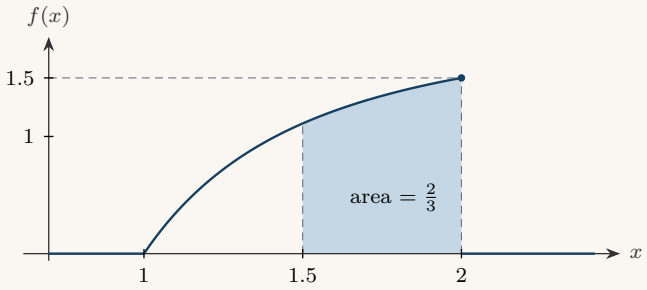
$$1 = \int_{-\infty}^{\infty} f(x)\,dx = \int_1^2 k\left(1 - \frac{1}{x^2}\right) dx = k\left[x + \frac{1}{x}\right]_1^2 = k\left[\left(2 + \frac{1}{2}\right) - (1 + 1)\right] = \frac{k}{2},$$

so $\boxed{k = 2}$. The other criterion also holds: for $1 \le x \le 2$ we have $x^2 \ge 1$, so $f(x) = 2\left(1 - \frac{1}{x^2}\right) \ge 0$. Notice that $f(2) = 1.5 > 1$; this is allowed, because $f(2)$ is a density and not a probability.

b) X is continuous, so a single value has probability zero: $P(X = 1.5) = \displaystyle\int_{1.5}^{1.5} f(x)\,dx = \boxed{0}$.
(The density there is $f(1.5) = \frac{10}{9}$, but that number is not $P(X = 1.5)$.)

c) This is the area under f to the right of 1.5. The density vanishes beyond $x = 2$, so the integral stops there:

$$P(X \ge 1.5) = \int_{1.5}^2 2\left(1 - \frac{1}{x^2}\right) dx = 2\left[x + \frac{1}{x}\right]_{1.5}^2 = 2\left[\left(2 + \frac{1}{2}\right) - \left(\frac{3}{2} + \frac{2}{3}\right)\right]$$
$$= 2\left(\frac{5}{2} - \frac{13}{6}\right) = \boxed{\frac{2}{3}}.$$



The pdf $f(x) = 2\left(1 - \frac{1}{x^2}\right)$ on $[1, 2]$. The shaded area is $P(X \ge 1.5)$.

Example 21.2 ▪ **Building and using the cdf** Worksheet 21, #4–5

For the propane demand X of Example 21.1, with pdf $f(x) = 2\left(1 - \frac{1}{x^2}\right)$ for $1 \le x \le 2$ and $f(x) = 0$ otherwise:

a) Give the cdf $F(x)$.
b) Use the cdf to calculate i) $P(X < 1.75)$ ii) $P(1.25 \le X \le 1.75)$ iii) $P(X > 1.6)$.

Solution.

a) By definition $F(x) = \displaystyle\int_{-\infty}^{x} f(t)\,dt$, and there are three cases, according to where x lies.

- If $x < 1$, no area has been collected yet: $F(x) = 0$.
- If $1 \le x \le 2$,

$$F(x) = \int_1^x 2\Big(1 - \frac{1}{t^2}\Big)\,\mathrm{d}t = 2\Big[t + \frac{1}{t}\Big]_1^x = 2\Big(x + \frac{1}{x}\Big) - 4 = 2x + \frac{2}{x} - 4.$$

- If $x > 2$, all of the area has been collected: $F(x) = 1$.

Therefore

$$\boxed{F(x) = \begin{cases} 0, & x < 1, \\ 2x + \dfrac{2}{x} - 4, & 1 \le x \le 2, \\ 1, & x > 2. \end{cases}}$$

Check. The middle formula gives $F(1) = 2 + 2 - 4 = 0$ and $F(2) = 4 + 1 - 4 = 1$, so the three pieces join up, and its derivative $2 - \frac{2}{x^2}$ is the pdf.

b) Each probability is now a matter of substituting into F.

i) $P(X < 1.75) = P(X \le 1.75) = F(1.75) = 3.5 + \dfrac{2}{1.75} - 4 = \dfrac{9}{14} \approx \boxed{0.6429}$.

ii) $F(1.25) = 2.5 + 1.6 - 4 = 0.1$, so

$$P(1.25 \le X \le 1.75) = F(1.75) - F(1.25) = \frac{9}{14} - \frac{1}{10} = \frac{19}{35} \approx \boxed{0.5429}.$$

iii) $P(X > 1.6) = 1 - F(1.6) = 1 - (3.2 + 1.25 - 4) = 1 - 0.45 = \boxed{0.55}$.

The cdf also gives back the answer of Example 21.1(c) without a new integral:

$$P(X \ge 1.5) = 1 - F(1.5) = 1 - \tfrac{1}{3} = \tfrac{2}{3}.$$

Example 21.3 ▪ Mean and variance

Worksheet 21, #6

The propane demand X of Example 21.1 has pdf $f(x) = 2\big(1 - \frac{1}{x^2}\big)$ for $1 \le x \le 2$ and $f(x) = 0$ otherwise. Compute $E(X)$ and $V(X)$.

Solution. *Mean.* Only the support $[1, 2]$ contributes to the integral:

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x\, f(x)\,\mathrm{d}x = \int_1^2 x \cdot 2\Big(1 - \frac{1}{x^2}\Big)\,\mathrm{d}x = 2\int_1^2 \Big(x - \frac{1}{x}\Big)\,\mathrm{d}x$$
$$= 2\Big[\frac{x^2}{2} - \ln x\Big]_1^2 = 2\Big[(2 - \ln 2) - \frac{1}{2}\Big] = \boxed{3 - 2\ln 2} \approx 1.6137.$$

Variance. Use the computational formula. By LOTUS,

$$E(X^2) = \int_1^2 x^2 \cdot 2\Big(1 - \frac{1}{x^2}\Big)\,\mathrm{d}x = 2\int_1^2 (x^2 - 1)\,\mathrm{d}x = 2\Big[\frac{x^3}{3} - x\Big]_1^2 = 2\Big[\frac{2}{3} - \Big(-\frac{2}{3}\Big)\Big] = \frac{8}{3},$$

so

$$\sigma_X^2 = V(X) = E(X^2) - \big[E(X)\big]^2 = \boxed{\tfrac{8}{3} - (3 - 2\ln 2)^2} \approx 2.66667 - 2.60405 \approx 0.0626,$$

and $\sigma_X = \sqrt{V(X)} \approx 0.2502$.

In context: the average weekly demand is about 1.61 thousand litres, with a standard deviation of about 0.25 thousand litres.

Example 21.4 ▪ Median and percentiles Worksheet 21, #7

Find the median and an expression for the $(100p)$th percentile (for example, $p = 0.85$ for the 85th percentile) of the distribution of the propane demand X of Example 21.1.

Solution. Use the cdf. From Example 21.2, $F(x) = 2x + \dfrac{2}{x} - 4$ for $1 \le x \le 2$.

Median. Solve $F(m) = 0.5$:

$$2m + \frac{2}{m} - 4 = \frac{1}{2} \implies 4m^2 + 4 - 8m = m \implies 4m^2 - 9m + 4 = 0 \implies m = \frac{9 \pm \sqrt{17}}{8}.$$

The root $\frac{9-\sqrt{17}}{8} \approx 0.61$ is outside the support $[1, 2]$, so the median is

$$\boxed{m = \frac{9 + \sqrt{17}}{8}} \approx 1.6404.$$

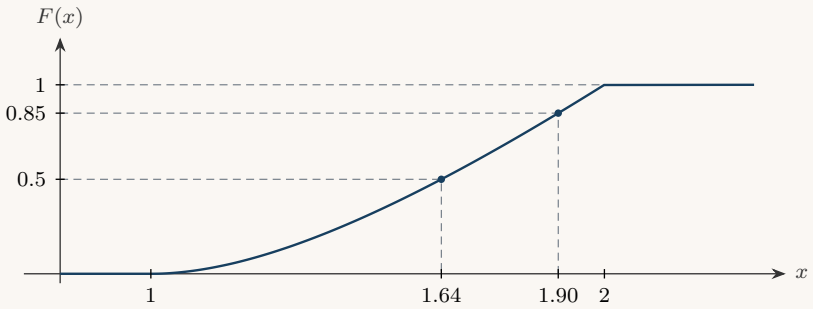
$(100p)$th percentile. Solve $F(x) = p$ in the same way:

$$2x + \frac{2}{x} - 4 = p \implies 2x^2 - (4 + p)\,x + 2 = 0$$
$$\implies x = \frac{(4 + p) \pm \sqrt{(4 + p)^2 - 16}}{4} = \frac{4 + p \pm \sqrt{8p + p^2}}{4}.$$

The product of the two roots is $\frac{2}{2} = 1$, so one root is at most 1 and the other is at least 1; the one in the support $[1, 2]$ is the larger one. The $(100p)$th percentile is therefore

$$\boxed{x = \frac{4 + p + \sqrt{8p + p^2}}{4}}.$$

Check. With $p = 0.5$ this gives $\frac{4.5+\sqrt{4.25}}{4} = \frac{9+\sqrt{17}}{8}$, the median. With $p = 0.85$ it gives $\frac{4.85+\sqrt{7.5225}}{4} \approx 1.8982$: in 85% of weeks the demand is at most about 1.90 thousand litres.



The cdf of the propane demand. Reading it backwards, from a height p to the value x, gives the percentiles: $F(1.64) \approx 0.5$ and $F(1.90) \approx 0.85$.

22 The Continuous Uniform Distribution

Like the standard discrete distributions of Sections 17–20, a few continuous distributions come up so often that their pdf, cdf, mean and variance are worth knowing by heart. This course uses three of these special continuous distributions: the uniform (this section), the exponential (Section 23) and the normal (Section 24). The uniform distribution is the simplest of the three: its density is flat.

22.1 The pdf and the cdf

Definition (continuous uniform distribution)

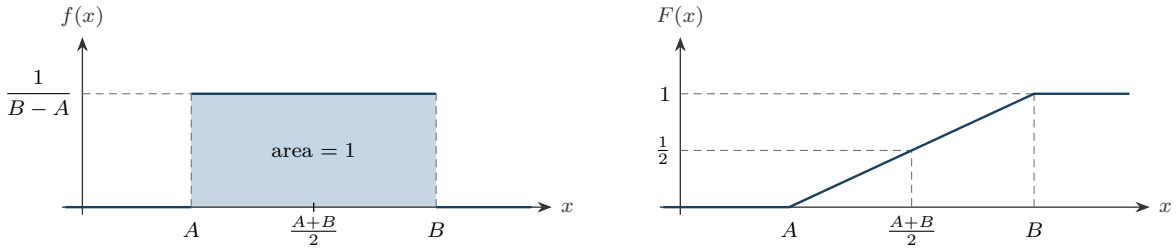
A random variable X has a *continuous uniform distribution* on the interval $[A, B]$, written $X \sim \mathrm{U}(A, B)$, if its pdf is constant on $[A, B]$ and zero everywhere else:

$$f(x) = \begin{cases} \dfrac{1}{B-A}, & A \le x \le B, \\ 0, & \text{otherwise.} \end{cases}$$

The height of the density is not a free choice. If $f(x) = k$ (a constant) on $[A, B]$ and $f(x) = 0$ otherwise, the region under the graph is a rectangle of width $B - A$ and height k, and a valid pdf must have total area 1:

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \int_{A}^{B} k\,\mathrm{d}x = k\,(B-A) = 1 \quad \Longrightarrow \quad k = \frac{1}{B-A}.$$

Because a single point has probability zero (Section 21), it makes no difference whether the endpoints are included: a uniform distribution on (A, B) is the same thing.



The pdf (left) and the cdf (right) of $X \sim \mathrm{U}(A, B)$.

Property (cdf of the uniform distribution)

If $X \sim \mathrm{U}(A, B)$, then

$$F(x) = P(X \le x) = \begin{cases} 0, & x < A, \\ \dfrac{x-A}{B-A}, & A \le x \le B, \\ 1, & x > B. \end{cases}$$

The middle piece comes from integrating the pdf: for $A \le x \le B$,

$$F(x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t = \int_{A}^{x} \frac{1}{B-A}\,\mathrm{d}t = \frac{x-A}{B-A},$$

which is the fraction of the interval $[A, B]$ that lies to the left of x. To the left of A no area has been collected, and to the right of B all of it has.

Formula (probabilities are proportions of length)

If $X \sim \mathrm{U}(A, B)$ and $A \le c \le d \le B$, then

$$P(c \le X \le d) = F(d) - F(c) = \frac{d-c}{B-A} = \frac{\text{length of } [c, d]}{\text{length of } [A, B]}.$$

So sub-intervals of $[A, B]$ of the same length are equally likely, wherever they sit. This is the continuous counterpart of the discrete uniform distribution (Section 17), in which all the values are equally likely. If an interval sticks out of $[A, B]$, only the part inside $[A, B]$ counts.

22.2 Mean and variance

Theorem (mean and variance of a uniform random variable)

If $X \sim \mathrm{U}(A, B)$, then

$$\mu_X = E(X) = \frac{A+B}{2}, \qquad \sigma_X^2 = V(X) = \frac{(B-A)^2}{12}, \qquad \sigma_X = \frac{B-A}{\sqrt{12}}.$$

Proof that $E(X) = \frac{A+B}{2}$ [Worksheet 22, #2]

By the definition of the expected value of a continuous random variable, and because f is zero outside $[A, B]$,

$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \int_A^B \frac{x}{B-A}\, \mathrm{d}x = \frac{1}{B-A}\left[\frac{x^2}{2}\right]_A^B = \frac{B^2-A^2}{2(B-A)}$$
$$= \frac{(B-A)(B+A)}{2(B-A)} = \frac{A+B}{2}.$$

The last step factors the difference of squares $B^2 - A^2 = (B-A)(B+A)$. □

Proof that $V(X) = \frac{(B-A)^2}{12}$

Use the computational formula $V(X) = E(X^2) - [E(X)]^2$. By LOTUS,

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\, \mathrm{d}x = \int_A^B \frac{x^2}{B-A}\, \mathrm{d}x = \frac{1}{B-A}\left[\frac{x^3}{3}\right]_A^B = \frac{B^3-A^3}{3(B-A)}$$
$$= \frac{(B-A)(B^2+AB+A^2)}{3(B-A)} = \frac{A^2+AB+B^2}{3},$$

where the difference of cubes was factored as $B^3 - A^3 = (B-A)(B^2+AB+A^2)$. Then, with $E(X) = \frac{A+B}{2}$ from the first proof,

$$V(X) = \frac{A^2+AB+B^2}{3} - \left(\frac{A+B}{2}\right)^2 = \frac{4\,(A^2+AB+B^2) - 3\,(A^2+2AB+B^2)}{12}$$
$$= \frac{A^2-2AB+B^2}{12} = \frac{(B-A)^2}{12}.$$

Taking the square root gives $\sigma_X = \frac{B-A}{\sqrt{12}}$. □

Remark (symmetry)

The pdf of $\mathrm{U}(A, B)$ is symmetric about the midpoint of the interval. This is why the mean is the midpoint $\frac{A+B}{2}$: it is the balance point of the rectangle. The median is the same number, since $F(m) = \frac{m-A}{B-A} = \frac{1}{2}$ gives $m = \frac{A+B}{2}$. The variance depends only on the length $B - A$ of the interval, not on where the interval is.

The pdf, the cdf, the mean and the variance of the uniform distribution are also listed on the formula sheet (Appendix A).

22.3 Worked example

Example 22.1 ▪ A coffee machine: uniform first, then discrete Worksheet 22, #1

A coffee vending machine advertises that it dispenses 250 mL of coffee, but in reality the volume of coffee dispensed is a random variable X that is distributed uniformly between 245 mL and 255 mL. The amounts dispensed in different cups are mutually independent.

a) What is the probability that any single cup of coffee bought from this machine contains exactly 252 mL of coffee?

b) What is the probability that any single cup of coffee bought from this machine contains at least 252 mL of coffee?

c) Ten people buy a cup of coffee from this machine in succession.

i) What is the probability that at least 3 of them receive a cup that contains 252 mL of coffee or more?

ii) What is the probability that the eighth cup bought is the second that contains 252 mL of coffee or more?

d) What is the probability that any single cup of coffee bought from this machine contains an amount that is more than one standard deviation below the mean, if we are told that the cup has been underfilled (contains less than the advertised amount)?

Solution. Here $X \sim \mathrm{U}(245, 255)$, so $B - A = 10$ and

$$f(x) = \frac{1}{10} \quad \text{and} \quad F(x) = \frac{x - 245}{10} \qquad \text{for } 245 \le x \le 255.$$

a) X is a continuous random variable, so a single value has probability zero: $P(X = 252) = \boxed{0}$.

b) Using the cdf,

$$P(X \ge 252) = 1 - P(X < 252) = 1 - F(252) = 1 - \frac{252 - 245}{10} = 1 - \frac{7}{10} = \boxed{\frac{3}{10}}.$$

The same answer comes from integrating the pdf, $\displaystyle\int_{252}^{255} \frac{1}{10}\,\mathrm{d}x = \frac{3}{10}$: the interval $[252, 255]$ is 3 of the 10 mL of the support.

c) Each cup is now a trial that is either a "success" (it contains at least 252 mL) or not. The cups are independent, and by part (b) the probability of success is $p = \frac{3}{10}$ for every cup, with $q = \frac{7}{10}$. The uniform distribution has done its job; from here on the problem is discrete.

i) *Binomial.* The number of trials is fixed ($n = 10$) and we count the successes. Let Y be the number of the 10 cups that contain at least 252 mL. Then $Y \sim \mathrm{Bin}\big(10, \frac{3}{10}\big)$ (Section 17), and through the complement

$$\begin{aligned} P(Y \ge 3) &= 1 - \big[P(Y = 0) + P(Y = 1) + P(Y = 2)\big] \\ &= 1 - \binom{10}{0}\Big(\frac{3}{10}\Big)^0\Big(\frac{7}{10}\Big)^{10} - \binom{10}{1}\Big(\frac{3}{10}\Big)^1\Big(\frac{7}{10}\Big)^{9} - \binom{10}{2}\Big(\frac{3}{10}\Big)^2\Big(\frac{7}{10}\Big)^{8} \\ &\approx 1 - 0.0282 - 0.1211 - 0.2335 \approx \boxed{0.6172}. \end{aligned}$$

ii) *Negative binomial.* Now the number of successes is fixed ($k = 2$) and the question is on which trial it is reached. Let Z be the number of the cup that is the second one to contain at least 252 mL. Then $Z \sim \mathrm{NB}\big(2, \frac{3}{10}\big)$ (Section 18), and

$$P(Z = 8) = \binom{8-1}{2-1} p^2 q^{8-2} = \binom{7}{1}\Big(\frac{3}{10}\Big)^2\Big(\frac{7}{10}\Big)^6 \approx \boxed{0.0741}.$$

In words: exactly one of the first seven cups is a success, and then the eighth cup is a success.

d) *Conditional probability with a uniform distribution.* First the mean and the standard deviation:

$$\mu = \frac{245+255}{2} = 250, \qquad \sigma = \frac{255-245}{\sqrt{12}} = \frac{10}{\sqrt{12}} \approx 2.887.$$

"Underfilled" means $X < 250$, that is, $X < \mu$; this is the given information. "More than one standard deviation below the mean" means $X < \mu - \sigma \approx 247.11$. By the definition of conditional probability (Section 13),

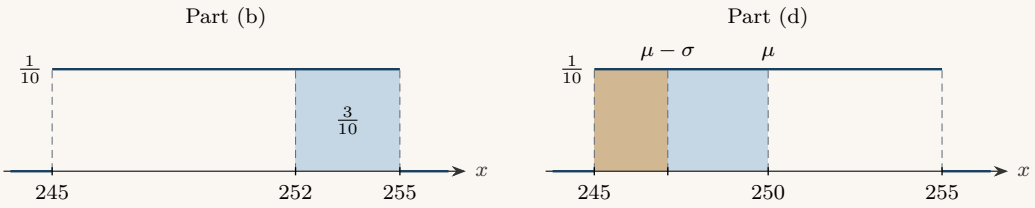
$$P(X < \mu - \sigma \mid X < \mu) = \frac{P(X < \mu - \sigma \text{ and } X < \mu)}{P(X < \mu)} = \frac{P(X < \mu - \sigma)}{P(X < \mu)},$$

because every cup with $X < \mu - \sigma$ automatically has $X < \mu$. Both probabilities come from the cdf:

$$P(X < \mu) = F(250) = \frac{250-245}{10} = \frac{1}{2},$$
$$P(X < \mu - \sigma) = F(250-\sigma) = \frac{(250-\sigma)-245}{10} = \frac{1}{2} - \frac{1}{\sqrt{12}} = \frac{3-\sqrt{3}}{6} \approx 0.2113,$$

using $\frac{\sigma}{10} = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$. Therefore

$$P(X < \mu - \sigma \mid X < \mu) = \frac{(3-\sqrt{3})/6}{1/2} = 1 - \frac{\sqrt{3}}{3} \approx \boxed{0.4226}.$$



The pdf of U(245, 255). Left: the shaded area is $P(X \geq 252)$. Right: among the underfilled cups (both shaded regions together, area $\frac{1}{2}$), the darker region is the event $X < \mu - \sigma$, where $\mu - \sigma \approx 247.11$; the answer to (d) is the ratio of the two areas.

Note (the continuous distribution supplies p, a discrete distribution does the counting)

Part (c) of Example 22.1 shows a pattern that returns often with continuous distributions: the continuous distribution gives the probability p of some event for *one* observation, and a discrete distribution then counts how often that event happens in several independent observations. So this is a good moment to review the standard distributions met so far:

- **uniform**: all values equally likely (the discrete version is in Section 17);
- **binomial**: the number of successes in a fixed number n of trials (Section 17);
- **negative binomial**: the number of the trial on which the kth success occurs; the **geometric** distribution is the special case $k = 1$ (Section 18);
- **hypergeometric**: the number of successes when drawing without replacement (Section 19);
- **Poisson**: the number of events in a fixed interval of time or space (Section 20).

23 The Exponential Distribution

The exponential distribution is the standard model for a *waiting time*: how long until the next bus comes, until the next customer arrives, or until a machine part fails. It is the continuous partner of the Poisson distribution of Section 20. A Poisson random variable counts *how many* events happen in a period of time; an exponential random variable measures *how long* we wait for one event to happen.

23.1 The pdf and the cdf

The distribution is defined through its pdf; the cdf is then calculated from the pdf.

Definition (exponential distribution)

A continuous random variable X has an *exponential distribution* with parameter $\lambda > 0$, written $X \sim \text{Exp}(\lambda)$, when its pdf is

$$f(x) = \begin{cases} 0, & \text{if } x \le 0, \\ \lambda e^{-\lambda x}, & \text{if } x > 0. \end{cases}$$

Its support is $\text{Supp}(X) = (0, \infty)$. In applications,

$$X = \text{the amount of time needed until an event occurs,}$$
$$\lambda = \text{the number of events in a unit of time.}$$

The parameter λ is a *rate*: the average number of events per unit of time (per minute, per hour, ...), exactly as for the Poisson distribution. The function f is a valid pdf (Section 21): it is never negative, and the total area under it is

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \int_{0}^{\infty} \lambda e^{-\lambda x}\,\mathrm{d}x = \Big[-e^{-\lambda x}\Big]_{0}^{\infty} = 0 - (-1) = 1.$$

The formula sheet (Appendix A) attaches the point $x = 0$ to the second line of f ($x \ge 0$ instead of $x > 0$). It makes no difference, because a single point carries no probability.

Theorem (cdf of the exponential distribution)

If $X \sim \text{Exp}(\lambda)$, then its cumulative distribution function is

$$F(x) = P(X \le x) = \begin{cases} 0, & x \le 0, \\ 1 - e^{-\lambda x}, & x > 0. \end{cases}$$

Proof of the cdf formula

The cdf is calculated from the pdf: by definition (Section 21), $F(x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$. The letter t is used for the variable of integration because x is already taken by the upper limit. Since f is given by two formulas, there are two cases.

Case $x \le 0$. Every $t \le x$ satisfies $t \le 0$, so $f(t) = 0$ on the whole range of integration and

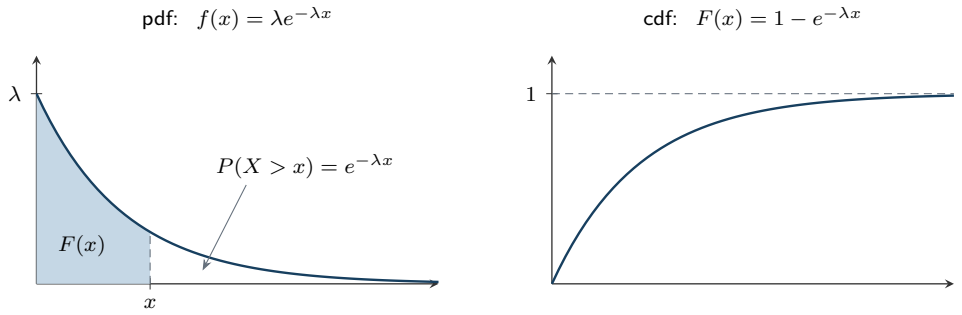
$$F(x) = \int_{-\infty}^{x} 0\,\mathrm{d}t = 0.$$

Case $x > 0$. Split the integral at 0, where the formula for f changes:

$$\begin{aligned} F(x) &= \int_{-\infty}^{0} 0\,\mathrm{d}t + \int_{0}^{x} \lambda e^{-\lambda t}\,\mathrm{d}t = 0 + \Big[-e^{-\lambda t}\Big]_{t=0}^{t=x} \\ &= -e^{-\lambda x} - \big(-e^{0}\big) = 1 - e^{-\lambda x}. \end{aligned}$$

The two cases are the two lines of the formula. □

Three quick checks that the formula is reasonable: both lines give $F(0) = 0$, so F is continuous; $F(x) \to 1$ as $x \to \infty$; and differentiating gives back the pdf, $F'(x) = \lambda e^{-\lambda x} = f(x)$ for $x > 0$.



The pdf starts at height λ and decays. The shaded area is $F(x) = 1 - e^{-\lambda x}$; the unshaded tail to its right is $e^{-\lambda x}$.

Recall (probabilities from a cdf)

For any continuous random variable X with cdf F (Section 21),

$$F(x) = P(X \le x), \qquad P(a \le X \le b) = F(b) - F(a), \qquad P(X > x) = 1 - F(x).$$

Because X is continuous, each $\le$ may be replaced by $<$ (and $>$ by $\ge$) without changing the probability.

Formula (exponential probabilities)

If $X \sim \text{Exp}(\lambda)$, then for $x \ge 0$ and $0 \le a \le b$,

$$P(X \le x) = 1 - e^{-\lambda x}, \qquad P(X > x) = e^{-\lambda x}, \qquad P(a \le X \le b) = e^{-\lambda a} - e^{-\lambda b}.$$

The middle formula is the one to remember: the probability of waiting *longer* than x is simply $e^{-\lambda x}$, and the other two follow from it.

23.2 Mean and variance

The mean and the standard deviation of an exponential random variable are both equal to the reciprocal of the rate λ.

Theorem (mean and variance of an exponential random variable)

If $X \sim \text{Exp}(\lambda)$, then

$$\mu_X = E(X) = \frac{1}{\lambda}, \qquad \sigma_X^2 = V(X) = \frac{1}{\lambda^2}, \qquad \sigma_X = \frac{1}{\lambda}.$$

Proof that $E(X) = 1/\lambda$

By the definition of the expected value of a continuous random variable, with the pdf $f(x) = 0$ for $x \le 0$ and $f(x) = \lambda e^{-\lambda x}$ for $x > 0$,

$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, \mathrm{d}x = \underbrace{\int_{-\infty}^{0} x \cdot 0 \, \mathrm{d}x}_{0} + \int_{0}^{\infty} x\, \lambda e^{-\lambda x}\, \mathrm{d}x = \int_{0}^{\infty} x\, \lambda e^{-\lambda x}\, \mathrm{d}x.$$

Integrate by parts with

$$u = x, \quad \mathrm{d}v = \lambda e^{-\lambda x}\, \mathrm{d}x \qquad \Longrightarrow \qquad \mathrm{d}u = \mathrm{d}x, \quad v = -e^{-\lambda x}:$$

$$E(X) = \Big[-x\, e^{-\lambda x}\Big]_0^{\infty} + \int_0^{\infty} e^{-\lambda x}\, \mathrm{d}x = (0 - 0) + \Big[-\frac{1}{\lambda}\, e^{-\lambda x}\Big]_0^{\infty} = 0 - \Big(-\frac{1}{\lambda}\Big) = \frac{1}{\lambda}.$$

The first bracket vanishes at its upper end because $\lim_{x\to\infty} x\, e^{-\lambda x} = \lim_{x\to\infty} \dfrac{x}{e^{\lambda x}} = \lim_{x\to\infty} \dfrac{1}{\lambda e^{\lambda x}} = 0$ by l'Hôpital's rule. □

Proof that $V(X) = 1/\lambda^2$

Use the computational formula $V(X) = E(X^2) - E(X)^2$. By LOTUS (Section 21) and integration by parts with $u = x^2$, $\mathrm{d}v = \lambda e^{-\lambda x}\,\mathrm{d}x$ (so $\mathrm{d}u = 2x\,\mathrm{d}x$ and $v = -e^{-\lambda x}$),

$$E(X^2) = \int_0^\infty x^2 \lambda e^{-\lambda x}\,\mathrm{d}x = \underbrace{\left[-x^2 e^{-\lambda x}\right]_0^\infty}_{0} + \int_0^\infty 2x\, e^{-\lambda x}\,\mathrm{d}x = \frac{2}{\lambda}\underbrace{\int_0^\infty x\,\lambda e^{-\lambda x}\,\mathrm{d}x}_{E(X)=1/\lambda} = \frac{2}{\lambda^2}.$$

Therefore $V(X) = E(X^2) - E(X)^2 = \dfrac{2}{\lambda^2} - \dfrac{1}{\lambda^2} = \dfrac{1}{\lambda^2}$. □

Note (rate and mean are reciprocals)

Since $E(X) = 1/\lambda$, a problem may describe the same distribution in two ways:

- by its **rate**: "one bus every 10 minutes" means $\lambda = \frac{1}{10}$ bus per minute, so the mean waiting time is $1/\lambda = 10$ minutes;
- by its **mean**: "the mean time until failure is 25 000 hours" means $1/\lambda = 25\,000$, so $\lambda = \frac{1}{25\,000}$ per hour.

The time x and the rate λ must be expressed in the same unit of time. Notice also that the standard deviation of an exponential random variable equals its mean.

23.3 The exponential and the Poisson distribution

The two distributions describe the same stream of random events from two points of view, and they share the parameter λ.

	Poisson, Pois(λ)	**Exponential**, Exp(λ)
Random variable	number of events in a given period of time	amount of time needed until an event occurs
It answers	"how many events happen?"	"how long until an event happens?"
Type	discrete, Supp $= \{0, 1, 2, \dots\}$	continuous, Supp $= (0, \infty)$
Parameter	λ = number of events in a unit of time	the *same* λ
Mean	λ events (λt events in a period of length t)	$1/\lambda$ units of time

Theorem (relationship between the Poisson and the exponential distribution)

Suppose that events occur over time in such a way that the number of events in a unit of time is $X \sim \text{Pois}(\lambda)$ (a *Poisson process* with rate λ). Let

$$T = \text{the time until the first event.}$$

Then $T \sim \text{Exp}(\lambda)$, with the same λ. In short: a Poisson process with rate λ gives rise to an exponential distribution with mean $E(T) = \dfrac{1}{\lambda}$.

Proof

Fix a time $t > 0$ and let Y be the number of events in the time interval from 0 to t. Rescaling the Poisson parameter to a period of length t (Section 20),

$$Y \sim \text{Pois}(\lambda_t) \qquad \text{with} \qquad \lambda_t = \lambda t.$$

The probability that *no* event occurs in time t comes from the Poisson pmf:

$$P(Y=0)=\frac{e^{-\lambda_t}\,\lambda_t^0}{0!}=e^{-\lambda t}.$$

But "no event occurs in time t" says exactly that the time it takes for the first event to occur is greater than t. Hence

$$P(T>t)=P(Y=0)=e^{-\lambda t},\qquad\text{so}\qquad F_T(t)=P(T\le t)=1-e^{-\lambda t}\quad(t>0).$$

A waiting time cannot be negative, so $F_T(t)=0$ for $t\le 0$. This is precisely the cdf of the $\mathrm{Exp}(\lambda)$ distribution, and differentiating it gives the pdf $f_T(t)=F_T'(t)=\lambda e^{-\lambda t}$ for $t>0$. Therefore $T\sim\mathrm{Exp}(\lambda)$ and $E(T)=1/\lambda$. □

Note (the same λ)

In both distributions λ is the number of events in a given unit of time. "Six buses per hour" and "a bus every 10 minutes on average" are the same statement: the mean number of events per unit of time (λ) and the mean time per event ($1/\lambda$) are reciprocals. Decide from the question which random variable is wanted:

- a *number of events* in a period of length t ⇒ Poisson with parameter λt;
- a *time* until the next event (or between two consecutive events) ⇒ exponential with parameter λ.

23.4 The memoryless property

For an exponential waiting time, the waiting already done has no influence on the waiting still to come.

Property (memoryless property of the exponential distribution)

If $X\sim\mathrm{Exp}(\lambda)$, then for all $t\ge 0$ and $h\ge 0$,

$$P(X\ge t+h\mid X\ge h)=P(X\ge t).$$

Proof

If $X\ge t+h$ then automatically $X\ge h$, so the event "$X\ge t+h$ and $X\ge h$" is just $X\ge t+h$. By the definition of conditional probability (Section 13) and the tail formula $P(X\ge x)=e^{-\lambda x}$,

$$P(X\ge t+h\mid X\ge h)=\frac{P(X\ge t+h)}{P(X\ge h)}=\frac{e^{-\lambda(t+h)}}{e^{-\lambda h}}=e^{-\lambda t}=P(X\ge t).$$

The time h already spent waiting has cancelled out of the answer. □

In words: if you have already waited a time h and nothing has happened, the *remaining* waiting time has the same $\mathrm{Exp}(\lambda)$ distribution as if you had only just started waiting. The process does not remember how long you have been there. For instance, if the wait for a bus is $X\sim\mathrm{Exp}(\frac{1}{10})$ (in minutes) and you have already waited 3 minutes, the probability that your total wait exceeds 10 minutes is $P(X>10\mid X>3)=P(X>7)=e^{-0.7}\approx 0.4966$.

Exponential distribution at a glance

Situation	waiting time until the next event, when events occur at a constant average rate
Variable	$X=$ time until an event occurs; $X\sim\mathrm{Exp}(\lambda)$; $\mathrm{Supp}(X)=(0,\infty)$
Parameter	$\lambda=$ number of events in a unit of time
pdf, cdf	$f(x)=\lambda e^{-\lambda x}$ and $F(x)=1-e^{-\lambda x}$ for $x>0$ (both are 0 for $x\le 0$)
Tail	$P(X>x)=e^{-\lambda x}$
Mean, variance	$\mu_X=\frac{1}{\lambda},\quad \sigma_X^2=\frac{1}{\lambda^2},\quad \sigma_X=\frac{1}{\lambda}$
Poisson link	the number of events in a period of length t is $\mathrm{Pois}(\lambda t)$

23.5 Worked example

The example below uses every formula of this section, and its last two parts cross over to the Poisson distribution.

Example 23.1 ▪ Waiting for a bus Lecture notes

A bus line runs a bus every 10 minutes on average. Let X be the time, in minutes, that you wait before the next bus comes, and suppose that X is exponentially distributed.

a) How long should you expect to wait?
b) What is the probability that you wait less than 10 minutes?
c) What is the probability that you wait between 9 and 11 minutes?
d) What is the probability that you wait exactly 10 minutes?
e) You stay at the stop for an hour. How many buses should you expect to pass?
f) What is the probability that 8 or 9 buses arrive in that hour?

Solution. The rate is $\lambda = 1$ bus per 10 minutes, that is, $\lambda = \frac{1}{10}$ bus per minute, so $X \sim \text{Exp}(\frac{1}{10})$ with cdf $F(x) = 1 - e^{-x/10}$ for $x > 0$.

a) "Expect" asks for the mean: $E(X) = \dfrac{1}{\lambda} = \boxed{10 \text{ minutes}}$.

b) X is continuous, so $<$ and $\leq$ give the same probability:

$$P(X < 10) = P(X \leq 10) = F(10) = 1 - e^{-\frac{1}{10}\cdot 10} = 1 - e^{-1} \approx \boxed{0.6321}.$$

The answer is not $\frac{1}{2}$: the mean of an exponential distribution is not its median, and most waits are shorter than the average wait.

c) Subtract the cdf at the two endpoints:

$$\begin{aligned} P(9 < X < 11) = F(11) - F(9) &= \left(1 - e^{-1.1}\right) - \left(1 - e^{-0.9}\right) = e^{-0.9} - e^{-1.1} \\ &\approx 0.4066 - 0.3329 = \boxed{0.0737}. \end{aligned}$$

d) $P(X = 10) = \boxed{0}$. This is a continuous distribution, and the probability at a single point is always 0.

Parts e) and f) show how the connection with the Poisson distribution can be asked. They are no longer about a waiting time but about a *number of buses*, so the random variable changes while the rate $\lambda = \frac{1}{10}$ bus per minute stays the same.

e) Let Y be the number of buses that pass in one hour. An hour is $t = 60$ minutes, so by rescaling

$$Y \sim \text{Pois}(\lambda_Y) \qquad \text{with} \qquad \lambda_Y = \lambda \cdot 60 = \tfrac{1}{10}(60) = 6.$$

The mean of a Poisson random variable is its parameter: $E(Y) = \lambda_Y = \boxed{6 \text{ buses}}$.

f) Y is discrete, so the event is made of the two values 8 and 9, and we add the Poisson pmf:

$$P(8 \leq Y \leq 9) = P(Y = 8) + P(Y = 9) = \frac{e^{-6}\,6^8}{8!} + \frac{e^{-6}\,6^9}{9!} \approx 0.1033 + 0.0688 = \boxed{0.1721}.$$

24 The Normal Distribution

The normal distribution is *the* bell curve. It models measurements that cluster symmetrically around a central value (heights, weights, filling volumes, measurement errors), and the sampling distributions and inference methods of the later sections are all built on it. Unlike the uniform and the exponential distributions, its cdf has no formula, so normal probabilities are read from a table. The skill to acquire here is to turn any question about a normal random variable into a table look-up, and to run the look-up backwards when a percentage is given and a value is wanted.

24.1 The normal density

A normal distribution is specified by two numbers: where its centre is, and how spread out it is.

Definition (normal distribution)

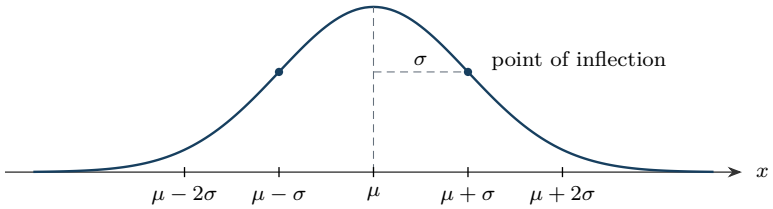
A continuous random variable X has a *normal distribution* with parameters μ and $\sigma > 0$, written $X \sim \mathrm{N}(\mu, \sigma)$, if the pdf for X is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \qquad -\infty < x < \infty.$$

The two parameters are the mean and the standard deviation of X: $E(X) = \mu$ and $V(X) = \sigma^2$.

Caution (the second parameter is σ)

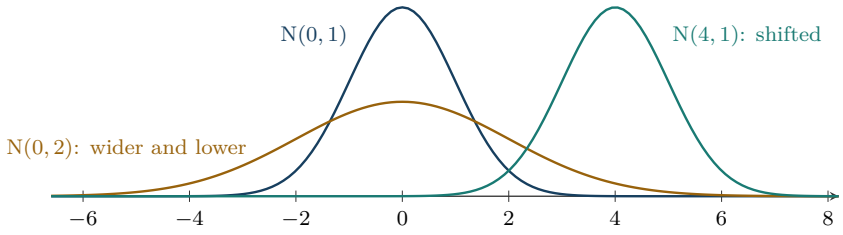
In this course $\mathrm{N}(\mu, \sigma)$ lists the *standard deviation*, not the variance: "mean 100 and variance 25" is $\mathrm{N}(100, 5)$. Many textbooks write $N(\mu, \sigma^2)$ instead, so always check which convention is being used.



Property (shape of the normal curve)

- **The bell curve.** The graph of f has a single peak, at $x = \mu$, and falls away on both sides without ever reaching the axis: $\mathrm{Supp}(X) = (-\infty, \infty)$.
- **Symmetric.** The curve is symmetric about the line $x = \mu$. So the mean, the median and the mode are all equal to μ, and $P(X \le \mu) = P(X \ge \mu) = 0.5$.
- **σ is the spread.** The curve changes from bending downwards to bending upwards at $x = \mu \pm \sigma$, so σ is the horizontal distance from the centre to a point of inflection.

Every normal curve is the basic bell e^{-x^2}, *stretched and shifted*. Replacing x by $(x - \mu)/(\sigma\sqrt{2})$ moves the centre from 0 to μ and stretches the curve horizontally in proportion to σ; the factor $1/(\sigma\sqrt{2\pi})$ in front then rescales the height so that the total area under the curve is 1. Changing μ slides the curve along the axis without changing its shape; increasing σ makes it wider and lower.



Remark (there is no formula for the cdf)

Showing that the total area $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$ equals 1 is already not routine: it takes a special technique (Feynman's trick of differentiating under the integral sign is one way). For the cdf the situation is worse. There is no way

to find

$$F(x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$$

by hand, because $e^{-t^2/2}$ has no antiderivative that can be written with elementary functions. Normal probabilities therefore come from a table of numerically computed values.

24.2 The standard normal distribution and its table

A separate table for every pair (μ, σ) is out of the question. Instead we create *one* table, for one standardized normal distribution, and convert every other normal distribution to it.

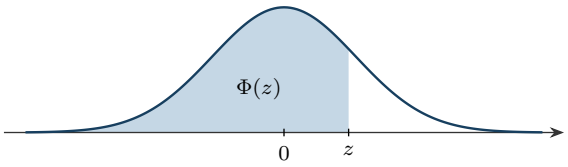
Definition (standard normal distribution)

The *standard normal distribution* is the normal distribution with $\mu = 0$ and $\sigma = 1$:

$$Z \sim \mathrm{N}(\mu = 0,\ \sigma = 1).$$

It is always centred at 0, with standard deviation 1. Its pdf is $f(z) = \dfrac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}z^2}$, and its cdf has its own symbol, Φ:

$$F(z) = \Phi(z) = P(Z \le z) = \text{the area under the standard normal curve to the left of } z.$$



Reading the table

Table B.1 lists $\Phi(z)$ to four decimal places for $z = -3.49, -3.48, \ldots, 3.49$. The row gives z up to its first decimal and the column supplies the second decimal. Three of its rows:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−1.7	0.0446	0.0436	0.0427	**0.0418**	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	**0.4013**	0.3974	0.3936	0.3897	0.3859
0.3	0.6179	**0.6217**	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517

For instance $\Phi(0.31)$ is in row 0.3, column 0.01: $\Phi(0.31) = 0.6217$. In the same way $\Phi(-0.25) = 0.4013$ (row −0.2, column 0.05) and $\Phi(-1.73) = 0.0418$ (row −1.7, column 0.03). Always round z to two decimal places before going to the table.

Property (evaluating probabilities with Φ)

Let $Z \sim \mathrm{N}(0, 1)$. The table gives areas to the *left*; every other area is obtained from those.

- **Left of z:** $P(Z \le z) = \Phi(z)$.
- **Right of z:** $P(Z > z) = 1 - \Phi(z) = \Phi(-z)$.
- **Between a and b:** $P(a \le Z \le b) = \Phi(b) - \Phi(a)$.

Z is continuous, so $P(Z = z) = 0$ and it never matters whether an endpoint is included ($<$ or $\le$).

The identity $1 - \Phi(z) = \Phi(-z)$ is the symmetry of the bell curve about 0: the tail to the right of z is the mirror image of the tail to the left of $-z$.



Note (beyond the ends of the table)

The table stops at $z = \pm 3.49$, where $\Phi(-3.49) = 0.0002$ and $\Phi(3.49) = 0.9998$. Practically all of the area lies between these values: for $z < -3.49$ take $\Phi(z) \approx 0$, and for $z > 3.49$ take $\Phi(z) \approx 1$.

24.3 Standardizing: from $\mathrm{N}(\mu, \sigma)$ to Z

The table only covers Z. Any other normal random variable is brought back to Z by measuring its values in standard deviations away from the mean.

Property (standardizing)

If $X \sim \mathrm{N}(\mu, \sigma)$, then $Z = \dfrac{X - \mu}{\sigma}$ has the standard normal distribution. Therefore

$$P(X \le x) = P\Big(Z \le \frac{x-\mu}{\sigma}\Big) = \Phi\Big(\frac{x-\mu}{\sigma}\Big),$$

$$P(X \ge x) = P\Big(Z \ge \frac{x-\mu}{\sigma}\Big) = 1 - \Phi\Big(\frac{x-\mu}{\sigma}\Big),$$

$$P(a \le X \le b) = \Phi\Big(\frac{b-\mu}{\sigma}\Big) - \Phi\Big(\frac{a-\mu}{\sigma}\Big).$$

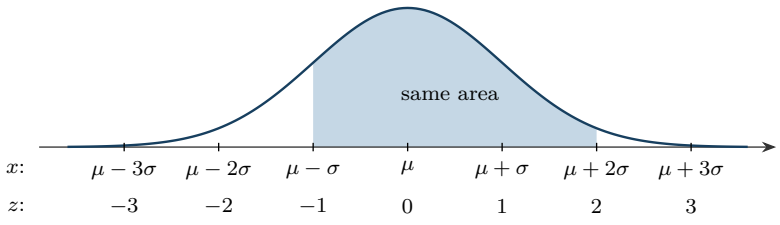
Proof

Write $P(X \le x)$ as the integral of the pdf and substitute $t = \dfrac{s-\mu}{\sigma}$, so that $\mathrm{d}s = \sigma\,\mathrm{d}t$ and the upper limit $s = x$ becomes $t = \dfrac{x-\mu}{\sigma}$:

$$P(X \le x) = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{s-\mu}{\sigma}\right)^2} \mathrm{d}s = \int_{-\infty}^{\frac{x-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}t^2}\,\mathrm{d}t = \Phi\Big(\frac{x-\mu}{\sigma}\Big).$$

The second line of the property is the complement of the first, and the third is $F(b) - F(a)$. □

The number $z = \dfrac{x-\mu}{\sigma}$ is the z-score of x (Section 7): it says how many standard deviations x lies above the mean ($z > 0$) or below it ($z < 0$). Standardizing only relabels the horizontal axis; the areas stay the same.



A normal probability in four steps

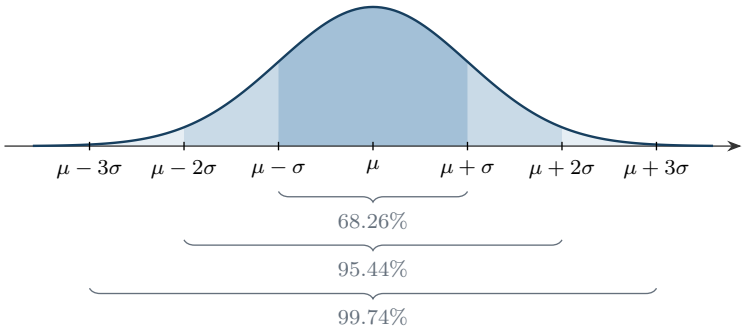
1. Name the random variable and write down its distribution, $X \sim \mathrm{N}(\mu, \sigma)$.
2. Sketch the bell curve, mark the given values and shade the area that is asked for.

3. Standardize each endpoint: $z = \dfrac{x-\mu}{\sigma}$, rounded to two decimal places.
4. Express the shaded area with Φ (left: $\Phi(z)$; right: $1-\Phi(z)$; between: $\Phi(z_2)-\Phi(z_1)$) and read the values from Table B.1.

Standardizing also explains the empirical rule of Section 7. Whatever the values of μ and σ, the probability of falling within k standard deviations of the mean is $P(\mu-k\sigma \le X \le \mu+k\sigma) = P(-k \le Z \le k) = \Phi(k)-\Phi(-k)$, and the table gives

$$\begin{aligned}
\text{within 1 standard deviation } (k=1): &\quad \Phi(1)-\Phi(-1) = 0.8413-0.1587 = 0.6826,\\
\text{within 2 standard deviations } (k=2): &\quad \Phi(2)-\Phi(-2) = 0.9772-0.0228 = 0.9544,\\
\text{within 3 standard deviations } (k=3): &\quad \Phi(3)-\Phi(-3) = 0.9987-0.0013 = 0.9974.
\end{aligned}$$

These are the "68%, 95%, 99.7%" of the empirical rule.



Remark (why μ and σ are the mean and the standard deviation)

For the standard normal variable, $E(Z) = \displaystyle\int_{-\infty}^{\infty} z\,\tfrac{1}{\sqrt{2\pi}}e^{-z^2/2}\,\mathrm{d}z = 0$, because the integrand is an odd function. Integrating by parts with $u = z$ and $\mathrm{d}v = z\,\tfrac{1}{\sqrt{2\pi}}e^{-z^2/2}\,\mathrm{d}z$ (so $v = -\tfrac{1}{\sqrt{2\pi}}e^{-z^2/2}$),

$$\begin{aligned}
V(Z) = E(Z^2) - 0^2 &= \int_{-\infty}^{\infty} z^2\,\tfrac{1}{\sqrt{2\pi}}e^{-z^2/2}\,\mathrm{d}z\\
&= \Big[-z\,\tfrac{1}{\sqrt{2\pi}}e^{-z^2/2}\Big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty}\tfrac{1}{\sqrt{2\pi}}e^{-z^2/2}\,\mathrm{d}z = 0+1 = 1.
\end{aligned}$$

Since $X = \mu+\sigma Z$, the rules $E(aZ+b) = aE(Z)+b$ and $V(aZ+b) = a^2V(Z)$ of Section 16 give $E(X) = \mu$ and $V(X) = \sigma^2$.

24.4 Working backwards: inverse look-ups

Many questions give a probability ("the top 5%", "the middle 75%") and ask for the value of x that goes with it. The table is then read from the inside out.

Notation (inverse look-up)

For a probability p with $0 < p < 1$, $\Phi^{-1}(p)$ is the z-value that has area p to its left:

$$z = \Phi^{-1}(p) \quad\Longleftrightarrow\quad \Phi(z) = p.$$

To find $\Phi^{-1}(p)$, look for p in the *body* of Table B.1 and read z from the margins. Two rows of the table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	**0.1251**	0.1230	0.1210	0.1190	0.1170
1.6	0.9452	0.9463	0.9474	0.9484	**0.9495**	**0.9505**	0.9515	0.9525	0.9535	0.9545

- $\Phi^{-1}(0.125)$: the entry closest to 0.125 is 0.1251, in row -1.1 and column 0.05. So $\Phi^{-1}(0.125) = -1.15$.
- $\Phi^{-1}(0.95)$: the value 0.95 lies exactly halfway between 0.9495 ($z = 1.64$) and 0.9505 ($z = 1.65$), so take the z halfway between: $\Phi^{-1}(0.95) = 1.645$.
- By symmetry, $\Phi^{-1}(1-p) = -\Phi^{-1}(p)$. For example $\Phi^{-1}(0.875) = +1.15$ and $\Phi^{-1}(0.05) = -1.645$.

In general, use the closest entry; when p falls between two entries, interpolating between their two z-values gives a third decimal.

Formula (from a probability back to a value of X)

If $X \sim \mathrm{N}(\mu, \sigma)$ and $P(X \le x) = p$, then

$$\frac{x-\mu}{\sigma} = z \quad \text{with} \quad z = \Phi^{-1}(p), \qquad \text{so} \qquad x = \mu + z\,\sigma.$$

This x is the $100p$-th percentile of X.

Caution (Φ^{-1} needs the area to the left)

Translate the wording into a left-hand area before the look-up. "The largest 5%" means an area of 0.05 to the *right*, so the area to the left is 0.95 and $z = \Phi^{-1}(0.95)$. Later sections write z_α for the z-value with area α to its right, that is, $z_\alpha = \Phi^{-1}(1-\alpha)$; for example $z_{0.05} = 1.645$.

The same equation $z = \dfrac{x-\mu}{\sigma}$ also handles problems in which μ or σ is the unknown. A given percentage fixes z through an inverse look-up, and the equation is then solved for the missing parameter. Two unknown parameters need two such equations, one for each percentage that is given.

24.5 Worked examples: from values to probabilities

In these examples the values of the variable are given and a probability is wanted: standardize, then look up.

Example 24.1 ▪ Reading the table Lecture notes

Let $Z \sim \mathrm{N}(0,1)$. Find a) $P(Z > 1.73)$ b) $P(-0.25 \le Z \le 0.31)$.

Solution. Sketch each area before touching the table.



a) The table only gives areas to the left, so use the symmetry of the curve: the area to the right of 1.73 equals the area to the left of -1.73.

$$P(Z > 1.73) = P(Z < -1.73) = \Phi(-1.73) = \boxed{0.0418}.$$

The complement gives the same answer: $1 - \Phi(1.73) = 1 - 0.9582 = 0.0418$.

b) An area between two values is the difference of two left-hand areas:

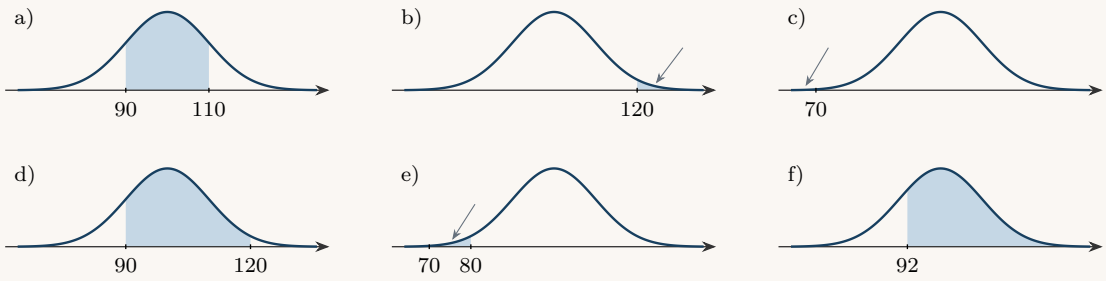
$$P(-0.25 \le Z \le 0.31) = \Phi(0.31) - \Phi(-0.25) = 0.6217 - 0.4013 = \boxed{0.2204}.$$

Example 24.2 ⋆ ▪ **Speeds on the autoroute: every type of look-up** Worksheet 24, #1

Suppose that the speeds of cars travelling on the Laurentian Autoroute are normally distributed with mean 100 km/h and standard deviation 10 km/h. Find the probability that a car selected at random on the Laurentian Autoroute will go at a speed

a) between 90 and 110 km/h?
b) greater than 120 km/h?
c) less than 70 km/h?
d) between 90 and 120 km/h?
e) between 70 and 80 km/h?
f) greater than 92 km/h?

Solution. Let X be the speed of the car, in km/h. Then $X \sim \mathrm{N}(100, 10)$ and each speed x is standardized by $z = \dfrac{x - 100}{10}$.



a) $P(90 \le X \le 110) = P(-1 \le Z \le 1) = \Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = \boxed{0.6826}$
b) $P(X > 120) = P(Z > 2) = 1 - \Phi(2) = 1 - 0.9772 = \boxed{0.0228}$
c) $P(X < 70) = P(Z < -3) = \Phi(-3) = \boxed{0.0013}$
d) $P(90 \le X \le 120) = P(-1 \le Z \le 2) = \Phi(2) - \Phi(-1) = 0.9772 - 0.1587 = \boxed{0.8185}$
e) $P(70 \le X \le 80) = P(-3 \le Z \le -2) = \Phi(-2) - \Phi(-3) = 0.0228 - 0.0013 = \boxed{0.0215}$
f) $P(X > 92) = P(Z > -0.8) = 1 - \Phi(-0.8) = 1 - 0.2119 = \boxed{0.7881}$

Use the sketches as a check. In f) the shaded region covers more than half of the curve, because 92 is below the mean, so the answer has to be larger than 0.5. In e) both z-values are negative, and the rule is still "upper endpoint minus lower endpoint".

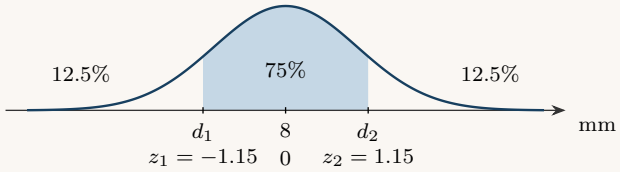
24.6 Worked example: from probabilities to values

Here the direction is reversed: a probability is given, and a value of the variable or a parameter of the distribution is wanted.

Example 24.3 ▪ **The middle 75%** Worksheet 25, #1

The diameter D of Goldy-Lox (yellow, fish-flavoured) candies produced is a normally distributed random variable with mean 8 mm and standard deviation 2 mm. Consumer research has shown that Goldy-Lox eaters are very particular about the size of their candies: they will only eat candies whose diameter is *just right*. The research shows that the middle 75% of the distribution of candy diameters will satisfy customers. So the company gives candies with extreme diameters to charity, i.e. candies with diameter $D < d_1$ or $D > d_2$ are given to charity. Find d_1 and d_2.

Solution. Here $D \sim \mathrm{N}(8, 2)$. The middle 75% leaves 25% for the two tails together, and the curve is symmetric, so each tail holds 12.5%.



Inverse look-up. The lower cut-off d_1 has area 0.125 to its left. In the table the closest entry is $\Phi(-1.15) = 0.1251$, so

$$z_1 = \Phi^{-1}(12.5\%) = -1.15, \qquad \text{and by symmetry} \qquad z_2 = +1.15.$$

Back to millimetres. The z-scores of the two cut-offs are $z_1 = \dfrac{d_1 - 8}{2}$ and $z_2 = \dfrac{d_2 - 8}{2}$, so

$$d_1 = 8 + 2(-1.15) = \boxed{5.7 \text{ mm}}, \qquad d_2 = 8 + 2(1.15) = \boxed{10.3 \text{ mm}}.$$

Candies narrower than 5.7 mm or wider than 10.3 mm go to charity.

25 The Normal Approximation to the Binomial

This section is an application of the normal distribution. A binomial probability such as $P(45 \le X \le 60)$ with $n = 200$ trials means adding sixteen terms of the binomial pmf (Section 17), each with an enormous binomial coefficient. When n is large enough, the bar chart of a binomial pmf is so close to a bell curve that an area under a normal curve can be used instead.

25.1 The approximating normal curve

The idea is to keep the two numbers that summarize the binomial distribution, its mean and its standard deviation, and to replace its bar chart by the normal curve that has those two parameters.

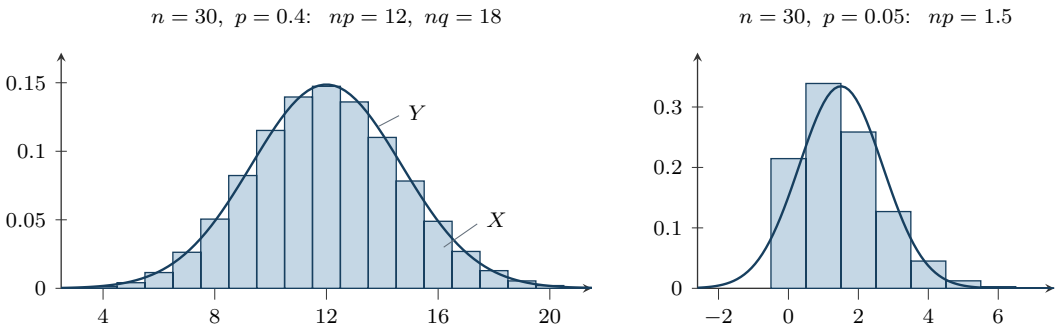
Normal approximation to the binomial

Let $X \sim \text{Bin}(n, p)$ and $q = 1 - p$. Replace X by the normal random variable that has the same mean and the same standard deviation as the binomial:

$$Y \sim \text{N}\big(\mu = np, \;\; \sigma = \sqrt{npq}\,\big).$$

Criterion. The approximation is valid if

$$np > 5 \qquad \text{and} \qquad nq > 5.$$



Bars: the pmf of $X \sim \text{Bin}(n, p)$. Curve: the pdf of $Y \sim \text{N}(np, \sqrt{npq})$. On the left the criterion holds and the curve follows the bars. On the right np is too small: the bars pile up against 0 and a symmetric bell curve cannot fit them.

The criterion asks for more than 5 expected successes (np) *and* more than 5 expected failures (nq). When both hold, the histogram of X has room on both sides of its mean and takes the symmetric bell shape; when one of them fails, it is skewed and the normal curve is the wrong model. (The bell shape is no accident: a binomial count is a sum of n independent trials, and the Central Limit Theorem of Section 28 says that such sums are approximately normal.)

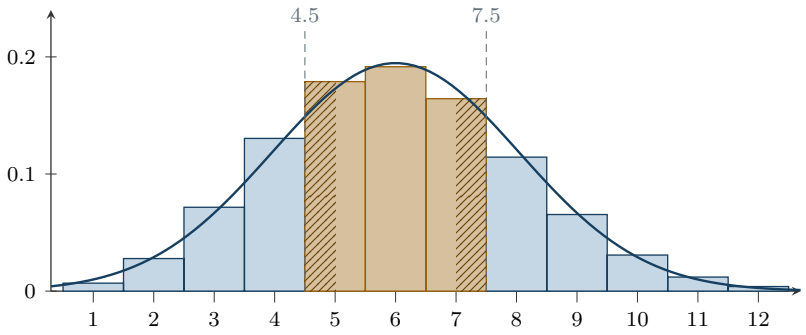
25.2 The continuity correction

The binomial variable X is discrete: it only takes whole-number values. The normal variable Y is continuous. Before Y can stand in for X, this mismatch has to be repaired.

In the histogram of X, the probability $P(X = k)$ is the area of the bar centred at k. The bar has width 1, so it stands on the interval from $k - 0.5$ to $k + 0.5$. Take $P(5 \le X \le 7)$: it is the total area of the three bars at 5, 6 and 7, and together these bars cover the interval from 4.5 to 7.5. If we took the area under the normal curve only from 5 to 7, we would miss two parts of the histogram: the left half of the bar at 5 and the right half of the bar at 7. So instead we say

$$P(5 \le X \le 7) \approx P(4.5 \le Y \le 7.5) = P\Big(\frac{4.5 - \mu}{\sigma} \le Z \le \frac{7.5 - \mu}{\sigma}\Big),$$

where the last step standardizes Y (Section 24) so that the table can be used. Moving each endpoint by half a unit in this way is called the *continuity correction*.



The event $5 \le X \le 7$ consists of the three coloured bars of the pmf of $X \sim \text{Bin}(20, 0.3)$, which reach from 4.5 to 7.5. The hatched half-bars are the two parts missed by the area under the curve of $Y \sim \text{N}(6,\ 2.05)$ from 5 to 7.

Formula (continuity correction)

Let $X \sim \text{Bin}(n, p)$ with $np > 5$ and $nq > 5$, and let $Y \sim \text{N}\big(\mu = np,\ \sigma = \sqrt{npq}\big)$. For whole numbers x and $a \le b$,

$$P(X \le x) \approx P(Y \le x + 0.5),$$
$$P(X \ge x) \approx P(Y \ge x - 0.5),$$
$$P(a \le X \le b) \approx P(a - 0.5 \le Y \le b + 0.5).$$

Each probability for Y is then evaluated by taking $Z = \dfrac{Y - \mu}{\sigma} \sim \text{N}(\mu = 0,\ \sigma = 1)$ and reading Table B.1.

One idea covers all three lines: every whole number that belongs to the event keeps its *entire* bar, so the interval is pushed outwards by 0.5 at each end.

Caution (strict inequalities)

For a continuous random variable $<$ and $\le$ are interchangeable, but for the discrete X they are not: "60 or more" includes the bar at 60 and "more than 60" does not. Rewrite a strict inequality with $\le$ or $\ge$ *first*, then correct:

$$P(X > 60) = P(X \ge 61) \approx P(Y \ge 60.5), \qquad P(X < 60) = P(X \le 59) \approx P(Y \le 59.5).$$

In words	Event for X	Event for Y
at most x	$X \le x$	$Y \le x + 0.5$
fewer than x	$X < x$, i.e. $X \le x - 1$	$Y \le x - 0.5$
at least x (x or more)	$X \ge x$	$Y \ge x - 0.5$
more than x	$X > x$, i.e. $X \ge x + 1$	$Y \ge x + 0.5$
exactly x	$X = x$	$x - 0.5 \le Y \le x + 0.5$
from a to b (inclusive)	$a \le X \le b$	$a - 0.5 \le Y \le b + 0.5$

How much does the correction matter?

Take $X \sim \text{Bin}(20, 0.3)$, the distribution in the figure above: $\mu = np = 6$, $\sigma = \sqrt{20(0.3)(0.7)} \approx 2.049$, and $np = 6$, $nq = 14$ are both greater than 5. Here are three ways of computing $P(5 \le X \le 7)$:

exact (pmf)	$p(5) + p(6) + p(7) = 0.1789 + 0.1916 + 0.1643$	$= 0.5348$
corrected	$P(4.5 \le Y \le 7.5) = P(-0.73 \le Z \le 0.73) = 0.7673 - 0.2327$	$= 0.5346$
uncorrected	$P(5 \le Y \le 7) = P(-0.49 \le Z \le 0.49) = 0.6879 - 0.3121$	$= 0.3758$

Forgetting the correction throws away the two half-bars, about 0.16 of probability here.

The approximation in four steps

1. Identify n and p, and check the criterion: $np > 5$ and $nq > 5$.
2. Compute $\mu = np$ and $\sigma = \sqrt{npq}$.
3. Write the event for X using $\le$ and $\ge$ only, then apply the continuity correction to get the event for Y.
4. Standardize the corrected endpoints, $z = \dfrac{y - \mu}{\sigma}$ rounded to two decimals, and use Table B.1.

25.3 Worked examples

The example contains the three kinds of event that have to be told apart: a range, "or more" and "more than".

Example 25.1 ▪ Guessing on a multiple-choice exam Worksheet 25, #3

A multiple choice exam has 200 questions, each with choices A, B, C, or D, and only one right answer. If an exam-taker guesses at the answer to all questions, estimate the probability that

a) 45 to 60 answers are right,

b) 60 answers or more are right,

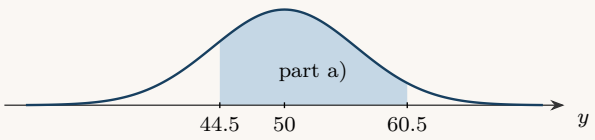
c) more than 60 answers are right.

Solution. *Identify the distribution.* Let X be the number of right answers. The 200 guesses are independent trials, each with probability $p = \frac{1}{4}$ of being right, so $X \sim \text{Bin}(200, 0.25)$ and $q = 0.75$.

Check the criterion. $np = 200(0.25) = 50 > 5$ and $nq = 200(0.75) = 150 > 5$, so the normal approximation is valid. Its parameters are

$$\mu = np = 50, \qquad \sigma = \sqrt{npq} = \sqrt{200(0.25)(0.75)} = \sqrt{37.5} \approx 6.124,$$

and we work with $Y \sim \text{N}(50,\ 6.124)$.



a) "45 to 60" includes both ends, so the bars at 45 and at 60 are both kept:

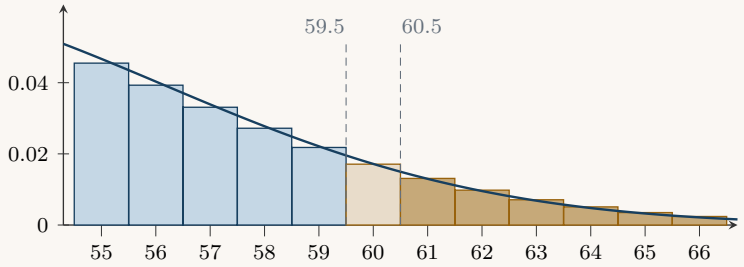
$$\begin{aligned} P(45 \le X \le 60) &\approx P(44.5 \le Y \le 60.5) = P\Big(\frac{44.5-50}{6.124} \le Z \le \frac{60.5-50}{6.124}\Big) \\ &= P(-0.90 \le Z \le 1.71) = \Phi(1.71) - \Phi(-0.90) \\ &= 0.9564 - 0.1841 = \boxed{0.7723} \approx 0.77. \end{aligned}$$

b) "60 or more" is $X \ge 60$; the bar at 60 starts at 59.5:

$$\begin{aligned} P(X \ge 60) &\approx P(Y \ge 59.5) = P\Big(Z \ge \frac{59.5-50}{6.124}\Big) = P(Z \ge 1.55) \\ &= 1 - \Phi(1.55) = 1 - 0.9394 = \boxed{0.0606} \approx 0.06. \end{aligned}$$

c) "More than 60" leaves out 60 itself, so rewrite it before correcting:

$$\begin{aligned} P(X > 60) = P(X \ge 61) &\approx P(Y \ge 60.5) = P\Big(Z \ge \frac{60.5-50}{6.124}\Big) = P(Z \ge 1.71) \\ &= 1 - \Phi(1.71) = 1 - 0.9564 = \boxed{0.0436} \approx 0.04. \end{aligned}$$



The right-hand tail of the histogram of X. Part b) uses every coloured bar and so starts at 59.5; part c) drops the pale bar at 60 and starts at 60.5.

The answers to b) and c) differ by $0.0606 - 0.0436 = 0.0170$, which is the normal estimate of the single probability $P(X = 60)$. For comparison, the exact binomial values (found with software) are 0.7694, 0.0625 and 0.0454: the approximation is reliable to about two decimal places, which is why its results are quoted as ≈ 0.77, ≈ 0.06 and ≈ 0.04.

Part VI

Sampling Distributions

A sample mean is itself a random variable. This part studies sums and averages of independent random variables and ends with the Central Limit Theorem, the bridge from probability to statistics.

26 Linear Combinations of Random Variables

Until now every problem has involved a single random variable. Many quantities of interest are built from several random variables at once: the total sales of three people, a day's revenue from three products, the combined lifetime of four light bulbs. This section shows how the mean and the variance of such a combination follow from the means and variances of its parts, and it identifies two cases, Poisson and normal, in which the whole distribution of the combination is known. These results are the foundation of the sampling distributions of Section 27.

26.1 The mean and variance of a linear combination

Definition (linear combination)

If $X_1, X_2, \dots, X_n$ are random variables and $a_1, a_2, \dots, a_n \in \mathbb{R}$ are constants, then

$$Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n = \sum_{i=1}^{n} a_iX_i$$

is a *linear combination* of $X_1, \dots, X_n$. A linear combination of random variables is again a random variable.

Sums, differences and averages are all linear combinations:

$$X_1 + X_2 \ \ (a_1 = a_2 = 1), \qquad X_1 - X_2 \ \ (a_1 = 1,\ a_2 = -1), \qquad \frac{X_1 + \cdots + X_n}{n} \ \ \left(\text{every } a_i = \tfrac{1}{n}\right).$$

Definition (independent random variables)

Random variables $X_1, \dots, X_n$ are *independent* if the value taken by any of them gives no information about the values taken by the others. Formally, events that concern different X_i are independent in the sense of Section 14:

$$P(X_1 \in A_1,\ \dots,\ X_n \in A_n) = P(X_1 \in A_1) \cdots P(X_n \in A_n)$$

for all sets of values $A_1, \dots, A_n$.

In exercises, independence is part of the given information: "their individual sales follow independent Poisson distributions", "the bulbs fail independently of each other".

Theorem (mean and variance of a linear combination)

Let $Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n$ be a linear combination of random variables.

1) For *any* random variables $X_1, \dots, X_n$, independent or not,

$$E(a_1X_1 + a_2X_2 + \cdots + a_nX_n) = a_1E(X_1) + a_2E(X_2) + \cdots + a_nE(X_n).$$

2) If $X_1, \dots, X_n$ are *independent*, then

$$V(a_1X_1 + a_2X_2 + \cdots + a_nX_n) = a_1^2\,V(X_1) + a_2^2\,V(X_2) + \cdots + a_n^2\,V(X_n).$$

With $\mu_i = E(X_i)$ and $\sigma_i^2 = V(X_i)$, this reads

$$\mu_Y = \sum_{i=1}^{n} a_i\mu_i \qquad \text{and, for independent } X_i, \qquad \sigma_Y^2 = \sum_{i=1}^{n} a_i^2\sigma_i^2, \quad \sigma_Y = \sqrt{a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2}\,.$$

In words: the mean of a linear combination is the same linear combination of the means. Variances also combine term by term, but only for independent random variables, and with the coefficients *squared*.

Proof for two discrete random variables

Let $p(x,y) = P(X_1 = x \text{ and } X_2 = y)$ be the joint pmf of X_1 and X_2, and let p_1 and p_2 be the pmfs of X_1 and X_2. Adding the joint pmf over all the values of one variable leaves the pmf of the other: $\sum_y p(x,y) = p_1(x)$ and $\sum_x p(x,y) = p_2(y)$. By LOTUS (Section 16), now applied to a function of both variables,

$$\begin{aligned} E(a_1X_1 + a_2X_2) &= \sum_x \sum_y (a_1x + a_2y)\,p(x,y) = a_1 \sum_x x \sum_y p(x,y) + a_2 \sum_y y \sum_x p(x,y) \\ &= a_1 \sum_x x\,p_1(x) + a_2 \sum_y y\,p_2(y) = a_1E(X_1) + a_2E(X_2). \end{aligned}$$

Independence was not used. For the variance, write $\mu_1 = E(X_1)$ and $\mu_2 = E(X_2)$. By the first part, $E(Y) = a_1\mu_1 + a_2\mu_2$, so $Y - E(Y) = a_1(X_1 - \mu_1) + a_2(X_2 - \mu_2)$. Squaring, and taking the expected value of each term,

$$\begin{aligned} V(Y) &= E\big[(Y - E(Y))^2\big] \\ &= a_1^2E\big[(X_1 - \mu_1)^2\big] + a_2^2E\big[(X_2 - \mu_2)^2\big] + 2a_1a_2\,E\big[(X_1 - \mu_1)(X_2 - \mu_2)\big] \\ &= a_1^2V(X_1) + a_2^2V(X_2) + 2a_1a_2\,E\big[(X_1 - \mu_1)(X_2 - \mu_2)\big]. \end{aligned}$$

If X_1 and X_2 are independent, then $p(x,y) = p_1(x)\,p_2(y)$ and the last expected value factors:

$$E\big[(X_1 - \mu_1)(X_2 - \mu_2)\big] = \Big[\sum_x (x - \mu_1)\,p_1(x)\Big]\Big[\sum_y (y - \mu_2)\,p_2(y)\Big] = (\mu_1 - \mu_1)(\mu_2 - \mu_2) = 0.$$

Hence $V(Y) = a_1^2V(X_1) + a_2^2V(X_2)$. For more than two random variables, apply the two-variable result repeatedly, for instance to $(a_1X_1 + a_2X_2) + a_3X_3$. For continuous random variables the sums become integrals. □

Remark (random variables that are not independent)

The quantity $\mathrm{Cov}(X_1, X_2) = E\big[(X_1 - \mu_1)(X_2 - \mu_2)\big]$ that appeared in the proof is called the *covariance* of X_1 and X_2; it is 0 when they are independent. Without independence the cross terms do not vanish, and the general formula is

$$V(a_1X_1 + \cdots + a_nX_n) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_j\,\mathrm{Cov}(X_i, X_j), \qquad \text{where } \mathrm{Cov}(X_i, X_i) = V(X_i).$$

All the problems of this section involve independent random variables, so the second formula of the theorem is the one to use.

Caution (variances add; standard deviations do not)

- Never add standard deviations. Add the *variances* (each multiplied by a_i^2) and take the square root at the end. For independent X_1 and X_2, $\sigma_{X_1+X_2} = \sqrt{\sigma_1^2 + \sigma_2^2}$, which is smaller than $\sigma_1 + \sigma_2$.
- Because the coefficients are squared, their signs disappear: $V(X_1 - X_2) = V(X_1) + (-1)^2 V(X_2) = V(X_1) + V(X_2)$. A difference of independent random variables is just as variable as their sum.
- A constant added to Y shifts the mean and leaves the variance alone: $E(Y + b) = E(Y) + b$ and $V(Y + b) = V(Y)$ (Section 16).

Remark (application: a short proof of $V(X) = npq$ for the binomial)

Let $X \sim \text{Bin}(n, p)$ be the number of successes in n independent trials (Section 17). For each trial $i = 1, \ldots, n$ let

$$I_i = \begin{cases} 1, & \text{if trial } i \text{ is a success,} \\ 0, & \text{if trial } i \text{ is a failure.} \end{cases}$$

Counting the successes is the same as adding up these zeros and ones, so $X = I_1 + I_2 + \cdots + I_n$. Each I_i takes the value 1 with probability p and 0 with probability $q = 1 - p$, so

$$E(I_i) = 1 \cdot p + 0 \cdot q = p, \qquad E(I_i^2) = 1^2 \cdot p + 0^2 \cdot q = p,$$
$$V(I_i) = E(I_i^2) - E(I_i)^2 = p - p^2 = p(1 - p) = pq.$$

The trials are independent, so the theorem gives

$$E(X) = E(I_1) + \cdots + E(I_n) = np, \qquad V(X) = V(I_1) + \cdots + V(I_n) = npq,$$

with no binomial coefficients to manipulate.

26.2 Two special cases in which the distribution is known

The theorem gives the mean and the variance of Y but not its distribution, and without the distribution we cannot compute a probability such as $P(Y > 2500)$. In two important cases the distribution of Y is known exactly.

Theorem (a sum of independent Poisson random variables)

If $X_1, \ldots, X_n$ are independent Poisson random variables with parameters $\lambda_1, \ldots, \lambda_n$ respectively, and if $Y = X_1 + X_2 + \cdots + X_n$, then

$$Y \sim \text{Pois}(\lambda) \qquad \text{with} \qquad \lambda = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$$

"A sum of independent Poisson random variables is also Poisson."

Proof for two random variables

Let $X_1 \sim \text{Pois}(\lambda_1)$ and $X_2 \sim \text{Pois}(\lambda_2)$ be independent, and let $k \geq 0$ be a whole number. The sum $Y = X_1 + X_2$ equals k exactly when $X_1 = j$ and $X_2 = k - j$ for one of $j = 0, 1, \ldots, k$. These cases are mutually exclusive, and X_1, X_2 are independent, so

$$P(Y = k) = \sum_{j=0}^{k} P(X_1 = j)\, P(X_2 = k - j) = \sum_{j=0}^{k} \frac{e^{-\lambda_1}\lambda_1^j}{j!} \cdot \frac{e^{-\lambda_2}\lambda_2^{k-j}}{(k-j)!}$$
$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{j=0}^{k} \binom{k}{j} \lambda_1^j \lambda_2^{k-j} = \frac{e^{-(\lambda_1+\lambda_2)}\,(\lambda_1 + \lambda_2)^k}{k!}$$

by the binomial theorem. This is the pmf of $\text{Pois}(\lambda_1 + \lambda_2)$. For n random variables, add them one at a time. □

Note (Poisson: only summing)

The Poisson result holds for plain *sums* only, where every coefficient is 1. Other linear combinations of Poisson random variables are not Poisson: $2X_1$ takes only even values, and $X_1 - X_2$ can be negative. Also, before adding the parameters, express every λ_i for the *same* interval of time or space (Example 26.1).

Theorem (a linear combination of independent normal random variables)

If $X_1, \dots, X_n$ are independent normal random variables with parameters $\mu_1, \dots, \mu_n$ and $\sigma_1, \dots, \sigma_n$ respectively, and if $Y = a_1X_1 + \cdots + a_nX_n$ with the a_i not all zero, then

$$Y \sim \mathrm{N}(\mu, \sigma) \qquad \text{where} \qquad \mu = a_1\mu_1 + \cdots + a_n\mu_n \quad \text{and} \quad \sigma = \sqrt{a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2}\,.$$

"A linear combination of independent normal random variables is also normal."

The values of μ and σ are nothing new: they are the mean and the standard deviation that the general theorem gives for any independent random variables. What this theorem adds is the *shape*: Y is again normal, so probabilities for Y are found by standardizing and reading Table B.1, exactly as in Section 24. Unlike the Poisson case, any coefficients a_i are allowed.

Caution (this does not work for other distributions)

The statement "a linear combination of independent *blah-blah* random variables is also *blah-blah*" is in general **false**. It is true for normal random variables (any linear combination) and for Poisson random variables (sums only). It fails, for example, for the uniform and the exponential distributions: the total of two rolls of a fair die is not uniform, since a total of 7 is six times as likely as a total of 2, and the sum of two independent exponential random variables is not exponential. For such a combination the theorem of the previous subsection still gives the mean and the variance, but the distribution has to be found some other way.

26.3 Independent copies: $X_1 + X_2$ is not $2X$

Definition (independent and identically distributed)

Random variables $X_1, \dots, X_n$ are *independent and identically distributed* (*i.i.d.*) if they are independent and all have the same distribution. When this common distribution is that of a random variable X, we write $X_i \sim X$ and call $X_1, \dots, X_n$ independent copies of X.

Having the same distribution does not mean taking the same value. The lifespans of four bulbs of the same model are four i.i.d. random variables: each follows the same distribution, but each bulb has its own outcome, and the four outcomes are usually different.

Caution (a sum of copies is not a multiple)

If X, X_1 and X_2 are random variables with the same distribution, then in general

$$X_1 + X_2 \neq 2X.$$

More generally, let $X_1, \dots, X_n$ be independent copies of a random variable X with mean μ and standard deviation σ. By the theorem on linear combinations, the sum of the n copies and n times a single copy have the same mean but different spreads:

	mean	variance	standard deviation
$X_1 + X_2 + \cdots + X_n$ (coefficients $1, 1, \dots, 1$)	$n\mu$	$n\sigma^2$	$\sqrt{n}\,\sigma$
nX (one variable, coefficient n)	$n\mu$	$n^2\sigma^2$	$n\,\sigma$

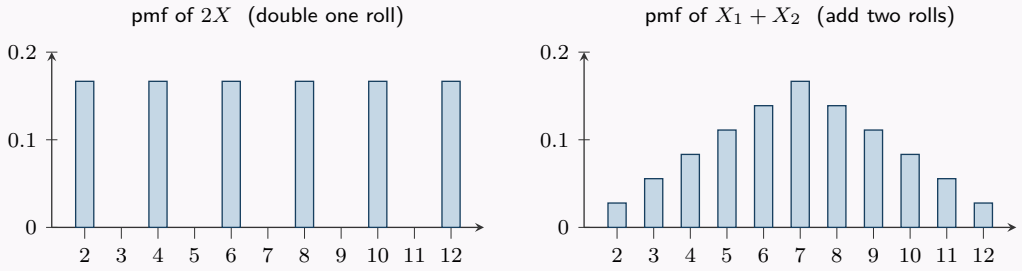
A sum of independent copies varies less than a multiple of one copy, because the high values of some copies are partly offset by the low values of others.

Why?

A random variable is a map $X\colon \Omega \to \mathbb{R}$: it attaches a number to every outcome of the experiment. To observe $2X$ we perform the experiment *once*, get an outcome x, and double its value: $2X(x)$. To observe $X_1 + X_2$ we perform it *twice*, get two outcomes x and y, and add their values: $X(x) + X(y)$. Unless the two outcomes happen to have the same value,

$$X(x) + X(y) \neq 2X(x) \qquad \text{and} \qquad X(x) + X(y) \neq 2X(y).$$

For instance, let X be the result of rolling a fair die. Then $2X$ can only be $2, 4, 6, 8, 10$ or 12, each with probability $\frac{1}{6}$, whereas the total $X_1 + X_2$ of two rolls can be any whole number from 2 to 12, with 7 the most likely.



Both have mean 7, but $V(2X) = 4 \cdot \frac{35}{12} \approx 11.67$ while $V(X_1 + X_2) = 2 \cdot \frac{35}{12} \approx 5.83$.

26.4 Worked examples

Example 26.1 ⋆ ▪ **A sum of Poisson random variables** Worksheet 26, #1

Three members of a high school band are selling chocolate bars at the mall to raise money for their trip. Their individual sales follow independent Poisson distributions. The flute player sells on average 2 chocolate bars every 15 minutes, the clarinet player 3 bars every 15 minutes, and the trumpet player 5 bars every half-hour. What is the probability that in any given hour they end up selling 28 chocolate bars in total?

Solution. *Put every rate on the same interval.* The question is about one hour, so we count sales per hour:

Player	Given rate	Rate per hour	Sales in one hour
flute	2 bars per 15 min	$\lambda_1 = 2 \times 4 = 8$	$X_1 \sim \text{Pois}(8)$
clarinet	3 bars per 15 min	$\lambda_2 = 3 \times 4 = 12$	$X_2 \sim \text{Pois}(12)$
trumpet	5 bars per 30 min	$\lambda_3 = 5 \times 2 = 10$	$X_3 \sim \text{Pois}(10)$

Identify the distribution of the total. The total number of bars sold in the hour is $Y = X_1 + X_2 + X_3$, a plain sum of independent Poisson random variables. Hence

$$Y \sim \text{Pois}(\lambda) \qquad \text{with} \qquad \lambda = 8 + 12 + 10 = 30.$$

Compute. With the Poisson pmf (Section 20),

$$P(Y = 28) = \frac{e^{-30}\, 30^{28}}{28!} \approx \boxed{0.0702}.$$

Example 26.2 ▪ Four bulbs, or four times one bulb? Worksheet 26, #3

The lifespan of a General Eclectic 13W LED bulb is a normally distributed random variable with mean 50 000 hours and standard deviation 2500 hours. Assume that the bulbs fail independently of each other.

a) What is the probability that a four-pack of these bulbs will last a total of less than 190 000 hours?

b) In Makeapointland, it costs \$4 per hour for the electricity required to run one of these bulbs. What is the probability that it costs less than \$190 000 in total to run one of these bulbs for its lifetime?

Solution. Let X be the lifespan, in hours, of one bulb: $X \sim \mathrm{N}(50\,000,\ 2500)$.

a) A four-pack contains four *different* bulbs. Their lifespans X_1, X_2, X_3, X_4 are four independent, identically distributed random variables, $X_i \sim X$: they follow the same distribution, but they are independent of each other and may have different outcomes. The total lifespan of the pack is

$$T = X_1 + X_2 + X_3 + X_4 \qquad (\text{not } 4X).$$

T is a linear combination of independent normal random variables with every $a_i = 1$, so T is normal with

$$\mu_T = 50\,000 + 50\,000 + 50\,000 + 50\,000 = 4(50\,000) = 200\,000,$$
$$\sigma_T = \sqrt{2500^2 + 2500^2 + 2500^2 + 2500^2} = \sqrt{4}\,(2500) = 5000.$$

Thus $T \sim \mathrm{N}(200\,000,\ 5000)$ and

$$P(T < 190\,000) = P\left(Z < \frac{190\,000 - 200\,000}{5000}\right) = P(Z < -2.00) = \boxed{0.0228}.$$

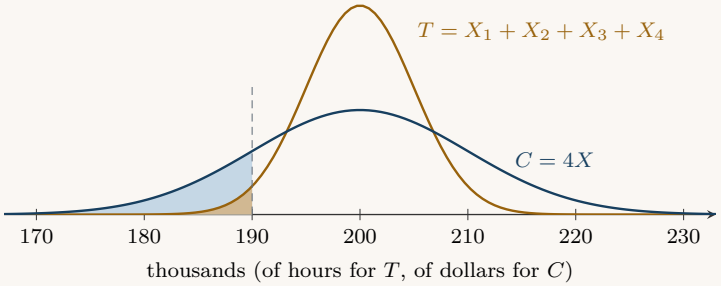
b) Now there is only *one* bulb, and its single lifespan X is multiplied by the price of an hour of electricity. The cost, in dollars, is $C = 4X$: a linear combination with one term and coefficient $a_1 = 4$. So C is normal with

$$\mu_C = 4(50\,000) = 200\,000, \qquad \sigma_C = \sqrt{4^2\,(2500)^2} = 4(2500) = 10\,000,$$

that is, $C \sim \mathrm{N}(200\,000,\ 10\,000)$, and

$$P(C < 190\,000) = P\left(Z < \frac{190\,000 - 200\,000}{10\,000}\right) = P(Z < -1.00) = \boxed{0.1587}.$$

The two parts have the same mean and the same cut-off, but different answers, because the standard deviations differ: $\sqrt{4}\,\sigma_X$ for the sum of four independent bulbs, $4\,\sigma_X$ for four times one bulb.



Left of 190 000: the area under the narrow curve (dark) is 0.0228; the area under the wide curve (dark and light together) is 0.1587.

27 Sampling Distributions

A sample mean $\bar{x}$ is computed from data, and a different sample would have given a different value. Before the sample is drawn, the sample mean is therefore a random variable with a distribution of its own. A *sampling distribution* describes how a sample statistic varies from one sample to the next, and so it examines the relationship

between a sample statistic and the population parameter it is meant to estimate. This section finds the mean and the variance of the sample mean and of the sample total; Section 28 deals with the shape of their distributions.

27.1 The sample mean and the sample total as random variables

Throughout, X is a random variable that describes one observation taken from a population, with mean μ_X and variance σ_X^2. A *random sample of size* n from X consists of n independent observations $X_1, X_2, \dots, X_n$, each with the same distribution as X. In the language of Section 26, they are independent and identically distributed copies of X: $X_i \sim X$.

Definition (sample mean, sample total, underlying distribution)

Let X be a random variable and let $X_1, \dots, X_n$ be a random sample of size n from X, so that $X_i \sim X$ for every i.

- The *sample mean*
$$\bar{X}_n = \frac{1}{n}\,(X_1 + X_2 + \cdots + X_n)$$
is also a random variable. It consists of all the possible sample means that samples of size n can produce.
- The *sample total*
$$T_n = X_1 + X_2 + \cdots + X_n$$
is a random variable consisting of all the possible sample totals from samples of size n.
- X is called the *underlying distribution* of $\bar{X}_n$ and T_n.

The distribution of $\bar{X}_n$ is called the *sampling distribution of the sample mean* for samples of size n.

Note (notation)

The lower-case $\bar{x}$ of Section 4 is a number: the mean of one particular sample that has already been observed. The upper-case $\bar{X}_n$ is the random variable of which $\bar{x}$ is one possible value. The two statistics carry the same information, since $T_n = n\,\bar{X}_n$; a question about an average can always be rephrased as a question about a total, and conversely.

The first example builds a sampling distribution by brute force, and its last lines point to a pattern.

Example 27.1 ▪ A sampling distribution built by hand Lecture notes

A population of white-tailed deer has tails that are either 4 cm or 6 cm long: 20% of the population has 4 cm tails and 80% has 6 cm tails. Construct the sampling distribution of $\bar{X}_3$, the sample mean for samples of size 3, and compare its mean and variance with those of the population.

Solution. Let X be the tail length of one deer chosen at random, so that $p(4) = 0.2$ and $p(6) = 0.8$. A sample of size 3 is a triple (X_1, X_2, X_3) of independent copies of X (the population is large, so one deer's tail tells us nothing about the next one's). There are $2^3 = 8$ possible samples. Because of independence, the probability of each sample is a product, for instance $P(X_1 = 4 \cap X_2 = 4 \cap X_3 = 6) = (0.2)(0.2)(0.8) = 0.032$.

X_1	X_2	X_3	$\bar{X}_3$	$P(X_1 \cap X_2 \cap X_3)$
4	4	4	4	$(0.2)^3 = 0.008$
4	4	6	14/3	$(0.2)^2(0.8) = 0.032$
4	6	4	14/3	0.032
4	6	6	16/3	$(0.2)(0.8)^2 = 0.128$
6	4	4	14/3	0.032
6	4	6	16/3	0.128
6	6	4	16/3	0.128
6	6	6	6	$(0.8)^3 = 0.512$

Adding the probabilities of the samples that give the same mean, for example $P\big(\bar{X}_3 = \frac{14}{3}\big) = 3(0.032) = 0.096$, gives the sampling distribution of $\bar{X}_3$. The population distribution is shown beside it.

Sampling distribution $\bar{X}_3$

x	4	$\frac{14}{3} = 4.\overline{6}$	$\frac{16}{3} = 5.\overline{3}$	6
$p(x)$	0.008	0.096	0.384	0.512

Population X

x	4	6
$p(x)$	0.2	0.8

Now compare the two distributions. For the population,

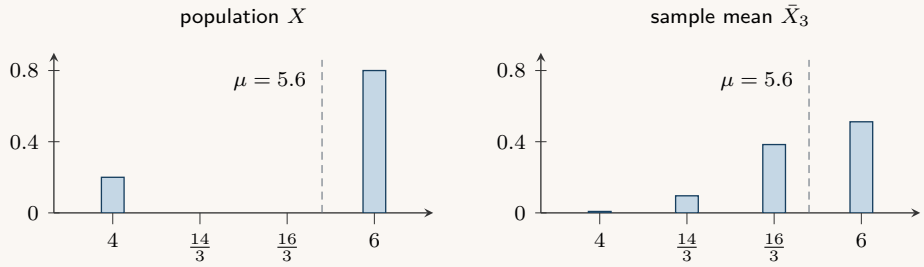
$$E(X) = 4(0.2) + 6(0.8) = 5.6,$$
$$V(X) = E(X^2) - E(X)^2 = \big[4^2(0.2) + 6^2(0.8)\big] - 5.6^2 = 32 - 31.36 = 0.64.$$

For the sample mean,

$$E(\bar{X}_3) = 4(0.008) + \tfrac{14}{3}(0.096) + \tfrac{16}{3}(0.384) + 6(0.512) = 5.6,$$
$$E(\bar{X}_3^2) = 4^2(0.008) + \big(\tfrac{14}{3}\big)^2(0.096) + \big(\tfrac{16}{3}\big)^2(0.384) + 6^2(0.512) = 31.57\overline{3},$$
$$V(\bar{X}_3) = 31.57\overline{3} - 5.6^2 = 0.21\overline{3}.$$

The two means are equal, and the variance of the sample mean is exactly one third of the population variance:

$$\boxed{E(\bar{X}_3) = E(X) = 5.6} \quad \text{and} \quad \boxed{3\,V(\bar{X}_3) = V(X)}, \quad \text{that is,} \quad V(\bar{X}_3) = \frac{V(X)}{3}.$$



Both distributions balance at 5.6, but the sample mean is less spread out around it.

This is not a coincidence: it is the case $n = 3$ of the proposition below.

27.2 The mean and variance of $\bar{X}_n$ and T_n

Proposition (mean and variance of the sample mean and the sample total)

Let X be an underlying distribution with mean μ_X and variance σ_X^2, and let $\bar{X}_n$ and T_n be the sample mean and the sample total of a random sample of size n from X. Then

$$\begin{cases} \mu_{\bar{X}_n} = \mu_X \\ \sigma^2_{\bar{X}_n} = \dfrac{\sigma_X^2}{n} \end{cases} \qquad \text{and} \qquad \begin{cases} \mu_{T_n} = n\,\mu_X \\ \sigma^2_{T_n} = n\,\sigma_X^2 \end{cases}$$

Taking square roots, the standard deviations are

$$\sigma_{\bar{X}_n} = \frac{\sigma_X}{\sqrt{n}} \qquad \text{and} \qquad \sigma_{T_n} = \sqrt{n}\,\sigma_X.$$

Proof

Both statistics are linear combinations of the independent random variables $X_1, \dots, X_n$, each of which has $E(X_i) = \mu_X$ and $V(X_i) = \sigma_X^2$. So the two formulas of Section 26 apply:

$$E(a_1X_1 + \cdots + a_nX_n) = a_1E(X_1) + \cdots + a_nE(X_n),$$
$$V(a_1X_1 + \cdots + a_nX_n) = a_1^2\,V(X_1) + \cdots + a_n^2\,V(X_n) \qquad \text{(because the } X_i \text{ are independent).}$$

Sample total. $T_n = X_1 + \cdots + X_n$ has every coefficient $a_i = 1$, so

$$\mu_{T_n} = E(X_1) + \cdots + E(X_n) = \underbrace{\mu_X + \cdots + \mu_X}_{n \text{ terms}} = n\,\mu_X,$$
$$\sigma^2_{T_n} = 1^2\,V(X_1) + \cdots + 1^2\,V(X_n) = \underbrace{\sigma_X^2 + \cdots + \sigma_X^2}_{n \text{ terms}} = n\,\sigma_X^2.$$

Sample mean. $\bar{X}_n = \frac{1}{n}X_1 + \cdots + \frac{1}{n}X_n$ has every coefficient $a_i = \frac{1}{n}$, so

$$\mu_{\bar{X}_n} = \frac{1}{n}E(X_1) + \cdots + \frac{1}{n}E(X_n) = n \cdot \frac{1}{n}\,\mu_X = \mu_X,$$
$$\sigma^2_{\bar{X}_n} = \left(\frac{1}{n}\right)^2 V(X_1) + \cdots + \left(\frac{1}{n}\right)^2 V(X_n) = n \cdot \frac{1}{n^2}\,\sigma_X^2 = \frac{\sigma_X^2}{n}.$$

The standard deviations are the square roots of the two variances. □

Note (how general this is)

The proof uses nothing about X except its mean and its variance. So the proposition is true for every underlying distribution, discrete or continuous, and it gives the mean and the standard deviation of $\bar{X}_n$ and T_n even when the distribution of $\bar{X}_n$ itself cannot be calculated directly.

Remark (what the formulas say)

- $\mu_{\bar{X}_n} = \mu_X$: sample means are centred on the population mean. An individual $\bar{x}$ will miss μ_X, but there is no systematic error in either direction.
- $\sigma_{\bar{X}_n} = \sigma_X/\sqrt{n}$: a sample mean varies less than a single observation, and the larger the sample, the more tightly the sample means cluster around μ_X. The gain is slow: to halve the standard deviation of the sample mean, the sample size must be multiplied by 4.
- $\sigma_{T_n} = \sqrt{n}\,\sigma_X$, not $n\,\sigma_X$: a total of n independent observations is not n times one observation (Section 26).

27.3 The shape of the sampling distribution

The proposition gives the centre and the spread of $\bar{X}_n$ and T_n, but not the shape of their distributions, and a probability such as $P(\bar{X}_n > 3.9)$ cannot be computed from a mean and a standard deviation alone. When the underlying distribution is normal, the shape is known.

Corollary (normal underlying distribution)

If $X \sim \mathrm{N}(\mu_X, \sigma_X)$, then for *every* sample size n

$$\bar{X}_n \sim \mathrm{N}\Big(\mu_X,\ \frac{\sigma_X}{\sqrt{n}}\Big) \qquad \text{and} \qquad T_n \sim \mathrm{N}\big(n\,\mu_X,\ \sqrt{n}\,\sigma_X\big).$$

This holds because $\bar{X}_n$ and T_n are linear combinations of independent normal random variables, which are again normal (Section 26); their parameters are those of the proposition.

For an underlying distribution that is not normal there is no such rule. In Example 27.1 the distribution of $\bar{X}_3$ could be found by listing all the samples, but this is rarely feasible, and Example 27.3 shows what can and cannot be said in that situation. Section 28 provides an approximate answer for large samples.

27.4 Worked examples

Example 27.2 ▪ The distribution of a total of two observations Worksheet 27, #1

There are two traffic lights on Pei's route to and from work. Let X be the number of traffic lights she gets stopped at in one direction. The pmf of X is given below.

x	0	1	2
$p(x)$	0.2	0.5	0.3

Let X_1 be the number of lights at which she gets stopped on the way to work and X_2 the number of lights at which she gets stopped on the way home.

a) Find the distribution of $T = X_1 + X_2$, the total number of lights she gets stopped at in her commute to work and home again.

b) Compute $E(T)$ and $V(T)$, and compare these values with $E(X)$ and $V(X)$ respectively. Do you notice something?

c) Draw the histograms for X and T. Compare their shapes.

Solution. The two trips are taken to be independent, each with the pmf of X. So X_1, X_2 is a random sample of size $n = 2$ from X, and $T = T_2$ is its sample total.

a) List the $3 \times 3 = 9$ possible pairs (x_1, x_2). By independence each pair has probability $p(x_1)\,p(x_2)$; the table shows the total $t = x_1 + x_2$ and this probability.

	$x_2 = 0$ (0.2)	$x_2 = 1$ (0.5)	$x_2 = 2$ (0.3)
$x_1 = 0$ (0.2)	$t = 0$: 0.04	$t = 1$: 0.10	$t = 2$: 0.06
$x_1 = 1$ (0.5)	$t = 1$: 0.10	$t = 2$: 0.25	$t = 3$: 0.15
$x_1 = 2$ (0.3)	$t = 2$: 0.06	$t = 3$: 0.15	$t = 4$: 0.09

Adding the probabilities of the cells with the same total, for instance $P(T = 2) = 0.06 + 0.25 + 0.06 = 0.37$, gives the distribution of T:

t	0	1	2	3	4
$p(t)$	0.04	0.20	0.37	0.30	0.09

b) For one trip,

$$E(X) = 0(0.2) + 1(0.5) + 2(0.3) = 1.1,$$
$$V(X) = E(X^2) - E(X)^2 = \left[0^2(0.2) + 1^2(0.5) + 2^2(0.3)\right] - 1.1^2 = 1.7 - 1.21 = 0.49.$$

For the total,

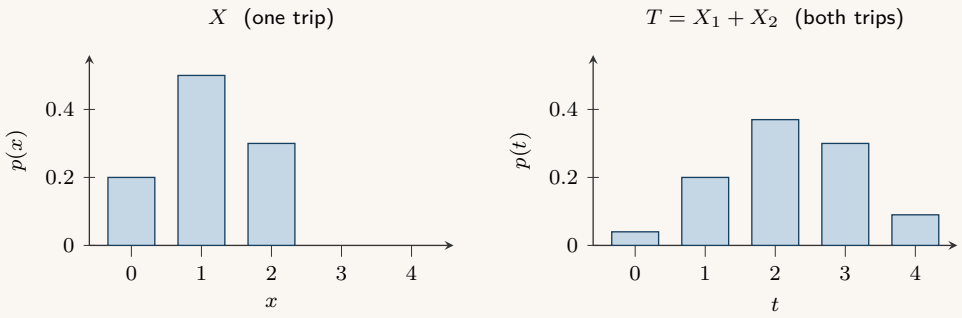
$$E(T) = 0(0.04) + 1(0.20) + 2(0.37) + 3(0.30) + 4(0.09) = \boxed{2.2},$$
$$V(T) = \left[1^2(0.20) + 2^2(0.37) + 3^2(0.30) + 4^2(0.09)\right] - 2.2^2 = 5.82 - 4.84 = \boxed{0.98}.$$

So

$$E(T) = 2\,E(X) \qquad \text{and} \qquad V(T) = 2\,V(X).$$

With $n = 2$ these are exactly $E(T) = n\,E(X)$ and $V(T) = n\,V(X)$, the formulas of the proposition. The proposition would have given 2.2 and 0.98 at once, without the distribution of T.

c) The two histograms are drawn below with the same scales.



The histogram of X has only three bars and is lopsided. The histogram of T is more symmetric and more spread out, with a single peak in the middle and small probabilities at both ends: adding just two observations already produces something closer to a bell shape.

Example 27.3 ▪ Mean and standard deviation known, shape unknown Worksheet 27, #2

Suppose that $X \sim \text{Exp}(\lambda = 4)$, and suppose that we are taking samples of size $n = 25$.

a) What is the expected mean of a sample?

b) Let $\bar{X}_{25}$ be the distribution of the sample means. What are $\mu_{\bar{X}_{25}}$ and $\sigma_{\bar{X}_{25}}$?

c) What distribution does $\bar{X}_{25}$ follow?

Solution. For an exponential random variable (Section 23), $\mu_X = \dfrac{1}{\lambda} = \dfrac{1}{4}$ and $\sigma_X = \dfrac{1}{\lambda} = \dfrac{1}{4}$.

a) The "expected mean of a sample" is the expected value of the sample mean, $E(\bar{X}_{25})$. By the proposition it equals the mean of the underlying distribution:

$$E(\bar{X}_{25}) = \mu_X = \boxed{\frac{1}{4}}.$$

b) By the proposition,

$$\mu_{\bar{X}_{25}} = \mu_X = \boxed{\frac{1}{4}} \qquad \text{and} \qquad \sigma_{\bar{X}_{25}} = \frac{\sigma_X}{\sqrt{n}} = \frac{1/4}{\sqrt{25}} = \boxed{\frac{1}{20}}.$$

c) We are not able to know this. X is exponential, not normal, so the corollary for a normal underlying distribution does not apply, and the proposition gives only the mean and the standard deviation of $\bar{X}_{25}$, not its shape. This is precisely the situation in which the proposition is still usable although the distribution of the sample mean is out of reach. (The sample size, 25, is also below the $n \geq 30$ required for the normal approximation of Section 28.)

28 The Central Limit Theorem

Section 27 gave the mean and the standard deviation of the sample mean $\bar{X}_n$ and of the sample total T_n for any underlying distribution, but their shape only when the underlying distribution is normal. The Central Limit Theorem fills this gap: when the sample is large, $\bar{X}_n$ and T_n are approximately normal, whatever the underlying distribution is. This is why the normal distribution is so central in statistics, and it is the basis of the inference methods that begin in Section 29.

28.1 The theorem

Theorem (Central Limit Theorem)

Let X be an underlying distribution with mean $\mu_X = E(X)$ and finite variance $\sigma_X^2 > 0$, and let $\bar{X}_n$ and T_n be the sample mean and the sample total of a random sample of size n from X. Then, as $n \to \infty$, the distributions of $\bar{X}_n$ and of T_n approach a normal distribution, whatever the distribution of X is.

Precisely, the standardized variable

$$Z_n = \frac{\bar{X}_n - \mu_X}{\sigma_X/\sqrt{n}} = \frac{T_n - n\,\mu_X}{\sqrt{n}\,\sigma_X}$$

satisfies

$$\lim_{n\to\infty} P(Z_n \le z) = \Phi(z) \qquad \text{for every real number } z,$$

where Φ is the cdf of the standard normal distribution. Consequently, when n is large,

$$\bar{X}_n \overset{\text{approx.}}{\sim} \mathrm{N}\Big(\mu_{\bar{X}_n} = \mu_X,\ \ \sigma_{\bar{X}_n} = \frac{\sigma_X}{\sqrt{n}}\Big) \qquad \text{and} \qquad T_n \overset{\text{approx.}}{\sim} \mathrm{N}\Big(\mu_{T_n} = n\,\mu_X,\ \ \sigma_{T_n} = \sqrt{n}\,\sigma_X\Big).$$

Remark (rule of thumb: $n \ge 30$)

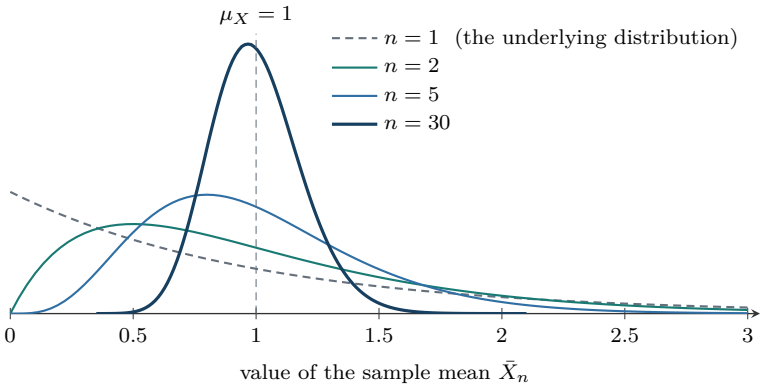
How large must n be? In practice, the normal approximation is considered good enough when

$$n \ge 30.$$

This is a rule of thumb, not part of the theorem. The further the underlying distribution is from a bell shape (strongly skewed, for instance), the larger n must be for the same accuracy. If the underlying distribution is itself normal, no condition on n is needed at all, because $\bar{X}_n$ and T_n are then exactly normal (Section 27).

Note (reading the theorem)

- *What is new is the shape.* The parameters $\mu_{\bar{X}_n}$, $\sigma_{\bar{X}_n}$, μ_{T_n} and σ_{T_n} are predetermined: the proposition of Section 27 gives them for every n and every underlying distribution. The theorem adds that the distribution with these parameters is approximately *normal*, so probabilities can be found with Table B.1.
- *Nothing is assumed about the shape of X.* It may be discrete or continuous, symmetric or skewed, and it may even be unknown. Only μ_X and σ_X are needed.
- *Why the precise statement is about Z_n.* As n grows, $\bar{X}_n$ piles up ever closer to μ_X and T_n spreads out ever more, so neither settles down to one fixed curve. The standardized Z_n has mean 0 and standard deviation 1 for every n, and it is the distribution of Z_n that tends to the standard normal. This is called convergence in distribution, written $Z_n \xrightarrow{d} \mathrm{N}(0,1)$.
- *The result is an approximation.* A probability obtained from the theorem is approximate, and should be reported as such.



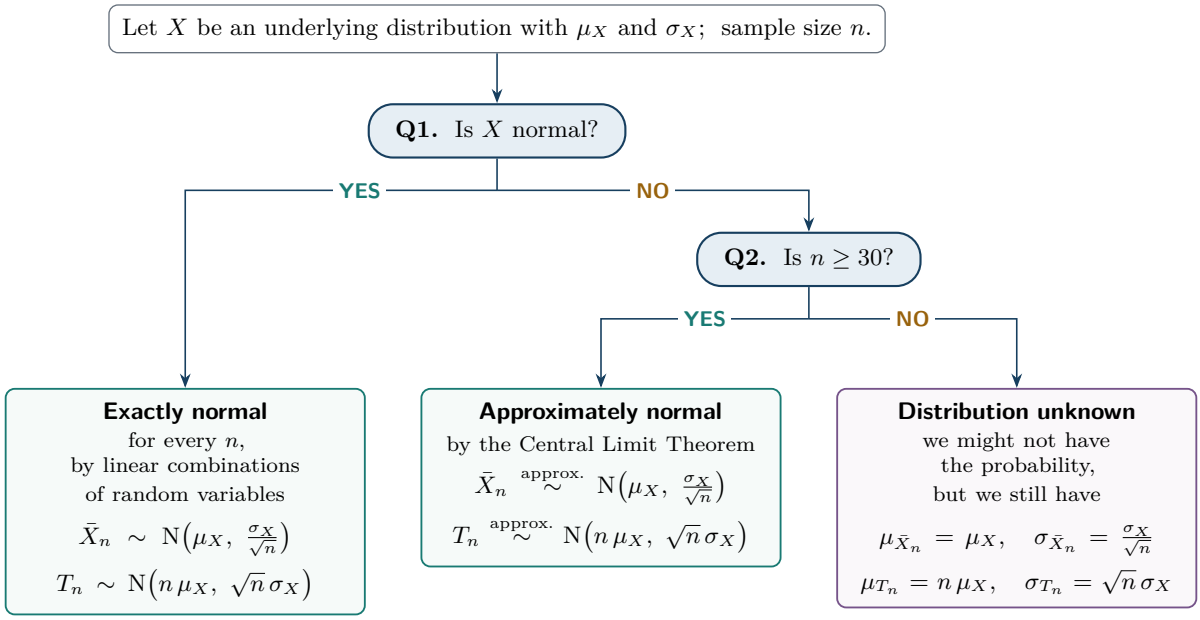
The distribution of the sample mean $\bar{X}_n$ when the underlying distribution is Exp(1), which is strongly skewed. As n grows the curve becomes bell-shaped and concentrates around $\mu_X = 1$, with standard deviation $\sigma_X/\sqrt{n} = 1/\sqrt{n}$.

Remark (where the theorem is used)

- *Totals and averages of many independent quantities:* the total sales of forty restaurants, the total lifetime of a stock of light bulbs, the average balance of a hundred bank accounts (see the worked examples).
- *The normal approximation to the binomial* (Section 25). A binomial random variable is the total of n independent trials, each worth 1 for a success and 0 for a failure (Section 26). It is therefore a sample total T_n with $\mu_X = p$ and $\sigma_X = \sqrt{pq}$, and the theorem says that for large n it is approximately $\mathrm{N}\big(np, \sqrt{npq}\big)$.
- *Statistical inference* (Sections 29 and 30). Confidence intervals and tests for a mean rest on the fact that, for a large sample, $\bar{X}_n$ is approximately normal even when the population is not.

28.2 Overview: which result applies?

Every problem on sampling distributions starts from an underlying distribution X with mean μ_X and standard deviation σ_X, and asks about $\bar{X}_n$ or T_n. Two questions decide which result may be used.



Note (using the overview)

- A question about *one* observation is a question about X itself, the case $n = 1$. It can be answered only if the distribution of X is known; the Central Limit Theorem says nothing about a single observation.
- In the left-hand box the answer is exact, in the middle box it is approximate.
- In the right-hand box a probability is out of reach in general, but the mean and the standard deviation of $\bar{X}_n$ and T_n are still available. For some particular underlying distributions the exact distribution of T_n is known whatever n is: a total of independent Poisson random variables is again Poisson (Section 26).

28.3 Worked examples

Example 28.1 ▪ An exact answer and its normal estimate Worksheet 28, #1

In Manhattan, one large downtown area has forty McBurgers restaurants, all operating independently. The sale of Big Mic burgers for each of them follows a Poisson distribution with average 1 burger sold every 5 minutes from 10 am to 10 pm.

a) The true probability that the total number of burgers sold in this area in a day from 10 am to 10 pm is at least 5800 is 0.3008. Give the expression in sigma notation that corresponds to this probability.

b) Check the conditions for the Central Limit Theorem, and use it to estimate this same probability.

Solution. Let X be the number of Big Mic burgers that one restaurant sells between 10 am and 10 pm. That period lasts 12 hours, or 720 minutes, or 144 intervals of 5 minutes with an average of one burger each, so

$$X \sim \text{Pois}(144), \qquad \mu_X = 144, \qquad \sigma_X = \sqrt{144} = 12$$

(for a Poisson random variable the variance equals the mean, Section 20). The $n = 40$ restaurants operate independently, so their sales $X_1, \ldots, X_{40}$ form a random sample from X, and the total number of burgers sold in the area is the sample total $T_{40} = X_1 + \cdots + X_{40}$.

a) A sum of independent Poisson random variables is again Poisson, and the parameters add (Section 26): $T_{40} \sim \text{Pois}(40 \times 144) = \text{Pois}(5760)$, exactly. Therefore

$$P(T_{40} \geq 5800) = \boxed{\sum_{x=5800}^{\infty} \frac{e^{-5760}\, 5760^x}{x!}} \approx 0.3008.$$

The sum has infinitely many terms and can only be evaluated with software.

b) *Conditions.* The forty sales figures are independent and have the same distribution, with a finite mean and variance. That distribution is Poisson, not normal, but the sample size is $n = 40 \geq 30$. So the Central Limit Theorem applies to the total T_{40}, whose mean and standard deviation are

$$\mu_{T_{40}} = n\,\mu_X = 40(144) = 5760, \qquad \sigma_{T_{40}} = \sqrt{n}\,\sigma_X = \sqrt{40}\,(12) = \sqrt{5760} \approx 75.89.$$

That is, T_{40} is approximately N(5760, 75.89).
Estimate. Standardizing,

$$P(T_{40} \geq 5800) \approx P\Big(Z \geq \frac{5800 - 5760}{75.89}\Big) = P(Z \geq 0.527).$$

The value $z = 0.527$ lies between the entries 0.52 and 0.53 of Table B.1. Taking the number halfway between $\Phi(0.52) = 0.6985$ and $\Phi(0.53) = 0.7019$, which is 0.7002, gives

$$P(T_{40} \geq 5800) \approx 1 - 0.7002 = \boxed{0.2998}.$$

Rounding z to 0.53 instead gives 0.2981, and adding a continuity correction because T_{40} is a count (Section 25) gives $P(Z \geq 0.52) = 0.3015$. All of these estimates are within 0.003 of the true value 0.3008: with a sample of 40 the normal approximation is already very good.

Example 28.2 ⋆ ▪ **One observation, or the mean of a sample?** Worksheet 28, #2

It is known that the weights of slices of roast beef served in the cafeteria are normally distributed with an average weight of 4 oz and a standard deviation of $\frac{1}{2}$ oz. Find, if possible, the probability that:

a) A slice selected at random will weigh more than 3.9 oz.

b) The average weight of a random sample of 36 slices will be more than 3.9 oz.

c) I took a sample of 36 slices of roast beef and calculated the average weight of the slices. I now tell you that the average weight is in the upper 20% of all expected average weights. At least how large is the sample mean?

d) Did you need to apply the Central Limit Theorem to answer any of the above questions? Explain your answer.

Solution. Let X be the weight, in ounces, of one slice: $X \sim \mathrm{N}(4,\, 0.5)$.

a) One slice is one observation, so we use the distribution of X itself:

$$P(X > 3.9) = P\Big(Z > \frac{3.9-4}{0.5}\Big) = P(Z > -0.20)$$
$$= 1 - \Phi(-0.20) = 1 - 0.4207 = \boxed{0.5793}.$$

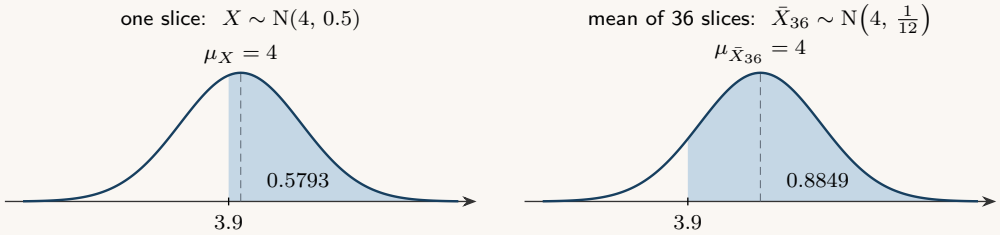
b) Now the question is about the sample mean $\bar{X}_{36}$. The underlying distribution is normal, so $\bar{X}_{36}$ is normal with

$$\mu_{\bar{X}_{36}} = \mu_X = 4, \qquad \sigma_{\bar{X}_{36}} = \frac{\sigma_X}{\sqrt{n}} = \frac{0.5}{\sqrt{36}} = \frac{1}{12} \approx 0.0833,$$

and

$$P(\bar{X}_{36} > 3.9) = P\Big(Z > \frac{3.9-4}{1/12}\Big) = P(Z > -1.20)$$
$$= 1 - \Phi(-1.20) = 1 - 0.1151 = \boxed{0.8849}.$$

The value 3.9 is only 0.2 standard deviations below the mean for a single slice, but 1.2 standard deviations below the mean for the average of 36 slices, because averages vary much less than individual slices do.



c) "In the upper 20% of all expected average weights" means that the sample mean is at least the value c that cuts off the top 20% of the distribution of $\bar{X}_{36}$: $P(\bar{X}_{36} \geq c) = 0.20$, so the area to the left of c is 0.80. In Table B.1 the entry closest to 0.8000 is 0.7995, at $z = 0.84$. Converting this z-value back to ounces with the standard deviation of the *sample mean*,

$$c = \mu_{\bar{X}_{36}} + z\,\sigma_{\bar{X}_{36}} = 4 + 0.84\Big(\frac{1}{12}\Big) = 4.07.$$

The sample mean is $\boxed{\text{at least 4.07 oz}}$.

d) No. The underlying distribution is normal, so the distribution of the sample means is also normal, exactly and for any sample size, because a linear combination of independent normal random variables is normal. The Central Limit Theorem is needed only when the underlying distribution is not normal. (Part (a) does not involve a sample mean at all.)

Example 28.3 ▪ A sample total, and an unknown sample size Worksheet 28, #4

A community centre wishes to leave on a single light continuously for security reasons. Light bulbs for this light last on average 1000 hours with a standard deviation of 250 hours. As soon as a light bulb burns out, it is replaced.

a) Estimate the probability that 45 lightbulbs are enough to last the community centre for 5 years (about 43 800 hours).

b) How many lightbulbs should they buy if they wished their chances of having enough bulbs to last 5 years to be 95%?

Solution. Let X be the lifetime, in hours, of one bulb: $\mu_X = 1000$ and $\sigma_X = 250$; its distribution is not given. The bulbs are used one after the other, so n bulbs last for a total of $T_n = X_1 + \cdots + X_n$ hours, and they are enough for 5 years when $T_n \geq 43\,800$.

a) Here $n = 45 \geq 30$, so the Central Limit Theorem gives an approximation:

$$T_{45} \overset{\text{approx.}}{\sim} \mathrm{N}\Big(\mu_{T_{45}} = n\,\mu_X = 45\,000, \quad \sigma_{T_{45}} = \sqrt{n}\,\sigma_X = \sqrt{45}\,(250) \approx 1677.05\Big).$$

Then

$$P(T_{45} \geq 43\,800) \approx P\Big(Z \geq \frac{43\,800 - 45\,000}{1677.05}\Big) = P(Z \geq -0.7155) = \Phi(0.7155) \approx \boxed{0.7627},$$

taking, as in Example 28.1, the value halfway between $\Phi(0.71) = 0.7611$ and $\Phi(0.72) = 0.7642$.

b) Now n is unknown. Let us assume that $n \geq 30$, so that the Central Limit Theorem can be used; the assumption will be checked at the end. Then

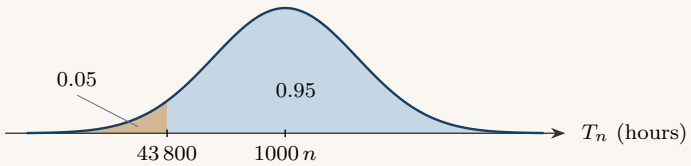
$$T_n \overset{\text{approx.}}{\sim} \mathrm{N}\big(\mu_{T_n} = n\,\mu_X = 1000\,n, \quad \sigma_{T_n} = \sqrt{n}\,\sigma_X = 250\sqrt{n}\,\big).$$

What we want is $P(T_n \geq 43\,800) = 0.95$, that is,

$$1 - 0.95 = 0.05 = P(T_n < 43\,800) = P\Big(Z < \frac{43\,800 - \mu_{T_n}}{\sigma_{T_n}}\Big).$$

The standard normal table turns this into an equation: the z-value with area 0.05 to its left is -1.645 (halfway between -1.64 and -1.65, whose areas are 0.0505 and 0.0495). So the standardized value must equal this fixed number:

$$\frac{43\,800 - \mu_{T_n}}{\sigma_{T_n}} = -1.645.$$



Now plug in $\mu_{T_n} = 1000\,n$ and $\sigma_{T_n} = 250\sqrt{n}$ and solve for n:

$$\frac{43\,800 - 1000\,n}{250\sqrt{n}} = -1.645 \quad \Longrightarrow \quad 1000\,n - 411.25\sqrt{n} - 43\,800 = 0.$$

This is a quadratic equation in $u = \sqrt{n}$, namely $1000u^2 - 411.25u - 43\,800 = 0$. By the quadratic formula,

$$u = \frac{411.25 \pm \sqrt{411.25^2 + 4(1000)(43\,800)}}{2(1000)} = \frac{411.25 \pm 13\,242.70}{2000},$$

and since $u = \sqrt{n}$ cannot be negative, $u \approx 6.827$ and $n = u^2 \approx 46.6$. The number of bulbs must be a whole number, and it must be rounded *up*: with fewer bulbs the probability would fall below 95%. They should buy $\boxed{47 \text{ lightbulbs}}$. Since $47 \geq 30$, the use of the Central Limit Theorem was justified. As a check, $n = 47$ gives $P(T_{47} \geq 43\,800) \approx 0.97$, whereas $n = 46$ gives only about 0.90.

Part VII

Statistical Inference

Inference uses a sample to say something about a whole population, with a stated level of confidence: confidence intervals for means and proportions, hypothesis tests, comparisons of two means, and chi-squared tests for goodness of fit and independence.

29 Confidence Intervals

Sections 27 and 28 started from a population whose mean was known and described how the mean of a sample behaves. Statistical inference runs the other way: the mean μ of a population is usually very difficult to obtain, whereas the mean $\bar{x}$ of a single sample is fairly simple to obtain. This section shows how to use $\bar{x}$ to predict μ with a given amount of confidence, and then how the same process uses the proportion $\hat{p}$ of successes in a sample to estimate the proportion p of successes in the population.

29.1 The idea of a confidence interval

Recall (the sampling distribution of the sample mean)

Let the underlying distribution (the population) X have mean μ and standard deviation σ, and let $\bar{X}_n$ be the mean of a random sample of size n. Then $\bar{X}_n$ has the same mean as X and a smaller standard deviation:

$$\mu_{\bar{X}_n} = \mu, \qquad \sigma_{\bar{X}_n} = \frac{\sigma}{\sqrt{n}}.$$

- If X is normally distributed, then $\bar{X}_n$ is also normally distributed, whatever the value of n (Section 27).
- If n is large ($n \geq 30$ as a rule of thumb), then $\bar{X}_n$ is approximately normally distributed, regardless of the distribution of the population X. This is the Central Limit Theorem (Section 28).

One sample gives one number $\bar{x}$, a single observed value of the random variable $\bar{X}_n$. Used alone as a guess for μ it is almost certainly a little off, and it does not say by how much. A confidence interval reports an interval around $\bar{x}$ instead, together with a measure of how far the method that produced it can be trusted.

Definition (confidence interval, confidence level)

A $100(1-\alpha)\%$ *confidence interval* for an unknown population parameter (in this section a mean μ or a proportion p) is an interval, computed from a sample, obtained by a method that captures the parameter for $100(1-\alpha)\%$ of all possible samples. The number $1-\alpha$ is the *confidence level*: it says how good the estimation is. For example, a 95% confidence interval has $1-\alpha = 0.95$ and $\alpha = 0.05$.

To see what such an interval must look like, take the 95% level and suppose that $\bar{X}_n$ is normally distributed. Under the standard normal curve the central 95% of the area lies between -1.96 and 1.96. The standard normal variable $Z = (\bar{X}_n - \mu)/\sigma_{\bar{X}_n}$ is obtained from $\bar{X}_n$ by shifting left by μ and compressing by $\sigma_{\bar{X}_n}$. Undoing these two steps, 95% of all possible sample means lie within the distance

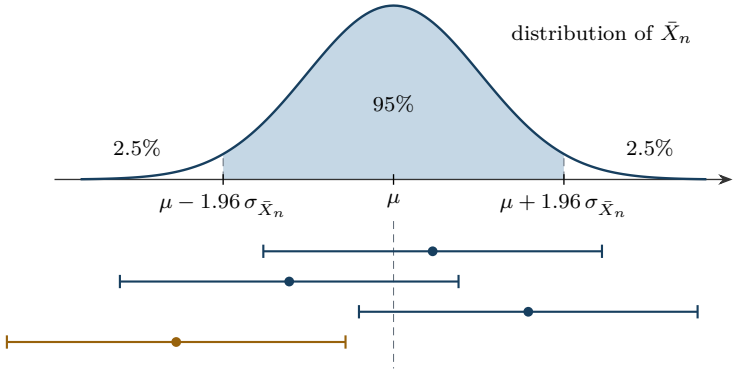
$$r = 1.96\,\sigma_{\bar{X}_n} = 1.96\,\frac{\sigma}{\sqrt{n}}$$

of the centre μ.

Now turn the statement around. "$\bar{x}$ is within r of μ" and "μ is within r of $\bar{x}$" say the same thing:

$$\mu - r \leq \bar{x} \leq \mu + r \qquad \Longleftrightarrow \qquad \bar{x} - r \leq \mu \leq \bar{x} + r.$$

So if we draw the interval $\bar{x} \pm r$ around whatever sample mean we happen to get, the interval contains μ exactly when our sample mean is one of the central 95%, and that is the case for 95% of all samples. The interval $\bar{x} \pm 1.96\,\sigma/\sqrt{n}$ is therefore a 95% confidence interval for μ.



The sampling distribution of $\bar{X}_n$, and the interval $\bar{x} \pm 1.96\,\sigma_{\bar{X}_n}$ drawn around four possible sample means (the dots). An interval reaches μ exactly when its $\bar{x}$ falls in the shaded central 95%; the last one, whose $\bar{x}$ lies in a tail, misses μ.

The number 1.96 belongs to the 95% level; Section 29.2 introduces the notation for the value that replaces it at any other level. The recipe also has to be adjusted when σ is unknown or the sample is small. The situations treated in this section are the following, and Section 29.6 collects them in a decision tree.

Estimating	Situation	The interval uses	Section
μ, case A	n large ($n \geq 30$), σ known (somewhat artificial)	$z_{\alpha/2}$ and σ	29.3
μ, case B	n large ($n \geq 30$), σ unknown (more realistic)	$z_{\alpha/2}$ and s	29.4
μ, case C	n small, σ unknown, X known to be normal	Student's t and s	29.4
p	n large	$z_{\alpha/2}$ and $\hat{p}$	29.5

29.2 The z_α notation

Notation (critical value z_α)

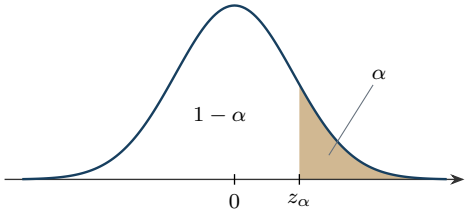
For $0 < \alpha < 1$, the symbol z_α denotes the z-value that has area α to its *right* under the standard normal curve:

$$P(Z > z_\alpha) = \alpha \quad \Longleftrightarrow \quad \Phi(z_\alpha) = 1 - \alpha \quad \Longleftrightarrow \quad z_\alpha = \Phi^{-1}(1-\alpha).$$

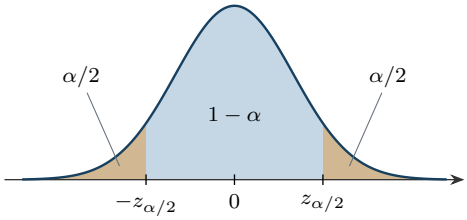
Here Φ is the standard normal cdf (Section 24), and Φ^{-1} means reading Table B.1 backwards, from an area to a z-value. By symmetry, $-z_\alpha$ has area α to its left: $\Phi(-z_\alpha) = \alpha$.

A confidence interval leaves out the same area in each tail. If the confidence level is $1-\alpha$, the area left over is α, so each tail receives $\alpha/2$ and the two cut-off values are $\pm z_{\alpha/2}$:

$$P\big(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\big) = 1 - \alpha.$$



$z_\alpha = \Phi^{-1}(1-\alpha)$



$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1-\alpha$

For the 95% level, $1-\alpha=0.95$, so $\alpha=0.05$ and $\alpha/2=0.025$. The cut-off values are $z_{0.025}=\Phi^{-1}(0.975)=1.96$ and $-z_{0.025}=\Phi^{-1}(0.025)=-1.96$, the numbers used in Section 29.1.

Most common confidence levels

Confidence level	$1-\alpha$	α	$\alpha/2$	$z_{\alpha/2}$
90%	0.90	0.10	0.05	$z_{0.05}=1.645$
95%	0.95	0.05	0.025	$z_{0.025}=1.96$
99%	0.99	0.01	0.005	$z_{0.005}=2.575$

Note (finding $z_{\alpha/2}$ for any level)

Work from the confidence level down to the tail area, for instance

$$1-\alpha=90\%=0.9 \quad\Longrightarrow\quad \alpha=0.1 \quad\Longrightarrow\quad \frac{\alpha}{2}=0.05,$$

then look for the area $1-\alpha/2$ in the body of Table B.1 and read off z. For 90% the area 0.9500 falls exactly halfway between $\Phi(1.64)=0.9495$ and $\Phi(1.65)=0.9505$, which is why $z_{0.05}=1.645$. In the same way 0.9950 is halfway between $\Phi(2.57)=0.9949$ and $\Phi(2.58)=0.9951$, giving $z_{0.005}=2.575$, while $\Phi(1.96)=0.9750$ exactly.

29.3 Estimating μ when σ is known

To begin with, make the somewhat artificial assumption that the population X has an unknown mean μ but a *known* standard deviation σ. The goal is to predict μ using $\bar{x}$.

Theorem (confidence interval for μ when σ is known)

Let $\bar{x}$ be the mean of a random sample of size n from a normally distributed population with unknown mean μ and known standard deviation σ. Then a $100(1-\alpha)\%$ confidence interval for μ is

$$\left(\bar{x}-z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \ \bar{x}+z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right), \qquad \text{for short} \qquad \bar{x}\pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Derivation

Step 1: the distribution of the sample mean. Because the population is normal, the sample mean is normal too, with mean μ and standard deviation $\sigma/\sqrt{n}$. Standardizing it gives a standard normal variable:

$$\bar{X}_n\sim \mathrm{N}\left(\mu,\ \frac{\sigma}{\sqrt{n}}\right) \qquad\Longrightarrow\qquad Z=\frac{\bar{X}_n-\mu}{\sigma/\sqrt{n}}\sim \mathrm{N}(0,1).$$

Step 2: the definition of $z_{\alpha/2}$. The area to the right of $z_{\alpha/2}$ is $\alpha/2$ and, by symmetry, so is the area to the left of $-z_{\alpha/2}$. The area between them is what we want:

$$P\left(-z_{\alpha/2}\le Z\le z_{\alpha/2}\right)=1-\alpha, \qquad \text{that is,} \qquad P\left(-z_{\alpha/2}\le \frac{\bar{X}_n-\mu}{\sigma/\sqrt{n}}\le z_{\alpha/2}\right)=1-\alpha.$$

Step 3: isolate μ. Solve the double inequality inside the probability for μ:

$$\begin{aligned} & -z_{\alpha/2} \le \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2} \\ \iff \quad & -z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \bar{X}_n - \mu \le z_{\alpha/2}\frac{\sigma}{\sqrt{n}} && \text{(multiply by } \sigma/\sqrt{n} > 0) \\ \iff \quad & -\bar{X}_n - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le -\mu \le -\bar{X}_n + z_{\alpha/2}\frac{\sigma}{\sqrt{n}} && \text{(subtract } \bar{X}_n) \\ \iff \quad & \bar{X}_n - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X}_n + z_{\alpha/2}\frac{\sigma}{\sqrt{n}} && \text{(multiply by } -1). \end{aligned}$$

Multiplying by -1 reverses both inequality signs, which is why the two ends change places. Hence

$$P\Big(\bar{X}_n - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X}_n + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\Big) = 1 - \alpha.$$

Step 4: conclusion. The random interval $\bar{X}_n \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ contains μ with probability $1-\alpha$. Replacing $\bar{X}_n$ by the mean $\bar{x}$ of the sample actually observed gives $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$, a $100(1-\alpha)\%$ confidence interval for μ. □

Note (large samples from any population)

Normality of the population was used only in Step 1. If the population is not normal, or its shape is unknown, but n is large ($n \ge 30$), the Central Limit Theorem says that $\bar{X}_n$ is *approximately* $\mathrm{N}(\mu, \sigma/\sqrt{n})$, and Steps 2 to 4 go through with "$= 1-\alpha$" replaced by "$\approx 1-\alpha$". In short, $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ forms a $100(1-\alpha)\%$ confidence interval for μ when X is normally distributed *or* n is large, and σ is known.

Definition (margin of error)

The quantity

$$E = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

is called the *margin of error*. It is the radius of the confidence interval: the interval is $\bar{x} \pm E$, its centre is the sample mean and its length is $L = 2E$. At the confidence level $100(1-\alpha)\%$, the estimate $\bar{x}$ is off from μ by at most E.

Remark (what "95% confident" means)

95% of the time when we build an interval using a sample mean in this way, the interval will contain the population mean.

The 95% belongs to the *method*, not to one particular interval. The population mean μ is a fixed number; what changes from sample to sample is $\bar{x}$, and the interval moves with it. Of all the intervals that could be built from samples of size n, 95% capture μ (19 samples out of 20), namely those whose $\bar{x}$ lies within E of μ.

Caution

Once the numbers are in, an interval such as $(79.35,\ 80.65)$ either contains μ or it does not, and we cannot tell which. "We are 95% confident that μ lies between 79.35 and 80.65" is shorthand for "this interval was produced by a method that captures μ for 95% of all samples". It does *not* say that 95% of the population lies in the interval, nor that 95% of all sample means do.

Choosing the sample size

The margin of error $E = z_{\alpha/2}\,\sigma/\sqrt{n}$ grows with the confidence level (a larger $z_{\alpha/2}$) and with σ, and it shrinks as n grows. Once the confidence level is fixed, the sample size is the ingredient we control, so knowing the desired margin of error we can choose the sample size wisely, before any data are collected. Let E represent the margin of

error and L the length of the confidence interval. The length is twice the margin of error,

$$L = 2E = 2z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

and rearranging this equation to isolate n gives $\sqrt{n} = z_{\alpha/2}\,\sigma/E$, hence the following.

Formula (sample size for estimating μ)

To estimate μ at the $100(1-\alpha)\%$ confidence level with margin of error E (an interval of length $L = 2E$) when σ is known, the sample size needed is

$$n = \left(z_{\alpha/2}\,\frac{\sigma}{E}\right)^2 = \left(2z_{\alpha/2}\,\frac{\sigma}{L}\right)^2,$$

rounded *up* to the next whole number.

Note (always round up)

A sample size is a whole number, and the margin of error is no larger than E only when n is at least the value given by the formula. Rounding down, even from 96.04 to 96, would leave a margin of error slightly larger than the one asked for. So round up every time, never to the nearest integer.

29.4 Estimating μ when σ is unknown

It is more realistic to know neither μ nor σ. The natural replacement for σ is the sample standard deviation s, computed from the same sample as $\bar{x}$. Whether this replacement is harmless depends on the sample size.

Large samples (case B)

Formula (large-sample confidence interval for μ)

Using the sample standard deviation s instead of σ introduces very little error when n is large ($n \geq 30$). In that case

$$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$$

is a large-sample confidence interval for μ with confidence level $100(1-\alpha)\%$, whatever the distribution of the population.

If only the sums $\sum x_i$ and $\sum x_i^2$ are given, first compute (Section 5)

$$\bar{x} = \frac{1}{n}\sum x_i, \qquad s = \sqrt{\frac{1}{n-1}\left(\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2\right)}.$$

Note

This interval is valid only as long as $n \geq 30$. When $n < 30$ the error made by replacing σ with s can no longer be ignored, and, if the population is normal, the t-distribution takes over.

Small samples from a normal population (case C)

In reality it is not always possible to ensure $n \geq 30$. With a small n the Central Limit Theorem does not apply, so for a general population nothing is known about the shape of the distribution of $\bar{X}_n$. If the underlying distribution X is normal, all is not lost.

Theorem (Student's t-distribution)

If $X_1, \ldots, X_n$ is a random sample from a *normally distributed* population X with mean μ, and $\bar{X}_n$ and s are the mean and the standard deviation of the sample, then the random variable

$$T = \frac{\bar{X}_n - \mu}{s/\sqrt{n}}$$

follows a *Student's t-distribution* with $\nu = n - 1$ *degrees of freedom*, written $T \sim t_\nu$.

Caution (this T is not the sample total)

T here is a new random variable with a new distribution. It is not the sample total $T_n = X_1 + \cdots + X_n$ of Sections 27 and 28.

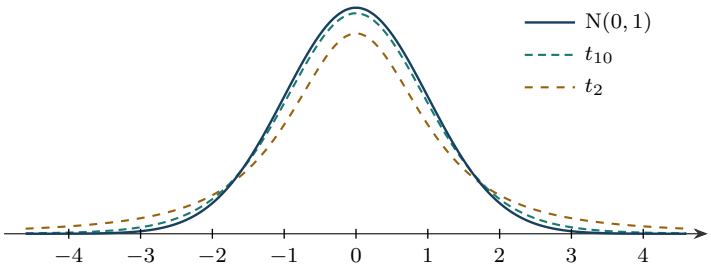
Compare it with the standardized sample mean

$$Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1).$$

The only change is that the constant σ has been replaced by s, which is computed from the sample and so varies from one sample to the next. This extra source of variation is what makes T more spread out than Z, most of all when n is small.

Property (properties of t_ν)

1) Each t_ν curve is bell-shaped and centred at 0.
2) Each t_ν curve is more spread out than the standard normal curve.
3) As $\nu \to \infty$, the t_ν curve approaches the standard normal curve.



The t_2 and t_{10} curves next to the standard normal curve: lower in the middle, heavier in the tails, and closer to the normal curve as ν grows.

Notation (critical value $t_{\alpha,\nu}$)

$t_{\alpha,\nu}$ denotes the value that has area α to its right under the t_ν curve. These values are listed in Table B.2: the row is given by the degrees of freedom ν and the column by the tail area α. For example, $t_{0.025,\,14} = 2.145$ and $t_{0.05,\,14} = 1.761$. Because the t_ν curves are more spread out, these are larger than $z_{0.025} = 1.96$ and $z_{0.05} = 1.645$.

Repeating the derivation of Section 29.3 with T in place of Z and $t_{\alpha/2,\,n-1}$ in place of $z_{\alpha/2}$ gives the interval for a small sample.

Formula (small-sample confidence interval for μ)

When the population is normally distributed and σ is unknown,

$$\bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}}$$

is a small-sample confidence interval for μ with confidence level $100(1-\alpha)\%$. The degrees of freedom are $\nu = n-1$.

Remark (when Student's t is needed)

- If σ were known, then $\bar{X}_n$ would be normally distributed (the population is normal) and we would not need the Student distribution: the interval $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ of Section 29.3 applies, however small the sample.
- If n is small and it is not known that the underlying distribution is normal, then we are essentially out of luck: none of these intervals can be used.
- A t-interval is wider than the z-interval built from the same numbers. The extra width is the price of not knowing σ. It fades as the sample grows, because $t_{\alpha/2,\,n-1}$ approaches $z_{\alpha/2}$, and this is why $z_{\alpha/2}$ is used once $n \geq 30$.

29.5 Estimating a proportion p

Let p represent the proportion of "successes" (individuals with a prescribed characteristic) in a population of size N, and let X be the number of successes in a sample of size n taken from this population. If $N = \infty$ or if $n \ll N$ (the large-population independence assumption), the n individuals behave like independent trials with the same probability of success, so $X \sim \mathrm{Bin}(n,p)$ (Section 17).

Definition (sample proportion)

The *sample proportion* is the random variable

$$\hat{P} = \frac{X}{n} = \frac{\text{number of successes}}{\text{total number of trials}}.$$

For a single sample with $X = x$ successes, its observed value is $\hat{p} = x/n$, and we write $\hat{q} = 1-\hat{p}$. Looking at all possible random samples of size n and calculating $\hat{p}$ for each of them produces the sampling distribution of $\hat{P}$, just like the sampling distributions of $\bar{X}_n$ and T_n.

Proposition (mean and variance of the sample proportion)

If $X \sim \mathrm{Bin}(n,p)$ and $q = 1-p$, then

$$\mu_{\hat{P}} = E(\hat{P}) = p, \qquad \sigma^2_{\hat{P}} = V(\hat{P}) = \frac{pq}{n}, \qquad \sigma_{\hat{P}} = \sqrt{\frac{pq}{n}}.$$

Proof

Use $E(X) = np$ and $V(X) = npq$ for a binomial random variable (Section 17), together with the rules $E(aX) = aE(X)$ and $V(aX) = a^2V(X)$ (Section 16) for $a = \frac{1}{n}$:

$$E(\hat{P}) = E\Big(\frac{1}{n}\cdot X\Big) = \frac{1}{n}E(X) = \frac{1}{n}\cdot np = p, \qquad V(\hat{P}) = V\Big(\frac{X}{n}\Big) = \frac{1}{n^2}V(X) = \frac{npq}{n^2} = \frac{pq}{n}.$$

The standard deviation is the square root of the variance, $\sigma_{\hat{P}} = \sqrt{pq/n}$. □

Theorem (sampling distribution of the sample proportion)

For large enough n ($n \geq 30$), $\hat{P}$ is approximately normally distributed with

$$\mu_{\hat{P}} = p \qquad \text{and} \qquad \sigma_{\hat{P}} = \sqrt{\frac{pq}{n}}, \qquad \text{where } q = 1 - p.$$

This is the Central Limit Theorem once more. Score each individual in the sample 1 for a success and 0 for a failure: then X is the total of the n scores and $\hat{P} = X/n$ is their sample mean.

Deriving the interval. Standardizing, $Z = (\hat{P} - p)/\sqrt{pq/n}$ is approximately standard normal, and the steps of the derivation in Section 29.3 give

$$P\left(\hat{P} - z_{\alpha/2}\sqrt{\frac{pq}{n}} \leq p \leq \hat{P} + z_{\alpha/2}\sqrt{\frac{pq}{n}}\right) \approx 1 - \alpha.$$

One obstacle remains. The goal is to use a sample proportion $\hat{p}$ to estimate the population proportion p, but the standard deviation $\sqrt{pq/n}$ contains p and q, which are of course unknown to us. As it turns out, using the sample statistics $\hat{p}$ and $\hat{q}$ in their place is a fairly good approximation. So in reality we work with

$$\hat{P} \overset{\text{approx}}{\sim} \mathrm{N}\left(\mu_{\hat{P}} = p, \quad \sigma_{\hat{P}} \approx \sqrt{\frac{\hat{p}\hat{q}}{n}}\right),$$

with an approximated standard deviation. This is the same move as in case B for a mean (n large, σ unknown), where s took the place of σ.

Formula (large-sample confidence interval for p)

A large-sample $100(1-\alpha)\%$ confidence interval for a population proportion p is

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}, \qquad \text{where} \quad \hat{p} = \frac{x}{n}, \quad \hat{q} = 1 - \hat{p}.$$

Its margin of error is $E = z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$.

Remark (how polls are reported)

Newspapers commonly state such an interval in the form "36%, with a margin of error of 2%, 19 times out of 20". Read it as $\hat{p} = 0.36$ and $E = 0.02$, so the interval is 0.36 ± 0.02, at a confidence level of $\frac{19}{20} = 95\%$.

Choosing the sample size for a proportion

As for a mean, if we want a specific margin of error we can find the sample size needed, by solving $E = z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$ for n:

$$E = z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} \qquad \Longrightarrow \qquad n = \hat{p}\hat{q}\left(\frac{z_{\alpha/2}}{E}\right)^2.$$

There is a catch. The formula asks for $\hat{p} = x/n$, the proportion of successes in the sample, and that depends on the very sample whose size we are trying to choose: at this stage we do not even have n. There are two ways out.

1) **Worst-case scenario.** Use $\hat{p} = \hat{q} = \frac{1}{2}$. This makes the product $\hat{p}\hat{q}$, and with it the sample size n, as large as it can be, so the margin of error E is guaranteed whatever $\hat{p}$ turns out to be. Indeed $\hat{p}\hat{q} = \hat{p}(1-\hat{p}) = \frac{1}{4} - \left(\hat{p} - \frac{1}{2}\right)^2 \leq \frac{1}{4}$, with equality only when $\hat{p} = \frac{1}{2}$.
2) **Preliminary study or historical data.** Use a value of $\hat{p}$ estimated from a small pilot sample or from earlier data. Since $\hat{p}\hat{q} < \frac{1}{4}$ as soon as $\hat{p} \neq \frac{1}{2}$, this gives a smaller sample size than the worst case.

Formula (sample size for estimating p)

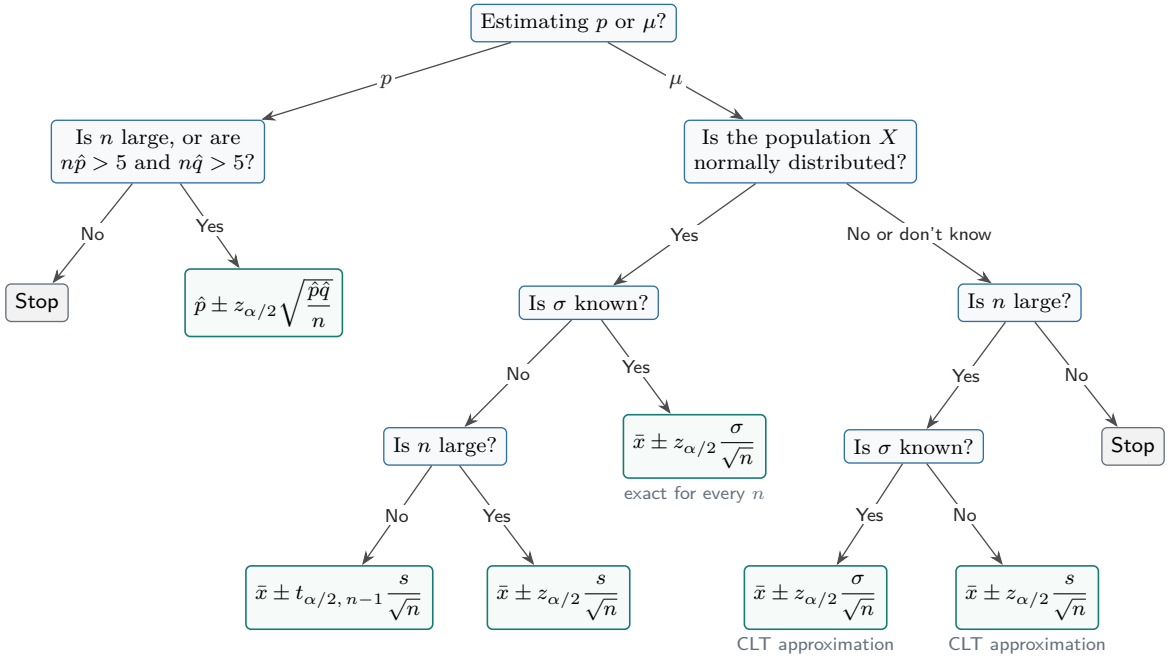
To estimate p at the $100(1-\alpha)\%$ confidence level with margin of error E, the sample size needed is

$$n = \hat{p}\hat{q}\left(\frac{z_{\alpha/2}}{E}\right)^2 \quad \text{(with a preliminary estimate } \hat{p}\text{)} \qquad \text{or} \qquad n = \frac{1}{4}\left(\frac{z_{\alpha/2}}{E}\right)^2 \quad \text{(worst case)},$$

rounded *up* to the next whole number. Here E is entered as a proportion: a margin of error of 2.5% means $E = 0.025$.

29.6 Decision tree for confidence intervals

Every interval of this section sits at the end of one branch of the following tree. Answer the questions from the top to find the interval that applies; "Stop" means that none of them does.



Remark (reading the tree)

- **Normal population, σ known.** Here $\bar{X}_n$ is exactly normally distributed even if n is not large, and that is what gives rise to the formula $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$. On this branch the formula is not an approximation: it is exact, for every sample size.
- **Population not normal, or shape unknown.** The only route to a normal curve is now the Central Limit Theorem, so n must be large, and the intervals on this branch are CLT approximations. With a small sample we have to stop.
- **The same formulas come back.** The same two formulas end both sides of the μ branch: $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ when σ is known, and $\bar{x} \pm z_{\alpha/2}\,s/\sqrt{n}$ when σ is unknown and n is large, whether the population is normal or not. Only the justification changes.
- **One place for t.** Student's t is needed in exactly one situation: normal population, σ unknown, n small.
- **Proportions.** The sample must be large. A quick check is that the sample contains more than 5 successes and more than 5 failures, that is, $n\hat{p} > 5$ and $n\hat{q} > 5$.

29.7 Worked examples

Example 29.1 ▪ Desk heights: building a 95% interval Worksheet 29, p. 1

To determine the preferred desk height of office professionals, EyeKia has done a study of $n = 36$ office workers. The sample average was $\bar{x} = 80.0$ cm. EyeKia believes the distribution of desk height preferences to have a standard deviation of $\sigma = 2.0$ cm. Build a 95% confidence interval for μ, the population average.

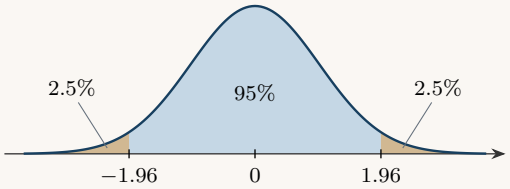
Solution. This first interval is built from scratch, to show where the formula comes from.

Distribution of the sample mean. Nothing is said about the shape of the population, but $n = 36 \geq 30$, so the Central Limit Theorem applies:

$$\bar{X}_{36} \overset{\text{approx}}{\sim} \mathrm{N}\Big(\mu_{\bar{X}_{36}} = \mu,\ \ \sigma_{\bar{X}_{36}} = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{36}} = \frac{1}{3}\Big).$$

What is the radius? A 95% confidence interval has to be built so that, for 95% of all possible sample means, the interval around the sample mean contains μ. So we need the distance from μ within which 95% of the sample means fall. Under the standard normal curve, the central 95% leaves 2.5% in each tail, so it ends at

$$\Phi^{-1}(0.975) = 1.96 \qquad \text{and} \qquad \Phi^{-1}(0.025) = -1.96.$$



To get 95% we therefore need

$$-1.96 \leq Z = \frac{\bar{X}_{36} - \mu}{\sigma_{\bar{X}_{36}}} \leq 1.96 \qquad \Longleftrightarrow \qquad \mu - 1.96\,\sigma_{\bar{X}_{36}} \leq \bar{X}_{36} \leq \mu + 1.96\,\sigma_{\bar{X}_{36}},$$

so the radius is $r = 1.96\,\sigma_{\bar{X}_{36}} = 1.96 \cdot \frac{1}{3} \approx 0.6533$ cm.

Write μ in the middle. In this case $\bar{x}_{36} = 80$. The sample mean is within r of μ exactly when μ is within r of the sample mean, so

$$80 - 0.6533 \leq \mu \leq 80 + 0.6533, \qquad \text{that is,} \qquad \boxed{79.35 \leq \mu \leq 80.65}\ \text{(in cm)}.$$

(79.35, 80.65) is a 95% confidence interval for μ about $\bar{x} = 80$. It contains μ provided our sample mean is one of the 95% that lie within r of μ. In context: EyeKia can be 95% confident that the average preferred desk height of office professionals is between 79.35 cm and 80.65 cm.

The formula of Section 29.3 gives the same interval in one line:

$$\bar{x} \pm z_{0.025}\,\frac{\sigma}{\sqrt{n}} = 80 \pm 1.96 \cdot \frac{2}{\sqrt{36}} \approx 80 \pm 0.6533.$$

Example 29.2 ▪ Affordable housing: a 90% interval Worksheet 29, p. 3

Housing is typically considered affordable if the monthly amount (rent or mortgage payment) is less than 30% of a person's take home pay. StatsCan is doing a study on a lower income neighbourhood to determine if the housing in the area is affordable. They take a sample of 35 people and calculate the percentage of their income that is spent on housing. The sample average is 31.6%. Assume that the standard deviation of the population is known to be 3% and create a 90% CI for the true population average.

Solution. Let X be the percentage of income that a resident spends on housing; we want an interval for $\mu = \mu_X$. The data are $n = 35$, $\bar{x} = 31.6$ and $\sigma = 3$ (known). Since $n \geq 30$, the sample mean is approximately normal and the interval is $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$.

Critical value. $1 - \alpha = 90\% = 0.9$, so $\alpha = 0.1$, $\alpha/2 = 0.05$ and $z_{\alpha/2} = z_{0.05} = 1.645$.

Interval.

$$\bar{x} \pm z_{0.05}\,\frac{\sigma}{\sqrt{n}} = 31.6 \pm 1.645 \cdot \frac{3}{\sqrt{35}} \approx 31.6 \pm 0.8342,$$

so $\boxed{30.77\% \leq \mu \leq 32.43\%}$ with 90% confidence. The whole interval lies above the 30% threshold, so at this confidence level the study indicates that housing in the neighbourhood is, on average, not affordable.

Example 29.3 ▪ Response time: how large a sample? Worksheet 29, p. 3

The response time of a particular computer component is normally distributed with $\sigma = 25$ milliseconds. What sample size is necessary to estimate μ at the 95% confidence level with a margin of error of at most 5 milliseconds?

Solution. The population is normal and σ is known, so the margin of error is $E = z_{\alpha/2}\,\sigma/\sqrt{n}$ for any sample size. We want $E = 5$ with $z_{\alpha/2} = z_{0.025} = 1.96$:

$$n = \left(z_{\alpha/2}\,\frac{\sigma}{E}\right)^2 = \left(1.96 \cdot \frac{25}{5}\right)^2 = (9.8)^2 = 96.04.$$

Round up: $\boxed{n = 97}$ components must be tested.

Rounding down would not do: with $n = 96$ the margin of error is $1.96 \cdot 25/\sqrt{96} \approx 5.001$ ms, just over the limit, while $n = 97$ gives $1.96 \cdot 25/\sqrt{97} \approx 4.975$ ms.

Example 29.4 ▪ SAT errors: a large sample, σ unknown Worksheet 29, p. 4

A sample of 75 students' tests from one section of the SAT was examined and the number of incorrect answers on each test was counted. If $\sum x_i = 1054$ and $\sum x_i^2 = 16576$, construct a 99% confidence interval for the mean number of questions answered incorrectly on this section of the SAT.

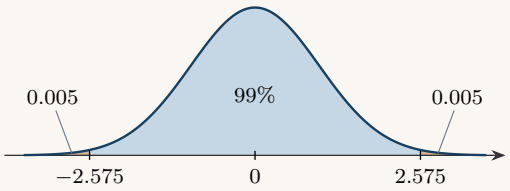
Solution. Let X be the number of incorrect answers on one test. Neither the distribution of X nor σ is known, but $n = 75 \geq 30$, so the large-sample interval $\bar{x} \pm z_{\alpha/2}\,s/\sqrt{n}$ applies.

Sample mean and standard deviation.

$$\bar{x} = \frac{1}{n}\sum x_i = \frac{1054}{75} \approx 14.0533,$$

$$s = \sqrt{\frac{1}{n-1}\left(\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2\right)} = \sqrt{\frac{1}{74}\left(16576 - \frac{1054^2}{75}\right)} \approx \sqrt{23.8350} \approx 4.8821.$$

Critical value. $1 - \alpha = 0.99$, so $\alpha = 0.01$ and $\alpha/2 = 0.005$: each tail holds an area of 0.005 and $z_{\alpha/2} = z_{0.005} = 2.575$.



Interval.

$$14.0533 \pm 2.575 \cdot \frac{4.8821}{\sqrt{75}} \approx 14.0533 \pm 1.4516 \quad \Longrightarrow \quad \boxed{12.60 \leq \mu \leq 15.50}.$$

We are 99% confident that the mean number of questions answered incorrectly on this section of the SAT is between about 12.6 and 15.5.

Example 29.5 ▪ A small normal sample: t-intervals Worksheet 29, p. 5

Assume that a random sample of size $n = 15$ from a normally distributed population with unknown μ and σ produced a sample mean of $\bar{x} = 25.31$ and standard deviation $s = 1.58$.

a) Construct a 95% CI for μ.
b) Construct a 90% CI for μ.
c) Now suppose that the sample size had been 43 and that $\bar{x}$ and s were the same. Construct a 90% CI for μ.

Solution. In parts (a) and (b) the sample is small ($n = 15 < 30$), σ is unknown and the population is normal, so the interval with confidence level $100(1-\alpha)\%$ is

$$\bar{x} \pm t_{\alpha/2,\,\nu}\,\frac{s}{\sqrt{n}}, \qquad \nu = n - 1 = 14.$$

a) $1 - \alpha = 0.95$ gives $\alpha/2 = 0.025$. In Table B.2, the row $\nu = 14$ and the column $t_{0.025}$ give $t_{0.025,\,14} = 2.145$, so

$$25.31 \pm 2.145 \cdot \frac{1.58}{\sqrt{15}} \approx 25.31 \pm 0.8751 \qquad \Longrightarrow \qquad \boxed{24.43 \le \mu \le 26.19}.$$

b) $1 - \alpha = 0.90$ gives $\alpha/2 = 0.05$, and the critical value is now $t_{0.05,\,14} = 1.761$:

$$25.31 \pm 1.761 \cdot \frac{1.58}{\sqrt{15}} \approx 25.31 \pm 0.7184 \qquad \Longrightarrow \qquad \boxed{24.59 \le \mu \le 26.03}.$$

c) Now $n = 43 \ge 30$, so we can use the normal distribution: the large-sample interval $\bar{x} \pm z_{\alpha/2}\, s/\sqrt{n}$ applies. For a 90% CI, $1 - \alpha = 0.90$, $\alpha/2 = 0.05$ and $z_{\alpha/2} = \Phi^{-1}(0.95) = 1.645$:

$$25.31 \pm 1.645 \cdot \frac{1.58}{\sqrt{43}} \approx 25.31 \pm 0.3964 \qquad \Longrightarrow \qquad \boxed{24.91 \le \mu \le 25.71}.$$

Compare the three margins of error, 0.8751, 0.7184 and 0.3964. Lowering the confidence level from 95% to 90% shortens the interval, because the critical value is smaller. Raising the sample size from 15 to 43 shortens it much more: the critical value drops from $t_{0.05,\,14} = 1.761$ to $z_{0.05} = 1.645$ and, above all, $s/\sqrt{n}$ drops from about 0.408 to about 0.241.

Example 29.6 ▪ An election poll: interval for a proportion Worksheet 29, p. 6

In a recent poll, 732 of 2032 people intended to vote Conservative in the next federal election. Give a 95% CI for p, the true population proportion.

Solution. A "success" is a person who intends to vote Conservative, and p is the proportion of such people among all voters. The sample is large ($n = 2032$, with 732 successes and 1300 failures), so the large-sample interval $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$ applies. Its ingredients are

$$\hat{p} = \frac{732}{2032} \approx 0.3602, \qquad \hat{q} = 1 - \hat{p} \approx 0.6398, \qquad n = 2032,$$

and, since $100(1-\alpha) = 95$, $z_{\alpha/2} = z_{0.025} = 1.96$. Therefore

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.3602 \pm 1.96\sqrt{\frac{(0.3602)(0.6398)}{2032}} \approx \boxed{0.3602 \pm 0.0209},$$

that is, $0.3394 \le p \le 0.3811$ (end points computed before rounding $\hat{p}$ and E).

A newspaper would report this result as "36%, with a margin of error of 2%, 19 times out of 20". Here "36%" is $\hat{p} \approx 0.3602$, "2%" is the margin of error $E \approx 0.0209$, and "19 times out of 20" is the 95% confidence level.

Example 29.7 ▪ Student fees: sample size for a proportion Worksheet 29, p. 7

Student Council wants to know what proportion of the student body would be willing to pay \$50 extra in fees each year to better fund the school's social events. They can accept to be 90% confident of their estimate, as long as the margin of error is no more than 2.5%.

a) What size sample should they take in the worst case scenario?

b) A preliminary study indicates that $p \approx 70\%$. How does this affect their choice of sample size?

Solution. The confidence level is 90%, so $z_{\alpha/2} = z_{0.05} = 1.645$, and the margin of error is $E = 0.025$. In both parts $n = \hat{p}\hat{q}\,(z_{\alpha/2}/E)^2$; only the value used for $\hat{p}$ changes.

a) Worst case: $\hat{p} = \hat{q} = \frac{1}{2}$.

$$n = (0.5)(0.5)\left(\frac{1.645}{0.025}\right)^2 = (0.25)(65.8)^2 = 1082.41,$$

and rounding up, they should survey 1083 students.

b) From the preliminary study, $\hat{p} = 0.7$ and $\hat{q} = 0.3$.

$$n = (0.7)(0.3)\left(\frac{1.645}{0.025}\right)^2 = (0.21)(65.8)^2 \approx 909.22,$$

and rounding up, 910 students are enough. The preliminary information lowers the required sample size by 173 students, because $\hat{p}\hat{q} = 0.21$ is smaller than the worst-case value 0.25.

30 Hypothesis Testing

A confidence interval (Section 29) uses a *sample* mean or proportion to predict the value of an unknown *population* mean or proportion. A hypothesis test starts from the other end. Some assertion has already been made about the value of a population mean or proportion; we wish to disagree with that statement, and we need statistical proof, taken from a sample, to back up our argument.

30.1 The idea of a hypothesis test

Every test in this section grows out of one question, which the first example asks in its simplest form.

Example 30.1 ▪ Tires: is the sample mean an outlier? Worksheet 30, Example 1

Michilyne claims that their tires last on average 75 000 km. A consumer protection group thinks that this is false advertising. Based on consumer reports, they think that the average is less than advertised. To verify the claim, they take a random sample of 50 tires and compute the sample mean as 72 051 km with standard deviation 8020 km. *If* μ really is equal to 75 000, how likely is it to obtain a sample mean of $\bar{x} = 72\,051$?

Solution. The sample mean is lower than the advertised value, but that alone proves nothing: sample means vary from sample to sample, and about half of all samples would fall below 75 000 km even if the claim were exactly true. What matters is whether 72 051 is an ordinary fluctuation or an outlier. So we suppose that the claim is true and ask, in terms of the distribution of the sample mean $\bar{X}_{50}$,

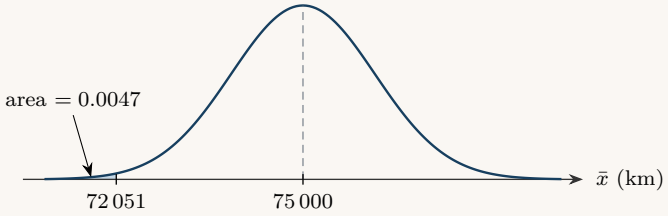
$$\text{what is } P(\bar{X}_{50} \le 72\,051)\,?$$

Let X be the lifetime of a tire, in km. Since $n = 50 \ge 30$, the Central Limit Theorem (Section 28) applies:

$$\bar{X}_{50} \overset{\text{approx}}{\sim} \mathrm{N}\Big(\mu_{\bar{X}} = 75\,000,\quad \sigma_{\bar{X}} = \frac{s}{\sqrt{n}} = \frac{8020}{\sqrt{50}} \approx 1134.2\Big).$$

The mean is 75 000 *as claimed*, because the probability is computed on the assumption that the claim is true. The unknown σ is replaced by the sample standard deviation $s = 8020$, which is acceptable for a large sample (Section 29). Then

$$P(\bar{X}_{50} \le 72\,051) = \Phi\Big(\frac{72\,051 - 75\,000}{8020/\sqrt{50}}\Big) = \Phi(-2.60) = \boxed{0.0047} = 0.47\%.$$



If the tires really lasted 75 000 km on average, fewer than 5 samples in 1000 would have a mean this low. Either the group drew an extraordinarily unlucky sample, or the claim is false. The second explanation is far more believable, so the sample is strong evidence that the true mean is less than 75 000 km.

The idea behind H_0 and H_A

A test sets two statements against each other: the claim, called H_0, and what we suspect is true instead, called H_A. We want to see whether H_A is true.

- Suppose that H_0 is true, and look at where the sample statistic falls ($\bar{x}$ for a mean, $\hat{p}$ for a proportion).
- A sample never reproduces the claimed value exactly, so some deviation has to be accepted. The *significance level* α, a small probability chosen in advance, fixes the most that we are prepared to accept.
- If $\bar{x}$ (or $\hat{p}$) falls outside what α tolerates, then H_A is justified: there is enough evidence to reject H_0.
- Otherwise the difference is just a normal deviation, of the kind that α tolerates, and H_0 is not rejected.

30.2 The language of hypothesis testing

Each term below is illustrated with the tire claim of Example 30.1.

Definition (null and alternate hypotheses)

The initial claim or generally accepted belief in a hypothesis testing problem is called the *null hypothesis* and is written H_0.

The counter-claim is called the *alternate hypothesis* or *research hypothesis* and is written H_A (some books write H_1).

In Example 30.1: $H_0 : \mu = 75\,000$ $H_A : \mu < 75\,000$.

Both hypotheses are statements about a *population* parameter (μ or p), never about a sample statistic: there is nothing to test about $\bar{x}$, whose value is known. The null hypothesis always says that the parameter *equals* one specific value; the inequality goes in H_A (Section 30.4).

Definition (test statistic)

The *test statistic* is the z- or t-score of the observed value: the sample result, standardized with the mean and standard deviation that it would have if H_0 were true,

$$\text{test statistic} = \frac{\text{observed value} - \text{value claimed by } H_0}{\text{standard deviation of the sample statistic}}.$$

In Example 30.1: $z_{\text{test}} = \dfrac{72\,051 - 75\,000}{8020/\sqrt{50}} \approx -2.60$: the sample mean lies 2.6 standard deviations below the

claimed mean.

Definition (p-value)

The *p-value* in a hypothesis test is the probability of getting a value *at least as extreme* (low or high) as the observed value from the sample, assuming that H_0 is true.

In Example 30.1: $p = P(\bar{X}_{50} \leq 72\,051) = 0.0047$.

"Extreme" is measured in the direction that supports H_A: low values when H_A says "less than" (as for the tires), high values when it says "greater than", and both when it says "not equal". The smaller the p-value, the harder it is to reconcile the sample with H_0.

Definition (significance level)

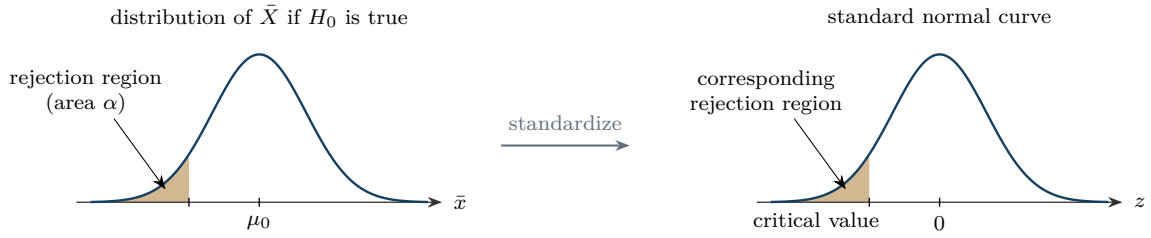
The *significance level*, usually written α, is the cut-off value for a given hypothesis test: a p-value less than or equal to α is considered *statistically significant*. Typical values for α are 0.1, 0.05 and 0.01. These values correspond to the probability of observing such an extreme value by chance.

In Example 30.1: no significance level was set, but the null hypothesis could have been rejected at a level of $\alpha = 0.01$, or 1%, because $0.0047 \leq 0.01$.

Definition (critical value and rejection region)

The *critical value* is the z- or t-score of the significance level: the score that cuts off a tail of area α under the curve (or two tails of total area α).

The *rejection region* is the region under the curve which lies beyond the critical value(s), as determined by the level of significance α. Rejection regions can be one- or two-tailed.



A left-tailed rejection region. Transforming to the standard normal distribution turns it into a tail of the z-curve; the z-value at its boundary is the critical value.

Caution (two different p's)

The "p" of a p-value is a probability computed from the sample. It is not the population proportion p that appears in hypotheses such as $H_0 : p = p_0$.

30.3 The standard procedure

There is a traditional procedure for setting up and solving a hypothesis testing problem. Every test worked out in this section and the next follows it step by step.

Standard procedure for hypothesis testing

1. Define any variables.
2. Identify and state clearly the null and alternate hypotheses.
3. Choose a level of significance α.

4. Calculate the test statistic (z-value or t-value).
5. Either find the rejection region (critical value) or the p-value.
6. Make the appropriate comparison.
7. Decide whether there is significant evidence to reject H_0 or not, and state your conclusion using one of the traditional wordings:

There is sufficient / insufficient evidence to warrant rejection of the claim that ... (H_0) ... at the significance level $\alpha = \ldots$

Decision rule (steps 5 and 6)

Reject H_0 in either of the following situations.

- **Rejection region.** The test statistic falls in the rejection region. For a left-tailed test this means $z_{\text{test}} \le z_{\text{crit}}$; the other cases are listed in Section 30.4.
- **p-value.** The p-value is at most α: $p \le \alpha$.

Otherwise, do not reject H_0.

The two routes always lead to the same decision. The critical value is the score whose tail area is exactly α, so a test statistic that lies beyond the critical value cuts off a tail of area smaller than α; that is, its p-value is smaller than α. The rejection region needs only a critical value from the tables. The p-value takes one more computation, but it also tells how strong the evidence is.

What the two possible conclusions mean

Reject H_0. If H_0 were true, a sample as extreme as ours would turn up with probability at most α. That is too unlikely to be put down to chance, so the sample counts as significant evidence in favour of H_A.

Fail to reject H_0. The sample lies within what chance alone produces when H_0 is true, so it gives no reason to abandon H_0. This does *not* prove that H_0 is true: the true value may differ from the claim by an amount too small for this sample to detect. For that reason we never say "accept H_0"; the verdict is "insufficient evidence", just as "not guilty" in court is not a proof of innocence.

Example 30.2 ▪ Tires: the complete test Worksheet 30, Example 2

In the situation described in Example 30.1, find the critical value and sketch the rejection region corresponding to a level of significance of $\alpha = 0.01$.

Solution. The critical value belongs to step 5 of the standard procedure, so we lay out the whole test.

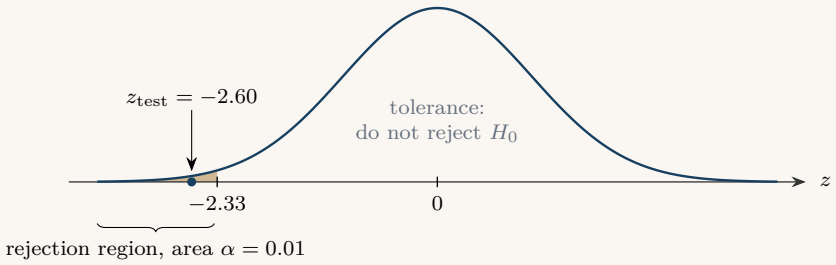
1. Let X be the lifetime of a tire, in km, and let μ be the mean lifetime of all the tires.
2. $H_0\colon \mu = 75\,000$ (the company's claim) $\qquad H_A\colon \mu < 75\,000$ (the consumer group's suspicion).
3. $\alpha = 0.01$.
4. $n = 50 \ge 30$, so $\bar{X}_{50}$ is approximately normal and the test statistic is a z-score:

$$z_{\text{test}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{72\,051 - 75\,000}{8020/\sqrt{50}} \approx -2.60.$$

5. H_A says "less than", so the rejection region is the left tail of area $\alpha = 0.01$ of the distribution of $\bar{X}_{50}$. Transformed to the standard normal distribution, it is the left tail of area 0.01 of the z-curve, and the z-value at its boundary is the critical value:

$$z_{\text{crit}} = \Phi^{-1}(0.01) = \boxed{-2.33} \qquad \text{(Table B.1: } \Phi(-2.33) = 0.0099\text{)}.$$

The rejection region is $z \le -2.33$. In kilometres, it consists of the sample means below $75\,000 - 2.33\,(1134.2) \approx 72\,357$.

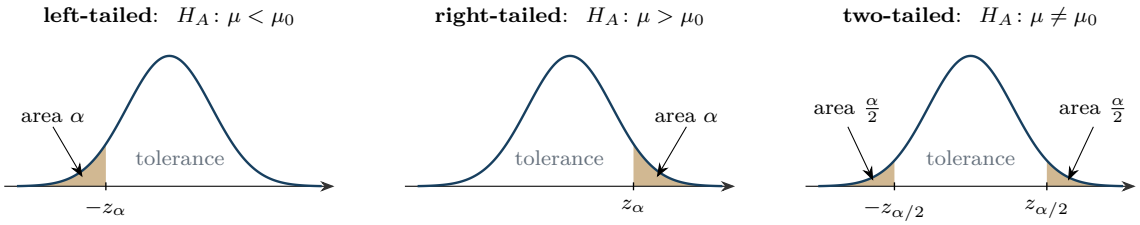


6. $z_{\text{test}} = -2.60 < -2.33 = z_{\text{crit}}$, so the test statistic falls in the rejection region. (Equivalently, the p-value 0.0047 found in Example 30.1 is less than $\alpha = 0.01$.)

7. There is sufficient evidence to warrant rejection of the claim that the tires last 75 000 km on average, at the significance level $\alpha = 0.01$. The data support the consumer group: the mean lifetime is less than advertised.

30.4 Left-tailed, right-tailed and two-tailed tests

The form of H_A decides where the rejection region lies: in the left tail, in the right tail, or in both.



The shaded tails are the rejection regions. The unshaded part of each picture is the *tolerance*: a test statistic that lands there is a deviation that the significance level tolerates, and H_0 is not rejected. In a two-tailed test the area α is shared equally between the two tails. That is why the critical values are $\pm z_{\alpha/2}$, and why the p-value is twice the tail area beyond the test statistic.

The table gives the rejection region and the p-value for each type of test. In it, μ_0 and p_0 are the values claimed by H_0 for a population mean μ or a population proportion p. The formulas are written for a z-test, that is, on the standard normal curve, and z_α is the value with area α to its right (Section 29).

Hypotheses	**Test**	**Rejection region**	**p-value**
$H_0: \mu = \mu_0$ or $p = p_0$ $H_A: \mu < \mu_0$ or $p < p_0$	left-tailed	$z_{\text{test}} \le z_{\text{crit}}$, where $z_{\text{crit}} = \Phi^{-1}(\alpha) = -z_\alpha$	$p = P(\bar{X}_n \le \bar{x})$ $= \Phi(z_{\text{test}})$
$H_0: \mu = \mu_0$ or $p = p_0$ $H_A: \mu > \mu_0$ or $p > p_0$	right-tailed	$z_{\text{test}} \ge z_{\text{crit}}$, where $z_{\text{crit}} = \Phi^{-1}(1-\alpha) = z_\alpha$	$p = P(\bar{X}_n \ge \bar{x})$ $= 1 - \Phi(z_{\text{test}})$
$H_0: \mu = \mu_0$ or $p = p_0$ $H_A: \mu \neq \mu_0$ or $p \neq p_0$	two-tailed	$z_{\text{test}} \le -z_{\text{crit}}$ or $z_{\text{test}} \ge z_{\text{crit}}$, where $z_{\text{crit}} = -\Phi^{-1}(\alpha/2) = z_{\alpha/2}$	$p = 2\big(1 - \Phi(\lvert z_{\text{test}}\rvert)\big)$

For a proportion, read $\hat{P}$ and $\hat{p}$ in place of $\bar{X}_n$ and $\bar{x}$. For a t-test, the critical values $t_{\alpha,\nu}$ and $t_{\alpha/2,\nu}$ of Table B.2 replace z_α and $z_{\alpha/2}$.

The critical values needed for the usual significance levels are read backwards from Table B.1:

Significance level α	0.10	0.05	0.01
One-tailed test: z_α	1.28	1.645	2.33
Two-tailed test: $z_{\alpha/2}$	1.645	1.96	2.575

Remark (reading $z_{0.01}$ from the table)

The area 0.99 falls between two entries of Table B.1: $\Phi(2.32) = 0.9898$ and $\Phi(2.33) = 0.9901$. These notes use the closer one, $z_{0.01} = 2.33$. You will also meet the midpoint 2.325, obtained in the same way as 1.645 and 2.575. Either value is fine; a conclusion that depended on the difference would be too close to call anyway.

Caution (H_0 must always be an equality)

The inequality ($<$, $>$ or $\neq$) always goes in H_A. The test statistic and the p-value are computed *assuming that* H_0 *is true*, and that requires one definite value μ_0 (or p_0) on which to centre the sampling distribution. Even when the claim being examined is itself an inequality, such as "the mean is at most 50", the null hypothesis is the boundary case $H_0 : \mu = 50$, tested against $H_A : \mu > 50$.

Note (choosing H_A)

H_A is the statement that we are looking for evidence *for*. Read it off the wording of the problem, before looking at the data:

- "less than", "lower", "has decreased": $H_A : \mu < \mu_0$ (left-tailed);
- "more than", "longer", "will increase", "exceeds": $H_A : \mu > \mu_0$ (right-tailed);
- "different", "has changed", "out of date": $H_A : \mu \neq \mu_0$ (two-tailed), because nothing says on which side of μ_0 the true value lies.

30.5 Which test statistic?

The test statistic standardizes the sample result under H_0, so it is compared with a standard normal variable $Z \sim \mathrm{N}(0,1)$ or with a Student t variable. The choice follows the same three cases as for confidence intervals (Section 29).

Testing	Situation	Test statistic	Compared with
mean μ	large sample ($n \geq 30$), any population	$z_{\text{test}} = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ or $\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	standard normal
mean μ	small sample ($n < 30$), normal population, σ unknown	$t_{\text{test}} = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	Student t, $\nu = n - 1$
proportion p	large sample	$z_{\text{test}} = \dfrac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$ $(q_0 = 1 - p_0)$	standard normal

- **Large sample.** By the Central Limit Theorem, $\bar{X}_n$ is approximately $\mathrm{N}\big(\mu_0,\ \sigma/\sqrt{n}\big)$ when H_0 is true, whatever the shape of the population. If σ is unknown, the sample standard deviation s takes its place; this introduces very little error when $n \geq 30$.
- **Small sample.** With $n < 30$ the Central Limit Theorem is not available. If the population is normally distributed, however,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad \text{follows a Student } t\text{-distribution with } \nu = n - 1 \text{ degrees of freedom}$$

  (Section 29). The test then runs exactly like a z-test, with critical values from Table B.2. (If the population is normal and σ happens to be known, $\bar{X}$ is exactly normal and the z statistic with σ is used, whatever the sample size.) If n is small and the population is not known to be normal, neither test applies.
- **Proportion.** For large n, the sample proportion $\hat{P}$ is approximately $\mathrm{N}\big(\mu_{\hat{P}} = p,\ \sigma_{\hat{P}} = \sqrt{pq/n}\big)$ (Section 29). When H_0 is true, $p = p_0$, so the standard deviation in the test statistic is computed from p_0 and q_0. This differs from a confidence interval, where no value of p is assumed and $\hat{p}$, $\hat{q}$ have to be used.

Note (the p-value of a t-test)

Table B.2 lists critical values for a few tail areas only, so the p-value of a t-test cannot be read off exactly. It can be bracketed: in the row of ν, find the two neighbouring entries that enclose t_{test}; the tail areas at the top of their columns enclose the p-value. This is enough to compare it with α (Example 30.5).

30.6 Types of errors

A test decides from a sample, so its conclusion can be wrong, and it can be wrong in two different ways.

Definition (type I and type II errors)

A *type I error* (or *α-error*) consists of rejecting the null hypothesis when in fact it is true.

A *type II error* (or *β-error*) consists of not rejecting the null hypothesis when in fact it is false.

	H_0 **is in fact true**	H_0 **is in fact false**
Reject H_0	type I error	correct decision
Do not reject H_0	correct decision	type II error

Property (probability of a type I error)

The probability of a type I error is the level at which the test is carried out:

$$P(\text{type I error}) = P(\text{reject } H_0 \mid H_0 \text{ is true}) = \alpha.$$

When H_0 is true, the test statistic follows the curve on which the rejection region was drawn, and that region was built to have area exactly α. Choosing α therefore means choosing the risk of a type I error that we are willing to run. The probability of a type II error, written β, depends on the true value of the parameter, which is unknown, and it is not computed in this course. For a fixed sample size the two risks pull against each other: a smaller α shrinks the rejection region, which makes it harder to reject H_0 when it is false.

To interpret the errors in a given situation, translate "H_0 is true" and "reject H_0" into the words of the problem. For the tire claim (H_0: $\mu = 75\,000$ against H_A: $\mu < 75\,000$):

- a type I error is concluding that the tires last less than 75 000 km on average when in fact the advertised mean is correct (the company is wrongly accused of false advertising);
- a type II error is failing to reject the advertised mean when in fact the tires last less than 75 000 km on average (false advertising goes undetected).

At $\alpha = 0.01$, a company whose claim is true runs a 1% risk of being wrongly contradicted by the test.

30.7 Worked examples

Each test below is laid out in the seven steps of the standard procedure. Together the examples cover a right-tailed z-test, a two-tailed test for a proportion and a small-sample t-test; the last example is about the two types of errors.

Example 30.3 ▪ Golf: a right-tailed z-test Worksheet 30, Example 3

Golf course designers have become concerned that older courses are becoming obsolete since new technologies have given golfers the ability to hit the ball further than before. A consortium of designers therefore has proposed that new golf courses need to be built with the expectation that golfers can hit the ball more than 250 yards on average. Suppose that a random sample of 121 golfers is chosen and that their mean driving distance is 250.6 yards, with a standard deviation of 42.4 yards. Test the consortium's hypothesis at the $\alpha = 0.1$ level by using a) the rejection region, and b) the p-value.

Solution.

1. Let X be a golfer's driving distance, in yards, and let μ be its mean over all golfers.
2. $H_0 : \mu = 250$ $\quad$ $H_A : \mu > 250$. The consortium's hypothesis, "more than 250 yards on average", is the statement that needs evidence, so it is H_A. The test is right-tailed.
3. $\alpha = 0.1 = 10\%$.
4. $n = 121 \geq 30$, so the Central Limit Theorem applies. If H_0 is true,

$$\bar{X}_{121} \overset{\text{approx}}{\sim} \mathrm{N}\Big(\mu_{\bar{X}} = \mu = 250, \quad \sigma_{\bar{X}} = \frac{s}{\sqrt{n}} = \frac{42.4}{\sqrt{121}} \approx 3.855\Big),$$

and the test statistic is

$$z_{\text{test}} = \frac{\bar{x} - \mu_{\bar{X}}}{s/\sqrt{n}} = \frac{250.6 - 250}{42.4/\sqrt{121}} \approx 0.156 \approx 0.16.$$

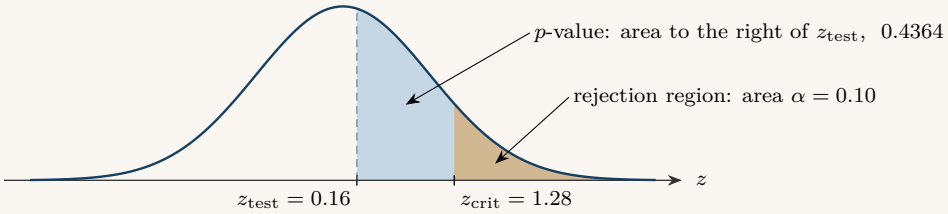
5. a) *Rejection region.* For a right-tailed test it is the right tail of area $\alpha = 0.1$:

$$z_{\text{crit}} = z_{0.10} = \Phi^{-1}(0.9) = 1.28,$$

and H_0 is rejected if $z_{\text{test}} \geq 1.28$.

b) *p-value.* Values at least as extreme as the observed one are sample means of 250.6 or more:

$$p = P(\bar{X}_{121} \geq 250.6) = P(Z \geq 0.16) = 1 - \Phi(0.16) = 1 - 0.5636 = 0.4364.$$



6. a) $z_{\text{test}} = 0.16 < 1.28 = z_{\text{crit}}$: the test statistic is *not* in the rejection region.

b) $p = 0.4364 > 0.1 = \alpha$: the p-value is not below the cut-off.

7. Both routes give the same decision: do not reject H_0. There is insufficient evidence to warrant rejection of the claim that the mean driving distance is 250 yards, at the significance level $\alpha = 0.1$.

The sample mean is above 250 yards, but it is still *within* what α tolerates. If μ were 250, about 44% of all samples of 121 golfers would have a mean of 250.6 yards or more, and only sample means of at least $250 + 1.28\,(3.855) \approx 254.9$ yards would fall in the rejection region. The sample does not support the consortium's proposal. It does not prove that $\mu = 250$ either.

Example 30.4 ▪ Emissions: a two-tailed test for a proportion Worksheet 30, Example 4

SAAQ records indicate that 70% of all vehicles pass emissions tests on the first try (without first undergoing a tune-up). The environment minister suspects that this information is out of date, so she orders a random sample of 200 cars tested; 158 of them passed. Is there enough evidence at the $\alpha = 0.01$ level to justify a more in-depth test to update the agency's records?

Solution.

1. Let p be the proportion of all vehicles that pass the emissions test on the first try.
2. $H_0 : p = 0.7$ $\quad$ $H_A : p \neq 0.7$. "Out of date" does not say in which direction the proportion has moved: we do not know whether $p < 0.7$ or $p > 0.7$, so the test is two-tailed.
3. $\alpha = 0.01$.

4. The sample proportion is $\hat{p} = \dfrac{158}{200} = 0.79$. Since $n = 200$ is large, the sample proportion $\hat{P}$ is approximately normal, with

$$\mu_{\hat{P}} = p \qquad \text{and} \qquad \sigma_{\hat{P}} = \sqrt{\frac{pq}{n}} \qquad \text{(Section 29)}.$$

If H_0 is true, then $p = p_0 = 0.7$ and $q = q_0 = 0.3$, so

$$z_{\text{test}} = \frac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}} = \frac{0.79 - 0.7}{\sqrt{(0.7)(0.3)/200}} = \frac{0.09}{0.0324} \approx 2.78.$$

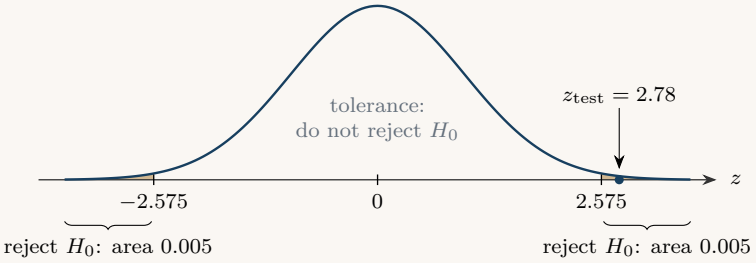
5. a) *Rejection region.* Because the test is two-tailed, α is shared between both tails, $\alpha/2 = 0.005$ in each. The critical values are $\pm z_{\text{crit}}$ with

$$z_{\text{crit}} = z_{0.005} = -\Phi^{-1}(\alpha/2) = -\Phi^{-1}(0.005) = 2.575,$$

and H_0 is rejected if $z_{\text{test}} \le -2.575$ or $z_{\text{test}} \ge 2.575$.

b) *p-value.* Both tails count:

$$p = 2\big(1 - \Phi(|z_{\text{test}}|)\big) = 2\big(1 - \Phi(2.78)\big) = 2(1 - 0.9973) = 0.0054.$$



6. $z_{\text{test}} = 2.78 > 2.575 = z_{\text{crit}}$: the test statistic is in the rejection region, out of tolerance. Equivalently, $p = 0.0054 < 0.01 = \alpha$.

7. Reject H_0. There is sufficient evidence to warrant rejection of the claim that 70% of vehicles pass the emissions test on the first try, at the significance level $\alpha = 0.01$. The records do appear to be out of date, and a more in-depth test is justified.

Example 30.5 ▪ Travel time: a small-sample t-test

Worksheet 30, Example 5

A cégep wishes to relocate its campus. One concern is that the travel time for students, faculty and staff will increase at the new campus. Currently, the average trip to the college takes 40 minutes. A random sample of 29 students, faculty and staff members was asked to measure travel times to the proposed site for the new campus. If $\bar{x} = 43$ min and $s = 11$, is there significant evidence at the $\alpha = 0.05$ level to conclude that the average travel time will increase? Assume that travel times are normally distributed, then find and use the rejection region to answer. Can you find the p-value?

Solution.

1. Let X be the travel time to the proposed site, in minutes, and let μ be its mean.

2. $H_0 : \mu = 40 \qquad H_A : \mu > 40$ ("will increase": a right-tailed test).

3. $\alpha = 0.05$.

4. The sample is small, $n = 29 < 30$, so the Central Limit Theorem cannot be used. But travel times are normally distributed and σ is unknown, so the test statistic follows a t-distribution with $\nu = n - 1 = 28$ degrees of freedom:

$$t_{\text{test}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{43 - 40}{11/\sqrt{29}} \approx 1.469.$$

5. a) *Rejection region.* It is the right tail of area 0.05 of the t curve with 28 degrees of freedom. From

Table B.2,

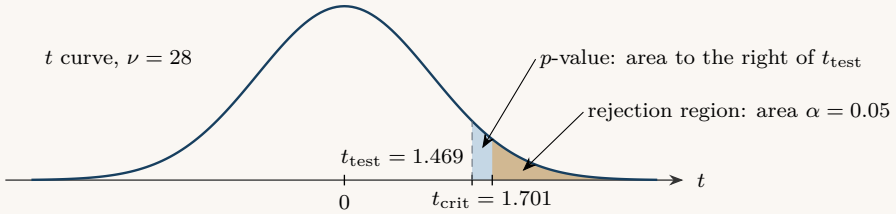
$$t_{\text{crit}} = t_{0.05,\,28} = 1.701,$$

and H_0 is rejected if $t_{\text{test}} \geq 1.701$.

b) *p-value.* $p = P(T \geq 1.469)$ for $\nu = 28$. Table B.2 does not give this area exactly, but in the row $\nu = 28$ the test statistic lies between two entries,

$$t_{0.10,\,28} = 1.313 < 1.469 < 1.701 = t_{0.05,\,28}, \qquad \text{so} \qquad 0.05 < p < 0.10.$$

(Software gives $p \approx 0.0765$.)



6. a) $t_{\text{test}} = 1.469 < 1.701 = t_{\text{crit}}$: the test statistic is not in the rejection region.

b) $p > 0.05 = \alpha$.

7. Do not reject H_0. There is insufficient evidence to warrant rejection of the claim that the mean travel time is 40 minutes, at the significance level $\alpha = 0.05$. A sample mean of 43 minutes is still tolerable at this level, so the data do not allow us to conclude that the average travel time will increase.

The evidence is not negligible, though: the p-value is below 0.10, so the same data would have led to rejection at the 10% level. This is the extra information that a p-value carries.

Example 30.6 * ▪ Type I and type II errors

Extra problems on hypothesis testing, #1–2

a) Before agreeing to purchase a large order of polyethylene sheaths for a particular type of high-pressure oil-filled submarine power cable, a company wants to see conclusive evidence that the true mean sheath thickness is less than 1 mm. Give the null and the alternate hypotheses. In this context, what are the type I and the type II errors?

b) Let the test statistic Z have a standard normal distribution when H_0 is true. Give the significance level for each of the following situations.

i) $H_A\colon \mu > \mu_0$, rejection region $z \geq 1.88$

ii) $H_A\colon \mu < \mu_0$, rejection region $z \leq -2.75$

iii) $H_A\colon \mu \neq \mu_0$, rejection region $z \geq 2.88$ or $z \leq -2.88$

Solution.

a) Let μ be the true mean sheath thickness, in mm. The company wants evidence *that* $\mu < 1$, so this is the alternate hypothesis, and the null hypothesis is the corresponding equality:

$$H_0\colon \mu = 1 \qquad H_A\colon \mu < 1.$$

- *Type I error* (rejecting a true H_0): concluding that the mean thickness is less than 1 mm when in fact it is not. The company buys sheaths that are not thin enough.
- *Type II error* (not rejecting a false H_0): failing to conclude that the mean thickness is less than 1 mm when in fact it is. The company turns down sheaths that are thin enough.

b) The significance level is the probability of a type I error: the probability that Z lands in the rejection region when H_0 is true. It is the area of the rejection region under the standard normal curve (Table B.1).

i) $\alpha = P(Z \geq 1.88) = 1 - \Phi(1.88) = 1 - 0.9699 = 0.0301$, so $\alpha \approx$ 0.03.

ii) $\alpha = P(Z \leq -2.75) = \Phi(-2.75) = 0.0030$, so $\alpha \approx$ 0.003.

iii) Both tails belong to the rejection region:

$$\alpha = P(Z \geq 2.88) + P(Z \leq -2.88) = 2\,\Phi(-2.88) = 2(0.0020) = 0.0040,$$

so $\alpha \approx \boxed{0.004}$.

31 Differences of Means

The confidence intervals of Section 29 and the tests of Section 30 deal with one population at a time. This section applies both methods to the situation where we wish to investigate statistically whether there is a *difference in mean values* between two different populations X and Y. Only large samples are considered.

31.1 The distribution of the difference of two sample means

Notation (the assumptions)

- $X_1, X_2, \ldots, X_{n_x}$ is an i.i.d. random sample of size n_x from a distribution with mean μ_X and variance σ_X^2.
- $Y_1, Y_2, \ldots, Y_{n_y}$ is an i.i.d. random sample of size n_y from a distribution with mean μ_Y and variance σ_Y^2.
- The X and Y samples are independent.

Here *i.i.d.* stands for *independent and identically distributed*. The two sample sizes do not have to be equal: in general $n_x \neq n_y$.

Let $\bar{X}_{n_x}$ and $\bar{Y}_{n_y}$ be the respective sample means. If n_x and n_y are large, the Central Limit Theorem (Section 28) tells us that

$$\bar{X}_{n_x} \overset{\text{approx}}{\sim} \mathrm{N}\Big(\mu_X,\ \frac{\sigma_X}{\sqrt{n_x}}\Big) \qquad \text{and} \qquad \bar{Y}_{n_y} \overset{\text{approx}}{\sim} \mathrm{N}\Big(\mu_Y,\ \frac{\sigma_Y}{\sqrt{n_y}}\Big).$$

To investigate the difference in mean values we create a new random variable, $W = \bar{X} - \bar{Y}$.

Theorem (distribution of $W = \bar{X} - \bar{Y}$)

Under the assumptions above, if n_x and n_y are large, then $W = \bar{X} - \bar{Y}$ is approximately normal:

$$W \overset{\text{approx}}{\sim} \mathrm{N}(\mu_W,\ \sigma_W),$$

where

$$\mu_W = E(W) = \mu_X - \mu_Y \qquad \text{and} \qquad \sigma_W = \sqrt{V(W)} = \sqrt{V(\bar{X}_{n_x}) + V(\bar{Y}_{n_y})} = \sqrt{\frac{\sigma_X^2}{n_x} + \frac{\sigma_Y^2}{n_y}}.$$

Proof

$W = (1)\,\bar{X} + (-1)\,\bar{Y}$ is a linear combination of two independent, approximately normal random variables, so W itself is approximately normal (Section 26). The rules for the mean and the variance of a linear combination of independent random variables give

$$E(W) = E(\bar{X}) - E(\bar{Y}) = \mu_X - \mu_Y,$$
$$V(W) = (1)^2\,V(\bar{X}) + (-1)^2\,V(\bar{Y}) = V(\bar{X}_{n_x}) + V(\bar{Y}_{n_y}) = \frac{\sigma_X^2}{n_x} + \frac{\sigma_Y^2}{n_y},$$

since a sample mean has the mean of its population and variance σ^2/n (Section 27). □

Caution (the variances add)

The means are subtracted, but the variances are *added*, because the coefficient -1 is squared. The difference of two sample means varies more than either sample mean does on its own. Add the variances, then take the square root; the standard deviations themselves do not add.

In practice σ_X and σ_Y are unknown. For large samples they are replaced by the sample standard deviations s_x and s_y, which introduces very little error (Section 29):

$$\sigma_W \approx \sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}.$$

31.2 Confidence interval for $\mu_X - \mu_Y$

Formula (confidence interval for the difference of two means)

For large samples, a $100(1-\alpha)\%$ confidence interval for $\mu_X - \mu_Y$ is

$$(\bar{x} - \bar{y}) \pm z_{\alpha/2}\sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}.$$

The derivation follows the one for a single mean. Since W is approximately $\mathrm{N}(\mu_W, \sigma_W)$,

$$P\Big(-z_{\alpha/2} \le \frac{W - \mu_W}{\sigma_W} \le z_{\alpha/2}\Big) \approx 1 - \alpha.$$

Isolating $\mu_W = \mu_X - \mu_Y$ in the middle of the inequality gives

$$P\big(W - z_{\alpha/2}\,\sigma_W \;\le\; \mu_X - \mu_Y \;\le\; W + z_{\alpha/2}\,\sigma_W\big) \approx 1 - \alpha.$$

Replacing W by its observed value $\bar{x} - \bar{y}$, and σ_W by its estimate, gives the formula.

Remark (the same pattern as for one mean)

Compare with the interval for a single mean, $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ (or the same with $s/\sqrt{n}$). Both intervals have the form

$$\text{estimate} \;\pm\; z_{\alpha/2} \times (\text{standard deviation of the estimator}).$$

	Estimate	**Standard deviation of the estimator**
One mean μ	$\bar{x}$	$\sigma_{\bar{X}} = \dfrac{\sigma}{\sqrt{n}}$ or $\dfrac{s}{\sqrt{n}}$
Difference $\mu_X - \mu_Y$	$\bar{x} - \bar{y}$	$\sigma_W = \sqrt{\dfrac{s_x^2}{n_x} + \dfrac{s_y^2}{n_y}}$

The interval estimates a *difference*, so its sign carries information. If the whole interval is negative, the data indicate that $\mu_X < \mu_Y$; if it is entirely positive, that $\mu_X > \mu_Y$. If the interval contains 0, the samples do not show that the two means differ. The formula sheet (Appendix A) writes the same formulas with subscripts 1 and 2 instead of x and y.

31.3 Hypothesis test for $\mu_X - \mu_Y$

Similarly, we can conduct hypothesis testing on the difference of two means. The null hypothesis claims a value Δ for the difference. The most common choice is $\Delta = 0$: "there is no difference between the two means".

Formula (large-sample test for the difference of two means)

$$H_0\colon\ \mu_X-\mu_Y=\Delta \qquad H_A\colon\ \begin{cases}\mu_X-\mu_Y<\Delta & \text{(left-tailed), or}\\ \mu_X-\mu_Y>\Delta & \text{(right-tailed), or}\\ \mu_X-\mu_Y\neq\Delta & \text{(two-tailed)}\end{cases}$$

$$z_{\text{test}}=\frac{(\bar{x}-\bar{y})-\Delta}{\sqrt{\dfrac{s_x^2}{n_x}+\dfrac{s_y^2}{n_y}}}$$

The test statistic has the usual form, (observed value − value claimed by H_0) divided by the standard deviation: the observed value of W is $\bar{x}-\bar{y}$, H_0 claims that its mean is Δ, and its standard deviation is σ_W. Because both samples are large, the Central Limit Theorem applies and the statistic is compared with the standard normal curve: this is a z-test, not a t-test. From here on the test runs exactly as in Section 30, with the same seven steps, the same critical values ($-z_\alpha$, z_α or $\pm z_{\alpha/2}$) and the same p-values.

Note

- The hypotheses are statements about the *population* means: μ_X is the mean of population X, which is also the mean of the distribution of the sample mean $\bar{X}_{n_x}$. The test statistic is computed from the *sample* means $\bar{x}$ and $\bar{y}$.
- H_0 must be an equality here as well: $\mu_X-\mu_Y=\Delta$, and usually $\mu_X-\mu_Y=0$.
- Decide at the start which population is X and which is Y, and keep to it. The sign of $\bar{x}-\bar{y}$ and the direction of H_A both depend on that choice.

31.4 Worked examples

The first example is a confidence interval. The second is a complete test, laid out in the seven steps of Section 30.

Example 31.1 ▪ Tensile strength: a confidence interval Worksheet 31, Example 1

The tensile strength (the pulling force required to snap an object) of two grades of wire rod are given:

grade X	$n_x=129$	$\bar{x}=107.6$ kg/mm^2	$s_x=1.3$
grade Y	$n_y=115$	$\bar{y}=123.6$ kg/mm^2	$s_y=2.0$

Give a 95% confidence interval for the difference in means of tensile strength for the two grades of wire.

Solution. Both samples are large ($n_x=129$ and $n_y=115$ are at least 30), and $\bar{x}$, $\bar{y}$, s_x, s_y are given, so the interval is

$$(\bar{x}-\bar{y})\pm z_{\alpha/2}\sqrt{\frac{s_x^2}{n_x}+\frac{s_y^2}{n_y}}.$$

Critical value. For 95% confidence, $\alpha=0.05$ and

$$z_{\alpha/2}=z_{0.025}=\Phi^{-1}(1-0.025)=\Phi^{-1}(0.975)=1.96.$$

Estimate and its standard deviation.

$$\bar{x}-\bar{y}=107.6-123.6=-16.0, \qquad \sqrt{\frac{1.3^2}{129}+\frac{2.0^2}{115}}=\sqrt{0.01310+0.03478}\approx 0.2188.$$

Interval.

$$-16.0\pm 1.96\,(0.2188)=-16.0\pm 0.43, \qquad \text{that is,} \qquad \boxed{(-16.43,\ -15.57)}\ \text{kg/mm}^2.$$

With 95% confidence, the mean tensile strength of grade X is between 15.57 and 16.43 kg/mm^2 *lower* than that of grade Y. The interval lies entirely below 0, so the data leave little doubt that the two grades differ.

Example 31.2 ▪ Married physicians: a right-tailed test Worksheet 31, Example 2

Of 215 male physicians who were Harvard graduates, 125 were married for 30 years or more and lived an average of 48.9 years after graduation with a (sample) standard deviation of 14.6 years. The other 90 were either unmarried or married for less than 30 years. They lived an average of 43.2 years beyond graduation with a (sample) standard deviation of 14.4 years. If all the physicians were the same age at graduation, is there enough evidence to suggest that those who are married for 30 years or more live longer? Test at the $\alpha = 0.01$ level of significance.

Solution.

1. Let X be the number of years lived after graduation by a physician married for 30 years or more, and Y the number of years lived after graduation by a physician who was unmarried or married for less than 30 years. Their means are μ_X and μ_Y. The data are

$$n_x = 125, \quad \bar{x} = 48.9, \quad s_x = 14.6 \qquad \text{and} \qquad n_y = 90, \quad \bar{y} = 43.2, \quad s_y = 14.4.$$

2. $H_0 \colon \mu_X - \mu_Y = 0$ (no difference, so $\Delta = 0$; H_0 must be an equality)
$H_A \colon \mu_X - \mu_Y > 0$ (those married for 30 years or more live longer: a right-tailed test).

3. $\alpha = 0.01 = 1\%$.

4. $n_x = 125$ and $n_y = 90$ are both greater than 30, so the Central Limit Theorem applies and the test statistic is a z-score, not a t-score:

$$z_{\text{test}} = \frac{(\bar{x} - \bar{y}) - \Delta}{\sqrt{\dfrac{s_x^2}{n_x} + \dfrac{s_y^2}{n_y}}} = \frac{(48.9 - 43.2) - 0}{\sqrt{\dfrac{14.6^2}{125} + \dfrac{14.4^2}{90}}} = \frac{5.7}{\sqrt{1.7053 + 2.3040}} = \frac{5.7}{2.0023} \approx 2.85.$$

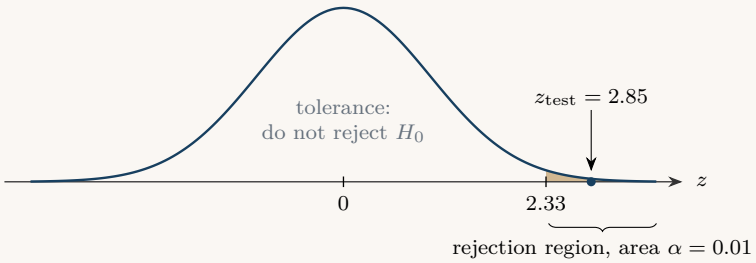
5. a) *Rejection region.* The test is right-tailed, so

$$z_{\text{crit}} = z_{0.01} = \Phi^{-1}(1 - 0.01) = 2.33,$$

and H_0 is rejected if $z_{\text{test}} \geq 2.33$.

b) *p-value.* Values at least as extreme as the observed one lie further to the right:

$$p = P(Z \geq z_{\text{test}}) = P(Z \geq 2.85) = 1 - P(Z < 2.85) = 1 - 0.9978 = 0.0022.$$



6. a) $z_{\text{test}} = 2.85 > 2.33 = z_{\text{crit}}$: the test statistic is in the rejection region.
b) $p = 0.0022 < 0.01 = \alpha$.

7. Reject H_0. There is sufficient evidence to warrant rejection of the claim that there is no difference between the mean lifetimes of the two populations, at the significance level $\alpha = 0.01$. The data support the statement that physicians married for 30 years or more live longer on average.

The conclusion does not depend on how $z_{0.01}$ is rounded: with the midpoint value 2.325 the comparison reads $2.85 > 2.325$. Note also what the test does and does not show. It shows that the difference between the two sample means is too large to be put down to chance; it does not show that a long marriage is the *cause* of the longer life.

32 Chi-Squared Tests: Goodness of Fit and Independence

The tests of Sections 30 and 31 deal with a claim about one number: a mean, a proportion, or a difference between two of them. This section tests claims that involve a whole table of counts, and it uses a new family of continuous distributions, the chi-squared distributions. They arise in many places, but here they serve two applications only:

I. determining the likelihood that a given sample comes from a given distribution (the *goodness of fit* of the sample to the distribution);

II. determining the likelihood that two variables are *independent*.

Both tests compare the counts that were observed with the counts that the claim leads us to expect.

32.1 The chi-squared distribution

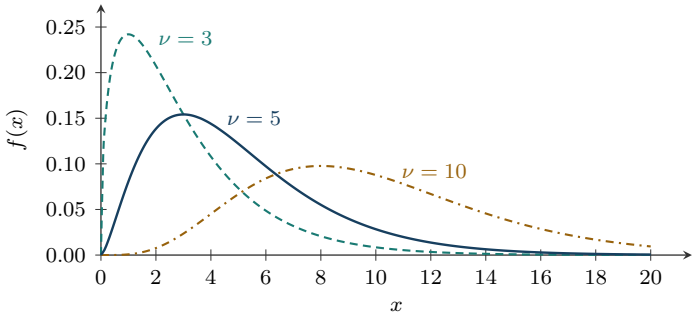
So far the common continuous distributions have been the uniform, the exponential, the normal and the Student t distributions. The χ^2-distributions are another family (χ is the Greek letter *chi*, pronounced to rhyme with "sky"). Like the Student t-distribution, a χ^2-distribution depends on a parameter ν called the *degrees of freedom*.

Definition (chi-squared distribution)

A continuous random variable X has a *chi-squared distribution with ν degrees of freedom* ($\nu = 1, 2, 3, \dots$) if its pdf is

$$f(x) = \begin{cases} \dfrac{x^{\nu/2-1}\, e^{-x/2}}{2^{\nu/2}\, \Gamma(\nu/2)}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

The denominator, in which Γ is the gamma function, is only the constant that makes the total area under the curve equal to 1. The formula is given for completeness: you do not need to know it, because every area that we need is read from a table.



Chi-squared densities for three values of ν. (For $\nu = 1$ and $\nu = 2$ the curve has no hump: it only decreases.)

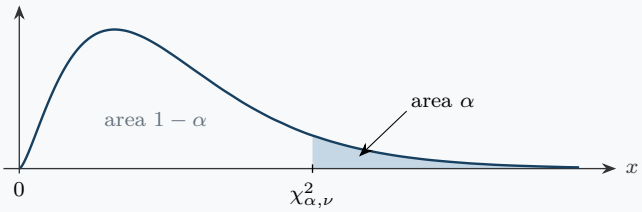
Property (shape of the chi-squared curves)

- A chi-squared random variable is never negative: all of the area lies over $x \ge 0$.
- The distributions are **not symmetric**: they are skewed right, with a long tail on the right.
- There is one curve for each value of ν. As ν increases the curve moves to the right and flattens out; in fact $E(X) = \nu$ and $V(X) = 2\nu$.
- The sum of the squares of ν independent standard normal random variables, $Z_1^2 + Z_2^2 + \cdots + Z_\nu^2$, has the chi-squared distribution with ν degrees of freedom. This is why the distribution turns up when squared standardized deviations are added together, as they are in the test statistics of this section.

Notation (critical values $\chi^2_{\alpha,\nu}$)

$\chi^2_{\alpha,\nu}$ is the value that has area α to its *right* under the chi-squared curve with ν degrees of freedom:

$$P\big(X \geq \chi^2_{\alpha,\nu}\big) = \alpha, \qquad \text{that is,} \qquad \alpha = 1 - F\big(\chi^2_{\alpha,\nu}\big).$$



The areas under the chi-squared curves are given in Table B.3, which is read like the Student t table (Table B.2): go to the row of the degrees of freedom ν and to the column of the right-tail area α. The entry found there is a value of the random variable, not a probability. Here is the corner of the table that is used most often.

ν	$\chi^2_{0.10}$	$\chi^2_{0.05}$	$\chi^2_{0.025}$	$\chi^2_{0.01}$
1	2.706	3.841	5.024	6.635
2	4.605	**5.991**	7.378	9.210
3	**6.251**	7.815	9.348	11.345
4	7.779	9.488	11.143	13.277
5	9.236	11.070	12.832	15.086

For example, $\chi^2_{0.05,2} = 5.991$ and $\chi^2_{0.10,3} = 6.251$: a chi-squared variable with 3 degrees of freedom exceeds 6.251 with probability 0.10.

Note (two more things about the table)

Because the curve is not symmetric, a left-hand critical value cannot be obtained by changing a sign, as it can for z and t. The table therefore also has columns with α close to 1 (for instance $\chi^2_{0.95,\nu}$ has area 0.95 to its right, hence 0.05 to its left); the tests in this section never need them. The table can also be read backwards to bracket a probability: with $\nu = 4$, the value 16.86 lies between $\chi^2_{0.005,4} = 14.860$ and $\chi^2_{0.001,4} = 18.466$, so $0.001 < P(X \geq 16.86) < 0.005$.

32.2 Goodness of fit: the chi-squared statistic

In its basic form, a goodness-of-fit problem is about several proportions at once. A population is divided into k different subgroups, and a claim is made about the proportion of the population in each subgroup. A random sample is then sorted into the same subgroups. If the claim is right, the number of sample members found in each subgroup should be close to what the claimed proportions predict.

Theorem (multiple proportions)

Suppose that a population is divided into k different subgroups with proportions $p_1, p_2, \ldots, p_k$ (necessarily $\sum p_i = 1$), and that a random sample of size n yields $n_1, n_2, \ldots, n_k$ individuals of each subgroup. Then the statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(n_i - np_i)^2}{np_i},$$

taken over all possible samples of size n, follows approximately a chi-squared distribution with

$$\nu = k - 1 \text{ degrees of freedom,} \qquad \text{provided that } np_i \geq 5 \text{ for every } i.$$

Formula (the chi-squared statistic)

The statistic is usually stated and remembered in this form:

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{e_i}, \qquad o_i = n_i, \quad e_i = np_i.$$

It is the same formula as in the theorem. The count n_i is what was *observed* in subgroup i, and np_i is what is *expected* there: each of the n sampled individuals belongs to subgroup i with probability p_i, so the number that do is a $\text{Bin}(n, p_i)$ random variable, and its expected value is np_i (Section 17).

What the statistic measures:

- Every term is a squared deviation, so $\chi^2 \geq 0$, and $\chi^2 = 0$ only when every observed count is exactly equal to its expected count (a perfect fit).
- The further the observed counts are from the expected counts, in either direction, the larger χ^2 becomes. A large value of χ^2 means a poor fit.
- Dividing by e_i puts the deviations on a common scale: being off by 10 is a great deal when 10 were expected and very little when 1000 were expected.

Caution (counts, not percentages)

The o_i and e_i are *counts*. Never put proportions or percentages into the formula. The expected counts do not have to be whole numbers (906.1875 is a perfectly good expected count), and they should not be rounded before they are used. As a check, the expected counts add up to n, just as the observed counts do.

Remark (why $\nu = k - 1$)

The observed counts and the expected counts both add up to n, so the k deviations $o_i - e_i$ always add up to 0. Once $k - 1$ of the deviations are known, the last one is forced. Only $k - 1$ of them are free to vary, and that is the number of degrees of freedom.

Remark (two subgroups: the z-test in disguise)

For $k = 2$ write $p_1 = p$, $p_2 = q = 1 - p$, and let x be the number of sample members in the first subgroup, so that $n - x$ are in the second. Since $(n - x) - nq = -(x - np)$,

$$\chi^2 = \frac{(x - np)^2}{np} + \frac{(x - np)^2}{nq} = \frac{(x - np)^2\,(q + p)}{npq} = \left(\frac{x - np}{\sqrt{npq}}\right)^2 = z^2,$$

the square of the test statistic for one proportion (Section 30). This is why $\nu = 1$ here, and why the critical values agree: $\chi^2_{0.05,1} = 3.841 \approx (1.96)^2 = z^2_{0.025}$. Squaring removes the sign, so both tails of the z-test land in the *right* tail of χ^2.

32.3 The goodness-of-fit test

The theorem turns the question "does the sample fit the claim?" into a hypothesis test of the kind studied in Section 30.

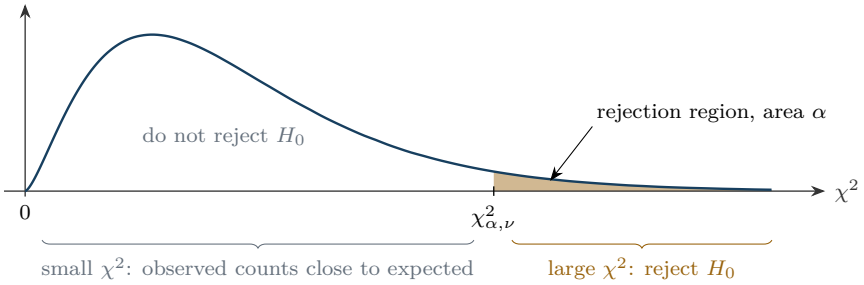
Hypothesis tests using χ^2: goodness of fit

H_0:	The population proportions (the p_i's) are as claimed.
H_A:	At least one of the p_i's is not as claimed.
Test statistic:	$\chi^2_{\text{test}} = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{e_i}$ with $e_i = np_i$ (every $e_i \geq 5$).
Rejection region:	$\chi^2 \geq \chi^2_{\alpha,\,k-1}$.

We write χ^2_{test} for the value of the statistic computed from the sample and $\chi^2_{\text{crit}} = \chi^2_{\alpha,\,k-1}$ for the critical value, so the decision rule reads: reject H_0 if $\chi^2_{\text{test}} \geq \chi^2_{\text{crit}}$.

Note (always right-tailed)

The rejection region of a chi-squared test is *always* the right tail. There is no left-tailed and no two-tailed version. This makes sense, because the statistic measures the deviation of the sample from the standard set by H_0, and the deviations are squared: too many or too few individuals in any subgroup both push χ^2 up. A small value of χ^2 says that the sample agrees with the claim, which is no reason to reject it. So the whole of α goes into the right tail: look up $\chi^2_{\alpha,\nu}$, never $\chi^2_{\alpha/2,\nu}$.



A chi-squared test is carried out with the standard seven steps of Section 30. Step 4 now consists of a table of observed and expected counts, a check that every expected count is at least 5, and the computation of χ^2_{test}. Step 5 needs the number of degrees of freedom. What changes from one problem to the next is what the subgroups are (from now on we call them *classes*) and where their expected counts come from.

Where the expected counts come from

Claimed proportions	The proportions p_i are given (percentages, ratios such as $9:3:3:1$, "equally likely"). Then $e_i = np_i$. (Examples 32.1 and 32.2)
Discrete distribution, finite support	The classes are the values x of the support, and $p_i = P(X = x)$ comes from the pmf of the claimed distribution. Then $e_i = n \cdot P(X = x)$. (Example 32.3)
Continuous distribution, or infinite support	There are no ready-made classes, so the support has to be split into intervals. Split it into k *equiprobable* intervals, each of probability $1/k$ under the claimed distribution; the cut points come from the inverse cdf. Then every $e_i = n/k$, and o_i is the number of observations that fall in the i-th interval. (Example 32.4)

In every case $\nu = k - 1$, where k is the number of classes, as long as the claimed distribution is completely specified in advance (as it always is in this section).

Remark (how many intervals?)

With a continuous distribution the number of intervals is our choice, and it matters. The more intervals we choose, the more finely the sample is compared with the claimed distribution, so the more closely the sample has to follow that distribution in order to pass the test. (With a single interval every sample would pass.) On the other hand, each interval must keep an expected count of at least 5, that is $n/k \geq 5$, so at most $n/5$

equiprobable intervals can be used.

Caution (check the expected counts)

The chi-squared curve is only an approximation to the true behaviour of the statistic, and the approximation is reliable only if $e_i \geq 5$ in *every* class. The condition is on the expected counts: an observed count may be small, even 0. If some expected counts are below 5, the usual remedy is to merge neighbouring classes until every expected count reaches 5 (each merger lowers k, and ν with it), or to take a larger sample.

Caution ("fail to reject" is not "H_0 is true")

When $\chi^2_{\text{test}} < \chi^2_{\text{crit}}$ we fail to reject H_0. That means the sample is *consistent* with the claimed distribution. It does not prove that the population follows that distribution, since the same sample would also be consistent with many slightly different distributions. Word the conclusion as "there is insufficient evidence to reject the claim that ...".

32.4 Testing independence with a contingency table

The second application of χ^2 is to test whether two variables are independent. The data are arranged in a table of a kind already met with conditional probability (Section 13).

Definition (two-way contingency table)

A *two-way contingency table* classifies every individual of a sample according to two variables at the same time. The categories of one variable label the rows, the categories of the other label the columns, and each cell contains the number of individuals with that combination. The margins contain the row totals, the column totals and the grand total.

The assumption behind the test is that if two variables are independent, then the proportions across any row are the same as across every other row, and the proportions down any column are the same as down every other column. The following table shows a sample in which the intention to vote Liberal or PQ is independent of age.

Age	Liberal	PQ	Total	Liberal share of the row
$[20, 30)$	40	60	100	$40/100 = 0.40$
$[30, 40)$	72	108	180	$72/180 = 0.40$
$[40, 50)$	92	138	230	$92/230 = 0.40$
$[50, 60)$	96	144	240	$96/240 = 0.40$
$[60, 70)$	80	120	200	$80/200 = 0.40$
$[70, 80)$	20	30	50	$20/50 = 0.40$
Total	400	600	1000	$400/1000 = 0.40$

So age is independent of voting intention. In the language of conditional probability, knowing a person's age group does not change the probability of either voting intention, and knowing the voting intention does not change the probability of any age group:

$$P(\text{Liberal} \mid \text{any given age group}) = 0.40 = P(\text{Liberal}),$$
$$P(\text{PQ} \mid \text{any given age group}) = 0.60 = P(\text{PQ}),$$
$$P(\text{age } [40, 50) \mid \text{PQ}) = \tfrac{138}{600} = 0.23 = \tfrac{230}{1000} = P(\text{age } [40, 50)), \quad \text{and so on.}$$

This is exactly the definition of independence, $P(A \mid B) = P(A)$ (Section 14).

A real sample hardly ever has exactly equal proportions, even when the variables are independent. The test measures how far the observed table is from the table with the *same totals* in which the proportions are exactly equal.

Formula (expected counts under independence)

In a contingency table with grand total N, the expected count of a cell, if the two variables are independent, is

$$\text{expected} = \frac{(\text{row total}) \times (\text{column total})}{\text{grand total}} = \frac{\text{column total}}{N} \times (\text{row total}).$$

There are two ways to see this. First, under independence every row must be shared out among the columns in the same proportions as the totals: if a fraction (column total)/N of the whole sample is in a column, the same fraction of each row should be in it. Second, by the multiplication rule for independent events, $P(\text{row and column}) = P(\text{row})\,P(\text{column}) \approx \frac{\text{row total}}{N} \cdot \frac{\text{column total}}{N}$, and multiplying by N gives the expected count. In the voting table every observed count already equals its expected count (for instance $\frac{400}{1000}(230) = 92$), so $\chi^2 = 0$.

Hypothesis tests using χ^2: independence

H_0:	The two variables are independent.
H_A:	The two variables are not independent.
Test statistic:	$\chi^2_{\text{test}} = \sum_{\text{all cells}} \frac{(\text{obs} - \text{exp})^2}{\text{exp}}$ (every expected count at least 5).
Degrees of freedom:	$\nu = (r-1)(c-1)$ for a table with r rows and c columns, totals not counted. For a table with two columns this is $\nu = \#\,\text{rows} - 1$.
Rejection region:	$\chi^2 \geq \chi^2_{\alpha,\nu}$ (the right tail, as always).

Remark (why $\nu = (r-1)(c-1)$)

The expected table has the same row and column totals as the observed table, so the deviations obs − exp add up to 0 along every row and down every column. Once the cells of the first $r-1$ rows and $c-1$ columns are filled in, all the others are forced by the totals. A table with 5 rows and 2 columns therefore has $\nu = 4 \cdot 1 = 4$, and a table with 2 rows and 5 columns also has $\nu = 1 \cdot 4 = 4$.

How to carry out a test of independence

1. Find the totals: every row total, every column total and the grand total.
2. Compute the expected counts for the first row. We want each row to match the proportions in the totals, because independence is what H_0 assumes: multiply the row total by (column total)/(grand total) for each column.
3. Do the same with the other rows. Each row of expected counts must add up to the same row total as before.
4. Calculate χ^2_{test}, adding $(\text{obs} - \text{exp})^2/\text{exp}$ over all the cells (not over the totals).
5. Compare with $\chi^2_{\text{crit}} = \chi^2_{\alpha,\nu}$, where $\nu = (r-1)(c-1)$.

32.5 Worked examples: goodness of fit

Example 32.1 ▪ EmmyNem candies: three claimed proportions Worksheet 32, Examples 1–2

EmmyNem candies claim that 30% of all candies produced are orange, 60% are brown and 10% are yellow. A randomly selected package with 100 candies has 50 orange, 45 brown and 5 yellow.

a) Calculate the sample statistic χ^2.

b) Use χ^2 at the $\alpha = 0.05$ level to test the likelihood that the sample was drawn from a population with the claimed distribution.

Solution.

a) The package is a sample of $n = 100$ candies sorted into $k = 3$ colours. The expected counts are $e_i = np_i$: $100(0.30) = 30$ orange, $100(0.60) = 60$ brown and $100(0.10) = 10$ yellow.

Colour	orange	brown	yellow	Total
Observed o_i	50	45	5	100
Expected $e_i = np_i$	30	60	10	100

$$\chi^2_{\text{test}} = \frac{(50-30)^2}{30} + \frac{(45-60)^2}{60} + \frac{(5-10)^2}{10} = 13.333 + 3.750 + 2.500 \approx \boxed{19.583}.$$

b) **1.** Let p_1, p_2, p_3 be the true proportions of orange, brown and yellow candies among all EmmyNem candies.

2. H_0: $p_1 = 0.30,\ p_2 = 0.60,\ p_3 = 0.10$ (the proportions are as claimed). H_A: at least one of the p_i's is not as claimed.

3. $\alpha = 0.05$.

4. $\chi^2_{\text{test}} = 19.583$, from part a). The expected counts 30, 60 and 10 are all at least 5, so the statistic can be compared with the chi-squared curve with $\nu = k - 1 = 2$ degrees of freedom.

5. $\chi^2_{\text{crit}} = \chi^2_{\alpha,\,k-1} = \chi^2_{0.05,\,3-1} = \chi^2_{0.05,\,2} = 5.991$ (Table B.3). The rejection region is $\chi^2 \geq 5.991$.

6. $\chi^2_{\text{test}} = 19.583 \gg 5.991 = \chi^2_{\text{crit}}$: the test statistic is far inside the rejection region.

7. $\boxed{\text{Reject } H_0}$. There is sufficient evidence to warrant rejection of the claim that 30% of the candies are orange, 60% are brown and 10% are yellow, at the significance level $\alpha = 0.05$. The orange term alone (13.333) is more than twice the critical value: the package has far more orange candies than the claim predicts.

Example 32.2 ▪ A dihybrid cross and Mendel's ratios

Worksheet 32, Example 3

In genetics, if a pure strain organism having genotype $AABB$ is crossed with another pure strain organism having genotype $aabb$, then the resulting offspring will necessarily have genotype $AaBb$. If these first-generation organisms are then crossed amongst themselves, the result is called a dihybrid cross.

a) How many different genotypes are possible from a dihybrid cross?

b) How many different phenotypes are possible from a dihybrid cross?

c) What are the Mendelian probabilities associated with each phenotype?

d) The following chart shows the number of organisms that resulted from a biology experiment. The biologist suspects that the original sample was not genetically pure. Test at the $\alpha = 0.1$ level the likelihood that these organisms originated from a dihybrid cross following Mendel's laws.

Phenotype	AB	Ab	aB	ab
Observed	926	288	293	104

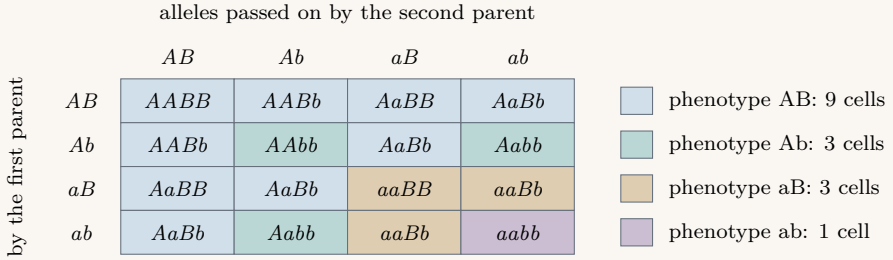
Solution. Some vocabulary first. Every organism carries two copies (*alleles*) of each gene, one from each parent. The *genotype* is the list of alleles, such as $AaBb$. The *phenotype* is the visible result: for each gene the dominant trait (capital letter) shows as soon as at least one of the two alleles is the dominant one, and the recessive trait (small letter) shows only when both alleles are recessive. In a dihybrid cross both parents are $AaBb$, and each parent passes on A or a, and independently B or b, each with probability $\frac{1}{2}$.

a) The offspring receives an allele of the first gene from each parent and an allele of the second gene from each parent, with 2 possibilities every time. By the multiplication rule (Section 8) there are $2 \times 2 \times 2 \times 2 = 16$ equally likely combinations. They are the 16 cells of the square below. Several cells hold the same genotype (for instance $AaBb$ appears four times, since it does not matter which parent supplied which allele). For

each gene the possible genotypes are AA, Aa and aa, so the number of different genotypes is

$$3 \times 3 = \boxed{9}.$$

For the probabilities in c) it is the 16 equally likely cells that count.



b) For each gene there are two possible appearances, dominant or recessive, so there are $2 \times 2 = \boxed{4}$ phenotypes: AB (both dominant traits show), Ab, aB, ab (both recessive traits show).

c) Look at one gene at a time. The cross $Aa \times Aa$ gives AA, Aa, aA, aa with probability $\frac{1}{4}$ each, so the dominant trait shows with probability $\frac{3}{4}$ and the recessive trait with probability $\frac{1}{4}$. The same holds for the second gene, and the two genes are passed on independently, so the probabilities multiply:

$$P(\text{AB}) = \tfrac{3}{4} \cdot \tfrac{3}{4} = \boxed{\tfrac{9}{16}}, \quad P(\text{Ab}) = \tfrac{3}{4} \cdot \tfrac{1}{4} = \boxed{\tfrac{3}{16}}, \quad P(\text{aB}) = \tfrac{1}{4} \cdot \tfrac{3}{4} = \boxed{\tfrac{3}{16}}, \quad P(\text{ab}) = \tfrac{1}{4} \cdot \tfrac{1}{4} = \boxed{\tfrac{1}{16}}.$$

They add up to 1, and they agree with the count of cells in the square. These are the *expected proportions* p_i for part d).

d) **1.** The sample consists of $n = 926 + 288 + 293 + 104 = 1611$ organisms in $k = 4$ classes. Let p_{AB}, p_{Ab}, p_{aB}, p_{ab} be the proportions of the four phenotypes in the population that the organisms come from.

2. H_0: the organisms follow Mendel's laws for a dihybrid cross, that is $p_{\text{AB}} = \frac{9}{16}$, $p_{\text{Ab}} = \frac{3}{16}$, $p_{\text{aB}} = \frac{3}{16}$, $p_{\text{ab}} = \frac{1}{16}$.
H_A: they do not follow them: at least one of these proportions is different (the biologist's suspicion).

3. $\alpha = 0.1$.

4. The expected counts are $e_i = np_i$, for instance $1611 \cdot \frac{9}{16} = 906.1875$.

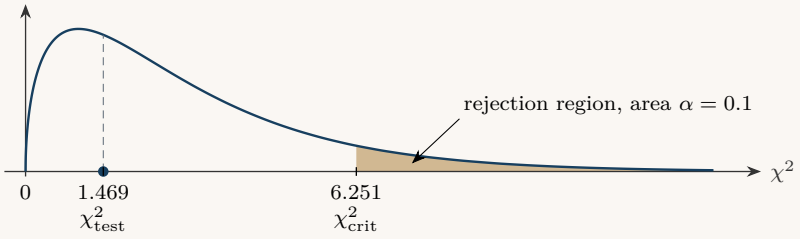
Phenotype	AB	Ab	aB	ab	Total
Observed o_i	926	288	293	104	1611
Expected $e_i = np_i$	906.1875	302.0625	302.0625	100.6875	1611

Every np_i is greater than 5, so the chi-squared approximation applies.

$$\begin{aligned}\chi^2_{\text{test}} &= \frac{(926 - 906.1875)^2}{906.1875} + \frac{(288 - 302.0625)^2}{302.0625} \\ &\quad + \frac{(293 - 302.0625)^2}{302.0625} + \frac{(104 - 100.6875)^2}{100.6875} \\ &\approx 0.433 + 0.655 + 0.272 + 0.109 \approx \boxed{1.469} \qquad \text{(more precisely 1.4687).}\end{aligned}$$

5. $\nu = k - 1 = 3$, so $\chi^2_{\text{crit}} = \chi^2_{0.1,\,4-1} = \chi^2_{0.1,\,3} = 6.251$ from Table B.3. The rejection region is $\chi^2 \geq 6.251$.

6. $\chi^2_{\text{test}} = 1.469 < 6.251 = \chi^2_{\text{crit}}$: the test statistic is not in the rejection region.



The chi-squared density with $\nu = 3$.

7. Fail to reject H_0. There is insufficient evidence to warrant rejection of the claim that the organisms originated from a dihybrid cross following Mendel's laws, at the significance level $\alpha = 0.1$. The observed counts are consistent with the ratio $9:3:3:1$, so the sample gives no support to the biologist's suspicion.

Example 32.3 ▪ Fitting a binomial distribution

Worksheet 32, Example 4

Use χ^2 to test at the 0.1 level the hypothesis that the following sample was drawn from a binomial distribution with $n = 4$ and $p = \frac{1}{3}$.

3 0 3 2 0 1 1 3 3 0

Solution. The claimed distribution is discrete with a finite support, so the classes are the values of the support and the expected proportions come from the pmf. Be careful with the letter n: here $n = 4$ is a parameter of the claimed distribution, and the sample size is 10.

1. Under the claim, each observation is a value of $X \sim \text{Bin}\left(4, \frac{1}{3}\right)$, with $\text{Supp}(X) = \{0, 1, 2, 3, 4\}$. The ten observations all belong to $\text{Supp}(X)$, and they are sorted into $k = 5$ classes, one for each value.

2. H_0: the sample is drawn from a $\text{Bin}\left(4, \frac{1}{3}\right)$ distribution. H_A: it is not drawn from that distribution.

3. $\alpha = 0.1$.

4. The observed counts are found by counting how many times each value occurs. The expected counts are $10 \cdot P(X = x)$, where $P(X = x) = \binom{4}{x}\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{4-x}$ (Section 17); for instance $10 \cdot P(X = 0) = 10\left(\frac{2}{3}\right)^4 = 10 \cdot \frac{16}{81} \approx 1.975$.

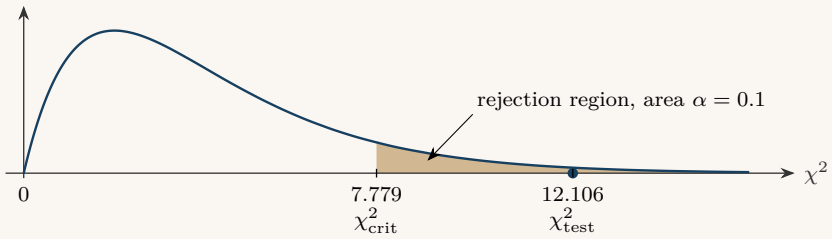
$x \in \text{Supp}(X)$	0	1	2	3	4	Total
Observed count o	3	2	1	4	0	10
$P(X = x)$	$\frac{16}{81}$	$\frac{32}{81}$	$\frac{24}{81}$	$\frac{8}{81}$	$\frac{1}{81}$	1
Expected count $e = 10 \cdot P(X = x)$	1.975	3.951	2.963	0.988	0.123	10
$(o - e)^2/e$	0.532	0.963	1.300	9.188	0.123	12.106

$$\chi^2_{\text{test}} = \sum \frac{(o - e)^2}{e} \approx 0.532 + 0.963 + 1.300 + 9.188 + 0.123 \approx \boxed{12.106}.$$

The terms are computed from the unrounded expected counts $\frac{160}{81}, \frac{320}{81}, \ldots$ (exactly, $\chi^2_{\text{test}} = \frac{1937}{160}$). If the three-decimal values of the table are used instead, the total comes out as 12.101; the conclusion is the same.

5. $\nu = k - 1 = 4$, so $\chi^2_{\text{crit}} = \chi^2_{0.1,\,4} = 7.779$. The rejection region is $\chi^2 \geq 7.779$.

6. $\chi^2_{\text{test}} = 12.106 > 7.779 = \chi^2_{\text{crit}}$: the test statistic is in the rejection region.



The chi-squared density with $\nu = 4$.

7. Reject H_0. There is sufficient evidence to warrant rejection of the claim that the sample was drawn from a binomial distribution with $n = 4$ and $p = \frac{1}{3}$, at the significance level $\alpha = 0.1$. Almost all of χ^2_{test} comes from one class: the value 3 was observed four times where about one was expected.

Caution (the condition $e_i \geq 5$ fails in Example 32.3)

Every expected count in Example 32.3 is below 5, so the condition of the theorem is not met and the chi-squared curve is only a rough guide there. The example shows how the test is set up for a discrete distribution. With real data you would merge classes until every expected count is at least 5, and with only 10 observations no grouping of the values $0, \dots, 4$ achieves that, so the honest remedy is a larger sample. (Listing all possible samples of size 10 shows that, if H_0 is true, a statistic of 12.106 or more has probability about 0.024. That is still below 0.1, so the decision to reject stands.)

Example 32.4 ▪ Fitting a normal distribution Worksheet 32, Example 5

Use χ^2 to test at the 0.05 level the hypothesis that the following sample was drawn from a normally distributed population with $\mu = 5$ and $\sigma = 2$.

0.3	0.5	1.7	1.9	2.0	3.7	4.1	4.2	4.2	4.3
4.4	4.7	5.5	5.7	5.8	5.9	7.7	7.8	8.0	8.1

Solution. The normal distribution is continuous (and its support is infinite), so the classes have to be built first: split the support into equiprobable intervals. There are $n = 20$ observations, and $20/4 = 5$, so $k = 4$ intervals is the largest number that keeps every expected count at 5.

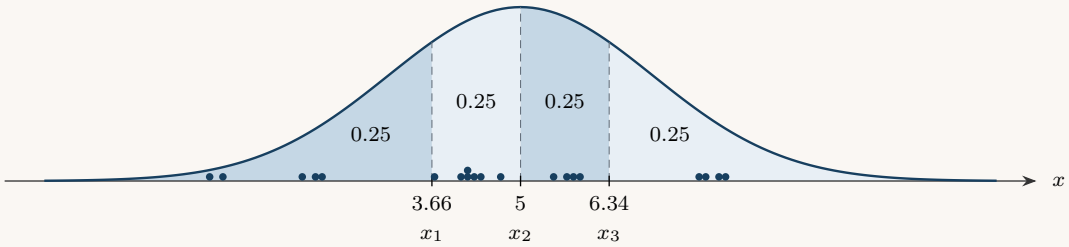
Finding the intervals. Let $X \sim \mathrm{N}(5, 2)$. We need the cut points $x_1 < x_2 < x_3$ that split the support into four intervals of the same probability:

$$P(X \leq x_1) = P(x_1 < X \leq x_2) = P(x_2 < X \leq x_3) = P(X > x_3) = 0.25,$$

that is, $F(x_1) = 0.25$, $F(x_2) = 0.5$ and $F(x_3) = 0.75$. Standardizing and using Table B.1 backwards (Section 24):

$$\begin{aligned}
\frac{x_1 - 5}{2} &= \Phi^{-1}(0.25) \approx -0.67 && \Longrightarrow \quad x_1 = 5 - 2(0.67) = 3.66,\\
\frac{x_2 - 5}{2} &= \Phi^{-1}(0.5) = 0 && \Longrightarrow \quad x_2 = 5,\\
\frac{x_3 - 5}{2} &= \Phi^{-1}(0.75) \approx 0.67 && \Longrightarrow \quad x_3 = 5 + 2(0.67) = 6.34.
\end{aligned}$$

(The table entries closest to 0.25 and 0.75 are $\Phi(-0.67) = 0.2514$ and $\Phi(0.67) = 0.7486$. A more exact value is $z = 0.6745$, which gives 3.65 and 6.35; no observation is close enough to a cut point for the difference to matter.)



The density of $\mathrm{N}(5, 2)$ cut into four intervals of probability 0.25. The dots on the axis are the 20 observations.

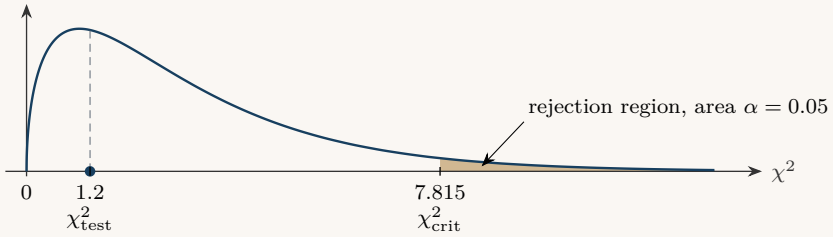
1. $X \sim \mathrm{N}(5, 2)$ under the claim; the classes are the $k = 4$ intervals $(-\infty, 3.66]$, $(3.66, 5]$, $(5, 6.34]$ and $(6.34, \infty)$.
2. H_0: the sample is drawn from a normally distributed population with $\mu = 5$ and $\sigma = 2$. H_A: it is not.
3. $\alpha = 0.05$.
4. Each interval has probability 0.25, so every expected count is $20(0.25) = 5$. Counting the observations: five are at most 3.66 (from 0.3 to 2.0), seven lie in $(3.66, 5]$ (from 3.7 to 4.7), four lie in $(5, 6.34]$ (from 5.5 to 5.9) and four are above 6.34 (from 7.7 to 8.1).

Interval	$(-\infty, 3.66]$	$(3.66, 5]$	$(5, 6.34]$	$(6.34, \infty)$	Total
Expected	5	5	5	5	20
Observed	5	7	4	4	20

$$\chi^2_{\text{test}} = \frac{(5-5)^2}{5} + \frac{(7-5)^2}{5} + \frac{(4-5)^2}{5} + \frac{(4-5)^2}{5} = 0 + 0.8 + 0.2 + 0.2 = \boxed{1.2}.$$

5. $\nu = k - 1 = 3$, so $\chi^2_{\text{crit}} = \chi^2_{0.05,\,4-1} = \chi^2_{0.05,\,3} = 7.815$. The rejection region is $\chi^2 \geq 7.815$.

6. $\chi^2_{\text{test}} = 1.2 < 7.815 = \chi^2_{\text{crit}}$: the test statistic is not in the rejection region.



The chi-squared density with $\nu = 3$.

7. $\boxed{\text{Fail to reject } H_0}$. There is insufficient evidence to warrant rejection of the claim that the sample was drawn from a normally distributed population with $\mu = 5$ and $\sigma = 2$, at the significance level $\alpha = 0.05$. The sample is consistent with that distribution.

32.6 Worked example: independence

Example 32.5 ▪ Germination and seed type Worksheet 32, Example 6

Five types of seeds were planted in an agricultural experiment. Whether the seeds had germinated or not after six weeks was recorded. Test at the $\alpha = 0.01$ level whether failure to germinate is independent of seed type.

Seed type	Germinated	Failed
1	31	7
2	57	33
3	87	60
4	52	44
5	10	19

Solution.

1. Each of the 400 seeds is classified by two variables: its seed type (5 categories, the rows of the table) and the outcome after six weeks (germinated or failed, the columns).

2. H_0: failure to germinate is independent of seed type. H_A: it is not independent of seed type.

3. $\alpha = 0.01$.

4. First find the totals: the row totals are 38, 90, 147, 96 and 29, the column totals are 237 and 163, and the grand total is 400. Overall, a fraction $\frac{237}{400}$ of the seeds germinated and $\frac{163}{400}$ failed. If the outcome is independent of the seed type, every seed type should show these same proportions. For type 1, with its 38 seeds, the expected counts are therefore $\frac{237}{400}(38) = 22.515$ and $\frac{163}{400}(38) = 15.485$. The same is done with the other rows: $\frac{237}{400}(90) = 53.325$ and $\frac{163}{400}(90) = 36.675$ for type 2, and so on.

	Germinated			Failed			
Seed type	observed	expected	term	observed	expected	term	Total
1	31	22.515	3.198	7	15.485	4.649	38
2	57	53.325	0.253	33	36.675	0.368	90
3	87	87.0975	0.000	60	59.9025	0.000	147
4	52	56.88	0.419	44	39.12	0.609	96
5	10	17.1825	3.002	19	11.8175	4.365	29
Total	237	237		163	163		400

Each row of expected counts adds up to the same row total as the observed counts, and every expected count is at least 5 (the smallest is 11.8175). The columns headed "term" contain $(\text{observed} - \text{expected})^2/\text{expected}$ for each cell. Adding the ten terms,

$$\begin{aligned}\chi^2_{\text{test}} &= \frac{(31-22.515)^2}{22.515} + \frac{(7-15.485)^2}{15.485} + \cdots + \frac{(19-11.8175)^2}{11.8175} \\ &\approx 3.198 + 4.649 + 0.253 + 0.368 + 0.000 + 0.000 + 0.419 + 0.609 + 3.002 + 4.365 \\ &\approx \boxed{16.86} \qquad \text{(or 16.85 if the expected counts are first rounded to two decimals).}\end{aligned}$$

5. The table has $r = 5$ rows and $c = 2$ columns, so $\nu = (5-1)(2-1) = 4$, the number of rows minus 1. Then $\chi^2_{\text{crit}} = \chi^2_{0.01,\,4} = 13.277$, and the rejection region is $\chi^2 \geq 13.277$.

6. $\chi^2_{\text{test}} = 16.86 > 13.277 = \chi^2_{\text{crit}}$: the test statistic is in the rejection region. (From the row $\nu = 4$ of Table B.3, $14.860 < 16.86 < 18.466$, so the p-value is between 0.001 and 0.005, below $\alpha = 0.01$.)

7. $\boxed{\text{Reject } H_0}$. There is sufficient evidence to warrant rejection of the claim that failure to germinate is independent of seed type, at the significance level $\alpha = 0.01$: the two are *not* independent. The largest contributions to χ^2_{test} come from type 1, which germinated much more often than expected (31 against 22.5), and from type 5, which germinated much less often (10 against 17.2).

A Formula Sheet and Summary of Distributions

A formula sheet is handed out with every quiz, test and the final exam. It grows during the term: the first version stops at the probability rules and $p(x) = P(X = x)$; the version for the second test adds random variables, linear combinations, the sampling distributions and the two tables of distributions; the final exam version adds the formulas for confidence intervals and hypothesis tests. Subsection A.1 reproduces the final version. Subsections A.2 and A.3 are a one-page review of all the standard distributions and of the facts about random variables that have to be known by heart.

A.1 The formula sheet provided in evaluations

Caution (what the sheet does not do)

The sheet is a memory aid. It is not an exhaustive list of the formulas needed in an evaluation, the formulas carry no labels, and nothing on it says when a formula applies: recognising the situation is your job. The labels in the left-hand columns below were added for these notes.

Descriptive statistics (Sections 4–7)

Mean of a sample, of a population	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \mu = \frac{1}{N}\sum_{i=1}^{N} x_i$
Median, odd number of data points	$m = x_{\frac{n+1}{2}}$
Median, even number of data points	$m = \frac{x_{\frac{n}{2}} + x_{(\frac{n}{2}+1)}}{2}$
Mean from a frequency table	$\bar{x} = \frac{1}{n}\sum_{j=1}^{np} f_j x_j, \qquad \mu = \frac{1}{N}\sum_{j=1}^{NP} f_j x_j$
Mean from class intervals (midpoints m_j)	$\bar{x} = \frac{1}{n}\sum_{j=1}^{nc} f_j m_j, \qquad \mu = \frac{1}{N}\sum_{j=1}^{NC} f_j m_j$
Weighted average	$\frac{\sum w_i x_i}{\sum w_i}$
Interquartile range; outlier fences	$\mathrm{IQR} = Q_3 - Q_1, \qquad [\,Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}\,]$
Sample variance (definition)	$s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2$
Sample variance (computational)	$s^2 = \frac{1}{n-1}\left(\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2\right)$
Population variance (definition)	$\sigma^2 = \frac{1}{N}\sum (x_i - \mu)^2$
Population variance (computational)	$\sigma^2 = \frac{1}{N}\left(\sum x_i^2 - \frac{1}{N}\left(\sum x_i\right)^2\right) = \frac{1}{N}\sum x_i^2 - \mu^2$
Standard deviations	$s = \sqrt{s^2}, \qquad \sigma = \sqrt{\sigma^2}$
Chebyshev's theorem (within k standard deviations)	at least $\left(1 - \frac{1}{k^2}\right) \cdot 100\%$

z-score	$z = \dfrac{x - \mu}{\sigma}$

Counting and probability (Sections 8–14)

Permutations	${}_nP_k = \dfrac{n!}{(n-k)!}$
Combinations	${}_nC_k = \binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$
Arrangements with repeated objects	$\dfrac{n!}{n_1!\,n_2!\cdots n_r!}$
De Morgan's laws	$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c$
Complement rule	$P(A^c) = 1 - P(A)$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Addition rule, three events	$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$
Conditional probability	$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$
Multiplication rule	$P(A \cap B) = P(A \mid B) \cdot P(B) = P(B \mid A) \cdot P(A)$
Independent events	$P(A \cap B) = P(A) \cdot P(B)$

Random variables (Sections 15–16 and 21)

Probability mass function	$p(x) = P(X = x)$
Cumulative distribution function	$F(x) = P(X \le x) = \displaystyle\int_{-\infty}^{x} f(t)\,\mathrm{d}t$
Expected value	$\mu_X = E(X) = \begin{cases} \sum x \cdot p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x \cdot f(x)\,\mathrm{d}x & \text{(continuous)} \end{cases}$
Function of a random variable, $Y = Y(X)$ (LOTUS)	$\mu_Y = E(Y) = \begin{cases} \sum y(x) \cdot p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} y(x) \cdot f(x)\,\mathrm{d}x & \text{(continuous)} \end{cases}$
Variance	$\sigma_X^2 = V(X) = E\big((X - \mu_X)^2\big) = E(X^2) - E(X)^2$
Linear change of variable	$E(aX + b) = aE(X) + b, \qquad V(aX + b) = a^2 V(X)$

Linear combinations and sampling distributions (Sections 26–28)

Mean of a linear combination	$E(a_1X_1 + \cdots + a_nX_n) = a_1E(X_1) + \cdots + a_nE(X_n)$
Variance of a linear combination (independent X_i)	$V(a_1X_1 + \cdots + a_nX_n) = a_1^2V(X_1) + \cdots + a_n^2V(X_n)$

Sample mean $\bar{X}_n$	$\mu_{\bar{X}_n} = \mu_X, \qquad \sigma^2_{\bar{X}_n} = \dfrac{\sigma^2_X}{n}$
Sample total T_n	$\mu_{T_n} = n \cdot \mu_X, \qquad \sigma^2_{T_n} = n \cdot \sigma^2_X$

Confidence intervals and sample sizes (Sections 29 and 31)

Mean, σ known	$\bar{x} \pm z_{\alpha/2}\,\dfrac{\sigma}{\sqrt{n}}$
Mean, large sample, σ unknown	$\bar{x} \pm z_{\alpha/2}\,\dfrac{s}{\sqrt{n}}$
Mean, small sample from a normal population	$\bar{x} \pm t_{\alpha/2,\,n-1}\,\dfrac{s}{\sqrt{n}}$
Proportion	$\hat{p} \pm z_{\alpha/2}\sqrt{\dfrac{\hat{p}\hat{q}}{n}}$
Difference of two means (large samples)	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$
Sample size for a mean, margin of error E	$n = \left(\dfrac{z_{\alpha/2}\,\sigma}{E}\right)^2$
Sample size for a proportion, margin of error E	$n = \left(\dfrac{z_{\alpha/2}}{E}\right)^2 \hat{p}\hat{q}$

Test statistics (Sections 30–32)

Mean, σ known	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$
Mean, large sample, σ unknown	$z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Mean, small sample from a normal population	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Proportion	$z = \dfrac{\hat{p} - p_0}{\sqrt{\dfrac{p_0 q_0}{n}}}$
Difference of two means, variances known; estimated	$z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}, \qquad z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$
Chi-squared statistic	$\chi^2 = \sum \dfrac{(\text{obs}_i - \text{exp}_i)^2}{\text{exp}_i}$

A.2 The standard distributions at a glance

The formula sheet also carries a table of the standard distributions. It is reproduced here in two layers: first how to recognise each distribution and what its parameters mean, then the formulas.

Discrete distributions: which one is it?

Distribution	Recognise it by	Parameters and what X counts
Uniform, $\mathrm{U}\{1,\dots,n\}$	n equally likely outcomes	X = the outcome of one trial
Binomial, $\mathrm{Bin}(n,p)$	two outcomes, fixed number of trials	n = number of trials (fixed); p = probability of success; X = number of successes
Negative binomial, $\mathrm{NB}(k,p)$	two outcomes, fixed number of successes	k = number of successes (fixed); p = probability of success; X = number of trials
Geometric, $\mathrm{NB}(1,p)$	two outcomes, one success	p = probability of success; X = number of trials until the first success
Hypergeometric, $\mathrm{H}(N,M,n)$	two outcomes, fixed number of trials, without replacement	N = population size; M = number of successes in the population; n = sample size (fixed); X = number of successes in the sample
Poisson, $\mathrm{Pois}(\lambda)$	events occurring over time (or space)	λ = number of events in a unit of time; X = number of events in that unit. Rescaling: over t units use $\lambda_t = \lambda \cdot t$

Discrete distributions: formulas

	Support	pmf $p(x)$	$\mu = E(X)$	$\sigma^2 = V(X)$
Uniform	$1,\dots,n$	$\frac{1}{n}$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
Binomial	$0,1,\dots,n$	$\binom{n}{x} p^x q^{n-x}$	np	npq
Neg. binomial	$k, k+1, \dots$	$\binom{x-1}{k-1} p^k q^{x-k}$	$\frac{k}{p}$	$\frac{kq}{p^2}$
Geometric	$1,2,\dots$	$p\,q^{x-1}$	$\frac{1}{p}$	$\frac{q}{p^2}$
Hypergeometric	$0,1,\dots,n$	$\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$	$n \cdot \frac{M}{N}$	$\frac{N-n}{N-1} \cdot n \cdot \frac{M}{N}\left(1-\frac{M}{N}\right)$
Poisson	$0,1,2,\dots$	$\frac{e^{-\lambda}\lambda^x}{x!}$	λ	λ

Throughout, $q = 1 - p$. On the printed sheet the number of successes in the population of the hypergeometric distribution is called k instead of M.

Continuous distributions: formulas

	pdf $f(x)$	**cdf** $F(x)$	$\mu = E(X)$	$\sigma^2 = V(X)$
Uniform, $\mathrm{U}(A,B)$	$\begin{cases} \dfrac{1}{B-A}, & x \in [A,B] \\ 0, & \text{otherwise} \end{cases}$	$\begin{cases} 0, & x < A \\ \dfrac{x-A}{B-A}, & x \in [A,B] \\ 1, & x > B \end{cases}$	$\dfrac{A+B}{2}$	$\dfrac{(B-A)^2}{12}$
Exponential, $\mathrm{Exp}(\lambda)$	$\begin{cases} 0, & x < 0 \\ \lambda e^{-\lambda x}, & x \geq 0 \end{cases}$	$\begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$	$\dfrac{1}{\lambda}$	$\dfrac{1}{\lambda^2}$
Normal, $\mathrm{N}(\mu,\sigma)$	$\dfrac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\Phi\left(\dfrac{x-\mu}{\sigma}\right)$ (Table B.1)	μ	σ^2

For the exponential distribution, λ = number of events in a unit of time and X = the time until an event occurs. The standard normal distribution is $Z \sim \mathrm{N}(\mu = 0, \sigma = 1)$, with cdf $\Phi(z)$.

A.3 Facts about random variables to know by heart

Discrete and continuous side by side

	Discrete X	**Continuous** X
Described by	pmf $p(x) = P(X = x)$	pdf $f(x)$; $P(X = x) = 0$ for every x
Valid when	$0 \leq p(x) \leq 1$ and $\sum_x p(x) = 1$	$f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$
cdf $F(x) = P(X \leq x)$	$\sum_{t \leq x} p(t)$	$\int_{-\infty}^{x} f(t)\,\mathrm{d}t$
Expected value $E(X)$	$\sum_x x\, p(x)$	$\int_{-\infty}^{\infty} x\, f(x)\,\mathrm{d}x$
LOTUS, $E\big(Y(X)\big)$	$\sum_x y(x)\, p(x)$	$\int_{-\infty}^{\infty} y(x)\, f(x)\,\mathrm{d}x$
Variance $V(X) = E\big[(X-\mu_X)^2\big]$	$\sum_x (x-\mu_X)^2\, p(x)$	$\int_{-\infty}^{\infty} (x-\mu_X)^2 f(x)\,\mathrm{d}x$
Computational formula	$V(X) = E(X^2) - [E(X)]^2$ (both cases)	

Property (rules that hold for every random variable)

- $P(X > x) = 1 - F(x)$.
- For a continuous random variable, $P(a \leq X \leq b) = F(b) - F(a)$, and it makes no difference whether the end points are included. For a discrete random variable the end points matter: $P(a < X \leq b) = F(b) - F(a)$.
- $E(aX + b) = aE(X) + b$ and $V(aX + b) = a^2 V(X)$.

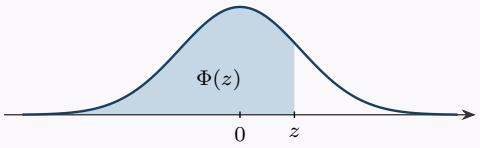
B Statistical Tables

These are the three tables supplied with the tests and the final exam. All probabilities for the normal, Student t and chi-squared distributions in this course are read from them.

B.1 Areas under the standard normal curve

How to read the table

The body of the table gives the cumulative probability $\Phi(z) = P(Z \le z)$ for $Z \sim \mathrm{N}(0,1)$: the area under the standard normal curve to the *left* of z. The row gives z to one decimal place and the column gives the second decimal. For example, $\Phi(-1.15)$ is in row -1.1, column 0.05: 0.1251.



- Area to the right: $P(Z > z) = 1 - \Phi(z) = \Phi(-z)$.
- Area between two values: $P(a \le Z \le b) = \Phi(b) - \Phi(a)$.
- Inverse look-up (Φ^{-1}): find the probability in the body of the table and read z from the margins. The critical value z_α has area α to its *right*, so $z_\alpha = \Phi^{-1}(1-\alpha)$.

Critical values used again and again:

α (area to the right)	0.10	0.05	0.025	0.01	0.005
z_α	1.28	1.645	1.96	2.33	2.575

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

B.2 Critical values of the Student t-distribution

How to read the table

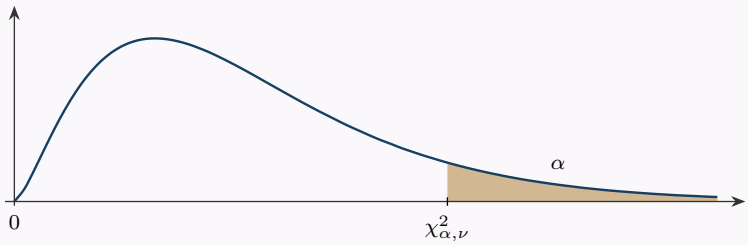
The entry in row ν and column t_α is the value $t_{\alpha,\nu}$ that has area α to its *right* under the t curve with ν degrees of freedom: $P(T > t_{\alpha,\nu}) = \alpha$. For a one-sample problem $\nu = n - 1$. The curve is symmetric, so the left-tail critical value is $-t_{\alpha,\nu}$, and a $100(1-\alpha)\%$ confidence interval uses $t_{\alpha/2,\nu}$. The last row ($\nu = \infty$) is the standard normal: $t_{\alpha,\infty} = z_\alpha$.

ν	$t_{0.20}$	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.0025}$	$t_{0.0005}$
1	1.376	3.078	6.314	12.706	31.821	63.657	127.321	636.619
2	1.061	1.886	2.920	4.303	6.965	9.925	14.089	31.599
3	0.978	1.638	2.353	3.182	4.541	5.841	7.453	12.924
4	0.941	1.533	2.132	2.776	3.747	4.604	5.598	8.610
5	0.920	1.476	2.015	2.571	3.365	4.032	4.773	6.869
6	0.906	1.440	1.943	2.447	3.143	3.707	4.317	5.959
7	0.896	1.415	1.895	2.365	2.998	3.499	4.029	5.408
8	0.889	1.397	1.860	2.306	2.896	3.355	3.833	5.041
9	0.883	1.383	1.833	2.262	2.821	3.250	3.690	4.781
10	0.879	1.372	1.812	2.228	2.764	3.169	3.581	4.587
11	0.876	1.363	1.796	2.201	2.718	3.106	3.497	4.437
12	0.873	1.356	1.782	2.179	2.681	3.055	3.428	4.318
13	0.870	1.350	1.771	2.160	2.650	3.012	3.372	4.221
14	0.868	1.345	1.761	2.145	2.624	2.977	3.326	4.140
15	0.866	1.341	1.753	2.131	2.602	2.947	3.286	4.073
16	0.865	1.337	1.746	2.120	2.583	2.921	3.252	4.015
17	0.863	1.333	1.740	2.110	2.567	2.898	3.222	3.965
18	0.862	1.330	1.734	2.101	2.552	2.878	3.197	3.922
19	0.861	1.328	1.729	2.093	2.539	2.861	3.174	3.883
20	0.860	1.325	1.725	2.086	2.528	2.845	3.153	3.850
21	0.859	1.323	1.721	2.080	2.518	2.831	3.135	3.819
22	0.858	1.321	1.717	2.074	2.508	2.819	3.119	3.792
23	0.858	1.319	1.714	2.069	2.500	2.807	3.104	3.768
24	0.857	1.318	1.711	2.064	2.492	2.797	3.091	3.745
25	0.856	1.316	1.708	2.060	2.485	2.787	3.078	3.725
26	0.856	1.315	1.706	2.056	2.479	2.779	3.067	3.707
27	0.855	1.314	1.703	2.052	2.473	2.771	3.057	3.690
28	0.855	1.313	1.701	2.048	2.467	2.763	3.047	3.674
29	0.854	1.311	1.699	2.045	2.462	2.756	3.038	3.659
30	0.854	1.310	1.697	2.042	2.457	2.750	3.030	3.646
40	0.851	1.303	1.684	2.021	2.423	2.704	2.971	3.551
60	0.848	1.296	1.671	2.000	2.390	2.660	2.915	3.460
120	0.845	1.289	1.658	1.980	2.358	2.617	2.860	3.373
∞	0.842	1.282	1.645	1.960	2.326	2.576	2.807	3.291

B.3 Critical values of the chi-squared distribution

How to read the table

The body of the table contains values of the random variable, not probabilities. The entry in row ν and column χ^2_α is the value $\chi^2_{\alpha,\nu}$ that has area α to its *right* under the chi-squared curve with ν degrees of freedom, so the subscript is $\alpha = 1 - F(\chi^2_{\alpha,\nu})$, one minus the cdf.



For a goodness-of-fit test with k categories, $\nu = k - 1$; for a test of independence, $\nu = (\text{rows} - 1)(\text{columns} - 1)$ (Section 32).

ν	$\chi^2_{0.995}$	$\chi^2_{0.99}$	$\chi^2_{0.98}$	$\chi^2_{0.975}$	$\chi^2_{0.95}$	$\chi^2_{0.90}$	$\chi^2_{0.80}$	$\chi^2_{0.75}$	$\chi^2_{0.70}$	$\chi^2_{0.50}$
1	0.0000393	0.000157	0.000628	0.000982	0.00393	0.0158	0.0642	0.102	0.148	0.455
2	0.0100	0.0201	0.0404	0.0506	0.103	0.211	0.446	0.575	0.713	1.386
3	0.0717	0.115	0.185	0.216	0.352	0.584	1.005	1.213	1.424	2.366
4	0.207	0.297	0.429	0.484	0.711	1.064	1.649	1.923	2.195	3.357
5	0.412	0.554	0.752	0.831	1.145	1.610	2.343	2.675	3.000	4.351
6	0.676	0.872	1.134	1.237	1.635	2.204	3.070	3.455	3.828	5.348
7	0.989	1.239	1.564	1.690	2.167	2.833	3.822	4.255	4.671	6.346
8	1.344	1.646	2.032	2.180	2.733	3.490	4.594	5.071	5.527	7.344
9	1.735	2.088	2.532	2.700	3.325	4.168	5.380	5.899	6.393	8.343
10	2.156	2.558	3.059	3.247	3.940	4.865	6.179	6.737	7.267	9.342
11	2.603	3.053	3.609	3.816	4.575	5.578	6.989	7.584	8.148	10.341
12	3.074	3.571	4.178	4.404	5.226	6.304	7.807	8.438	9.034	11.340
13	3.565	4.107	4.765	5.009	5.892	7.042	8.634	9.299	9.926	12.340
14	4.075	4.660	5.368	5.629	6.571	7.790	9.467	10.165	10.821	13.339
15	4.601	5.229	5.985	6.262	7.261	8.547	10.307	11.037	11.721	14.339
16	5.142	5.812	6.614	6.908	7.962	9.312	11.152	11.912	12.624	15.338
17	5.697	6.408	7.255	7.564	8.672	10.085	12.002	12.792	13.531	16.338
18	6.265	7.015	7.906	8.231	9.390	10.865	12.857	13.675	14.440	17.338
19	6.844	7.633	8.567	8.907	10.117	11.651	13.716	14.562	15.352	18.338
20	7.434	8.260	9.237	9.591	10.851	12.443	14.578	15.452	16.266	19.337
21	8.034	8.897	9.915	10.283	11.591	13.240	15.445	16.344	17.182	20.337
22	8.643	9.542	10.600	10.982	12.338	14.041	16.314	17.240	18.101	21.337
23	9.260	10.196	11.293	11.689	13.091	14.848	17.187	18.137	19.021	22.337
24	9.886	10.856	11.992	12.401	13.848	15.659	18.062	19.037	19.943	23.337
25	10.520	11.524	12.697	13.120	14.611	16.473	18.940	19.939	20.867	24.337
26	11.160	12.198	13.409	13.844	15.379	17.292	19.820	20.843	21.792	25.336
27	11.808	12.879	14.125	14.573	16.151	18.114	20.703	21.749	22.719	26.336
28	12.461	13.565	14.847	15.308	16.928	18.939	21.588	22.657	23.647	27.336
29	13.121	14.256	15.574	16.047	17.708	19.768	22.475	23.567	24.577	28.336
30	13.787	14.953	16.306	16.791	18.493	20.599	23.364	24.478	25.508	29.336
40	20.707	22.164	23.838	24.433	26.509	29.051	32.345	33.660	34.872	39.335
50	27.991	29.707	31.664	32.357	34.764	37.689	41.449	42.942	44.313	49.335
60	35.534	37.485	39.699	40.482	43.188	46.459	50.641	52.294	53.809	59.335

ν	$\chi^2_{0.30}$	$\chi^2_{0.25}$	$\chi^2_{0.20}$	$\chi^2_{0.10}$	$\chi^2_{0.05}$	$\chi^2_{0.025}$	$\chi^2_{0.02}$	$\chi^2_{0.01}$	$\chi^2_{0.005}$	$\chi^2_{0.001}$
1	1.074	1.323	1.642	2.706	3.841	5.024	5.412	6.635	7.879	10.828
2	2.408	2.773	3.219	4.605	5.991	7.378	7.824	9.210	10.597	13.816
3	3.665	4.108	4.642	6.251	7.815	9.348	9.837	11.345	12.838	16.266
4	4.878	5.385	5.989	7.779	9.488	11.143	11.668	13.277	14.860	18.467
5	6.064	6.626	7.289	9.236	11.070	12.833	13.388	15.086	16.750	20.515
6	7.231	7.841	8.558	10.645	12.592	14.449	15.033	16.812	18.548	22.458
7	8.383	9.037	9.803	12.017	14.067	16.013	16.622	18.475	20.278	24.322
8	9.524	10.219	11.030	13.362	15.507	17.535	18.168	20.090	21.955	26.124
9	10.656	11.389	12.242	14.684	16.919	19.023	19.679	21.666	23.589	27.877
10	11.781	12.549	13.442	15.987	18.307	20.483	21.161	23.209	25.188	29.588
11	12.899	13.701	14.631	17.275	19.675	21.920	22.618	24.725	26.757	31.264
12	14.011	14.845	15.812	18.549	21.026	23.337	24.054	26.217	28.300	32.909
13	15.119	15.984	16.985	19.812	22.362	24.736	25.472	27.688	29.819	34.528
14	16.222	17.117	18.151	21.064	23.685	26.119	26.873	29.141	31.319	36.123
15	17.322	18.245	19.311	22.307	24.996	27.488	28.259	30.578	32.801	37.697
16	18.418	19.369	20.465	23.542	26.296	28.845	29.633	32.000	34.267	39.252
17	19.511	20.489	21.615	24.769	27.587	30.191	30.995	33.409	35.718	40.790
18	20.601	21.605	22.760	25.989	28.869	31.526	32.346	34.805	37.156	42.312
19	21.689	22.718	23.900	27.204	30.144	32.852	33.687	36.191	38.582	43.820
20	22.775	23.828	25.038	28.412	31.410	34.170	35.020	37.566	39.997	45.315
21	23.858	24.935	26.171	29.615	32.671	35.479	36.343	38.932	41.401	46.797
22	24.939	26.039	27.301	30.813	33.924	36.781	37.659	40.289	42.796	48.268
23	26.018	27.141	28.429	32.007	35.172	38.076	38.968	41.638	44.181	49.728
24	27.096	28.241	29.553	33.196	36.415	39.364	40.270	42.980	45.559	51.179
25	28.172	29.339	30.675	34.382	37.652	40.646	41.566	44.314	46.928	52.620
26	29.246	30.435	31.795	35.563	38.885	41.923	42.856	45.642	48.290	54.052
27	30.319	31.528	32.912	36.741	40.113	43.195	44.140	46.963	49.645	55.476
28	31.391	32.620	34.027	37.916	41.337	44.461	45.419	48.278	50.993	56.892
29	32.461	33.711	35.139	39.087	42.557	45.722	46.693	49.588	52.336	58.301
30	33.530	34.800	36.250	40.256	43.773	46.979	47.962	50.892	53.672	59.703
40	44.165	45.616	47.269	51.805	55.758	59.342	60.436	63.691	66.766	73.402
50	54.723	56.334	58.164	63.167	67.505	71.420	72.613	76.154	79.490	86.661
60	65.227	66.981	68.972	74.397	79.082	83.298	84.580	88.379	91.952	99.607